goxpyriment

Complete Documentation

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goxpyriment

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  Goxpyriment is a software framework for programming behavioral and cognitive experiments using the Go programming language. It was designed to address some limitations of existing Python-based experiment tools, particularly the runtime environment complexity that frequently complicates deployment across laboratories. Because Go is a compiled language that can natively embed assets (e.g., graphics, audio files, and stimulus lists), Goxpyriment compiles entire experiments into single, self-contained executable binaries with zero runtime dependencies. This drastically simplifies distribution to collaborators and testing computers. The library includes an array of visual stimuli (text, shapes, images, Gabor patches, motion clouds, …) and audio capabilities (WAV playback and tone generation). A lot of effort was devoted to ensure timing reliability. For instance, input events are timestamped by the operating system at hardware-interrupt time, so reaction times are computed by subtracting two OS-level timestamps rather than relying on continuous polling. The framework also supports sending and receiving TTL signals to synchronize with various external equipment. Go's garbage collector can be disabled, greatly reducing the probability of unpredictable pauses that could corrupt stimulus timing. Finally, a set of over forty psychology experiments implemented in Goxpyriment are provided that promote not only learning by humans but also improve the ability of modern AI-assisted coding tools to help program experiments. The framework is released under the Apache License 2.0.

Remark: If you are looking for a simple experiment generator - with fixed stimulus presentation schedule, check out Gostim2 which does not require any coding.

1.1 Documentation

The top navigation bar provides access to various documentation guides.

Everything in one file: goxpyriment-docs.pdf — every page below as a single searchable PDF, one chapter each, with a full table of contents and bookmarks. Use it when you want to search the whole documentation at once rather than a page at a time. The individual PDFs linked beside each guide are the same content, split.

Start here — install and build your first experiment

1. **Install** Go and build the bundled examples. [PDF]
2. **Creating Your Own Experiment** — a step-by-step beginner guide from an empty folder to a shareable executable.
3. **Getting Started** — worked tutorials: trials, data logging, RSVP streams, reaction times. [PDF]

User Manual — core concepts explained in depth: rendering model, timing, input, data, streams, audio, and experimental design. [PDF]

API Reference — complete function and type reference, organized by package. [PDF]

Misc — focused how-to guides: migrating from PsychoPy / Expyriment / Psychtoolbox, checking your computer's timing, video playback, packaging & sharing binaries, giving your experiment real-time priority on Linux, and more, all found under the **Misc** tab in the navigation bar at the top of the page.

In addition, if you clone goxpyriment, you can run `pkgsite` from the root folder to view the source code documentation.

1.2 Gallery of ready-to-run experiments

Browse the gallery of over fifty example experiments and demos (pre-built binaries (ready-to-run apps) are available for Windows, macOS, and Linux).

1.3 Support

- Google group — Forum
- Report bugs at https://github.com/chrplr/goxpyriment/issues

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1.4 Background

The framework is described in a short paper: *Goxpyriment: A Go Framework for Behavioral and Cognitive Experiments*.

Goxpyriment is written in Go and relies on the SDL3 library through the go-sdl3 bindings.

The original inspiration was expyriment.org, a lightweight Python library for cognitive and neuroscientific experiments (Krause & Lindemann, 2014. *Behavior Research Methods*, 46(2), 416–428. https://doi.org/10.3758/s13428-013-0390-6).

1.5 License & citation

Apache License 2.0 — see LICENSE and NOTICE.

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Chapter 2

Goxpyriment installation instructions

There are four ways to use goxpyriment, depending on what you already have and what you want to do. Find your situation below and jump to the matching section — each choice builds on the previous one, so you only need to read as far as your goal requires.

Your situation	What you need	Go to
You want to **try the experiments from the gallery**	Nothing	Pre-built binaries
You were given a **ready-to-run app** (a compiled file) and just want to launch it	Nothing	No installation
You have the **source code** of an experiment (`.go` files) and want to run or build it	Go	Minimal installation
You want to **write your own experiments** or explore the framework	Go + Git + a code editor	Full installation

2.1 No installation is needed to run a goxpyriment app

If you were given a ready-to-run goxpyriment app (a compiled file, such as the ones provided in the compiled examples), you do not need to install anything to use it. Just launch the app by double-clicking its icon in your file folder, or by typing its name in your command line (Terminal or Command Prompt).

! WARNING One potential issue can arise from the antivirus or protection system of your computer if these binaries are unsigned: macOS Gatekeeper and Windows Defender will show security warnings or worse, *misleading messages* such as 'this

program is damaged'. Your antivirus may quarantine the files. Don't let them intimidate you:

- macOS: Right-click the app → **Open**, or run `xattr -dr com.apple.quarantine <AppName>.app` in Terminal. See macOS installation and security for step-by-step instructions.
- Windows: Just click on "More info" then "Run anyway". (Note: These warnings will only pop out the first time you try to execute a given program).

2.2 Minimal installation to execute a goxpyriment source code

Let us consider the case where you have the *source code* of a goxperiment program.

If Go is not already installed on your computer (you can check if Go installed by typing `go version` on a command line. If an error message is displayed, then Go is not installed), download and install it following the instructions at https://go.dev/doc/install

Then there are two possibilities. You have either:

1. A folder containing at least the files `go.mod`, `go.sum` and the source code, say `experiment.go` (it can be any file with the `.go` extension).

   Then, execute the command line:

   ```
   go run experiment.go    # replace experiment by the actual name
   ```

   Or build the app, then run it, by typing:

   ```
   go build experiment.go
   ./experiment
   ```

2. A single `.go` file, say `experiment.go`.

   Then, you need to copy this file into a folder, e.g. `experiment` and create `go.mod` and `go.sum` by typing

   ```
   go mod init experiment
   go mod tidy
   ```

   This will create the missing files and you are in the situation of the previous paragraph: you can run the program with the command `go run .` or `go build .`.

Remark: the very first time you run or build a goxpyriment program, it will take a while as Go needs to download a number of modules from the Internet. Afterwards, running or building an app should be near instantaneous.

2.3 Full installation

If you want to do serious development, in addition to Go, you need to install Git, a code editor and, probably, an AI-coding agent like Claude Code, Gemini cli or Aider. If you are new to this, consult the detailed instructions

Then:

1. Clone goxpyriment Github repository, by opening a shell (e.g. launch `Git Bash` under Windows), and executing the command-line

   ```
   git clone https://github.com/chrplr/goxpyriment.git
   ```

   This will create a folder `goxpyriment` contain the entire source code, examples and documentation of the framework. In the future, you will be able run the `git pull` command within this folder to update to the latest version.

2. Navigate to the `goxpyriment` folder and type the command-line:

   ```
   ./build-all.sh
   ```

   (Note: On Linux/macOS you can instead use `make all`)

   If all goes well, this will compile codes from examples/* and tests/* (Note: The first time, this will take a while because Go needs to download several modules. Once this is done, future compilations will be fast.)

   After this operation, a new `_build` folder contains executable apps for many experiments. You can either run them from the command line, or launch them by clicking on their icon in the folder.

3. If you would like to read the documentation locally — both the API reference (via pkgsite) and this site (via zensical) — see Viewing the documentation locally.

2.3.1 One extra step on a Linux machine used for data collection

Nothing above requires special privileges, and you can write and run experiments without doing any of this. But an experiment that has to present stimuli on time competes with everything else on the machine, and the operating system has no reason to prefer it — so on a machine that will actually collect data, grant your user real-time scheduling before you start trusting its timing.

The procedure is in Setting priority under Linux: create a group, add a file to `/etc/security/limits.d/`, add yourself to the group, then **log out and log back in**.

Check the file afterwards with `grep rtprio /etc/security/limits.d/*.conf`, and check the grant is live with `ulimit -r` after logging back in — it should print your chosen priority, not `0`.

Once that is in place you can verify the machine's timing end to end with Timing Tests, which also lists the other Linux tuning worth doing (disabling the compositor is the single largest improvement).

2.3.2 Program your own experiments

Follow the step-by-step guide in Creating Your Own Experiment, which walks you through writing, running, building, and sharing your own experiment.

For background and reference, see Getting Started, the examples' source codes, and the available functions.

2.3.3 Use an AI coding agent

If you launch `claude` (Claude Code) from the `goxpyriment` folder, the many `CLAUDE.md` files will help the AI coding agent to create experiments for you. You can describe the experimental paradigm in plain language (see the `description.md` files in `examples` subfolders) and ask claude code to program it using the goxpyriment framework.

Chapter 3

Creating Your Own Experiment (a beginner's guide)

This guide is for people who have **never written Go before**. It walks you through making a brand-new experiment from scratch: creating a folder, writing a `main.go` file, running it, building a standalone program, embedding your own images/sounds/CSV files, and finally producing programs you can hand to colleagues on Windows, macOS, or Linux.

You do **not** need to learn the whole Go language. For most experiments you only copy an example, change a few lines, and run it.

> **Tip:** The easiest way to start is to copy an existing example from the `examples/` folder that resembles your paradigm, then change a few lines. That is far easier than writing from a blank page.
>
> * **Vibe-coding:** You can also let an AI coding agent (Claude, Gemini, etc.) write the experiment for you. Open the agent inside the `goxpyriment` folder and describe your experiment (stimuli, design, etc.) in plain language — asking it to add a new experiment to the `examples` folder leads it to read the existing examples for context. Recommendation: save your prompt in a `description.md` file.

3.1 Before you start

Make sure goxpyriment is installed first — follow Installation, which walks you through installing Git and Go, cloning the repository, and building the examples. (If you are new to setting up Go, that page links to a more detailed development-environment guide.)

Once that is done, open a terminal and confirm the Go compiler is available:

```
go version
```

If you see something like `go version go1.25 ...`, you are ready.

> **What is a terminal?** A text window where you type commands. On Windows, use **Git Bash** (installed with Git). On macOS use **Terminal**, on Linux any terminal app.

3.2 Step 1 — Create a folder for your experiment

Each experiment lives in its own folder. Pick a short name with no spaces:

```
mkdir my_experiment
cd my_experiment
```

Everything you do from now on happens inside this folder.

3.3 Step 2 — Write `main.go`

Create a file called `main.go` in that folder with the following content. This is the smallest complete experiment: it shows a message and waits for a key press.

```go
package main

import (
    "github.com/chrplr/goxpyriment/control"
    "github.com/chrplr/goxpyriment/stimuli"
)

func main() {
    // Title, background color, foreground color, default font size.
    exp := control.NewExperimentFromFlags("My Experiment", control.Black,
 ↪  control.White, 32)
    defer exp.End()

    hello := stimuli.NewTextBox("Hello, World!\n\nPress any key to quit.",
 ↪  600, control.FPoint{}, control.White)

    exp.Run(func() error {
        exp.Show(hello)
        exp.Keyboard.Wait()
        return control.EndLoop // leaving the loop ends the experiment
    })
}
```

You don’t need to understand every line yet. The pattern is always the same:

- `control.NewExperimentFromFlags(...)` sets up the screen, audio, keyboard, and data file.
- `defer exp.End()` makes sure everything is closed cleanly when the program stops.
- `exp.Run(func() error { ... })` is your experiment body. Return `control.EndLoop` to finish.

Remember: you rarely start from a blank page like this. The copy-an-example and vibe-coding tips at the top of this page are the easiest way in — this section just shows what the generated code looks like.

3.4 Step 3 — Tell Go about its dependencies

Your code uses the `goxpyriment` library, so Go needs to know how to fetch it. Run these two commands **once**, inside your folder:

```
go mod init my_experiment   # creates go.mod, the project's identity card
go mod tidy                 # downloads goxpyriment and everything it needs
```

`go mod tidy` reads your `import` lines, downloads the missing packages, and records exact versions in `go.mod` / `go.sum`. Run it again any time you add a new `import` or when Go complains it can't find a package.

You should now have these files:

```
my_experiment/
├── go.mod
├── go.sum
└── main.go
```

3.5 Step 4 — Run it while you work

To compile *and* run in one step:

```
go run .
```

The `.` means "the program in this folder". A window opens, your experiment runs, and the window closes when it ends.

This is your everyday loop: **edit `main.go` → `go run .` → look → edit again.** You don't need to "build" anything during development.

Useful command-line flags that every experiment understands automatically:

```
go run . -w          # windowed mode (1024×768) instead of fullscreen —
 ↪  great for development
go run . -s 007      # set the subject/participant ID to 007
```

```
go run . -d 1        # use monitor #1 (for a second screen)
```

When the experiment ends, two files are written to `~/goxpy_data/`: a `.csv` with your data and a `-info.txt` with session metadata.

3.6 Step 5 — Build a standalone program

When you are happy with the experiment, turn it into a single executable file that runs **without Go installed**:

```
go build .
```

This produces a file named after your folder (`my_experiment` on macOS/Linux, `my_experiment.exe` on Windows). It is fully self-contained — SDL3 and all assets are baked in — so you can copy that one file to another machine and run it:

```
./my_experiment          # macOS / Linux
my_experiment.exe        # Windows
```

You can rename the output or put it somewhere specific:

```
go build -o builds/my_experiment .
```

3.7 Step 6 — A more advanced example

Here is a script implementing a parity decision task (inspired from https://docs.expyriment.org/0.6.3/Examples.html)

```
package main

import (
    "slices"
    "strconv"

    "github.com/chrplr/goxpyriment/control"
    "github.com/chrplr/goxpyriment/design"
    "github.com/chrplr/goxpyriment/stimuli"
)

func main() {
    exp := control.NewExperimentFromFlags("Parity Decision",
        control.Black, control.White, 32)
    defer exp.End()

    Instructions := "When you'll see a number, your task to decide, " +
```

```
        "as quickly as possible, whether it is even or odd.\n\n" +
        "if it is even, press 'F'\n\nif it is odd, press 'J'"

    Targets := []int{0, 1, 2, 3, 4, 5, 6, 7, 8, 9}
    NTrialsPerTarget := 2
    trials := slices.Repeat(Targets, NTrialsPerTarget)
    design.ShuffleList(trials)

    EvenResponse := control.K_F
    OddResponse := control.K_J

    cue := stimuli.NewFixCross(25, 2, control.DefaultTextColor)

    // creates a map number -> image
    Image := make(map[int]*stimuli.TextLine)
    for _, num := range Targets {
        Image[num] = stimuli.NewTextLine(strconv.Itoa(num),
            0, 0, control.DefaultTextColor)
    }

    exp.AddDataVariableNames([]string{"number", "key", "rt",
        "correct"})

    exp.Run(func() error {
        exp.ShowInstructions(Instructions)

        for _, t := range trials {
            exp.Blank(1000)
            exp.ShowTimed(cue, 500)
            key, rt, _ := exp.ShowAndGetRT(Image[t],
                []control.Keycode{EvenResponse, OddResponse},
                -1)
            correct := (t%2 == 0) == (key == EvenResponse)
            exp.Data.Add(t, key, rt, correct)
            if !correct {
                exp.Audio.PlayBuzzer()
            }
            exp.Wait(500)
        }

        return control.EndLoop
    })

    exp.ShowEndMessage("Experiment complete. Thank you!\n\n" +
        "Press any key to exit.")
}
```

Save this script in a folder ‘parity-decision’, open a Terminal, cd to this folder and type:

```
go mod init parity-decision
go mod tidy
go run .
```

This script may look a bit intinmidating. This is because especially the lines:

```
// creates a map number -> image
Image := make(map[int]*stimuli.TextLine)
for _, num := range Targets {
    Image[num] = stimuli.NewTextLine(strconv.Itoa(num),
        0, 0, control.DefaultTextColor)
}
```

Their purpose is to prepare in advanced the images (stimuli.Textline) that will displayed on the screen during the main loop: One image is created for each number and saved in a map associating a number to an image (Go maps are similar to Python's dict).

Note: it is a generally a good habit to prepare the stimuli in advance, but we could have written the script without creating the map, and calling NewTextLine in the main loop, like this:

```
package main

import (
    "slices"
    "strconv"

    "github.com/chrplr/goxpyriment/control"
    "github.com/chrplr/goxpyriment/design"
    "github.com/chrplr/goxpyriment/stimuli"
)

func main() {
    exp := control.NewExperimentFromFlags("Parity Decision",
        control.Black, control.White, 32)
    defer exp.End()

    Instructions := "When you'll see a number, your task to decide, " +
        "as quickly as possible, whether it is even or odd.\n\n" +
        "if it is even, press 'F'\n\nif it is odd, press 'J'"

    Targets := []int{0, 1, 2, 3, 4, 5, 6, 7, 8, 9}
    NTrialsPerTarget := 2
    trials := slices.Repeat(Targets, NTrialsPerTarget)
    design.ShuffleList(trials)

    EvenResponse := control.K_F
    OddResponse := control.K_J

    cue := stimuli.NewFixCross(25, 2, control.DefaultTextColor)
```

```
    exp.AddDataVariableNames([]string{"number", "key", "rt",
        "correct"})

    exp.Run(func() error {
        exp.ShowInstructions(Instructions)

        for _, t := range trials {
            exp.Blank(1000)
            exp.ShowTimed(cue, 500)
            key, rt, _ :=
             ↪  exp.ShowAndGetRT(stimuli.NewTextLine(strconv.Itoa(t), 0, 0,
             ↪  control.White),
                []control.Keycode{EvenResponse, OddResponse},
                -1)
            correct := (t%2 == 0) == (key == EvenResponse)
            exp.Data.Add(t, key, rt, correct)
            if !correct {
                exp.Audio.PlayBuzzer()
            }
            exp.Wait(500)
        }

        return control.EndLoop
    })

    exp.ShowEndMessage("Experiment complete. Thank you!\n\n" +
        "Press any key to exit.")
}
```

The function `stimuli.NewTextLine()` converts a string into displayable image. The `stimuli` package provides many other functions, such as `NewPicture()` which loadds an imapge, `NewSound()` which loads a sound file, etc. All stimuli can be presented with the method `Present()` (in the above example, the function `ShowAndGetRT()` called `Present()` behind the scene)

If you have installed pkgsite and launched it from the `goxperiment` folder, you can check the list of available functions in `stimuli` by pointing your browser to http://localhost:8080/github.com/chrplr/goxpyriment@v0.0.0/stimuli

---

3.8 Step 7 — Embed your stimuli (images, sounds, CSV)

A built program is just one file. If your experiment loads images, sounds, or a list of trials from a CSV, those files would normally be missing on someone else's computer (unless the user copied them at the expected location). The solution is to **embed** them inside the program with Go's `//go:embed` directive. Embedded files travel inside the executable — nothing to copy alongside it.

Put your asset files in the folder (or a subfolder), then declare them at the top of `main.go`.

3.8.1 A single CSV file of trials

```
import (
    "bytes"
    "encoding/csv"
    _ "embed" // needed for //go:embed
)

//go:embed trials.csv
var trialsCSV []byte

func loadTrials() [][]string {
    r := csv.NewReader(bytes.NewReader(trialsCSV))
    rows, _ := r.ReadAll()
    return rows
}
```

3.8.2 A single image or sound

```
//go:embed apple.png
var appleImg []byte

//go:embed beep.wav
var beepSnd []byte

// ... later, inside your experiment:
picture := stimuli.NewPictureFromMemory(appleImg, 0, 0) // x=0, y=0 →
 ↪  centered
sound := stimuli.NewSoundFromMemory(beepSnd)
```

3.8.3 A whole folder of files

When you have many files, embed the folder as a virtual filesystem:

```
import "embed"

//go:embed images
var imagesFS embed.FS

// ... read one file by name:
data, err := imagesFS.ReadFile("images/apple.png")
if err != nil {
    exp.Fatal("could not load image: %v", err)
}
```

```
pic := stimuli.NewPictureFromMemory(data, 0, 0)
```

Rules to remember:

- The `//go:embed` line must sit **directly above** the variable, with no blank line.
- Paths are **relative to the `.go` file** and use forward slashes `/` on every OS.
- Add `import "embed"` (for `embed.FS`) or the blank import `_ "embed"` (for a `[]byte` / `string`).
- After adding embeds, run `go run .` again to recompile.

  When *not* to embed: very large files such as long videos. Embedding a 500 MB video makes a 500 MB program and slows builds. For those, ship the file next to the program instead — see Packaging Your Experiment.

For the full reference on embedding — folders as virtual filesystems, trial-list CSVs, binary size, and cross-compilation of embedded assets — see User Manual §13, *Embedding Stimuli and Trial Lists*.

3.9 Step 8 — Build for other operating systems (to share with colleagues)

Here is where Go shines. Because `goxpyriment` is pure Go (no C compiler needed), you can build a Windows program **from a Mac**, a Mac program **from Linux**, and so on — just by setting two variables: `GOOS` (operating system) and `GOARCH` (processor type).

```
# Windows (Intel/AMD 64-bit) → produces my_experiment.exe
GOOS=windows GOARCH=amd64 go build -o my_experiment-windows.exe .

# macOS (Apple Silicon: M1/M2/M3/M4)
GOOS=darwin  GOARCH=arm64 go build -o my_experiment-macos-arm .

# macOS (older Intel Macs)
GOOS=darwin  GOARCH=amd64 go build -o my_experiment-macos-intel .

# Linux (Intel/AMD 64-bit)
GOOS=linux   GOARCH=amd64 go build -o my_experiment-linux .
```

Common values:

Target machine	`GOOS`	`GOARCH`
Windows PC	`windows`	`amd64`
Mac with Apple Silicon	`darwin`	`arm64`
Older Intel Mac	`darwin`	`amd64`
Linux PC	`linux`	`amd64`
Raspberry Pi (64-bit)	`linux`	`arm64`

A small script that builds all of them at once (save as `cross-build.sh`):

```
#!/bin/bash
set -e
mkdir -p dist
GOOS=windows GOARCH=amd64 go build -o dist/my_experiment-windows.exe .
GOOS=darwin  GOARCH=arm64 go build -o dist/my_experiment-macos-arm .
GOOS=darwin  GOARCH=amd64 go build -o dist/my_experiment-macos-intel .
GOOS=linux   GOARCH=amd64 go build -o dist/my_experiment-linux .
echo "Done — see the dist/ folder."
```

Then:

```
bash cross-build.sh
```

Each file in `dist/` is a complete, standalone program. Email it, drop it on a USB stick, or share it — your colleague just double-clicks (or runs it from a terminal). No Go, no Python, no SDL install required.

> **On macOS**, a binary downloaded from the internet may be blocked by Gatekeeper. The recipient can allow it once via **System Settings → Privacy & Security**, or run `xattr -d com.apple.quarantine my_experiment-macos-arm` in a terminal.

For a more polished distribution — `.deb`/`.rpm` packages, Windows installers, macOS `.app` bundles, and automated GitHub releases — see Packaging Your Experiment.

3.10 Quick reference

Task	Command
Create the project	`go mod init my_experiment && go mod tidy`
Run while developing	`go run .`
Run in a window	`go run . -w`
Add a new imported package	`go mod tidy`
Build a standalone program	`go build .`
Build for Windows from another OS	`GOOS=windows GOARCH=amd64 go build -o app.exe .`
Embed a file	`//go:embed file.png` above a variable

3.11 Where to go next

- Getting Started — worked tutorials (trials, data logging, RSVP, reaction times).
- Gallery of Examples — dozens of complete experiments to copy from.

- User Manual — the concepts behind the framework.
- API Reference — every function and type.
- Packaging Your Experiment — installers and automated releases.

Happy experimenting!

Chapter 4

Getting Started with goxpyriment

Welcome! If you are a psychologist or neuroscientist used to building experiments in Python (with **Expyriment** or **PsychoPy**), you are in the right place.

`goxpyriment` is a alternative framework to build experiments using the **Go** programming language, offering significant advantages in timing precision and ease of sharing your work.

This document provides a few examples for you to get acquainted with the framework (if you want to run the examples, you will need goxpyriment installed and a working Go toolchain. See Installation for instructions).

4.1 Tutorial 1: Your First Trial

A classic sequence: **Fixation Cross** (500 ms) → **Target Circle** → **Wait for Spacebar**.

```
package main

import (
    "log"

    . "github.com/chrplr/goxpyriment/control"
    "github.com/chrplr/goxpyriment/stimuli"
)

func main() {
    exp := NewExperimentFromFlags("SimpleTrial", Black, White, 32)
    defer exp.End()

    fixation := stimuli.NewFixCross(20, 3, control.White)
    target   := stimuli.NewCircle(50, control.White)

    exp.ShowInstructions("Press SPACE when you see the circle.")
```

```
    exp.ShowTimed(fixation, 500)  // hold fixation for 500 ms

    exp.Show(target)
    exp.Keyboard.WaitKey(K_SPACE)
}
```

The following example goesa bit furhter: it runs a loops where a white square and a short tone are presented every second.

```
package main

import (
    . "github.com/chrplr/goxpyriment/control"
    "github.com/chrplr/goxpyriment/stimuli"
)

func main() {
    exp := NewExperimentFromFlags("Test", Black, White, 32)
    defer exp.End()

    w, h, _ := exp.Screen.Size()
    rect := stimuli.NewRectangle(0, 0, float32(w)/2.0, float32(h)/2.0,
 ↪  White)

    sound := stimuli.NewTone(440, 120, 0.5)
    sound.PreloadDevice(exp.AudioDevice)    // generates the sound

    instr := stimuli.NewTextBox(
        "A white rectangle will be displayed periodicallay\n\nPress any key
         ↪  to start\n\n'Esc' to interrupt",
        600, FPoint{}, White)

    exp.AddDataVariableNames([]string{"onset", "duration"})

    exp.Run(func() error {
        exp.Show(instr)
        exp.Keyboard.Wait()

        for {
            onsetNS, _ := exp.ShowTS(rect)
            sound.Play()
            exp.Wait(200)
            exp.Screen.Clear()
            offsetNS, _ := exp.Screen.FlipTS()
            duration_ms := (offsetNS - onsetNS) / 1_000_000
            exp.Wait(800)
```

```
            exp.Data.Add(onsetNS/1_000_000, duration_ms)
        }
    })

}
```

4.2 Tutorial 2: Blocks, Trials, and Data Logging

For real research you need multiple trials and a CSV result file.

```
package main

import (
    "fmt"
    "log"

    "github.com/chrplr/goxpyriment/control"
    "github.com/chrplr/goxpyriment/design"
    "github.com/chrplr/goxpyriment/stimuli"
)

func main() {
    exp := control.NewExperimentFromFlags("ParityTask", control.Black,
↪   control.White, 32)
    defer exp.End()

    exp.AddDataVariableNames([]string{"number", "is_even", "rt_ms",
↪   "correct"})

    err := exp.Run(func() error {
        exp.ShowInstructions(
            "Is the number EVEN or ODD?\n\nPress F for Even, J for Odd.",
        )

        numbers := []int{1, 2, 3, 4, 5, 6, 7, 8}
        trials  := design.MakeMultipliedShuffledList(numbers, 4) // 32
↪   trials, randomized

        for _, n := range trials {
            exp.Blank(1000)

            stim := stimuli.NewTextLine(fmt.Sprint(n), 0, 0, control.White)
            exp.Show(stim)

            // WaitKeysRT returns the key pressed, the reaction time in ms,
             ↪  and any error.
```

```
            key, rt, _ := exp.Keyboard.WaitKeysRT(
                []control.Keycode{control.K_F, control.K_J}, -1,
            )

            isEven := (n % 2 == 0)
            correct := (isEven && key == control.K_F) || (!isEven && key ==
↪   control.K_J)

            // Subject ID and timestamp are prepended automatically.
            exp.Data.Add(n, isEven, rt, correct)
        }
        return control.EndLoop
    })
    if err != nil && !control.IsEndLoop(err) {
        log.Fatalf("experiment error: %v", err)
    }
}
```

`exp.Run()` handles the detection of the ESC keypress to interrupt the experiment. Two files are written to `~/goxpy_data/` automatically when the experiment ends:

- `<expName>_sub-<NNN>_date-<YYYYMMDD>-<HHMMSS>.csv` — pure CSV data rows, directly importable by Excel, R, or pandas.
- `<expName>_sub-<NNN>_date-<YYYYMMDD>-<HHMMSS>-info.txt` — session metadata (start/end time, hostname, OS, framework version, display and audio configuration, participant info).

4.3 Tutorial 3: Rapid Serial Visual Presentation (RSVP Streams)

Many paradigms — RSVP, Attentional Blink, priming — need stimuli flashed at high speed with precise timing. `goxpyriment` has dedicated stream functions that:

- Disable GC during the stream (no garbage-collection pauses).
- Lock every onset and offset to a VSYNC boundary.
- Record a `TimingLog` (predicted vs. actual onset) for every stimulus.
- Capture any key or mouse event that occurs during the stream.

4.3.1 Text stream

The simplest entry point is `stimuli.PresentStreamOfText`:

```
package main

import (
    "fmt"
```

```
    "log"
    "time"

    "github.com/chrplr/goxpyriment/control"
    "github.com/chrplr/goxpyriment/stimuli"
)

func main() {
    exp := control.NewExperimentFromFlags("RSVP Demo", control.Black,
↪   control.White, 36)
    defer exp.End()

    exp.AddDataVariableNames([]string{"target_pos", "detected", "rt_ms"})

    // A stream of words; "TIGER" is the target at position 3 (0-indexed).
    words     := []string{"CHAIR", "RIVER", "LAMP", "TIGER", "CLOCK",
↪   "STONE", "BREAD", "HOUSE"}
    targetPos := 3

    err := exp.Run(func() error {
        exp.ShowInstructions(
            "Words will flash rapidly on screen.\n\n" +
            "Press SPACE as quickly as possible when you see TIGER.\n\n" +
            "Press any key to start.",
        )

        // 200 ms on, 50 ms off per word → ~4 words per second
        on  := 200 * time.Millisecond
        off :=  50 * time.Millisecond

        events, logs, err := stimuli.PresentStreamOfText(
            exp.Screen, words, on, off,
            0, 0,           // centred on screen
            control.White,
        )
        if err != nil {
            return err
        }

        // Did the participant press SPACE during the stream?
        ev, detected := stimuli.FirstKeyPress(events, control.K_SPACE)
        rtMs := ev.Timestamp.Milliseconds()

       // Optional: inspect timing quality. Read the authoritative SDL-clock
        // fields (OnsetNS/OffsetNS), not the Go-clock
         ↪  ActualOnset/ActualOffset
        // diagnostics. Achieved on-screen duration vs the target is the
         ↪  jitter.
```

```
        for i, l := range logs {
            shownMs := int64(l.OffsetNS-l.OnsetNS) / 1_000_000
            jitterMs := shownMs - l.TargetOn.Milliseconds()
            fmt.Printf("word %d: target %d ms  shown %d ms  jitter %d ms\n",
                i, l.TargetOn.Milliseconds(), shownMs, jitterMs)
        }

        exp.Data.Add(targetPos, detected, rtMs)
        return control.EndLoop
    })
    if err != nil && !control.IsEndLoop(err) {
        log.Fatalf("experiment error: %v", err)
    }
}
```

Similar functions exist to present streams of images or sounds:

- Images:

```
stims := []stimuli.VisualStimulus{pic1, pic2, fixation, pic3}
on  := 100 * time.Millisecond
off :=   0 * time.Millisecond

elements := stimuli.MakeRegularVisualStream(stims, on, off)
events, logs, err := stimuli.PresentStreamOfImages(exp.Screen,
 ↪  elements, 0, 0)
```

- Sounds

```
tones     := []stimuli.AudioPlayable{tone440, tone880, tone440}
onsets    := []int{0, 500, 1000}   // ms from stream start
durations := []int{200, 200, 200}

elements, _ := stimuli.MakeSoundStream(tones, onsets, durations)
events, logs, err := stimuli.PlayStreamOfSounds(elements)
```

All stream functions return `([]UserEvent, []TimingLog, error)` to analyse timing quality and log participant responses.

4.4 Tutorial 4: Hardware-Precision RT with Event Timestamps

`WaitKeysRT` measures reaction time from the moment the function is called. That works well for a single-stimulus trial, but breaks down when several stimuli appear in sequence and you need RT from a specific onset.

Consider a **masked priming** paradigm: a prime word appears briefly, then a tar-

get appears 500 ms later, and you want RT measured from the prime onset — not from when `GetKeyEventTS` was called. With the standard approach you would need to record a timestamp before the prime and do arithmetic afterward. The event-timestamp API handles this directly.

SDL3 stamps every keyboard event with a hardware-interrupt time (`KeyboardEvent.Timestamp`, nanoseconds). `exp.ShowTS(stim)` returns the SDL nanosecond time captured immediately after the VSYNC flip. Because both values are on the same clock, their difference is hardware-precision RT — no arithmetic needed:

```
package main

import (
    "fmt"
    "log"

    "github.com/chrplr/goxpyriment/control"
    "github.com/chrplr/goxpyriment/stimuli"
)

func main() {
    exp := control.NewExperimentFromFlags("PrimingRT", control.Black,
↪   control.White, 36)
    defer exp.End()

    exp.AddDataVariableNames([]string{"prime", "target", "key", "rt_prime_
↪   ns", "rt_target_ns"})

    primes  := []string{"DOCTOR", "NURSE", "BREAD", "TABLE"}
    targets := []string{"NURSE", "DOCTOR", "TABLE", "BREAD"}  // related /
↪   unrelated pairs
    responseKeys := []control.Keycode{control.K_F, control.K_J}

    err := exp.Run(func() error {
        exp.ShowInstructions(
            "A word will flash, then a second word will appear.\n\n" +
            "F = Living thing   J = Non-living thing\n\n" +
            "Respond to the SECOND word as quickly as possible.\n\n" +
            "Press SPACE to start.",
        )

        for i := range primes {
            exp.Blank(500)   // inter-trial interval

            // 1. Show prime — record its VSYNC flip timestamp
            prime := stimuli.NewTextLine(primes[i], 0, 0, control.Gray)
            primeOnset, _ := exp.ShowTS(prime)
            prime.Unload()
```

```
            exp.Wait(500)  // prime–target SOA

            // 2. Show target — record its VSYNC flip timestamp
            target := stimuli.NewTextLine(targets[i], 0, 0, control.White)
            targetOnset, _ := exp.ShowTS(target)
            target.Unload()

            // 3. Wait for response — get hardware event timestamp
           key, eventTS, _ := exp.Keyboard.GetKeyEventTS(responseKeys, 3000)

            // 4. Compute RTs from each stimulus onset
            rtFromPrime  := int64(eventTS - primeOnset)   // nanoseconds
            rtFromTarget := int64(eventTS - targetOnset)  // nanoseconds

            exp.Data.Add(primes[i], targets[i], key, rtFromPrime,
↪   rtFromTarget)
            fmt.Printf("prime RT: %.1f ms   target RT: %.1f ms\n",
                float64(rtFromPrime)/1e6, float64(rtFromTarget)/1e6)
        }
        return control.EndLoop
    })
    if err != nil && !control.IsEndLoop(err) {
        log.Fatalf("experiment error: %v", err)
    }
}
```

Key observations:

- `exp.ShowTS(stim)` does the same thing as `exp.Show(stim)` — clear → draw → flip — and additionally returns the nanosecond timestamp of the flip.
- `GetKeyEventTS` returns the SDL3 event timestamp (not a polling delta), so subtracting any previously recorded `ShowTS` onset gives a physically meaningful RT.
- Both timestamps are in SDL nanoseconds (divide by `1e6` for milliseconds). Storing raw nanoseconds in the data file and converting offline is the recommended practice.
- `GetPressEventTS` provides the same capability for mouse responses.

4.5 Tutorial 5 — Test AI Coding

1. Take the PDF of a scientific paper about a psychology experiment and ask Gemini or Claude to read it and generate a detailed description (stimuli, design, timings) to give to n ai coder to replicate it. Save the answer in a `description.md` file.
2. Open Gemini or Claude (command-line tool), and ask it to implement the experiment detailed in `description.md`, using the goxpyriment library (provide the path of your local clone of the repository) and taking inspiration from the examples provided in the `examples/` folder.

4.6 Next Steps

- Browse the `examples/` folder for complete, documented paradigms (Stroop, Simon, Posner, Sperling, QUEST threshold estimation, and more).
- Read the User Manual for a deeper explanation of every concept, or the API Reference for complete function signatures.
- Report bugs and suggestions at https://github.com/chrplr/goxpyriment/issues.
- Happy experimenting!

Chapter 5

Goxpyriment Example Experiments

5.1 Psychological Experiments

Attention-Posner-Task Arrow cue directs covert attention; measure cost/benefit on reaction time to a peripheral target*Posner (1980)*	**Attentional-Blink**RSVP stream; participant detects two targets embedded in a stream of distractors — the second target is often missed within ~500 ms of the first*Raymond et al. (1992)*	**Change-Blindness**Flicker paradigm: alternating original and modified scenes separated by blanks; participant detects what changed*Rensink et al. (1997)*
Classification-Posner-Mitchell Classify letter pairs at three levels (physical, name, rule identity); RT increases with depth of processing required*Posner & Mitchell (1967)*	**Contrast-Detection-QUEST**2-IFC adaptive staircase estimating the contrast detection threshold for a Gabor patch; converges on the 82 % correct point*Watson & Pelli (1983)*	**Finger-Tapping**Patterned finger-tapping: memorise a key sequence then reproduce it 6 times consecutively as fast as possible; only error-free runs recorded*Povel & Collard (1982)*
Finger-Tracking Number-to-position task: drag the finger (or mouse, button held) from a bottom box up to an unmarked 0-40 number line, aiming at a target number; the full trajectory, endpoint, bias, and movement time are recorded*Dotan & Dehaene (2013)*	**Go-NoGo**Stop-signal task: respond to letters on go-trials; withhold response when a stop-signal tone is played at variable delays*Logan et al. (1984)*	**Hemispheric-differences-word-processing**Lateralised recognition memory: words studied in LVF or RVF, tested centrally with old/new judgements*Federmeier & Benjamin (2005)*

Language-Localizer Bilingual auditory language localizer for fMRI; spoken sentences in French, Mandarin, and Wolof are played on a fixed onset schedule started by the scanner trigger, while the participant fixates and listens*Lin, E. (2016), after Pallier et al. (2011) auditory language localizer*	**Language-Localizer-French-audio**auditory language localizer; spoken sentences in French, and their temporally reversed versions, are played on a fixed onset schedule started by the scanner trigger, while the participant fixates and listens*Pallier C. (unpublished)*	**Language-Localizer-Reading-English**EvLab visual language localizer for fMRI: the participant reads 12-word sentences and matched nonword sequences presented word by word (RSVP, 450 ms per word, no gap) on a fixed onset schedule started by the scanner ('t') trigger. 48 trials in 16 blocks of 3 (8 sentence, 8 nonword), separated by five 14 s fixation periods; a disc at the end of each trial prompts a button press that keeps the participant alert. The sentences > nonwords contrast localizes the high-level language network. Port of the Psychtoolbox evlab_ langloc_2conds.m, using its stimulus tables unchanged (2 runs x 5 sets, selected with -run and -set).*Fedorenko, E., Hsieh, P.-J., Nieto-Castañón, A., Whitfield-Gabrieli, S., & Kanwisher, N. (2010). New method for fMRI investigations of language: defining ROIs functionally in individual subjects. Journal of Neurophysiology, 104(2), 1177-1194. Psychtoolbox implementation by T. Scott (MIT, 2013).*

Letter-size-illusion Compare heights of letters vs. mirror/pseudo-letters; replicates the letter height superiority illusion (two experiments)*New et al. (2015)*	**lexical_decision**Decide whether a letter string is a word or a non-word (F / J keys); stimuli loaded from a CSV file	**LoT-geometry**Comprehension of geometric primitives and rules; reproduces Amalric et al. (2017)*Amalric et al. (2017)*
Magnitude-Estimation-Luminosity Stevens' magnitude estimation of luminance: assign a number to perceived brightness of grey disks*Stevens (1957)*	**MEG-localizer**Multimodal MEG localizer — table-driven RSVP of faces, houses, words, consonant strings, equations, retinotopic wedges/rings/disks, and natural sounds	**Memory-for-binary-sequences**Memory and reproduction of auditory binary sequences of varying complexity*Planton et al. (2021)*
Memory-Iconic-Sperling Partial-report procedure measuring capacity and duration of iconic (visual sensory) memory*Sperling (1960)*	**Memory-Scanning**Hold a set of digits in memory; decide whether a probe was in the set — RT scales with set size*Sternberg (1966)*	**Memory_span**Adaptive staircase measuring immediate serial recall span for digits, letters, or words
Mental-Logic-Card-Game Mental logic and inference task using a card-game paradigm	**Mental-Rotation-2D**Decide whether two 3-D figures are identical or mirror images; RT increases linearly with angular disparity*Shepard & Metzler (1971)*	**Mental-Rotation-3D**Decide whether two 3D figures (procedurally generated assemblies of cubes) are identical or mirror images; RT increases linearly with angular disparity.*Shepard & Metzler (1971)*
Mouse-tracking Mouse-trajectory paradigm (written-word adaptation of Spivey et al.): click the labelled box matching a written target word that appears 500 ms after onset, while the cursor path is sampled at ~36 Hz, revealing continuous attraction toward phonological cohort competitors*Spivey et al. (2005)*	**Multiple-Object-Tracking**Track a subset of identical moving targets among distractors; evidence for a parallel tracking mechanism*Pylyshyn & Storm (1988)*	**Naming-Stroop**Vocal-response Stroop task — a colour word is named aloud while a microphone voice key measures RT from word onset; per-trial WAV files are saved for offline verification*Stroop (1935)*

Number-Change-Detection Two concurrent dot-array streams (5 vs 20 dots) test infants' preference for numerosity change; experimenter codes looking direction in real time*Decarli, Piazza & Izard (2023)*	**Number-Comparison**Compare numerical magnitudes of digits and dot patterns; stimulus group (digits / regular / irregular / random) selected via GetParticipantInfo UI*Buckley & Gillman (1974)*	**Number-Double-Digits-Comparison**Compare two-digit numbers against a fixed standard (55 or 65); two experiments with different response-key mappings*Dehaene et al. (1990)*
parity_decision Classify single digits (0–9) as even or odd (F / J keys)	**Perception-of-Temporal-Patterns**Reproduction of isochronous and non-isochronous rhythmic patterns; tests internal clock induction and coding complexity*Povel & Essens (1985)*	**picture_naming**Vocal naming-latency task: a picture is named aloud while a microphone voice key measures RT from image onset; per-trial WAV files are saved for offline verification
Posner-ANT Attention Network Task (vertical variant): flanker arrows above/below fixation measure alerting, orienting, and executive attention networks*Fan et al. (2009)*	**Psychological-Refractory-Period**Two tasks presented in rapid succession; the second response is delayed when the SOA is short*Welford (1952)*	**Retinotopy**HCP retinotopic mapping paradigm (ported from Python); flickering wedge/ring/bar stimuli for visual cortex mapping; run type selected via GetParticipantInfo UI

RSVP-ImagesRSVP oddball-detection task replicating Hebart et al. (2023, THINGS-data). Object images are shown in rapid serial visual presentation on a mid-gray background with a central black fixation dot; the participant presses SPACE whenever a pixelated oddball appears. Timings are read from a design CSV (onset,duration,trial_type,file_path); companion tool cmd/gen-design produces the MEG (SOA 1500±200 ms) and fMRI (SOA 4.5 s) designs. Oddballs are pixelated at runtime; results add a reaction_time column.*Hebart, M. N. et al. (2023). THINGS-data, a multimodal collection of large-scale datasets for investigating object representations in human brain and behavior. eLife 12:e82580.*	**RSVP-Images-OneBack**One-back RSVP picture task: images from images/ flash by in random order (200 ms on, 100 ms blank); ~10% are immediate repeats and the participant presses SPACE on repeats, with a buzzer on misses. VSYNC-locked, with per-image onset/offset, response, and hit/false-alarm logging	**RSVP-MathLang**MathLang RSVP paradigm for fMRI: sentences are presented word by word (300 ms per word, no blank between words) on a fixed onset schedule started by the scanner ('t') trigger. Nine sentence types (pseudoword lists, word lists, meaningless sentences, factual/contextual knowledge, theory of mind, arithmetic principles, calculation, geometry) in true and false variants contrast the language and mathematical reasoning networks. The participant judges each statement's veracity (1 = vrai, 2 = faux); percent correct and median RT are reported per condition at the end of the bloc. Onsets are read from the per-subject table stim_lists/sub-NNN_task-MathLang.csv; each of the 5 blocs is a separate run, selected with -b.*Amalric, M. & Dehaene, S. (2016). Origins of the brain networks for advanced mathematics in expert mathematicians. PNAS 113(18), 4909-4917.*

RSVP-multimodal Multimodal fMRI localizer driven by a stimulus table: rapid serial visual presentation of words, flashing checkerboards, and spoken sentences are presented on a fixed absolute onset schedule started by the scanner ('t') trigger. The bundled protocols implement the French Pinel functional localizer — reading/listening to sentences, reading/listening to mentally-solved subtractions, reading/hearing an instruction to press the left or right button three times, and passively viewing horizontal/vertical checkerboards — contrasting the language, number, motor and visual networks in one five-minute run. The schedule is not hardcoded: protocols/*.tsv give onset_time, duration, type (TEXT, BOX, IMAGE, SOUND, TEXT_STREAM, IMAGE_STREAM, SOUND_STREAM), cond and stimuli, so the same program plays any multimodal script. Select a run with -p (instructions, run1..run4). Scheduled and achieved onsets are both written to the data file.*Pinel, P., Thirion, B., Meriaux, S., Jobert, A., Serres, J., Le Bihan, D., Poline, J.-B., & Dehaene, S. (2007). Fast reproducible identification and large-scale databasing of individual functional cognitive networks. BMC	**Rush-Hour**Rush Hour sliding-block puzzle as a problem-solving task: on each trial the participant frees the red car from a 6x6 traffic jam by clicking the blocking vehicles one cell at a time, and every mouse click is logged, so the whole solution path — moves, dead ends, and latencies — can be reconstructed offline. The embedded library holds 49 boards ordered from 3 to 51 moves, each carrying the exact length of its shortest solution, so excess moves over the optimum is a per-trial measure; -n limits a session to the first N puzzles*Fogleman (2018), Rush Hour puzzle database; ThinkFun "Rush Hour" (Nob Yoshigahara, 1996)*	**Sensory-Threshold-Estimation-Auditory**1-up/2-down adaptive staircase with 2-IFC to estimate pure-tone hearing thresholds across multiple frequencies*Levitt (1971)*

shadowing Vocal shadowing-latency task: repeat a played sound aloud while a microphone voice key measures the delay between audio onset and speech onset	**Simon_task** Identify colour of a square regardless of its screen position; congruent trials are faster *Simon (1969)*	**simple_reaction_times** 20-trial simple RT task: press any key as quickly as possible when a target appears
SNARC-effect Parity judgment on digits 0–9 with reversed key mappings across blocks; demonstrates the SNARC effect *Dehaene, Bossini & Giraux (1993)*	**Statistical-Learning-Auditory** Statistical learning of tone sequences: exposure to a continuous tone stream with structured transitional probabilities, probed with 2AFC or head-turn preference *Saffran et al. (1999)*	**Statistical-Learning-Community-Structure** Implicit learning of community structure in a continuous visual sequence, with random-walk exposure and spacebar-based event segmentation *Schapiro et al. (2013)*
Statistical-Learning-Visual Implicit learning of statistical regularities in a shape stream, probed with forced-choice and RT tests *Turk-Browne et al. (2005)*	**Stroop_task** Name the ink colour of colour words; incongruent trials (e.g. RED in blue ink) are slower *Stroop (1935)*	**Subliminal-Priming** Masked word priming: words rendered invisible by surrounding masks still influence processing *Dehaene et al. (2004)*

Temporal-Integration-Word-Recognition Alternating odd/even letter components at variable SOA; Exp 1 (subjective report: 0/1/2 words perceived) and Exp 2 (lexical decision with RT); experiment selected via GetParticipantInfo UI*Forget et al. (2010)*	**Trubutschek_Unconscious_Working_Memory**Probe access to briefly presented stimuli below and above the threshold of consciousness*Trübutschek et al. (2017)*	**Typing-Speed**Typing-performance task: on each trial a target sentence is shown and the participant copies it into a framed input box below (with a blinking cursor). Every keystroke onset is recorded with a hardware-precision timestamp — its time from target onset, the interval since the previous key, and one of three categories (correct character, incorrect character, or movement/editing key). A trial advances only on an exact copy followed by ENTER. At the end, percent correct keystrokes and typing speed in characters/second (mean, P10, P50, P90) for correct keystrokes are reported. Built-in sentences can be overridden with -file.
Visual-Illusion-Lilac-Chaser Lilac chaser illusion: a ring of disappearing disks produces a rotating green afterimage	**Visual-Search**Feature vs. conjunction visual search across set sizes (4/12/24), measuring how reaction time scales with the number of distractors*Treisman & Gelade (1980)*	

Programs that open a **GetParticipantInfo** dialog collect all setup interactively (subject ID, monitor dimensions, fullscreen toggle, and any experiment-specific options). Pass `-headless` on the command line to skip the dialog and use field defaults — useful for scripted runs and automated testing. Programs that do not use the dialog still accept `-w` for windowed mode, `-d N` to select a monitor, and `-s <id>` for a subject ID.

Note: These experiments record and save behavioural data to an `.csv` file in `goxpy_data/`.

5.2 Demonstrations

Visual illusions, interactive showcases, and minimal templates. Most do not write a data file.

demo_bouncing_gv_movies Two .gv movies bounce around the screen, additively blend on overlap, fire beep tones at frame markers, and log full state to the data file on every key press; demonstrates every runtime mutator media.Movie exposes (Pause, PauseWithLoop, SetRate, SeekFrame, SetPosition, SetSize, SetBlendMode) with the multi-movie synchrony invariant preserved via BeginBurst/EndBurst	**demo_canvas**Drawing on an off-screen `Canvas` surface before presenting it in one frame	**demo_follow_mouse**A white dot follows the mouse cursor every frame — minimal real-time input loop
demo_getinfo Demonstrates the `GetParticipantInfo` dialog: collects participant demographics and monitor characteristics before the experiment window opens	**demo_hello_world**Simplest possible goxpyriment program — good starting point for new users	**demo_images**Loads every PNG file in the current directory and displays each one centered for one second
demo_images_embedded Displays embeded images sequentially, each one centered for one second	**demo_joystick**Move a cursor with a controller's analog stick (gamepad API with raw-joystick fallback, dead zone, live axis readout) — analog input handling	**demo_keyboard**Demonstrates every keyboard input method provided by the framework, section by section
demo_menu Demonstrates the stimuli.Menu widget	**demo_Motion-Blur**Motion blur vs. phantom array demo: animated bar demonstrates retinal blur and the strobe effect at 60 Hz	**demo_mouse_audio_feedback**Left/right mouse clicks trigger ping/buzzer audio; useful for testing sound output

demo_MovingGrating Presents 10 drifting sinusoidal gratings, orientation +45°/-45° in random order	**demo_naming**Vocal digit-naming task (“one”..”ten”) with a live VU-meter microphone calibration screen that auto-sets the voice-key threshold from observed min/max levels	**demo_play_gv_videos**Plays a single .gv video through the control facade
demo_play_two_gvvideos Plays two .gv videos side by side, synchronised, logging keypresses with their time relative to video onset	**demo_play_two_videos**Plays two MPEG videos side by side, synchronised, logging keypresses with their time relative to video onset	**demo_play_videos**Plays a sequence of MPEG video files found in the assets folder
demo_random-dot-stereogram Random-dot stereogram that reveals a 3-D shape when fused binocularly	**demo_simple-rt-example**Minimal 10-trial reaction-time loop using ShowTS / GetKeyEventTS for hardware-timestamped RT	**demo_spritesheet**Cut one image into a grid of stimuli sharing a single GPU texture, drawn by source rectangle (Psychtoolbox’s DrawTexture sourceRect)
demo_stimuli_extras Showcase of advanced stimuli: visual mask, Gabor patch, dot cloud, stimulus circle, thermometer	**demo_stream_callback**Per-frame FrameCallback for stream overlays and real-time feedback (PresentStreamOfStimuliFunc)	**demo_stream_images**High-precision RSVP loop streaming a sequence of images (PresentStreamOfImages)
demo_stream_mixed Heterogeneous stream mixing visual and audio stimuli via PresentStreamOfStimuli	**demo_stream_text**High-precision RSVP loop streaming a sequence of text strings	**demo_stream_tones**Sequential auditory stream playing a regular sequence of pure tones with timing logs
demo_sync_two_gv_movies Two .gv movies played side by side under a shared MasterClock; demonstrates media.MovieManager multi-movie sync, burst-pause, Movie[At]/Movie[AtDisplay], and Display[Onset/Offset] events with optional TTL-trigger wiring	**demo_text_input**Demonstration of the `TextInput` stimulus collecting free-text keyboard input	**demo_Visual-Angle-Calibration**Draws concentric rings at 2°, 5°, and 10° of visual angle for a quick sanity-check of the `units.Monitor` calibration

demo_Visual-Illusion-Ebbginghaus Animated Ebbinghaus (Titchener circles) size-contrast illusion	**demo_Visual-Illusion-Kanizsa**Kanizsa illusory-contour square: a square is perceived where none is drawn	**demo_Visual-Persistence**Persistence-of-vision illusion: an image seen only through moving vertical slits integrates into a whole when the eyes track a moving dot

---

5.3 Technical Tests

Hardware, timing, and feature tests live in the `tests/` directory at the repository root (separate Go module). Build them with `make tests`.

GvFilesPlays sequences of .gv movies while flashing a white square and pulsing DLP-IO8 line 0 at frame markers, for photodiode+oscilloscope display/TTL synchrony checks (port of PsyScope WhiteSquareSync)	**set_fullscreen**Minimal SDL program that toggles exclusive fullscreen mode — sanity check for the fullscreen video path	**test_audio_drift**Measures the audio clock's rate error against the system clock in ppm, by regressing frames consumed on monotonic time over a long tone
test_av_sync Verifies PlaySyncedWithFlip aligns audio onset with the VSYNC flip (icon flash + 1 kHz tone each second)	**test_clear_only_frames**Presents frames whose only content is a clear (no draw calls) to verify the library defence against the compositor clear-only presentation bug	**test_dlpio20**Exercises a DLP-IO20 as an 8-bit TTL trigger device (output + input loopback), with static hold/set modes for multimeter probing
test_dlpio8 Square-wave output on a DLP-IO8-G pin with rising/falling edge jitter statistics; characterises the trigger box against an oscilloscope	**test_ft232h**Exercises an Adafruit FT232H breakout as an 8-bit TTL trigger device (output + input loopback)	**test_fullscreen**Sanity check for fullscreen rendering through the control facade
test_gv_sync Plays a synthesised .gv flash train with a TTL pulse at each bright onset; measures whether PlayGvFunc puts frames on screen when it claims to	**test_labjackt4**Exercises a LabJack T4 as an 8-bit TTL trigger device (output + input loopback)	**test_linuxgpio**Exercises a Linux GPIO character-device (v2) trigger as an 8-bit TTL device (output + input loopback)

test_megttlbox Exercises the NeuroSpin Arduino Mega TTL box — firmware identification, TTL outputs on D30–D37, response inputs on D22–D29, and an optional jumpered loopback check.	**test_netstation**Exercises an EGI/NetStation EEG host over TCP/IP (ECI) — connect, synchronize, record, and send event markers	**test_parallel_port**Reads and writes a Linux LPT parallel port via the ppdev kernel interface
test_photodiode_latency Measures the delay between the timestamp ShowTS returns and light actually leaving the screen, by firing a DLP-IO8 TTL synchronously on the flip thread and letting an AD3 or BBTK time the TTL against a photodiode	**test_stream_trigger**Post-flip onset hooks fire a per-stimulus TTL aligned to TimingLog.OnsetNS (PresentStreamOfStimuli-Hooks + PlayStreamOfSound-sHook)	**test_tearing**Full-height white bar sweeping horizontally to reveal screen tearing; prints frame-interval statistics on exit
test_triggers Fires the same pulse train on two or more TTL output devices at once, on one absolute schedule, so an oscilloscope (or a MEG TTL box acting as witness) can measure the skew between their edges	**test_vblank_drift**Measures whether Screen.FlipTS drifts against the display, using the kernel's DRM vblank timestamps as a reference instead of a photodiode	**test_videorecorder**Exercises the BEL_video networked video recorder over TCP/IP — start/stop recording and send overlay labels
test_vsync_blocking Reports nominal, unaided and paced frame periods to show whether the platform/driver blocks on VSYNC or drops frames	**Timing-Tests**Hardware timing verification suite: six sub-tests selected with -test ; av (visual+audio+TTL, five photodiode squares) is the default	

5.4 Binaries

If you have followed the installation instructions, running the script `build-all.sh` has created binaries for all examples in the subfolder `_build`. You can just open this folder and launch the executables.

Else, you can download pre-built executable programs for all examples.

Alternatively, you can read the next section if you want to compile these experiments on your computer.

5.5 Building from source

If you have installed go on your computer, you can run any example directly from a local clone of the goxpyriment repository:

```
go run ./examples/parity_decision/ -w -s 1
```

Or build and run from inside the example directory:

```
cd examples/hello_world
go run .            # fullscreen by default
go run . -w         # windowed 1024×768
go run . -w -s 1    # windowed, subject ID = 1
go run . -d 1       # fullscreen on monitor 1
go run . -w -d 1    # windowed on monitor 1
go build .          # build a standalone binary
```

To build all examples (and tests) at once (binaries go to `_build/`):

```
./build-all.sh      # from the repo root — works on Linux, macOS, and
↪  Windows (Git Bash)
```

(On Linux/macOS you can also use `make examples` or `make all`.)

Chapter 6

Pre-built binaries of examples and tests for goxpyriment

If you prefer not to install Go and compile the examples yourself, you can still run them by downloading *pre-built executables* ready to run on your computer.

! WARNING These binaries are unsigned. macOS Gatekeeper and Windows Defender will show security warnings or worse, *misleading messages* such as 'this program is damaged'. Your antivirus may quarantine the files. Don't let them intimidate you. These warnings pop out because I am not willing to pay third parties to sign the executables. The instructions below explain how to run these programs anyway.

6.0.1 Windows (x86-64)

1. Download `goxpyriment-examples-windows-x86_64.zip`;

2. Before unzipping it: right-click the downloaded `.zip` → **Properties** → at the bottom of the **General** tab tick **Unblock** → **OK**. (You can achieve the same thing in PowerShell with `Unblock-File .\goxpyriment-examples-windows-x86_64.zip`.)
3. Unzip it. You should be able to run the `.exe` files directly, either by clicking on them, or from a command line.

If a security dialog window pops out, click on **More info**, then **Run anyway**. You can unblock all the `.exe` files with `Get-ChildItem -Recurse <folder> | Unblock-File`.

6.0.2 macOS (Apple Silicon)

1. Download `goxpyriment-examples-macos-arm64.zip`, unzip it and move the folder to the location of your choice, e.g., `$HOME/goxpy`

2. Open a Terminal at that location (`cd $HOME/goxpy`) and remove the protection with `xattr -rd com.apple.quarantine .`, and set the execute bit (`chmod +x *`) (See macOS installation and security.)

6.0.3 Linux

- **(x86-64):** Download goxpyriment-examples-linux-x86_64.tar.gz, extract the binaries with `tar xzf`, and run them directly (potentially, you may need to do `chmod +x *` to set their execute permission bit).
- **(arm64 / Raspberry Pi):** Download goxpyriment-examples-linux-arm64.tar.gz, extract the binaries with `tar xzf`, and run the binaries directly.

Note: if you do not find a version for your hardware (e.g. Mac/Intel or Windows/ARM), do not despair: it is very easy to compile these examples
following the instructions in Installation.md *** Good programs to start: `Memory_span`, `Change-Blindness`, `Simon_task`, `LoT_geometry`, `Retinotopy`...

When launched from the command line, most programs accept `-w` (windowed 1024×768 mode), `-d N` (open on monitor N, 0 = primary), and `-s <id>` (subject ID written to the `.csv` data file).

Results are saved in a folder `goxpy_data` in your Home (User) folder

Chapter 7

goxpyriment User's Manual

This manual explains the key concepts of the library. It assumes you have read the Getting Started guide and want to understand the framework well enough to write experiments confidently.

7.1 Table of Contents

7.2 1. The Experiment Object

Every experiment revolves around a single `*control.Experiment` value. It is a façade that owns the SDL window, renderer, default font, audio device, keyboard and mouse handlers, and data file. You never create these separately.

```
exp := control.NewExperimentFromFlags("My Experiment", control.Black,
 ↪  control.White, 32)
defer exp.End()
```

`NewExperimentFromFlags` parses these optional command-line flags:

Flag	Effect
`-w`	Windowed mode: opens a 1024×768 window instead of going fullscreen
`-d N`	Display ID: open on monitor N (-1 = primary display)
`-s N`	Set subject ID to N (integer); written to the data file automatically
`-headless`	Skip the setup dialog (below) and use field defaults — for batch/automated runs

7.2.1 Automatic setup dialog (launch without `-s`)

If `-s` is **not** passed — most importantly when the binary is launched by **double-clicking its icon** rather than from a terminal — a small setup dialog opens before the experiment starts, letting the experimenter enter the **subject code** and confirm the **display**, **fullscreen/windowed** mode, and **results folder**. It seeds its defaults from any `-w`/`-d` you did pass and remembers the display/fullscreen/folder choices across sessions (the subject code is always asked fresh). Cancelling the dialog exits the program.

The dialog is skipped when `-s N` is given (that value is used directly, as before), when `-headless` is passed, or when the program already collected participant information via `GetParticipantInfo` itself — so no program is ever prompted twice.

`defer exp.End()` must appear immediately after construction. It saves the data file, releases fonts, destroys the window, and shuts down SDL — even if the experiment panics or returns early.

7.2.2 Key fields

```
exp.Screen       *apparatus.Screen           // window + renderer
exp.Keyboard     *apparatus.Keyboard         // keyboard input
exp.Mouse        *apparatus.Mouse            // mouse input
exp.AudioDevice  sdl.AudioDeviceID    // SDL audio device for Sound/Tone
exp.Audio        *control.AudioManager // high-level audio helpers
exp.Data         *results.DataFile         // output data file
```

```
exp.Design       *design.Experiment   // block/trial structure
exp.SubjectID    int
exp.DefaultFont  *ttf.Font
exp.Info         map[string]string    // values collected by
 ↪  GetParticipantInfo, if used
```

7.3 2. Collecting Participant Information

Before the experiment window opens, you can display a graphical setup dialog that lets the experimenter fill in participant demographics, monitor characteristics, and display mode. The dialog returns the collected values as a `map[string]string`.

```
fields := append(control.StandardFields, control.FullscreenField)
info, err := control.GetParticipantInfo("My Experiment", fields)
if err != nil {
    log.Fatalf("setup cancelled: %v", err)
}
```

Call `GetParticipantInfo` **before** `NewExperiment` or `NewExperimentFromFlags`. It initializes SDL internally, shows the window, and shuts SDL down again before returning, so the subsequent experiment initialization starts from a clean state.

7.3.1 Pre-built field sets

Variable	Fields included
`control.ParticipantFields`	`subject_id`, `age`, `gender`, `handedness`
`control.MonitorFields`	`screen_width_cm`, `viewing_distance_cm`, `refresh_rate_hz`
`control.StandardFields`	`ParticipantFields` + `MonitorFields` combined
`control.FullscreenField`	Checkbox: `fullscreen` (`"true"` / `"false"`)

7.3.2 Defining custom fields

```
fields := []control.InfoField{
    {Name: "subject_id", Label: "Subject ID",      Default: ""},
    {Name: "session",    Label: "Session (1/2/3)", Default: "1"},
    {Name: "room",       Label: "Testing room",    Default: "Lab A"},
    {Name: "fullscreen", Label: "Fullscreen mode", Default: "true",
     Type: control.FieldCheckbox},
}
info, err := control.GetParticipantInfo("My Experiment", fields)
```

Fields of type `FieldText` (the default) render as text input boxes. Fields of type `FieldCheckbox` render as tick-boxes; their value is always `"true"` or `"false"`.

7.3.3 Dialog interaction

Action	Effect
Click a field	Focus it
Type	Append text to the focused field
Backspace	Delete last character
Tab / Shift-Tab	Move focus to next / previous text field
Enter	Confirm (same as clicking OK)
Escape / close window	Cancel — `ErrCancelled` is returned

7.3.4 Session persistence

All values except `subject_id` are saved to `~/.cache/goxpyriment/last_session.json` when the experimenter confirms. They are pre-filled on the next run. `subject_id` is always reset — the experimenter must enter it fresh each session.

7.3.5 Using the results

```
info, err := control.GetParticipantInfo("My Experiment", fields)
if err != nil {
    log.Fatalf("setup cancelled: %v", err)
}

// Use the fullscreen checkbox to choose the window mode
fullscreen := info["fullscreen"] == "true"
width, height := 0, 0
if !fullscreen {
    width, height = 1024, 768
}

exp := control.NewExperiment("My Experiment", width, height, fullscreen,
    control.Black, control.White, 32)

// Set Info and SubjectID BEFORE Initialize so they are written to the .csv
 ↪  header
exp.SubjectID, _ = strconv.Atoi(info["subject_id"])
exp.Info = info

if err := exp.Initialize(); err != nil {
    log.Fatalf("failed to initialize: %v", err)
}
defer exp.End()
```

When `exp.Info` is non-nil at the time `Initialize()` is called, the collected key/value pairs are automatically written as a `--PARTICIPANT INFO` block in the `.csv` metadata header — no extra call is required.

> **Note:** When using `GetParticipantInfo` you will typically call the lower-level `NewExperiment` + `Initialize()` instead of `NewExperimentFromFlags`, so you can pass the fullscreen/windowed choice from the dialog directly.

7.4 3. The Run Loop and Error Handling

7.4.1 The `exp.Run` wrapper

All experiment logic runs inside a callback passed to `exp.Run`:

```
err := exp.Run(func() error {
    // your experiment here
    return control.EndLoop
})
if err != nil && !control.IsEndLoop(err) {
    log.Fatalf("experiment error: %v", err)
}
```

`exp.Run` does two important things:

1. **Ensures everything runs on the SDL main thread.** SDL requires that all rendering and event calls happen on the thread that created the window. `exp.Run` guarantees this.
2. **Installs a panic/recover guard.** When the participant presses ESC, the library immediately aborts the trial loop by calling `panic` with an internal sentinel value. `exp.Run` catches this panic, converts it back to an error, and returns it cleanly. Your data is saved.

7.4.2 Returning from the callback

Return value	Meaning
`control.EndLoop`	Normal exit; experiment is done
`nil`	Continue — call the callback again on the next frame
any other `error`	Propagated as the return value of `exp.Run`

In a typical experiment you run the full trial loop inside a single callback invocation and return `control.EndLoop` at the end. The `nil` / "keep looping" pattern is used for frame-by-frame animation; in most cases you will never return `nil`.

7.4.3 You don't need to check every error

Because pressing ESC triggers the panic/recover mechanism, any call that would have returned an error (e.g., `exp.Show`, `exp.Wait`, `exp.Keyboard.WaitKey`) will instead abort the loop immediately. The error never reaches your code.

This means the following two styles are both correct:

```
// Style A — check errors explicitly (safer for library/production code)
if err := exp.Show(fixation); err != nil {
    return err
}
if err := exp.Wait(500); err != nil {
    return err
}

// Style B — omit error checks inside exp.Run (fine for experiment scripts)
exp.ShowTimed(fixation, 500)
```

Style B works because if something goes wrong (ESC pressed, window closed), the panic/recover mechanism unwinds the call stack before your code can observe an error. Use Style A if you have cleanup that must run on abort; otherwise Style B keeps experiment scripts readable.

7.5 4. The Coordinate System

All stimulus positions use a **center-origin** system:

```
            +Y (up)
             |
  -X --------+-------- +X
             |
            -Y (down)
```

- `(0, 0)` is the screen center.
- Positive Y is **up** (opposite to SDL's default, which is top-down).
- Units are pixels at the logical resolution.

```
// Center of screen
stimuli.NewTextLine("Hello", 0, 0, control.White)

// 200 px to the right
stimuli.NewCircle(30, control.Red).SetPosition(control.Point(200, 0))

// Bottom-left area
stimuli.NewTextLine("Score", -400, -250, control.White)
```

The conversion to SDL coordinates (top-left origin) is handled internally by `screen.CenterToSDL(x, y)`. You never call this yourself unless you are writing a custom stimulus

type — in which case `screen.CenteredRect(pos, w, h)` returns the SDL destination rectangle for a `w×h` texture centered at `pos`, factoring out the convert-and-offset step.

7.5.1 Logical size

If you call `exp.SetLogicalSize(width, height)`, the coordinate system scales to that virtual resolution regardless of the actual window or screen size. This is useful for making experiments resolution-independent:

```
if err := exp.SetLogicalSize(1920, 1080); err != nil {
    log.Printf("warning: %v", err)
}
// Now (960, 540) is the center-right area, regardless of actual screen
 ↪  size.
```

7.5.2 Display scaling (HiDPI / fractional scaling)

> **Recommendation: set your OS display scaling to 100% before running experiments.**

Most desktop environments (GNOME, KDE, Windows, macOS) allow the user to scale the entire UI — e.g. 125%, 133%, 150% — to make text and icons larger on high-density screens. Goxpyriment does not currently compensate for fractional OS-level scaling. Running with a scale factor other than 100% can cause:

- stimulus positions and sizes to be wrong (coordinates are in logical pixels, but the OS maps them to physical pixels at the scale factor),
- video playback and canvas rendering to appear cropped or misaligned,
- the window to not fill the screen correctly in fullscreen mode.

To disable scaling before running an experiment:

- **GNOME (Ubuntu):** *Settings → Displays → Scale → 100%*
- **KDE:** *System Settings → Display and Monitor → Scale → 100%*
- **Windows:** *Settings → Display → Scale → 100%*
- **macOS:** *System Settings → Displays → Resolution → Default* (or choose a non-HiDPI mode)

You can restore your preferred scaling after the session.

7.6 5. The Rendering Model

SDL uses a **double-buffered** rendering model. There is an off-screen backbuffer where you draw, and the visible display. You draw to the backbuffer; calling `screen.Update()` (also called a "flip") presents it to the screen, synchronized to the vertical retrace (VSYNC).

The three-step cycle for showing one stimulus is:

```
exp.Screen.Clear()       // fill backbuffer with background color
myStim.Draw(exp.Screen)  // draw stimulus onto backbuffer
exp.Screen.Update()      // present to display (VSYNC-blocks)
```

`exp.Show(stim)` does all three in one call and is the right choice for single-stimulus presentations. For cases where you need to draw multiple stimuli simultaneously — so they appear on screen at the same time — call each `Draw` separately before the single `Update`:

> **Triple/mailbox buffering — handled for you.** On many systems (NVIDIA + compositor, Wayland, and more besides) `SDL_RenderPresent` does **not** block until the retrace: it queues the frame in a non-blocking mode — mailbox ("triple buffering"), where a newer frame replaces the pending one — or hands the buffer to a compositor that accepts it rather than blocking on scanout. The display stays tear-free, but the *call* no longer paces one frame per call. A per-frame loop then completes without consuming wall-clock time and the stimulus is replaced too early, its on-screen duration unpredictably short (often, but not always, zero frames).
>
> `screen.Update()` therefore presents **and then holds to the frame boundary**, so one call always occupies one display frame. You do not have to select a special method or detect the platform: where the driver blocks correctly the wait runs zero iterations and costs nothing. This is why there is no `PacedFlip` — it was removed once pacing became unconditional.
>
> The one thing pacing cannot do is recover a frame that was *dropped* on the way to the panel: it enforces a minimum frame time, not a maximum. See §Calibration below.

```
// Show fixation cross and stimulus simultaneously
exp.Screen.Clear()
fixation.Draw(exp.Screen)
targetCircle.Draw(exp.Screen)
exp.Screen.Update()
```

7.6.1 The blank screen

`exp.Blank(ms)` clears the screen, flips it (showing a blank), and then waits:

```
exp.Blank(1000) // 1-second blank inter-trial interval
```

This is equivalent to `exp.Screen.Clear() + exp.Screen.Update() + exp.Wait(ms)`.

7.6.2 Never draw outside exp.Run

All rendering must happen inside the `exp.Run` callback — equivalently, on the SDL main thread. Drawing from a goroutine will silently do nothing or crash.

7.7 6. Timing Architecture

7.7.1 Frame cadence

`screen.Update()` returns at the display's vertical retrace (VSYNC). On a 60 Hz monitor this is ~16.67 ms; on a 120 Hz monitor ~8.33 ms. This is the fundamental clock of the visual display: every stimulus change is aligned to a frame boundary automatically.

On some systems (NVIDIA proprietary driver + compositor, Wayland) the system selects a non-blocking presentation mode (mailbox/"triple buffering") or routes the swap through a compositor, so the underlying `SDL_RenderPresent` returns before the frame is scanned out. The display is still tear-free, but the call cannot be relied on to block for one frame. (This is not VSYNC being disabled: a blocking mode such as FIFO exists; the system just isn't guaranteed to use it. Wayland paces via frame callbacks rather than a blocking swap.)

`screen.Update()` closes this gap itself by holding to the frame boundary after presenting, so a tight frame loop is correct on every driver without you choosing anything. Measured on Intel i915 + Wayland with a 120 Hz panel, unaided presents returned as little as 6.95 ms apart against an 8.33 ms frame; through `Update` the same loop held 8.333 ms with SD 0.06 ms.

To know your frame duration at runtime:

```
frameDur := exp.Screen.FrameDuration()  // e.g., 16.666ms on a 60 Hz display
fmt.Printf("Frame: %.2f ms\n", frameDur.Seconds()*1000)
```

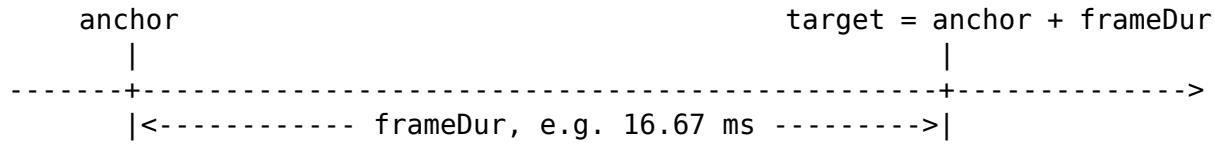
7.7.2 What `Update()` actually does with a frame

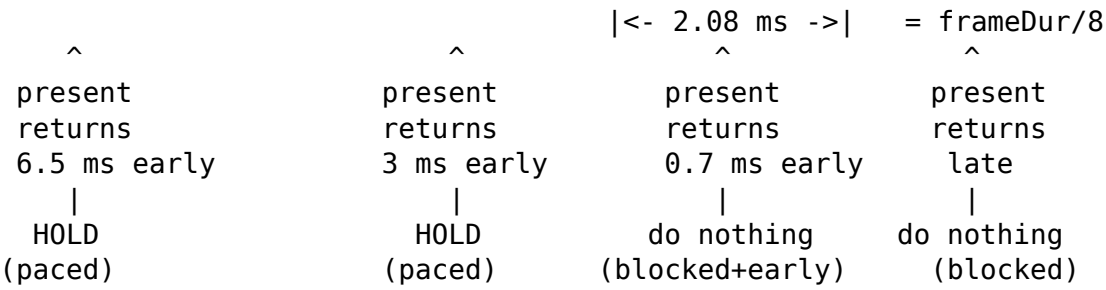
You rarely need this, but when a duration comes out one frame long, or you are deciding whether to trust an onset timestamp, this is the machinery underneath. Each `Update()` does four things in order (sketch, not compilable Go):

```
anchor := previous frame's onset      1. sampled BEFORE this frame
 ↪  overwrites it
SDL_RenderPresent(); presentNS = now  2. submit, and stamp the return
paceToFrame(anchor)                   3. maybe hold to the frame boundary
recordOnset()                         4. decide this frame's onset
```

The deadline is measured from the previous frame's *onset*, not from the previous `Update()` call. That onset comes from the best source available, in this order: the kernel's vblank timestamp (`GOXPY_VBLANK=on`), the scheduled boundary if pacing held the frame, or the moment `SDL_RenderPresent` returned. Whichever it was, `target = anchor + FrameDuration()`, and the run's `onset_source` column records which.

Now let `now` be the clock read the instant `SDL_RenderPresent` returned. Where it falls decides everything:





Early by more than an eighth of a frame — hold. The driver queued the frame and returned (triple/mailbox buffering). `Update` sleeps until 2 ms before the boundary, then spins the last 2 ms, and reports the *scheduled* boundary as the onset. It is split that way for a reason in each direction: spinning the whole wait puts a real-time thread at 100 % duty and trips the kernel's throttle (measured: 24 stalls of 51 ms in 25 s, three dropped frames a second at 60 Hz), while sleeping the whole wait overshoots the boundary by up to 0.83 ms and misses the frame.

Early by an eighth of a frame or less — do nothing. This is a driver that *did* block on the retrace and simply came back a little inside the *nominal* boundary, because the panel's grid and the nominal grid sit at a fixed phase offset. Its return is a hardware instant, and holding would throw that away and substitute a synthesised time — on one machine that happened on 98.8 % of frames. The two cases are far apart in practice, which is what makes a threshold workable: a blocking present came back 0.68 ms early on average and 1.14 ms at worst, against 6.5–15 ms for a non-blocking one. Nothing has been observed in between.

Late — also nothing, and the missed time is not made up. If `now` is already past `target`, the target is *abandoned*, not chased. `Update` never shortens a later frame to compensate, and the next deadline is measured from the present that just returned, so the schedule quietly re-phases onto wherever the display actually is. The consequence is the one you see in the data: a loop of twelve `Update()` calls that hits one stall takes thirteen frames, and its duration column reads one frame long. That is the honest report — a compressed frame afterwards would be worse — but it does mean a stall shows up as a long interval, never as a short one.

Two things follow that are worth knowing:

- A late present is tallied as `blocked`, the same as a healthy blocking one, so the pacing counters alone do not reveal lateness. What reveals it is the measured-vs-nominal comparison in Startup calibration below, and `sequence_gaps` when the vblank clock is enabled.
- Pacing is skipped entirely when VSync is off, on the grounds that a caller who turned it off wants frames as fast as the GPU can make them.

The tallies land in the run's `-info.txt`, and `early` is a **subset** of `blocked` rather than a fourth category:

```
# sys pacing: presents=30000 blocked=30000 early=29610 paced=0
 ↪  vblank_held=0 … class=blocking
```

So `presents = blocked + paced + vblank_held`. `blocked` with a high `early` and a sub-millisecond `early_mean` is a healthy blocking driver; `paced` with a multi-millisecond

`wait_mean` is a driver that does not block, being held; `vblank_held` is the kernel-vblank path. Timing Tests reads these out in full.

7.7.3 Startup calibration

`FrameDuration()` is the *nominal* period — what SDL derives from the display mode. It is not proof that your frames actually took that long. `Initialize()` therefore measures the display over 60 frames before the experiment starts and records both numbers in the `-info.txt` file:

```
# sys refresh_nominal_hz: 120.0000
# sys refresh_measured_hz: 119.9820
```

If the two disagree by more than 10%, a warning is logged. Take it seriously — it means this session's stimulus durations were not what your analysis will assume. The two directions mean different things:

Measured vs nominal	Meaning	Does pacing help?
≈ **nominal**	Normal.	—
Faster than nominal	`SDL_RenderPresent` is not blocking to the retrace (triple/mailbox buffering).	Yes — `Update`'s pacing is exactly this case.
Slower than nominal	Frames are being *dropped* before reaching the panel: a compositor throttling an unfocused or occluded window, or a GPU that cannot keep up.	**No.** Pacing enforces a minimum frame time, not a maximum.

The third row is the one that catches people out. A compositor will happily throttle a windowed experiment to a fraction of the refresh rate while every timestamp your program records still looks plausible. Run fullscreen, keep the window focused, and check the measured rate in the info file afterwards.

To measure it yourself at any point — it bypasses the pacing, so it reports the unaided driver behaviour:

```
measured, err := exp.Screen.CalibrateRefresh(120)  // median over 120 frames
```

`tests/test_vsync_blocking` reports nominal, unaided and paced side by side, with a verdict; run it once on any machine you plan to collect data on.

7.7.4 `exp.Wait` vs `clock.Wait`

Function	Pumps SDL events?	Detects ESC?	Use when
`exp.Wait(ms)`	Yes	Yes	Inside `exp.Run`, between stimuli
`clock.Wait(ms)`	No	No	Short busy-waits, inside animation loops

Always use `exp.Wait` for inter-trial and inter-stimulus intervals — it keeps the OS responsive and responds to ESC. Use `clock.Wait` only inside VSYNC-locked loops that handle events themselves (streams, animation loops).

7.7.5 Sub-millisecond timing is not guaranteed for `exp.Wait`

`exp.Wait` sleeps in 1 ms increments. For coarse delays (ISIs, fixation durations) this is perfectly fine. For frame-accurate stimulus timing — e.g., showing a stimulus for exactly 2 frames — use the stream functions described in Section 10.

7.7.6 Nanosecond event timestamps

SDL3 timestamps every input event **at hardware-interrupt time**, before any application code runs. The timestamp is stored with the event in the SDL event queue, where it remains untouched until your code reads it.

This has a crucial consequence: **it does not matter when you call `GetKeyEventTS`**. Any keypress that occurred — even during `exp.Wait`, between two `ShowTS` calls, or while other computation was running — is waiting in the queue with its exact hardware timestamp. Nothing is lost. This is fundamentally different from a polling-based approach, where a keypress can only be measured from the moment the polling function is called, and any code running before that call inflates the apparent RT.

`exp.ShowTS(stim)` records the SDL nanosecond time of the VSYNC flip. `GetKeyEventTS` retrieves the event's own hardware timestamp. Subtracting the two gives hardware-precision RT regardless of how much code ran in between:

```
onset, _ := exp.ShowTS(stim1)   // record flip timestamp
exp.Wait(500)                    // keypress here is NOT lost
exp.ShowTS(stim2)                // keypress here is also NOT lost
key, keyTS, _ := exp.Keyboard.GetKeyEventTS(responseKeys, -1)
rtToStim1 := int64(keyTS - onset)   // correct even if pressed during Wait
```

This is especially useful when you need RT relative to a specific stimulus in a multi-stimulus sequence — something that `WaitKeysRT` cannot express directly, since it measures from the call site rather than from a recorded onset.

7.7.7 Two clocks: the Go clock and the SDL clock

goxpyriment exposes time through **two independent clocks with different zero points**. A value read from one must never be subtracted from a value read from the other — the difference would silently include the unknown, machine-dependent offset between their origins.

Clock	Read it via	Zero point	Use it for
Go monotonic clock	`clock.GetTime()`, `clock.GetTimeNS()`, `Clock.Now()`, `Clock.NowMillis()`, `Clock.NowNanos()`	First use of the `clock` package, or `clock.NewClock()`	*Scheduling and logging in your own code*: inter-trial and inter-stimulus intervals, drift-free trial pacing (`SleepUntil`), and "time since start of block" columns in your data file
SDL event clock	`exp.ShowTS()`, `Screen.FlipTS()`, `Keyboard.GetKeyEventTS()`, `WaitAnyEventTS()`, and the `Timestamp` field of any SDL event (all on `sdl.TicksNS()`)	SDL initialisation	**Reaction-time measurement**: whenever you subtract a stimulus onset from a response, *both* values must come from this clock

The rule. Measure reaction times entirely on the SDL clock — subtract a `FlipTS`/`ShowTS` onset from a `GetKeyEventTS` response. Use the Go clock only for things you *schedule* rather than *measure*: ISIs, fixation durations, block pacing, and log timestamps. Crossing the two — e.g. `keyTS - clock.GetTimeNS()` — is meaningless.

Why two clocks? The SDL clock is the one SDL stamps onto hardware input events at interrupt time (see the previous subsection), so it is the only clock that can measure RT without inflation from intervening code. The Go clock needs no SDL context, is what `exp.Wait`, `clock.Wait`, and `Clock.SleepUntil` are built on, and is the natural choice for scheduling and for log timestamps that just need to be internally consistent.

The two roles coexist cleanly in a single trial loop — Go clock for pacing and logging, SDL clock for the RT:

```
c := clock.NewClock()                                    // Go clock: pacing +
 ↪  logging
for i, trial := range block.Trials {
    c.SleepUntil(time.Duration(i) * 2 * time.Second)   // Go clock:
 ↪  drift-free onset cadence
```

```
    onset, _ := exp.ShowTS(trial.Stimulus)            // SDL clock:
↪   stimulus onset
    key, keyTS, _ := exp.Keyboard.GetKeyEventTS(responseKeys, -1)
    rtMs := float64(keyTS-onset) / 1e6                 // SDL − SDL  ✓
↪   valid RT
    exp.Data.Add(trial.Label, key, rtMs, c.NowMillis())// log time on the Go
↪   clock
}
```

Here `keyTS - onset` stays within the SDL clock (a real RT), while `c.SleepUntil` and `c.NowMillis()` stay within the Go clock (scheduling and a log column). The two are never subtracted from each other.

7.7.8 Disabling garbage collection

Go's garbage collector can pause execution for several milliseconds at unpredictable times. The stream and animation functions disable it during the critical loop:

```
old := debug.SetGCPercent(-1)
defer debug.SetGCPercent(old)
```

You do not need to do this yourself for ordinary trial loops; only for high-speed RSVP or animation. The stream and animation functions handle it automatically.

7.7.9 Choosing a timing approach

Use this decision guide before committing to a timing strategy.

Situation	Recommended approach
Coarse ISIs and fixation durations (≥ 100 ms)	`exp.Wait(ms)` — adequate precision, keeps the event loop alive
Show stimulus + timed hold (fixation, cue, mask)	`exp.ShowTimed(stim, ms)` — Show + Wait in one call
Single-stimulus trial, RT relative to onset	`key, rt, err := exp.ShowAndGetRT(stim, keys, timeout)` — hardware-precise RT, clears stale events automatically
Stimulus duration measured in frames (e.g. 2-frame mask)	Per-frame loop with `screen.Flip()` (paced, safe on all drivers), or `PresentStreamOfImages` (Section 10)
Multi-stimulus sequence, RT relative to a specific stimulus	`exp.ShowTS(stim)` + `exp.Keyboard.GetKeyEventTS(...)` — nanosecond timestamps on the same SDL clock; subtract directly
RSVP or rapid animation (frame-accurate presentation of many items)	`PresentStreamOfImages` / `PresentStreamOfText` (Section 10) — GC disabled, VSYNC-locked, full timing log

Situation	Recommended approach
EEG/MEG synchronisation required	Add a TTL pulse via `triggers` immediately after `exp.ShowTS` (Section 17)
Stimulus duration not a multiple of the frame period (e.g. 10 ms, 17 ms)	Variable Refresh Rate mode — see below

OS considerations. Presentation latency depends heavily on whether the display compositor is active — see the next section. For EEG work, always verify onset timing with a photodiode regardless of OS.

When in doubt, use `ShowAndGetRT`. It clears stale events, records the VSYNC flip timestamp, waits for the key with hardware-precision timing, and returns `(key, rtMs, error)` — the canonical single-stimulus RT call. Use raw `ShowTS` + `GetKeyEventTS` only when composing multiple stimuli before a single response.

`ShowTS` vs `FlipTS`. `exp.ShowTS(stim)` is the standard high-level call: it clears the screen, draws the stimulus, flips, and returns the flip timestamp — one line for the common case. `exp.Screen.FlipTS()` is the lower-level primitive that only flips and timestamps, leaving clearing and drawing to you. Both hold to the frame boundary, so both are safe in a per-frame loop; there is no third variant to choose between.

7.7.10 Display compositor bypass

A compositing window manager blends every application frame with other on-screen elements before sending the result to the display. This adds latency and jitter that can be significant for millisecond-accurate stimulus presentation. How much overhead you incur — and whether you can avoid it — depends on the operating system.

Linux / X11 (fullscreen). SDL3 automatically sets the `_NET_WM_BYPASS_COMPOSITOR` window property when the application enters fullscreen mode. Compliant compositing WMs (KWin, Mutter, Compton) respond by *unredirecting* the fullscreen window: the application's framebuffer is routed directly to the display scan-out pipeline without compositing. Presentation latency drops below 1 ms — comparable to a compositor-free session. Nothing special needs to be done in goxpyriment; running fullscreen (omit `-w`) is sufficient.

Linux / Wayland (fullscreen). The Wayland compositor is always the intermediary for frame delivery, but modern compositors (KWin 5.21+, GNOME Mutter 44+) support *direct scan-out* for fullscreen applications, which routes the framebuffer to the display hardware with near-zero overhead. In practice the latency is similar to X11 bypass, though this depends on the compositor version and GPU driver.

Linux / KMS–DRM (no display server). The lowest-latency configuration on Linux is to run without any display server at all, from a virtual terminal (`Ctrl+Alt+F2`). Setting the environment variable `SDL_VIDEODRIVER=kmsdrm` before launching the experiment directs SDL3 to use the Linux kernel's KMS/DRM subsystem directly, bypassing both X11 and Wayland entirely:

```
SDL_VIDEODRIVER=kmsdrm go run myexperiment/main.go
```

This is the recommended configuration for the most demanding timing requirements, such as single-frame subliminal stimulation or high-frequency RSVP.

Windows (fullscreen). SDL3's fullscreen mode creates an exclusive fullscreen surface that bypasses the Desktop Window Manager (DWM), yielding frame-interval jitter typically below 0.3 ms (one standard deviation), comparable to Linux.

macOS. Apple's Metal pipeline always delivers frames through the WindowServer compositor, and no third-party application can bypass it — even in fullscreen.

These platform differences can be measured empirically with the `display` and `av` subtests of the bundled `Timing-Tests` suite (see TimingTests.md).

7.7.11 Variable Refresh Rate (VRR) — arbitrary stimulus durations

On a standard 60 Hz monitor, every stimulus duration must be a multiple of 16.67 ms. You cannot present a stimulus for 10 ms or 20 ms — the display simply cannot change in between frame boundaries. Variable Refresh Rate technology (AMD FreeSync, NVIDIA G-Sync, VESA Adaptive-Sync) removes this constraint: the panel holds each frame for exactly as long as the software asks, refreshing only when the next `Present()` call arrives.

7.7.11.1 Enabling VRR in goxpyriment

goxpyriment opens the renderer with VSYNC enabled by default, which locks every flip to a frame boundary. To enter VRR mode, disable VSYNC before the timing-critical section and restore it afterwards:

```
// Switch to VRR mode — Present() now returns immediately.
if err := exp.Screen.SetVSync(0); err != nil {
    log.Fatalf("VRR not supported: %v", err)
}
defer exp.Screen.SetVSync(1) // always restore

// Pre-load stimulus textures before disabling GC.
stimuli.PreloadVisualOnScreen(exp.Screen, myStim)

old := debug.SetGCPercent(-1)
defer debug.SetGCPercent(old)

// Draw stimulus and present (returns immediately with VSync=0).
exp.Screen.Renderer.SetDrawColor(0, 0, 0, 255)
exp.Screen.Clear()
myStim.Draw(exp.Screen)
onsetNS, _ := exp.Screen.FlipTS()

// Hold for exactly targetDur using a busy-wait (sub-ms precision).
```

```
deadline := time.Now().Add(targetDur)
for time.Now().Before(deadline) { /* spin */ }

// Present blank — the display switches immediately on VRR.
exp.Screen.Renderer.SetDrawColor(0, 0, 0, 255)
exp.Screen.Clear()
offsetNS, _ := exp.Screen.FlipTS()

actualDurMs := float64(offsetNS-onsetNS) / 1e6
```

`onsetNS` and `offsetNS` are both `sdl.TicksNS()` timestamps on the SDL nanosecond clock, captured right after each `Present()` returns. Their difference measures the software-controlled stimulus duration. On a VRR display this equals the actual on-screen duration plus a small, constant hardware pipeline latency (GPU scan-out + panel response, typically 1–5 ms) that can be measured independently with the `frames` test and a photodiode.

7.7.11.2 VRR range and self-diagnosis

Every VRR panel has a supported refresh-rate range (e.g. 48–144 Hz on a typical gaming monitor). Requesting a duration shorter than `1000/max_Hz` ms or longer than `1000/min_Hz` ms takes the panel outside its VRR window; the display then falls back to fixed-rate behaviour and durations are again quantised to frame multiples.

You can characterise your panel's VRR window empirically with the `Timing-Tests` suite:

```
go run ./tests/Timing-Tests -test vrr -vrr-max-ms 50 -cycles 5
```

This sweeps target durations from 1 ms to 50 ms in 1 ms steps and reports the actual vs. target duration at each step. Duration errors below 0.5 ms across the sweep confirm that VRR is working; large periodic errors confirm that VRR is absent or the requested duration is outside the supported range. See TimingTests.md — VRR section for detailed interpretation, including how to read the VRR boundary from the output and how to enable FreeSync on Linux.

7.7.11.3 Requirements

- A VRR-capable monitor (FreeSync, G-Sync Compatible, or Adaptive-Sync).
- VRR enabled in the OS display settings (Linux: DRM/KMS or compositor; Windows: Display Settings → Advanced display → Variable refresh rate).
- The application does not need any special SDL hints; `SetVSync(0)` is sufficient.

7.7.11.4 Note on tearing

On a monitor without VRR, `SetVSync(0)` disables the frame-boundary synchronisation entirely and the GPU writes to the framebuffer while the display is reading it, producing a horizontal tear line. On a VRR monitor this does not happen: the display waits

for the GPU's signal before starting each new refresh. If your VRR test shows tearing artefacts, VRR is either not enabled in your OS or not supported by your hardware.

7.8 7. Input Handling

7.8.1 Keyboard

The `exp.Keyboard` object provides blocking and non-blocking access:

```
// Block until any key — returns the keycode
key, err := exp.Keyboard.Wait()

// Block until a specific key
err := exp.Keyboard.WaitKey(control.K_SPACE)

// Block until one of several keys, with optional timeout (-1 = no timeout)
// Returns 0 on timeout, sdl.EndLoop on ESC/quit
key, err := exp.Keyboard.WaitKeys([]control.Keycode{control.K_F, control.K_
↪  J}, 3000)

// Same, but also returns reaction time in milliseconds from the call site
key, rt, err := exp.Keyboard.WaitKeysRT([]control.Keycode{control.K_F,
↪  control.K_J}, 3000)

// Hardware-precision version: returns the SDL event's nanosecond timestamp
key, eventTS, err :=
↪  exp.Keyboard.GetKeyEventTS([]control.Keycode{control.K_F, control.K_J},
↪  3000)

// Non-blocking poll — returns 0 if nothing pressed
key, err := exp.Keyboard.Check()

// Query whether a key is physically held down right now (no event queue
↪  involvement)
held := exp.Keyboard.IsPressed(control.K_SPACE)

// Wait for KEY_UP on a specific key; returns its SDL hardware timestamp
↪  (nanoseconds)
upTS, err := exp.Keyboard.WaitKeyReleaseTS(key, -1)

// Drain all pending key events (use before a new trial to discard stale
↪  presses)
exp.Keyboard.Clear()
```

`GetKeyEventTS` "get" semantics. Unlike the `Wait*` functions, `GetKeyEventTS` reads from the SDL event queue and returns *immediately* if a matching event is already there — it only blocks when the queue has no matching event. This means a keypress

that happened during `exp.Wait` or any other intervening code is consumed instantly on the next `GetKeyEventTS` call, with its original hardware timestamp intact.

When to call `Clear()`. Despite living on `exp.Keyboard`, `Clear()` drains the **entire** SDL event queue — keyboard, mouse, gamepad, and everything else — because SDL uses a single shared queue. Call it at the start of a trial (before `ShowTS`) to discard stale events from any device. Do **not** call it after `ShowTS`: the participant may have already responded, and `Clear()` would silently discard that event before `GetKeyEventTS` or `GetPressEventTS` ever sees it.

`WaitKeysRT` vs `GetKeyEventTS`:

`WaitKeysRT` measures elapsed time from the moment the call is made (using `sdl.Ticks()`, millisecond granularity). In a single-stimulus trial this is a good approximation of RT from stimulus onset, but it accumulates any delay between `Show()` returning and `WaitKeysRT` being called.

`GetKeyEventTS` returns the SDL3 event's own `Timestamp` field — the nanosecond time at which the hardware key-down event was generated. Combined with `exp.ShowTS(stim)`, which records the nanosecond time of the VSYNC flip, reaction time is simply:

```
key, rtMs, _ := exp.ShowAndGetRT(target, responseKeys, 3000)
```

Internally `ShowAndGetRT` calls `exp.Keyboard.Clear()`, then `exp.ShowTS` to get the VSYNC flip timestamp, then `GetKeyEventTS`, and returns the difference in milliseconds. When you need the raw nanosecond timestamps (e.g. for EEG trigger alignment), you can call the primitives directly:

```
onset, _ := exp.ShowTS(target)   // nanoseconds at VSYNC flip
key, eventTS, _ := exp.Keyboard.GetKeyEventTS(responseKeys, 3000)
rtNS := int64(eventTS - onset)   // nanoseconds; divide by 1e6 for ms
```

This form is also the only correct approach when multiple stimuli are presented in sequence and you need RT relative to a specific one:

```
onset1, _ := exp.ShowTS(prime)   // prime appears
exp.Wait(500)
exp.ShowTS(target)               // target appears 500 ms later
key, eventTS, _ := exp.Keyboard.GetKeyEventTS(responseKeys, 3000)
rtToPrime := int64(eventTS - onset1)  // RT from prime onset
```

Collecting multiple simultaneous responses — `GetKeyEventsTS`:

In rare cases you may want to capture *all* key events, not just the first. `GetKeyEventsTS` is designed for detecting bilateral responses (e.g. both hands pressing at once) and works in two phases:

1. **Phase 1** — blocks until the first matching key arrives, respecting the timeout exactly like `GetKeyEventTS`.
2. **Phase 2** — waits up to 50 ms for any additional matching keys.

The second phase is necessary because human "simultaneous" presses are rarely truly

simultaneous — the two KEY_DOWN events typically arrive 10–50 ms apart. Without this window, a non-blocking drain after the first key would miss the second key almost every time.

```
events, err := exp.Keyboard.GetKeyEventsTS(responseKeys, 3000)
if len(events) > 0 {
    firstKey := events[0].Key
    firstTS  := events[0].TimestampNS
    rtNS     := int64(firstTS - onset)
}
if len(events) == 2 {
    lagNS := int64(events[1].TimestampNS - events[0].TimestampNS)
}
```

In the common single-response case only one event is returned, at the cost of at most 50 ms of extra latency before the function returns. For the typical single-response trial, `GetKeyEventTS` avoids that overhead entirely.

Recording all presses over a fixed duration — `CollectKeyEventsTS`:

When the goal is to record *everything* the participant presses within a known time window (finger-tapping, free-response periods, stimulus-stream logging), use `CollectKeyEventsTS`. Unlike `GetKeyEventsTS`, it always runs for the full duration — it does not return early on the first keypress:

```
// Record every tap of F or J over 5 seconds
exp.Keyboard.Clear()
onset, _ := exp.ShowTS(stim)
events, err := exp.Keyboard.CollectKeyEventsTS(
    []control.Keycode{control.K_F, control.K_J}, 5000)
for _, ev := range events {
    rtNS := int64(ev.TimestampNS - onset)
    fmt.Printf("key %s at %d ms\n", ev.Key.KeyName(), rtNS/1_000_000)
}
```

Pass `keys = nil` to capture any key. Pass `durationMS = 0` to do a non-blocking drain of whatever is already in the queue. Returns an empty (non-nil) slice if nothing was pressed.

Querying whether a key is currently held — `IsPressed`:

`IsPressed` queries SDL's internal scancode state array and returns `true` if the key is physically depressed at the instant of the call, regardless of the event queue. It is useful in animation loops that need to react continuously to a held key, or to verify that the participant has released a key before starting a new trial:

```
for exp.Keyboard.IsPressed(control.K_SPACE) {
    // do something while SPACE is held
    time.Sleep(10 * time.Millisecond)
}
```

Because `IsPressed` does not consume queue events, it can be called freely alongside

`GetKeyEventTS` without interfering with the event stream.

Measuring keypress duration — `WaitKeyReleaseTS`:

`WaitKeyReleaseTS` blocks until a KEY_UP event arrives for the specified key and returns its SDL3 hardware timestamp in nanoseconds. Together with the KEY_DOWN timestamp from `GetKeyEventTS`, this gives nanosecond-precision keypress duration:

```
key, downTS, _ := exp.Keyboard.GetKeyEventTS(responseKeys, -1)
// ... optionally update display while key is held ...
upTS, _ := exp.Keyboard.WaitKeyReleaseTS(key, 5000)
durationNS := upTS - downTS           // nanoseconds
durationMS := durationNS / 1_000_000  // milliseconds
```

This is the recommended approach for paradigms where press duration is a dependent variable (e.g., force-choice hold-to-respond, finger-tapping, or hold-triggered responses).

7.8.2 Mouse

```
// Block until any button
btn, err := exp.Mouse.WaitPress()

// RT in milliseconds from call site (mirrors Keyboard.WaitKeysRT)
btn, rt, err := exp.Mouse.WaitPressRT(3000)

// Hardware-precision version: returns the SDL event's nanosecond timestamp
btn, eventTS, err := exp.Mouse.GetPressEventTS(3000)

// Non-blocking poll
btn, err := exp.Mouse.Check()

// Query whether a button is physically held down right now
held := exp.Mouse.IsPressed(sdl.BUTTON_LEFT)

// Wait for MOUSE_BUTTON_UP; returns its SDL hardware timestamp
 ↪  (nanoseconds)
upTS, err := exp.Mouse.WaitButtonReleaseTS(btn, 5000)

// Current cursor position in center-based coordinates
x, y := exp.Mouse.Position()

// Show / hide cursor
exp.Mouse.ShowCursor(false)
```

The cursor is hidden by default. `Initialize()` (and therefore `NewExperimentFromFlags`) hides the mouse pointer, so it never sits on top of your stimuli. Mouse-driven paradigms must ask for it back, right after the experiment is created:

```
exp := control.NewExperimentFromFlags("Mouse Tracking", control.Black,
 ↪  control.White, 36)
defer exp.End()
exp.ShowCursor()   // participants need to see what they are pointing at
```

Setting `exp.CursorVisible = true` before `Initialize()` does the same thing. The participant-info dialog always shows the cursor while it is open, whatever the experiment's setting, since it is clicked.

Button values: `sdl.BUTTON_LEFT`, `sdl.BUTTON_MIDDLE`, `sdl.BUTTON_RIGHT`, `sdl.BUTTON_X1`, `sdl.BUTTON_X2`.

`IsPressed` and `WaitButtonReleaseTS` mirror the keyboard's `IsPressed` and `WaitKeyReleaseTS` and can be used to measure click-hold duration:

```
btn, downTS, _ := exp.Mouse.GetPressEventTS(-1)
upTS, _        := exp.Mouse.WaitButtonReleaseTS(btn, 5000)
durationMS     := int64(upTS-downTS) / 1_000_000
```

7.8.3 Multi-device input — `WaitAnyEventTS`

If you want to accept a response from *either* the keyboard or the mouse (or both), use `exp.WaitAnyEventTS` instead of calling `Keyboard` and `Mouse` separately:

```
onset, _ := exp.ShowTS(stim)

// Accept F or J key, or any mouse button click, up to 3 s
ev, err := exp.WaitAnyEventTS(
    []control.Keycode{control.K_F, control.K_J},
    true,   // catchMouse
    3000,
)

rtNS := int64(ev.TimestampNS - onset)  // hardware-precision RT, nanoseconds
rtMs := rtNS / 1_000_000

switch ev.Device {
case apparatus.DeviceKeyboard:
    fmt.Printf("key %d, RT %d ms\n", ev.Key, rtMs)
case apparatus.DeviceMouse:
    fmt.Printf("mouse button %d, RT %d ms\n", ev.Button, rtMs)
}
```

Pass `keys = nil` to accept any key. Pass `catchMouse = false` to ignore the mouse. On timeout, returns a zero `InputEvent` with `nil` error.

7.8.4 Key codes

Key codes are re-exported in `control` so experiment code never needs to import `go-sdl3` just to name a key. Coverage includes the full alphabet (`K_A` … `K_Z`), the digit row (`K_0` … `K_9`), the numeric keypad (`K_KP_0` … `K_KP_9`, `K_KP_ENTER`, `K_KP_PLUS`, `K_KP_MINUS`), navigation/control keys (`K_SPACE`, `K_ESCAPE`, `K_RETURN`, `K_BACKSPACE`, `K_TAB`, the arrow keys), and common punctuation (`K_MINUS`, `K_PLUS`, `K_EQUALS`, `K_LEFTBRACKET`, `K_RIGHTBRACKET`):

```
key, _ := exp.Keyboard.Wait()
if key == control.K_A { ... }
```

For a rarely-used code not in `defaults.go`, fall back to `go-sdl3/sdl` directly (e.g. `sdl.K_SEMICOLON`).

7.8.5 Response Devices

All of the input methods above are device-specific: `GetKeyEventTS` for keyboards, `GetPressEventTS` for the mouse, and so on. If the experiment should run equally well with different input hardware — a keyboard in one lab, a response box in another — device-specific calls scatter conditional logic throughout the trial loop.

`ResponseDevice` is a single interface that abstracts over all of them:

```
// in package io
type ResponseDevice interface {
    WaitResponse(ctx context.Context) (Response, error)
    DrainResponses(ctx context.Context) error
    Close() error
}
```

`WaitResponse` blocks until a response is detected and returns a `Response`:

```
type Response struct {
    Source  apparatus.DeviceKind  // which device fired
    Code    uint32         // key code, button index, or TTL bitmask
    RT      time.Duration  // elapsed from WaitResponse call
    Precise bool           // timing quality (see below)
}
```

Wrap any input device with the matching adapter:

```
// Keyboard (SDL hardware timestamps → Precise: true)
var rd apparatus.ResponseDevice = &apparatus.KeyboardResponseDevice{KB:
 ↪  exp.Keyboard}

// Mouse (SDL hardware timestamps → Precise: true)
var rd apparatus.ResponseDevice = &apparatus.MouseResponseDevice{M:
 ↪  exp.Mouse}

// TTL response box, e.g. MEGTTLBox (software poll → Precise: false)
```

```
box, _ := triggers.NewMEGTTLBox("/dev/ttyACM0")
var rd apparatus.ResponseDevice = apparatus.NewTTLResponseDevice(box,
 ↪  5*time.Millisecond)
```

All three look identical inside a trial loop:

```
onset, _ := exp.ShowTS(stim)
_ = rd.DrainResponses(ctx)        // clear stale presses from previous trial
resp, err := rd.WaitResponse(ctx)
if err != nil { /* ESC, timeout, etc. */ }

rtMs := resp.RT.Milliseconds()     // always valid
```

7.8.5.1 Timing precision and the `Precise` flag

The `Precise` field tells you whether `RT` came from a nanosecond hardware event timestamp or a software poll:

Device	`Precise`	RT accuracy
Keyboard, Mouse, Gamepad	`true`	SDL3 hardware event timestamp, nanosecond resolution — identical to using `GetKeyEventTS` directly
MEGTTLBox, DLPIO8	`false`	`time.Now()` at poll detection; accuracy bounded by poll interval (default 5 ms)

When `Precise` is `true`, the RT in the `Response` is computed from `sdl.TicksNS()` captured at the `WaitResponse` call versus the SDL3 hardware event timestamp — the same nanosecond clock used by `Screen.FlipTS`. It is therefore suitable for stimulus-onset-locked RT:

```
onset, _ := exp.ShowTS(stim)
resp, _ := rd.WaitResponse(ctx)
if resp.Precise {
    // resp.RT was computed from SDL hardware timestamps: nanosecond
     ↪  accuracy
    rtFromOnset := time.Duration(int64(onset) + resp.RT.Nanoseconds())  //
 ↪  approximate
}
```

When `Precise` is `false`, `RT` is still a valid elapsed duration — it just has ~poll-interval jitter (5 ms by default). For typical behavioral RT tasks this is acceptable; for EEG/MEG experiments requiring sub-millisecond synchrony, use the TTL output path for stimulus marking and treat RT as an approximate measure or use a hardware response box with sub-ms timestamping.

7.9 8. Data Collection

7.9.1 Output files

When `exp.End()` is called, two files are written to `~/goxpy_data/`:

File	Contents
`<expName>_sub-<NNN>_date-<YYYYMMDD>-<HHMMSS>.csv`	Pure CSV data rows — directly importable by Excel, R, or pandas
`<expName>_sub-<NNN>_date-<YYYYMMDD>-<HHMMSS>-info.txt`	Session metadata: start/end time, hostname, framework version, participant info, the machine and OS it ran on (model, kernel, desktop/compositor, CPU, every GPU and its driver, sound server and version), and the display and audio configuration SDL actually opened

Both files open automatically on `Initialize()` and are flushed to disk when `exp.End()` is called. Any spaces in `<expName>` are replaced by `-` so the resulting filenames never contain whitespace.

7.9.2 Choosing the output directory

By default, data files are written to `~/goxpy_data/` (the folder name is the `results.DataFileDirectory` constant, appended to the user's home directory). To write them elsewhere, call `exp.SetOutputDirectory()` **before** `Initialize()` opens the files:

```
exp := control.NewExperiment("My Experiment", control.Black, control.White,
 ↪  32)
exp.SetOutputDirectory("/path/to/mydata")  // must come before Initialize()
if err := exp.Initialize(); err != nil { log.Fatal(err) }
defer exp.End()
```

The directory is created if it does not exist.

Because `NewExperimentFromFlags` calls `Initialize()` for you, overriding the directory requires the lower-level `NewExperiment(...) + Initialize()` path shown above rather than `NewExperimentFromFlags`.

7.9.3 Declaring column names

Call this once, before the trial loop:

```
exp.AddDataVariableNames([]string{"condition", "response", "rt_ms",
 ↪  "correct"})
```

Subject ID is always prepended automatically — do not include it in the name list.

7.9.4 Adding rows

```
exp.Data.Add(condition, response, rt, correct)
```

Fields are written in the same order as the variable names. Any type that has a meaningful `fmt.Sprint` representation works: `int`, `float64`, `bool`, `string`. Fields containing commas or quotes are escaped per RFC 4180.

7.9.5 Example CSV output

```
subject_id,condition,response,rt_ms,correct
3,"congruent","F",412,true
3,"incongruent","J",538,false
```

Numbers and booleans are unquoted; strings are always double-quoted.

7.9.6 Saving mid-experiment

For long experiments it is good practice to call `exp.Data.Save()` after each block. This flushes both files to disk so that data up to that point is not lost if the experiment crashes.

7.10 9. Stimuli: Lifecycle and Preloading

7.10.1 GPU textures are lazily allocated

Creating a stimulus (e.g., `stimuli.NewTextLine(...)`) is cheap — it allocates a Go struct and stores the parameters, but does no GPU work. The SDL texture is created on the **first call to `Draw`**. This means:

- You can safely create all your stimuli before the run loop.
- The first presentation of each stimulus will be slightly slower due to texture upload.

For timing-sensitive presentations, preload explicitly:

```
stimuli.PreloadVisualOnScreen(exp.Screen, myStim)
// or, for a slice:
stimuli.PreloadAllVisual(exp.Screen, []stimuli.VisualStimulus{stim1, stim2,
 ↪  stim3})
```

Call this after `exp.Run` starts (the renderer is ready), but before the critical section:

```
err := exp.Run(func() error {
    // Preload everything while showing instructions
    exp.ShowInstructions("Loading, please wait...")
    stimuli.PreloadAllVisual(exp.Screen, stimSlice)
```

```
    // Now timing-sensitive trials
    for _, stim := range stimSlice { ... }
    return control.EndLoop
})
```

7.10.2 Releasing textures

If you generate many stimuli dynamically (e.g., hundreds of unique text strings), call `stim.Unload()` after each trial to free the GPU memory:

```
for _, word := range wordList {
    stim := stimuli.NewTextLine(word, 0, 0, control.White)
    exp.Show(stim)
    exp.Keyboard.Wait()
    stim.Unload()  // release GPU texture
}
```

For a fixed, small set of stimuli created once and reused throughout the experiment, you never need to call `Unload` — `exp.End()` handles cleanup.

7.10.3 Writing a custom stimulus

Any type that implements `VisualStimulus` can be passed to `exp.Show`:

```
type VisualStimulus interface {
    Present(screen *apparatus.Screen, clear, update bool) error
    Preload() error
    Unload() error
    Draw(screen *apparatus.Screen) error
    GetPosition() sdl.FPoint
    SetPosition(pos sdl.FPoint)
}
```

Embed `stimuli.BaseVisual` to get the position methods and no-op `Preload`/`Unload` for free, then implement only `Draw` and `Present`:

```
type MyStimulus struct {
    stimuli.BaseVisual
    // your fields
}

func (m *MyStimulus) Draw(screen *apparatus.Screen) error {
    // draw using screen.Renderer SDL calls
    return nil
}

func (m *MyStimulus) Present(screen *apparatus.Screen, clear, update bool)
 ↪  error {
    return stimuli.PresentDrawable(m, screen, clear, update)
```

```
}
```

7.10.4 Interactive widgets

Two stimuli present a UI and collect a structured response from the participant in a blocking loop. Neither is timing-critical, so they use `sdl.WaitEvent` rather than VSYNC polling.

`Menu` — a numbered, keyboard-navigable list:

```
m := stimuli.NewMenu([]string{"Beginner", "Expert", "Practice run"})
// Optional customisation:
m.HighlightColor = sdl.Color{R: 255, G: 220, B: 0, A: 255}

idx, err := m.Get(exp.Screen, exp.Keyboard, 0)
// idx is 0-based; -1 + sdl.EndLoop on ESC/quit
```

The selected item is shown with a `>` prefix in `HighlightColor`; all others use `TextColor`. Navigation: UP/DOWN arrows move the highlight; ENTER or SPACE confirms; digit keys 1–9 (0 for tenth) select directly without requiring confirmation.

`ChoiceGrid` — a grid of labelled buttons, activated by mouse click or matching key:

```
cg := stimuli.NewChoiceGrid([]string{"B","C","D","F","G"}, 3, "Pick 3
↪  letters:")
selections, err := cg.Get(exp.Screen, exp.Keyboard)
// selections is []string in press order
```

7.11 10. High-Precision Streams (RSVP)

For paradigms that present stimuli at high speed — RSVP, attentional blink, priming — the standard `exp.Show` / `exp.Wait` cycle is not precise enough. The stream functions provide frame-accurate presentation:

- GC is **disabled** for the duration of the stream.
- Every onset and offset is aligned to a **VSYNC boundary**.
- A `TimingLog` records the predicted and actual onset for each stimulus.
- All keyboard and mouse events that occur during the stream are captured.

7.11.1 Text stream

The simplest entry point:

```
words := []string{"CHAIR", "RIVER", "TIGER", "CLOCK", "STONE"}
on  := 150 * time.Millisecond
off :=  50 * time.Millisecond
```

```
events, logs, err := stimuli.PresentStreamOfText(
    exp.Screen, words, on, off,
    0, 0,          // center of screen
    control.White,
)
```

7.11.2 Image / mixed stimulus stream

For images or any `VisualStimulus`:

```
stims := []stimuli.VisualStimulus{pic1, fixation, pic2, fixation}

// Regular cadence (all stimuli share the same on/off duration)
elements := stimuli.MakeRegularVisualStream(stims, 100*time.Millisecond, 0)

// Irregular cadence (individual onset times and durations, in ms)
elements, err := stimuli.MakeVisualStream(stims, onsetMs, durationMs)

events, logs, err := stimuli.PresentStreamOfImages(exp.Screen, elements, 0,
 ↪  0)
```

7.11.3 Reading events from the stream

Each `UserEvent` carries two timestamps:

- `Timestamp` — a `time.Duration` relative to stream start, Go monotonic clock, millisecond precision. Useful for quick inspection.
- `TimestampNS` — the SDL3 hardware event timestamp in nanoseconds, same clock as `Screen.FlipTS`. Use this for sub-millisecond RT computation.

```
for _, ev := range events {
    if ev.Event.Type == sdl.EVENT_KEY_DOWN {
        key := ev.Event.KeyboardEvent().Key
        t   := ev.Timestamp.Milliseconds()  // ms from stream start (coarse)
        fmt.Printf("key %d pressed at %d ms\n", key, t)
    }
}
```

Use `stimuli.FirstKeyPress` to find the first matching key-down event without writing a loop:

```
if ev, ok := stimuli.FirstKeyPress(events, sdl.K_SPACE); ok {
    fmt.Printf("Space pressed at %d ms\n", ev.Timestamp.Milliseconds())
}
```

Both `ev.Timestamp` (Go clock, ms precision) and `ev.TimestampNS` (SDL3 hardware, nanoseconds) are available on the returned `UserEvent`.

7.11.4 Computing RT from stream events

Because `UserEvent.TimestampNS` and `TimingLog.OnsetNS` are on the same SDL nanosecond clock, reaction time from a specific stimulus is exact:

```
for _, ev := range events {
    if ev.Event.Type == sdl.EVENT_KEY_DOWN {
        for _, l := range logs {
            if ev.TimestampNS >= l.OnsetNS && (l.OffsetNS == 0 ||
             ↪  ev.TimestampNS < l.OffsetNS) {
                rtNS := int64(ev.TimestampNS - l.OnsetNS)
                fmt.Printf("RT from stimulus %d: %d ms\n", l.Index, rtNS/1_
 ↪  000_000)
            }
        }
    }
}
```

7.11.5 Reading the timing log

```
for i, l := range logs {
    // Read the authoritative SDL-clock fields, not the Go-clock
     ↪  ActualOnset/
    // ActualOffset diagnostics. Achieved on-screen duration vs target =
     ↪  jitter.
    shownMs := int64(l.OffsetNS-l.OnsetNS) / 1_000_000
    jitter := shownMs - l.TargetOn.Milliseconds()
    fmt.Printf("stimulus %d: target %d ms, shown %d ms, jitter %d ms\n",
        i, l.TargetOn.Milliseconds(), shownMs, jitter)
}
```

`OnsetNS` and `OffsetNS` give the SDL3 nanosecond timestamps of the actual VSYNC flips that turned each stimulus on and off — the authoritative timing record, on the same clock as input events. `ActualOnset`/`ActualOffset` are Go-clock diagnostics on a different timebase; never subtract them from an event timestamp or use them as onsets. `OnsetNS` is zero only for a stimulus that was never displayed (zero on-frames); `OffsetNS` is the takedown flip, synthesised from the onset for the final element of a contiguous stream (which has no ISI flip of its own) so it stays on the SDL clock rather than reading a frame early.

Jitter below ±1 frame (±8–17 ms depending on monitor) is normal and expected. Larger jitter indicates system load or GPU driver issues.

7.11.6 Platform timing notes

`PresentStreamOfImages` provides the best timing accuracy that the operating system and display driver allow. The characteristics differ by platform:

Linux

- Without a compositing window manager (e.g. a plain X11 session with no compositor), SDL3's `Present()` blocks until the next VSYNC boundary. Onset jitter is typically **< 1 ms**.
- With a Wayland compositor or an X11 compositing WM (KWin, Mutter, Picom), the compositor controls buffer swaps. Onset jitter is typically **1–3 ms**; the compositor may add one frame (~17 ms at 60 Hz) of fixed latency.
- For the most reliable timing on Linux, disable the compositor or use a plain X11 session.

macOS (Metal)

- The macOS WindowServer compositor is **always active** — exclusive fullscreen does not bypass it. SDL3 submits each frame to the compositor, which forwards it to the display at the next VSYNC.
- `FlipTS` captures `sdl.TicksNS()` immediately after `Present()` returns; this reflects when the frame was *submitted to the compositor*, not when photons reached the screen. The additional compositor latency is typically **1–2 frames** (~17-33 ms at 60 Hz).

Windows

- In **exclusive fullscreen** mode (`fullscreen=true`), the DWM (Desktop Window Manager) compositor is bypassed. Timing behaviour is similar to Linux without a compositor — jitter is typically **< 1 ms**.
- In **windowed mode**, DWM is always active and adds approximately one frame of compositor latency. Jitter is typically **1–3 ms**.
- For the most reliable timing on Windows, always run in fullscreen mode.

What `OnsetNS` measures

`TimingLog.OnsetNS` is recorded immediately after `screen.FlipTS()` returns, not at the moment photons reach the screen. On top of the compositor latency noted above, there is additional hardware pipeline latency (GPU scan-out, cable propagation, display panel response) of typically **0–2 frames**. This latency is constant across trials and does not affect within-experiment RT precision, but it must be accounted for if absolute (photodiode-verified) onset times are required.

Minimum duration

Because presentation is VSYNC-locked, durations shorter than one frame period (e.g. < 16.7 ms at 60 Hz) are rounded up to the nearest whole frame. A stimulus requested for 50 ms on a 60 Hz display is shown for exactly **3 frames (50.0 ms)**; one requested for 60 ms is shown for **4 frames (66.7 ms)**.

7.11.7 Audio stream

The same model applies to sounds:

```
// All tones must be pre-loaded before the stream
for _, t := range tones {
    t.PreloadDevice(exp.AudioDevice)
}
```

```
elements := stimuli.MakeRegularSoundStream(tones, 200*time.Millisecond,
 ↪  100*time.Millisecond)
events, logs, err := stimuli.PlayStreamOfSounds(elements)
```

7.11.8 Mixed stream

`PresentStreamOfStimuli` presents a single **sequential** stream that mixes visual and audio stimulus types. It reuses the VSYNC-locked visual loop, so visual elements are frame-accurate; audio elements are triggered once at the slot's first flip and the previous frame is **held** for the slot (the last stream visual is re-rendered every frame, so it stays rock-steady rather than relying on GPU buffer persistence) — place a visual just before a sound to keep it visible while the sound plays.

```
tone.PreloadDevice(exp.AudioDevice) // audio elements must be device-bound
 ↪  first

elements := []stimuli.StreamElement{
    {Stimulus: stimuli.NewTextLine("Ready?", 0, 0, control.White),
     ↪  DurationOn: 600 * time.Millisecond, DurationOff: 200 *
     ↪  time.Millisecond},
    {Stimulus: pic,  DurationOn: 800 * time.Millisecond}, // stays on screen
     ↪  during the tone
    {Stimulus: tone, DurationOn: 400 * time.Millisecond, DurationOff: 200 *
     ↪  time.Millisecond},
}
events, logs, err := stimuli.PresentStreamOfStimuli(exp.Screen, elements,
 ↪  0, 0)
```

Concurrent audio-visual overlap (a sound and an animating visual at the same instant) is not yet supported; the stream is strictly one slot at a time.

7.11.9 Per-stimulus onset triggers

To emit a hardware TTL — or run any action — aligned to each stimulus onset, use `PresentStreamOfStimuliHooks(screen, elements, x, y, onFrame, onOnset)`. The `onOnset` `OnsetCallback` (`func(index int, onsetNS uint64) error`) fires once per element **immediately after** its onset VSYNC flip, with `onsetNS == TimingLog[index].OnsetNS`. This is the *post-flip* counterpart to `FrameCallback`, which runs one frame earlier (pre-flip): a trigger emitted from `onOnset` coincides with the logged onset instead of leading it by a frame — the right hook for EEG/MEG synchronisation. Keep it short and non-blocking (emit an edge, clear it from a later `FrameCallback`; never sleep or call a blocking `Pulse`). The pure-audio equivalent is `PlayStreamOfSoundsHook(elements, onOnset)`, firing at the `Play()` instant. See the API reference and the `test_stream_trigger` example.

7.12 11. Audio

7.12.1 Sounds from files or embedded bytes

```
// From a file
snd := stimuli.NewSound("ping.wav")
snd.PreloadDevice(exp.AudioDevice)
snd.Play()
snd.Wait()   // block until playback finishes

// From embedded bytes (go:embed)
//go:embed assets/beep.wav
var beepWav []byte

snd := stimuli.NewSoundFromMemory(beepWav)
snd.PreloadDevice(exp.AudioDevice)
snd.Play()
```

7.12.2 Procedural tones

```
tone := stimuli.NewTone(1000.0, 200*time.Millisecond, 200) // 1 kHz, 200
 ↪  ms, medium volume
tone.PreloadDevice(exp.AudioDevice)
tone.Play()
```

Volume is 0–255. Use `PreloadDevice` once; then call `Play` repeatedly.

7.12.3 Segment playback with fade

Play only part of a longer sound, with smooth fade-in and fade-out:

```
snd.PlaySegment(1.5, 3.0, 0.05) // play 1.5–3.0 s, 50 ms ramps
```

This is useful for stimuli like vowels extracted from longer recordings.

7.12.4 Built-in feedback sounds

```
exp.Audio.PlayBuzzer()   // play the embedded error/incorrect sound
 ↪  asynchronously
exp.Audio.PlayCorrect()  // play the embedded correct/reward sound
 ↪  asynchronously
```

Both return immediately; the sound plays in the background.

7.12.5 Synchronous vs asynchronous

`snd.Play()` starts playback and returns immediately. `snd.Wait()` blocks until it finishes. For trial timing where you need to know when a sound ended, call both:

```
snd.Play()
doSomethingElse()
snd.Wait()  // wait here if needed
```

For fire-and-forget feedback sounds, call `snd.Play()` without `snd.Wait()`.

7.12.6 Microphone recording

goxpyriment can capture audio from the system microphone via SDL3's recording API. The primary use cases are:

- **Picture naming** — measure the latency from image onset to the participant's first vocalization.
- **Vocal shadowing** — measure the latency between a spoken or tonal stimulus and the participant's repetition of it.

Open the microphone once, before the trial loop:

```
// nil uses the default recording spec: F32LE mono 44100 Hz
if err := exp.OpenMicrophone(nil); err != nil {
    log.Fatalf("microphone unavailable: %v", err)
}
// exp.Microphone is now ready; it is closed automatically by exp.End()
```

To select a custom format (e.g. higher sample rate or stereo):

```
spec := &sdl.AudioSpec{Format: sdl.AUDIO_F32LE, Channels: 1, Freq: 48000}
exp.OpenMicrophone(spec)
```

To list all connected recording devices and open a specific one:

```
devices, _ := apparatus.GetRecordingDevices()
for _, d := range devices {
    name, _ := d.Name()
    fmt.Println(d, name)
}
mic, _ := apparatus.NewMicrophoneFromDevice(devices[0], nil)
exp.Microphone = mic
```

7.12.7 Voice key — detecting vocal onset

A `VoiceKey` monitors the microphone and fires when the amplitude exceeds a threshold. It is the goxpyriment equivalent of a hardware voice key or PsychoPy's `OnsetVoiceKey`.

```
vk := apparatus.NewVoiceKey(exp.Microphone, 0.03, 128)
// threshold = 0.03 (RMS amplitude 0–1 for F32LE)
// windowSz  = 128 samples ≈ 2.9 ms at 44100 Hz
```

Per-trial pattern:

```
// 1. Arm: flush the mic buffer and start capturing.
//    Returns the SDL3 nanosecond timestamp of capture start.
captureStartNS, _ := vk.Arm()

// 2. Present the stimulus.
imageOnsetNS, _ := exp.ShowTS(picture)   // picture naming
// – or –
stimulusNS, _ := tone.PlaySyncedWithFlip(exp.Screen)  // shadowing

// 3. Wait for the vocal response (2-second window).
onsetNS, pcm, err := vk.WaitOnset(captureStartNS, 2000)
if errors.Is(err, apparatus.ErrVoiceTimeout) {
    // participant did not respond in time
}

// 4. Compute RT (both timestamps are on the SDL3 nanosecond clock).
rtMs := int64(onsetNS - imageOnsetNS) / 1_000_000
```

`WaitOnset` also returns `pcm` — the raw PCM bytes captured during the trial. Save them as WAV for offline verification (see below).

Threshold calibration. A value of 0.02–0.05 works in a quiet room. If you observe:

- **Spuriously short RTs (< 100 ms)** — the threshold is too low; it is triggering on breath noise, lip smacks, or microphone bleed from the speakers. Raise the threshold or use headphones.
- **`detected = false` (missed responses)** — the threshold is too high for soft or breathy voices. Lower it.

Always verify using the saved WAV files rather than adjusting the threshold blindly.

7.12.8 Timing: microphone and screen on the same clock

Both `imageOnsetNS` (from `exp.ShowTS` or `screen.FlipTS`) and the voice onset computed by `WaitOnset` use `sdl.TicksNS()` — SDL3's monotonic nanosecond clock. No cross-clock conversion is needed:

```
vk.Arm()  ──────────────────────────── captureStartNS  (sdl.TicksNS())
                │
                │  exp.ShowTS(picture)
                │  ↳ VSYNC flip ──────── imageOnsetNS   (sdl.TicksNS())
                │
                │  [participant starts speaking at sample N]
                │
```

```
                └── onsetNS = captureStartNS + N × 1e9 / sampleRate
                                                    (sdl.TicksNS() domain)

RT = (onsetNS − imageOnsetNS) / 1 000 000  [ms]
```

Call `vk.Arm()` just before showing the stimulus so the microphone is already capturing when the image appears. There is no need to arm it long in advance.

7.12.9 Saving WAV files for verification

Always save a WAV file for each trial and spot-check a random subset after the session. Voice keys can fire on breaths, lip smacks, or room noise; the WAV is the ground truth.

```
apparatus.WriteWAV(
    fmt.Sprintf("sub-%03d_trial-%02d.wav", exp.SubjectID, trial),
    exp.Microphone.Spec,
    pcm,
)
```

`WriteWAV` supports F32LE, S16LE, S32LE, and U8 formats and writes a standard RIFF/WAV file that any audio editor can open.

For post-hoc re-analysis of saved files (e.g. applying a different threshold), use `apparatus.ScanOnset`:

```
sampleIndex := apparatus.ScanOnset(pcm, newThreshold, 128)
if sampleIndex >= 0 {
    revisedOnsetNS = apparatus.SampleOnsetNS(captureStartNS, sampleIndex,
 ↪  int(exp.Microphone.Spec.Freq))
}
```

7.12.10 Acoustic feedback (shadowing and AV tasks)

When audio plays through laptop built-in speakers, the microphone picks up the playback signal and the voice key triggers on it rather than on the participant's voice. This produces spuriously short latencies (sometimes < 0 ms if the mic captures the speaker before the amp threshold is crossed).

Always use headphones for playback in any task that simultaneously records and plays audio.

If headphones are unavailable, a simple analysis-side fix is to reject any onset that falls within a guard interval after stimulus onset (e.g. discard `onsetNS - stimulusNS < 100 ms`).

7.13 12. Experimental Design and Randomization

7.13.1 When to use `design.Block` / `design.Trial`

The `design` package provides a structured `Experiment → Block → Trial` hierarchy with string-keyed factors. Use it when:

- You have a multi-block experiment and want `exp.Design.Blocks` to drive the loop.
- You want between-subjects Latin-square counterbalancing (`AddBWSFactor`).

For simple single-block experiments, a plain Go slice of a custom struct is often more readable (and type-safe). Both approaches are valid.

7.13.2 Building a design

```
block := design.NewBlock("main")
for _, cond := range []string{"congruent", "incongruent"} {
    t := design.NewTrial()
    t.SetFactor("condition", cond)
    t.SetFactor("soa", 200)
    block.AddTrial(t, 20, false)  // 20 copies, appended in order
}
block.ShuffleTrials()
exp.AddBlock(block, 1)
```

Iterate at runtime:

```
for _, blk := range exp.Design.Blocks {
    for _, trial := range blk.Trials {
        cond := trial.GetFactor("condition").(string)
        soa  := trial.GetFactor("soa").(int)
        // ...
    }
}
```

7.13.3 Randomization helpers

The `design` package provides randomization functions that work on any slice type:

```
// In-place shuffle (generic — works on any []T)
design.ShuffleList(mySlice)

// Random integer in [a, b] inclusive
design.RandInt(500, 1500)

// Random element from a slice
word := design.RandElement(wordList)

// Weighted coin flip
```

```
if design.CoinFlip(0.75) { /* 75% chance */ }

// Truncated normal sample in [a, b]
design.RandNorm(800.0, 1200.0)

// Shuffled integer range [first, last]
order := design.RandIntSequence(0, len(stimuli)-1)
```

7.13.4 Between-subjects counterbalancing

```
// Register once during setup
exp.AddBWSFactor("mapping", []interface{}{"F=left/J=right",
 ↪  "F=right/J=left"})

// At runtime — returns the condition assigned to this subject's ID
mapping := exp.GetPermutedBWSFactorCondition("mapping").(string)
```

The assignment follows a balanced Latin square so that conditions rotate across subjects (subject 1 → condition A, subject 2 → condition B, subject 3 → condition A, …).

7.13.5 Constrained shuffling

For designs where you need to prevent the same condition from appearing more than N times in a row, use block-level `AddTrial` with `randomPosition: true`, which inserts each trial at a random position rather than appending:

```
for _, cond := range conditions {
    t := design.NewTrial()
    t.SetFactor("condition", cond)
    block.AddTrial(t, 5, true)  // 5 copies each, randomly interleaved
 ↪  during construction
}
```

7.14 13. Embedding Stimuli and Trial Lists in the Executable

One of Go's most useful deployment features is `//go:embed`: a compiler directive that copies files from disk into the binary at build time. The result is a **single self-contained executable** that carries all its images, sounds, and trial-list CSV files with it — no separate `assets/` folder, no installer, no missing-file errors on lab computers.

> **When to embed.** Embedding is the right choice for stimuli that are fixed at design time: images, sounds, and pre-generated trial lists. It is not appropriate for stimuli you generate procedurally at runtime (textures drawn

with `Canvas`, tones created with `NewTone`, text rendered from `NewTextLine`), or for data files that are written during the experiment.

7.14.1 13.1 Project layout

Collect all static assets in a subdirectory alongside `main.go`:

```
myexperiment/
  main.go
  assets/
    stim_A.png
    stim_B.png
    feedback_correct.wav
    trials.csv
```

7.14.2 13.2 Embedding individual image and sound files

Import `_ "embed"` (blank identifier, needed to activate the directive) and declare a `[]byte` variable per file:

```go
package main

import _ "embed"

//go:embed assets/stim_A.png
var stimAImg []byte

//go:embed assets/stim_B.png
var stimBImg []byte

//go:embed assets/feedback_correct.wav
var feedbackWav []byte
```

Pass the byte slices to the `FromMemory` constructors:

```go
import "github.com/chrplr/goxpyriment/stimuli"

stimA := stimuli.NewPictureFromMemory(stimAImg, 0, 0)
stimB := stimuli.NewPictureFromMemory(stimBImg, 0, 0)
feedback := stimuli.NewSoundFromMemory(feedbackWav)
```

These constructors behave identically to `NewPicture` and `NewSound`; textures are still allocated lazily on first `Draw`, and `PreloadAllVisual` / `PreloadDevice` work the same way.

7.14.3 13.3 Embedding a whole stimulus directory

When you have many images (e.g. one per trial) it is impractical to declare a variable for each one. Embed the whole directory as an `embed.FS` and iterate over it at runtime:

```
package main

import "embed"

//go:embed assets/stimuli
var stimuliFS embed.FS
```

Load every PNG in the directory into a map:

```
import (
    "fmt"
    "github.com/chrplr/goxpyriment/stimuli"
)

pics := map[string]*stimuli.Picture{}

entries, err := stimuliFS.ReadDir("assets/stimuli")
if err != nil {
    log.Fatal(err)
}
for _, e := range entries {
    if e.IsDir() {
        continue
    }
    data, err := stimuliFS.ReadFile("assets/stimuli/" + e.Name())
    if err != nil {
        log.Fatal(err)
    }
    pics[e.Name()] = stimuli.NewPictureFromMemory(data, 0, 0)
}

// Use by filename:
exp.Show(pics["face_happy.png"])
```

Or load a specific file by name:

```
data, _ := stimuliFS.ReadFile(fmt.Sprintf("assets/stimuli/face_%s.png",
 ↪  condition))
stim := stimuli.NewPictureFromMemory(data, 0, 0)
```

7.14.4 13.4 Embedding a CSV trial list

A CSV file that defines trial order and parameters is a natural fit for embedding. The workflow is: embed → read into `[]byte` → wrap in `bytes.NewReader` → parse with `encoding/csv`.

Example `assets/trials.csv`:

```
condition,soa_ms,correct_key
congruent,200,F
incongruent,200,J
congruent,400,F
incongruent,400,J
```

Embedding and parsing:

```
package main

import (
    "bytes"
    _ "embed"
    "encoding/csv"
    "log"
    "strconv"
)

//go:embed assets/trials.csv
var trialsCSV []byte

type trial struct {
    condition  string
    soaMs      int
    correctKey string
}

func loadTrials() ([]trial, error) {
    r := csv.NewReader(bytes.NewReader(trialsCSV))
    records, err := r.ReadAll()
    if err != nil {
        return nil, err
    }
    var trials []trial
    for i, rec := range records {
        if i == 0 {
            continue // skip header row
        }
        soa, _ := strconv.Atoi(rec[1])
        trials = append(trials, trial{
            condition:  rec[0],
            soaMs:      soa,
            correctKey: rec[2],
        })
    }
    return trials, nil
}
```

For **tab-separated** files, set `r.Comma = '\t'` immediately after creating the reader.

For **semicolon-separated** files (common in European locales), set `r.Comma = ';'`.

Use the loaded trials in the experiment loop:

```
trials, err := loadTrials()
if err != nil {
    log.Fatalf("loading trials: %v", err)
}

err = exp.Run(func() error {
    exp.ShowInstructions("Press F or J to respond.")
    for _, t := range trials {
        exp.Blank(500)
        stim := stimuli.NewTextLine(t.condition, 0, 0, control.White)
        key, rt, _ := exp.ShowAndGetRT(stim, []control.Keycode{control.K_F,
↪   control.K_J}, 3000)
        correct := string(key.KeyName()) == t.correctKey
        exp.Data.Add(t.condition, t.soaMs, rt, correct)
    }
    return control.EndLoop
})
```

7.14.5 13.5 Mixing embedded and runtime-generated stimuli

Embedding is selective — embed only what is fixed, generate the rest at runtime:

```
//go:embed assets/mask.png
var maskImg []byte

// Fixed stimulus: loaded from embedded bytes
mask := stimuli.NewPictureFromMemory(maskImg, 0, 0)

// Dynamic stimulus: generated from trial parameters
target := stimuli.NewTextLine(word, 0, 0, control.White)
```

7.14.6 13.6 Build and deployment

Nothing changes in the build command:

```
go build -o myexperiment .
```

The `assets/` directory must exist on disk at build time but is **not needed at runtime**. Copy the single binary to any lab computer running the same OS:

```
# Cross-compile for Windows from Linux
GOOS=windows GOARCH=amd64 go build -o myexperiment.exe .

# Cross-compile for macOS (Apple Silicon)
```

```
GOOS=darwin GOARCH=arm64 go build -o myexperiment-mac .
```

The embedded assets are included in all cross-compiled outputs automatically.

> **Binary size.** Each embedded file adds its uncompressed size to the binary. A typical experiment with a few dozen PNG images and WAV files will add 5–50 MB — acceptable for a self-contained research tool. For very large stimulus sets (hundreds of high-resolution images, video files), consider distributing them as a separate archive alongside the binary instead of embedding.

7.15 14. Animated Stimuli

Three functions run self-contained VSYNC-locked animation loops. All three:

- Disable GC for the duration of the loop.
- Drain the SDL event queue before the first frame.
- Return a `MotionResult` with the response key, mouse button, and reaction time.

```
type MotionResult struct {
    Key    sdl.Keycode  // interrupt key pressed (0 if none)
    Button uint8        // mouse button (0 if none)
    RTms   int64        // ms from first frame to response; total elapsed on
 ↪  timeout
}
```

7.15.1 Moving dot cloud

```
result, err := stimuli.PresentMovingDotCloud(
    exp.Screen,
    100,                          // number of dots
    3.0,                          // dot radius (px)
    150.0,                        // cloud radius (px)
    control.Origin(),             // center at (0, 0)
    150.0,                        // speed in px/sec
    5000,                         // max duration ms (0 = infinite)
    []control.Keycode{control.K_SPACE}, // interrupt keys
    false,                        // catch mouse?
    control.White,                // dot color
    control.Color{A: 0},          // background (transparent)
)
fmt.Printf("RT: %d ms\n", result.RTms)
```

7.15.2 Drifting grating

```
result, err := stimuli.PresentMovingGrating(
    exp.Screen,
    400, 400,              // width, height (px)
    control.Origin(),      // center
    0.0,                   // orientation (degrees; 0 = vertical bars
    ↪  drifting right)
    0.05,                  // spatial frequency (cycles per pixel)
    2.0,                   // temporal frequency (Hz)
    0.8,                   // contrast [0, 1]
    0.5,                   // mean luminance [0, 1]
    3000,                  // max duration ms
    []control.Keycode{control.K_SPACE},
    false,
)
```

7.15.3 Drifting Gabor patch

```
result, err := stimuli.PresentMovingGabor(
    exp.Screen,
    200,                   // bounding box size (px); at least 6×sigma
    30.0,                  // sigma (Gaussian SD in px)
    control.Origin(),      // center
    45.0,                  // orientation
    0.05,                  // spatial frequency (cycles/px)
    3.0,                   // temporal frequency (Hz)
    0.9,                   // contrast
    0.5,                   // mean luminance
    0,                     // max duration (0 = until response)
    []control.Keycode{control.K_SPACE},
    false,
)
```

7.16 15. Video

goxpyriment plays video in two formats, and the choice is a deliberate trade between **timing determinism** and **file size / convenience**. Neither is a general-purpose media player: both exist to put moving images on screen at a known time.

	`.gv`	MPEG-1 (`.mpg`)
Type	`stimuli.GvVideo`, `stimuli.PlayGv`, `media.Movie`	`stimuli.Video`

	`.gv`	MPEG-1 (`.mpg`)
Frame cadence	VSYNC-locked, GC disabled	wall-clock, drops frames under load
Decode cost	constant per frame	varies with I/P/B frame type
Audio	none	MP2 soundtrack
Seeking	O(1), every frame independent	sequential only
Size (10 s, 960×400)	~41 MB	~2 MB
Use for	measured stimuli	instructions, fillers, long clips

Rule of thumb: if the onset time of a frame ends up in your data file, use `.gv`. If the participant is just watching something between trials, MPEG-1 is far more convenient.

7.16.1 Start here

Almost everyone arrives with an `.mp4` and one of two goals.

"I want to play a movie, with its sound." You need MPEG-1, because `.gv` has no audio track. Convert once, then play:

```
ffmpeg -i movie.mp4 -c:v mpeg1video -b:v 1500k \
                    -c:a mp2 -b:a 192k -ar 44100 -ac 2 \
                    -f mpeg movie.mpg
```

```
v, _ := stimuli.NewVideo(exp.Screen, "movie.mpg")
defer v.Close()
v.PreloadDevice(exp.AudioDevice)   // without this it plays silently
v.Play()
```

Full walkthrough in 15.2. Frame onsets are approximate — fine for instructions and filler, not for measured stimuli.

"I need frame-accurate timing, and maybe a TTL trigger." You need `.gv`. Convert once, then play:

```
make gv-convert
_build/gv-convert movie.mp4 movie.gv    # needs ffmpeg for non-MPEG-1 input
```

```
events, logs, err := stimuli.PlayGv(exp.Screen, "movie.gv", 0, 0)
```

`logs` tells you afterwards whether any frame was late. Full walkthrough, including how to fire a trigger on a chosen frame, in 15.1. If you also need sound, play a separate `stimuli.Sound` alongside it — which additionally gives you independent control of audio onset (see Section 11).

7.16.2 15.1 `.gv` files — the timing-critical format

`.gv` is the *Extreme GPU Friendly Video Format*, originally from ofxExtremeGpuVideo. Each frame is an independent, LZ4-compressed block of raw RGBA, followed by an index table of `(address, size)` pairs at the end of the file.

That layout is what buys the timing guarantee. There is no inter-frame prediction, so presenting frame *n* costs a seek, one LZ4 decompression, and a texture upload — the same work for every frame, whatever the content. Compare this with H.264, where an I-frame costs several times a B-frame and the resulting jitter is *periodic*, aligned to the GOP boundary. That is the pathological case for frame-accurate onsets, and it is exactly what `.gv` avoids.

You pay for it in disk space. A frame is `width × height × 4` bytes before compression; LZ4 is fast rather than thorough, so expect files roughly an order of magnitude larger than an equivalent MPEG-1. The 10 s 960×400 clip in the table above is 366 MB raw, 41 MB as `.gv`, and 2 MB as MPEG-1.

`.gv` carries **no audio track**. Play a separate `stimuli.Sound` alongside it, which also gives you independent control over audio onset (see Section 11).

7.16.2.1 Creating a `.gv` file

`cmd/gv-convert` converts a video directly:

```
make gv-convert                          # builds _build/gv-convert

gv-convert clip.mpg clip.gv              # MPEG-1 input: pure Go, no ffmpeg
gv-convert movie.mp4 movie.gv            # anything else: needs ffmpeg on
 ↪  PATH
gv-convert frames/ anim.gv               # a directory of numbered images
gv-convert -fps 30 movie.mov movie.gv    # resample to 30 fps
gv-convert -max 100 long.mp4 preview.gv  # first 100 frames only
```

MPEG-1 program streams are decoded in pure Go, so that path needs nothing installed. Every other video format is piped through `ffmpeg`, which must be on `PATH`; if it is missing, the tool says so and names both decoders it tried.

7.16.2.2 From an image sequence

Passing a **directory** converts the `.png`/`.jpg`/`.jpeg`/`.gif` files it contains. This is the right route when frames are generated programmatically (drifting gratings, custom animations) rather than filmed. Two behaviours differ from a naive image-sequence converter, both deliberately:

- **Frames are ordered numerically, not lexicographically.** With unpadded names a plain string sort puts `frame_10.png` before `frame_2.png`, silently playing the sequence out of order. `gv-convert` sorts by numeric value and prints a note when the two orders disagree, so you can see that it mattered.

- **All frames must be the same size.** A mismatch is an error, not a silent crop: a wrongly-sized stimulus is the kind of fault that surfaces only at analysis. Pass `-force-size` to crop/pad against the first frame instead.

A directory has no inherent frame rate, so `-fps` sets it (default 30).

The standalone images2gv does the same job without requiring the goxpyriment tree, and targets other `.gv` players such as `ofxExtremeGpuVideo` and ebiten.

7.16.2.3 Inspecting a `.gv` file

`cmd/gv-getinfo` prints what a `.gv` file actually contains — useful to confirm a conversion produced the frame rate and dimensions you expected, and to check that a file will play at all:

```
make gv-getinfo                  # builds _build/gv-getinfo

$ gv-getinfo stim.gv
file        : stim.gv
frame size  : 960 x 400 px  (1.5 MB raw per frame)
frame count : 250
frame rate  : 25.000 fps   (duration 10.00 s)
pixel format: raw RGBA
file size   : 40.8 MB
frame data  : 40.8 MB compressed from 366.2 MB  (9.0x, 11.2% of raw)
frame sizes : min 5.9 KB, median 104.3 KB, max 611.2 KB
```

`-frames` lists every frame's offset and compressed size; `-q` prints one tab-separated line per file for scripting. The tool also cross-checks the header against the index table and warns about inconsistencies — a `.gv` whose header disagrees with its index loads but plays incorrectly, which is otherwise hard to diagnose from a black screen.

Files written by other tools may use DXT1/DXT3/DXT5 or BC7 block compression instead of raw RGBA. `gv-getinfo` flags these: goxpyriment's reader LZ4-decompresses straight into a texture and never DXT-decodes, so only `raw RGBA` is playable.

7.16.2.4 Playing a `.gv` file

One-shot, for when the video *is* the trial:

```
events, logs, err := stimuli.PlayGv(exp.Screen, "stim.gv", 0, 0)
```

`PlayGv` disables GC, locks to VSYNC, and returns a per-frame `TimingLog` you can write to your data file to verify that no frame was late. It exits on ESC.

7.16.2.4.1 The clip's frame rate must divide the refresh rate

`PlayGv` plays at the rate stored in the `.gv` header, holding each video frame for however many refreshes that takes — a 30 fps clip on a 60 Hz display is shown for 2 refreshes per frame. Both rates are rounded first, so a display reporting 59.94 Hz and a clip authored at 29.97 fps behave as 60 and 30.

Only exact integer ratios are supported. A 24 fps clip on a 60 Hz display would need 2.5 refreshes per frame, so `PlayGv` refuses it rather than play it at the wrong speed or with uneven onsets:

```
display refresh 60 Hz is not an integer multiple of the video's 24 fps
(2.500 frames per refresh); re-encode the clip at a divisor of 60 fps,
e.g. gv-convert -fps 20
```

The 3:2 pulldown that would otherwise be required makes frame onsets uneven, which defeats the reason for using `.gv` at all. **Choose the clip's frame rate to divide the refresh rate of the machine it will run on** — at 60 Hz that means 60, 30, 20, 15, 12, 10 ...; `gv-convert -fps` sets it, and `gv-getinfo` reports what a file actually claims.

Interactive, when you need to draw around the video or respond mid-clip:

```
v, err := stimuli.NewGvVideo("stim.gv")
if err != nil {
    log.Fatal(err)
}
defer v.Close()

v.Play()
exp.Run(func() error {
    if err := v.Update(exp.Screen); err == io.EOF {
        return control.EndLoop
    }
    exp.Screen.Clear()
    v.DrawAt(exp.Screen, &control.FRect{X: 100, Y: 50, W: 640, H: 480})
    return exp.Screen.Flip()   // presents and holds one frame: see Section 5
})
```

`Flip` is an alias for `Update`; both hold one frame. On drivers that present without blocking (triple/mailbox buffering), `Update` returns immediately and the loop runs faster than the display, swallowing frames.

`Width`, `Height`, `FrameCount` and `FPS` are read from the header at construction; the GPU texture is allocated lazily on the first `Update` or `Draw`. For playing several clips back to back without a gap, use `media.MovieManager` (see `docs/MediaMovies.md`).

7.16.2.5 Firing a trigger on a chosen frame

`PlayGvFunc` is `PlayGv` with a per-frame callback, invoked immediately after the flip that puts each frame on screen:

```
events, logs, err := stimuli.PlayGvFunc(exp.Screen, "stim.gv", 0, 0,
    func(ctx stimuli.GvFrameContext) error {
        if ctx.Frame == onsetFrame {
            trig.SetHigh(0)              // rising edge tied to this flip
        }
        if ctx.Frame == onsetFrame+3 {   // ~50 ms later at 60 Hz
            trig.SetLow(0)
```

```
        }
        return nil                          // control.EndLoop to stop early
    })
```

`ctx.OnsetNS` is the SDL timestamp of that frame's **first** flip, in the same reference frame as event timestamps. `ctx.Frame` is the 0-based video frame index and `ctx.Hold` the number of refreshes it is shown for. The callback runs inside the GC-disabled VSYNC loop, so it must not allocate heavily or sleep — see the warning below.

`tests/test_gv_sync` is a complete worked example: it plays a generated flash train and fires a TTL pulse at each bright onset for photodiode verification.

If you need to draw around the video as well, drive the loop yourself instead. `CurrentFrame()` returns the frame now in the GPU texture — the one the next flip will show — so raise the line immediately *after* the flip that displays it:

```
const onsetFrame = 120        // the frame whose onset you want marked
const pulseFrames = 3         // ~50 ms at 60 Hz

// Falls back to a no-op device when no hardware is present, so this code
// still runs on a machine without a trigger box.
trig, _, err := triggers.AutoDetectDLPIO8()
if err != nil {
    log.Fatal(err)
}
defer trig.Close()

lowAt := -1
v.Play()
exp.Run(func() error {
    if err := v.Update(exp.Screen); err == io.EOF {
        return control.EndLoop
    }
    exp.Screen.Clear()
    v.Draw(exp.Screen)

    flipNS, err := exp.Screen.FlipTS()
    if err != nil {
        return err
    }
    if v.CurrentFrame() == onsetFrame {
        trig.SetHigh(0)                     // rising edge, right after the
↪   flip
        lowAt = v.CurrentFrame() + pulseFrames
        exp.Data.Add(trialNo, onsetFrame, flipNS)   // trialNo: your own
↪   counter
    }
    if v.CurrentFrame() == lowAt {
        trig.SetLow(0)
```

```
    }
    return nil
})
```

Do not call `trig.Pulse` here. `Pulse` is `SetHigh`, `time.Sleep`, `SetLow` — it blocks for the whole pulse width, which inside a VSYNC-locked loop costs you frames. Splitting it across iterations as above sleeps not at all, and ties the rising edge (the thing your amplifier timestamps) to a known flip. See Section 17 for why wrapping `Pulse` in a goroutine is not the fix.

`flipNS` shares its reference frame with SDL event timestamps, so it subtracts directly from a keypress timestamp to give a hardware-precision RT relative to the video frame (see Section 6).

7.16.2.6 Verifying that no frame was late

`PlayGv` returns a `[]VideoFrameLog`, one entry per frame:

```
type VideoFrameLog struct {
    Frame         int    // 0-based video frame index
    FlipNS        uint64 // SDL nanosecond timestamp after the flip
    SkippedFrames int    // display frames late (0 = on time)
}
```

Scan `SkippedFrames` after a trial and record the worst value, or abort the session if it exceeds what your paradigm tolerates. A clip that drops frames on the experiment machine is a data-quality problem you want to detect during piloting, not discover in the analysis.

7.16.3 15.2 MPEG-1 files — the convenient format

`stimuli.Video` decodes MPEG-1 in pure Go via gen2brain/mpeg. No ffmpeg, no cgo, no conversion step — you point it at a `.mpg` and it plays, with sound.

The cost is that playback is driven by the wall clock, not by VSYNC. `Update` computes the target frame from elapsed time and *drops* frames to catch up if decoding falls behind. Playback stays the right length, but the onset of any particular frame is only approximate. **Do not put a frame onset from `stimuli.Video` in a data file** — use `.gv` for that.

7.16.3.1 Converting an mp4 (or anything else) to MPEG-1

`stimuli.Video` plays only MPEG-1, so anything else has to be converted once, up front. This is the command:

```
ffmpeg -i input.mp4 -c:v mpeg1video -b:v 1500k \
                    -c:a mp2 -b:a 192k -ar 44100 -ac 2 \
                    -f mpeg output.mpg
```

Reading it piece by piece:

Part	Why
`-c:v mpeg1video`	MPEG-**1** video. MPEG-2 in a `.mpg` will not play
`-b:v 1500k`	video bitrate; raise for large frames, lower for smaller files
`-c:a mp2`	MP2 is the only audio codec an MPEG-1 program stream carries
`-ar 44100 -ac 2`	44.1 kHz stereo; safe defaults
`-f mpeg`	**the load-bearing flag** — selects the MPEG-1 system muxer

`-f mpeg` is the one people get wrong. `-f vob` looks equivalent and produces an MPEG-2 program stream that is rejected at load. If you omit `-f` entirely, ffmpeg guesses from the `.mpg` extension and usually gets it right, but being explicit costs nothing.

Drop the three audio flags for a silent clip. To check the result before wiring it into an experiment, see the next section.

7.16.3.2 Format requirements

The decoder is strict, and the two most common failures are worth knowing:

- It must be an **MPEG-1 program stream** — the file starts with the pack header `00 00 01 BA`. A raw elementary stream (starting `00 00 01 B3`) is rejected with `invalid MPEG-PS`. An elementary stream also cannot carry audio at all, whatever the filename suggests.
- The video must be **MPEG-1**, not MPEG-2. A `.mpg` containing MPEG-2 is out of scope for the library even though the extension is the same.

Check a file before blaming your code:

```
file clip.mpg     # want: "MPEG sequence, v1, system multiplex"
ffprobe clip.mpg  # want: format_name=mpeg, codec_name=mpeg1video (+ mp2
 ↪  audio)
```

If `file` says *"MPEG sequence, v2"* the video is MPEG-2; if it does not say *"system multiplex"* you have an elementary stream. Either way, re-run the conversion above. A file with only one stream listed by `ffprobe` has no soundtrack, whatever its name promises.

7.16.3.3 Playing an MPEG-1 file

```
v, err := stimuli.NewVideo(exp.Screen, "clip.mpg")
if err != nil {
    log.Fatal(err)
}
```

```
defer v.Close()

// Enable the MP2 soundtrack. Safe to call unconditionally: it is a no-op
// returning nil when the file has no audio stream.
if err := v.PreloadDevice(exp.AudioDevice); err != nil {
    log.Printf("audio unavailable: %v", err)
}

v.Play()
exp.Run(func() error {
    if err := v.Update(); err == io.EOF {   // note: no screen argument
        return control.EndLoop
    }
    exp.Screen.Clear()
    v.Draw(exp.Screen, 0, 0)
    exp.Screen.Update()
    return nil
})
```

Audio is off until `PreloadDevice` is called. Without it the video plays silently, and audio packets are discarded at the demuxer — leaving decoding enabled with nothing draining the buffer would grow it for the length of the clip. Check `v.HasAudio` if you need to know whether a soundtrack exists, and use `v.SetVolume(gain)` to scale it.

`Update` keeps 100 ms of decoded audio queued on the SDL device. Audio is free-running once queued, so it stays aligned with the video to within that depth. Two consequences: `Pause` clears the queue so sound stops with the image (resuming leaves audio up to 100 ms ahead), and `Rewind` clears it too so a restarted clip does not play over its own tail.

7.17 16. Gamma Correction and Luminance Linearity

7.17.1 The gamma problem

Standard monitors apply a power-law transfer function when converting digital RGB values to physical luminance:

`L(V) = k · (V/255)^γ` $\gamma \approx 2.2$ for sRGB/LCD

This means equal steps in RGB values do **not** produce equal steps in physical luminance (cd/m²). For example, with $\gamma = 2.2$:

RGB value	Physical luminance (% of max)
0	0%
64	4.9%
128	21.6%
192	52.2%

RGB value	Physical luminance (% of max)
255	100%

This matters in psychophysics experiments where luminance values are specified as physical quantities (e.g. contrast masks, threshold estimation, magnitude estimation).

7.17.2 Inverse-gamma LUT

goxpyriment corrects for this by pre-computing a 256-entry look-up table (LUT) per channel. For each desired linear luminance value v (0–255), the LUT stores the digital value required to achieve it:

`LUT[v] = 255 · (v/255)^(1/γ)`

Applying the LUT before sending a color to the renderer means the monitor's forward gamma converts it back to the intended linear value.

7.17.3 Enabling gamma correction

```
// After exp.Initialize() and before the trial loop:
exp.SetGamma(2.2)   // typical sRGB monitor

// Or with per-channel calibration (from photometer measurements):
exp.GammaCorrector = apparatus.NewGammaCorrector(2.1, 2.2, 2.3)
```

Then wrap every color with `exp.CorrectColor`:

```
// Specify colors in linear luminance space (0–255).
// exp.CorrectColor maps them to the physical digital values.
disk := stimuli.NewFilledCircle(exp.CorrectColor(control.RGB(128, 128,
 ↪  128)), radius)
```

When `SetGamma` has not been called, `CorrectColor` is a no-op that returns the color unchanged, so it is safe to always call it.

7.17.4 Measuring your monitor's gamma

The most accurate approach is photometer-based: measure luminance at several RGB levels, fit the model L(V) = k·(V/255)^γ, and pass the resulting γ to `SetGamma`. Without a photometer, γ = 2.2 is a reasonable default for modern sRGB displays.

7.17.5 Practical example

The `examples/Magnitude-Estimation-Luminosity` example accepts a `-gamma` flag:

```
go run main.go -gamma 2.2
```

With the flag set, the 7 luminance levels (10, 25, 50, 100, 150, 200, 255) are treated as linear targets and corrected before display, so they produce physically uniform luminance steps.

7.18 17. Hardware Triggers and TTL Devices

EEG and MEG recordings require a synchronisation signal — a short TTL pulse sent at the exact moment a stimulus is presented — so that electrophysiological data can be time-locked to experimental events. The `triggers` package provides this, along with the ability to read TTL inputs from response hardware such as fiber-optic response pads.

7.18.1 Concepts

All trigger devices in goxpyriment share two interfaces:

- **`OutputTTLDevice`** — send trigger codes (set lines HIGH/LOW, generate pulses)
- **`InputTTLDevice`** — read response button states (poll or block)

Lines are **0-indexed (0–7)**. Bit N of a bitmask drives line N.

7.18.2 Sending triggers (EEG event codes)

```
import (
    "github.com/chrplr/goxpyriment/triggers"
    "time"
)

// Auto-detect a DLP-IO8-G; falls back to NullOutputTTLDevice (no-op) if
 ↪  absent.
out, _, err := triggers.AutoDetectDLPIO8()
if err != nil { log.Fatal(err) }
defer out.Close()

// In the trial loop:
onset, _ := exp.ShowTS(stim)
out.Pulse(0, 10*time.Millisecond)  // 10 ms pulse on line 0 = EEG event
 ↪  marker
```

For the NeuroSpin MEGTTLBox:

```
box, err := triggers.NewMEGTTLBox("/dev/ttyACM0")
defer box.Close()

box.Pulse(0, 5*time.Millisecond)   // single line
box.PulseMask(0b00000011, 5*time.Millisecond)  // lines 0 and 1
 ↪  simultaneously
```

7.18.3 Timing advice

The output pulse should be sent as close as possible to `exp.ShowTS(stim)`. Because both `Screen.FlipTS` and the trigger output use the system's real-time clock, the latency between the VSYNC flip and the TTL edge is typically under 1 ms. The EEG amplifier records this edge, and the exact onset can be recovered by subtracting `int64(triggerEdgeNS - onsetNS)` if the amplifier's sample clock is synchronised.

7.18.4 `Pulse` blocks — and a goroutine is not the fix

`Pulse(line, d)` is `SetHigh`, `time.Sleep(d)`, `SetLow`. It blocks for the full width. That is fine in an ordinary trial loop, where the sleep overlaps a stimulus you were going to display anyway — but inside a per-frame loop (`.gv` playback, `PresentStreamOfStimuliFunc`, any VSYNC-locked animation) it costs you frames.

The tempting fix is `go trig.Pulse(0, 5*time.Millisecond)`. Don't:

- **It races.** `ParallelPort.SetHigh` is an unsynchronised read-modify-write on a shadow register, so overlapping calls can lose an update and leave a line stuck HIGH or send a wrong code. `LabJackT4` shares one Modbus TCP connection; the serial devices give no ordering guarantee across goroutines. Only `FT232HTrigger` and `LinuxGPIOTrigger` hold a mutex. These failures are silent and intermittent, and they corrupt the event record in your EEG file.
- **It moves the edge onto the Go scheduler.** The amplifier timestamps the *rising* edge — that is your event marker. Blocking `Pulse` raises the line immediately and only the *width* costs time; `go Pulse` defers the `SetHigh` to whenever the scheduler runs it, adding jitter to the one edge that must not have any.
- **It outlives its device.** If the trial ends and the device closes while the goroutine sleeps, `SetLow` fires on a closed port.

In a per-frame loop, split the pulse across iterations instead — no sleeping at all, and the rising edge is locked to a known flip:

```
if thisIsTheOnsetFrame {
    trig.SetHigh(0)              // rising edge, right after the flip
    lowAt = frameIndex + 3       // ~50 ms at 60 Hz
}
if frameIndex == lowAt {
    trig.SetLow(0)
}
```

The width quantises to whole frames, which is well above the ~1 ms most amplifiers need. Section 15 shows this applied to `.gv` playback.

7.18.5 Reading response inputs (FORP pads)

The MEGTTLBox wires a fiber-optic response pad (fORP) to its 8 TTL input lines. Use `WaitForInput` directly or via the `ResponseDevice` wrapper (section 7):

```
// Via InputTTLDevice directly
_ = box.DrainInputs(ctx)                    // clear stale presses
mask, rt, err := box.WaitForInput(ctx)
buttons := triggers.DecodeMask(mask)
for _, b := range buttons {
    fmt.Println(b)  // e.g. "left blue", "right red"
}

// Via ResponseDevice
rd := apparatus.NewTTLResponseDevice(box, 5*time.Millisecond)
_ = rd.DrainResponses(ctx)
resp, _ := rd.WaitResponse(ctx)
// resp.Code == bitmask, resp.Precise == false (software poll)
```

fORP button mapping (NeuroSpin wiring):

FORPButton constant	Line	Arduino pin	STI channel
`FORPLeftBlue`	0	D22	STI007
`FORPLeftYellow`	1	D23	STI008
`FORPLeftGreen`	2	D24	STI009
`FORPLeftRed`	3	D25	STI010
`FORPRightBlue`	4	D26	STI012
`FORPRightYellow`	5	D27	STI013
`FORPRightGreen`	6	D28	STI014
`FORPRightRed`	7	D29	STI015

7.18.6 FT232H (Adafruit FT232H breakout, Linux)

A USB-to-MPSSE bridge. AD0–AD7 are TTL output lines; AC0–AC7 are TTL input lines. No libftdi or D2XX installation needed — the driver uses Linux usbfs directly.

```
box, err := triggers.NewFT232H()
defer box.Close()

box.Pulse(0, 5*time.Millisecond)
mask, rt, _ := box.WaitForInput(ctx)

// Auto-detect (returns NullOutputTTLDevice if no device found):
out, _ := triggers.AutoDetectFT232H()
```

Prerequisites: unload the `ftdi_sio` kernel module (`sudo rmmod ftdi_sio`) and ensure the user has rw access to the USB device node (udev rule or `plugdev` group membership).

7.18.7 GPIO (Raspberry Pi and other Linux SBCs)

Drives any 8 GPIO pins as TTL outputs and reads any other 8 pins as TTL inputs, using the Linux GPIO character device v2 API. Works on Raspberry Pi (BCM pin numbers),

Rock Pi, BeagleBone, Jetson, and any Linux SBC with kernel $\geq$ 5.10.

```
box, err := triggers.NewLinuxGPIOTrigger(
    triggers.WithGPIOOutputPins([8]int{17, 27, 22, 5, 6, 13, 19, 26}),
    triggers.WithGPIOInputPins([8]int{12, 16, 20, 21, 4, 25, 24, 23}),
)
defer box.Close()

box.Pulse(0, 5*time.Millisecond)
mask, rt, _ := box.WaitForInput(ctx)
```

Output-only and input-only configurations are both valid. Prerequisites: kernel $\geq$ 5.10; user in the `gpio` group.

7.18.8 LabJack T4 (Modbus TCP)

Communicates over Ethernet via Modbus TCP — no LabJack SDK or driver required. The eight output lines are DIO4–DIO11 (FIO4–FIO7 + EIO0–EIO3) and the eight input lines are DIO12–DIO19 (EIO4–EIO7 + CIO0–CIO3); `WithT4OutputBase` / `WithT4InputBase` move those groups. DIO0–DIO3 are the T4's dedicated analog inputs and are unavailable as digital lines.

```
box, err := triggers.NewLabJackT4("192.168.1.100")
defer box.Close()

box.Pulse(0, 5*time.Millisecond)
mask, rt, _ := box.WaitForInput(ctx)
```

The device must be reachable on the local network. Default Modbus port 502 is used; append `:port` to the host string to override.

7.18.9 DLP-IO20 (USB-CDC serial)

A 20-channel USB module, 5 V logic. It has more digital channels (17) than the 8-bit TTL interfaces can express, so interface lines 0–7 address a *window* of 8 channels — `AN0`–`AN7` for output and `AN8`–`AN13` + `RB6`/`RB7` for input by default. Every channel remains reachable through the channel-level methods.

```
d, err := triggers.NewDLPIO20("/dev/ttyUSB0")   // or
 ↪  triggers.AutoDetectDLPIO20()
defer d.Close()

d.Send(0b00000101)                    // lines 0 and 2 → AN0, AN2
d.SetChannelHigh(triggers.IO20_RA4)   // a channel outside the window
v, _ := d.ReadChannel(triggers.IO20_AN12)
```

Remap the windows with `WithIO20OutputChannels` / `WithIO20InputChannels` (8 channels each, no duplicates, no overlap between the two groups).

Two caveats. The device has no write-all command, so `Send` issues one packet per line

and the 8 lines change over ~3.5 ms rather than together — use `SetHigh` on a single line when trigger onset matters. And the driver is **untested on real hardware**: it was written from the datasheet, so verify it with `tests/test_dlpio20` before an experiment depends on it.

7.18.10 Supported devices

Device	Type	Output	Input	Connection	Notes
DLP-IO8-G	`DLPIO8`	✓	✓	USB-CDC serial	ASCII protocol
DLP-IO20	`DLPIO20`	✓	✓	USB-CDC serial	Packet protocol; 17 digital channels via 8-line windows; 5 V; untested on hardware
NeuroSpin MEGTTL-Box	`MEGTTLBox`	✓	✓	USB serial	Arduino Mega; binary protocol; fORP input
Adafruit FT232H	`FT232HTrigger`	✓	✓	USB (usbfs)	Linux only; no libftdi needed
Linux GPIO	`LinuxGPIOTrigger`	✓	✓	GPIO pins	Any Linux SBC; kernel $\geq$ 5.10
LabJack T4	`LabJackT4`	✓	✓	Ethernet (Modbus TCP)	No SDK needed
LPT parallel port	`ParallelPort`	✓	—	LPT	Linux only; ppdev ioctl
None / fallback	`NullOutputTTLDevice`	✓ (no-op)	—	—	Returned by `AutoDetectDLPIO8`, `AutoDetectFT232H`

7.18.11 Networked recording control (non-TTL)

The `triggers` package also hosts two devices that are **not** TTL bitmask hardware: they mark events in — and control — an *external recorder* over TCP/IP. They share the package because their purpose is the same (annotate a recording so it can be aligned with your paradigm), but they implement neither `OutputTTLDevice` nor `InputTTLDevice`; each has its own richer API. `SerialPort` (a generic UART wrapper) is the third non-TTL member.

7.18.11.1 NetStation (EGI EEG event markers over TCP/IP)

`NetStation` talks to an EGI / Philips NetStation EEG host using the ECI (Experiment Control Interface) protocol. Instead of an electrical pulse it sends *named events* — a 4-character code with an onset time, a duration, and optional key/value payloads — directly into the EEG stream, and it starts/stops recording and synchronises the host clock remotely.

```
import "github.com/chrplr/goxpyriment/triggers"

ns, err := triggers.NewNetStation("134.225.198.12") // default ECI port
 ↪  55513
if err != nil { log.Fatal(err) }
defer func() {
    // Always check this: a failed stop can leave an unreadable .mff on the
     ↪  host.
    if err := ns.Close(); err != nil { log.Printf("netstation: %v", err) }
}()                          // stops recording + disconnects (blocks ~2 s to
 ↪  flush)

ns.Synchronize()            // align the host clock to ours (call once after
 ↪  connect)
ns.StartRecording()

// In the trial loop — send the marker right after the VSYNC flip:
onset, _ := exp.ShowTS(stim)
ns.SendEvent("STIM")        // now, 1 ms, no keys

// Full form: explicit onset, duration and key/value metadata:
ns.SendEventFull(triggers.Event{
    Code:     "RESP",
    Start:    time.Now(),   // zero value = now; pass the flip time to mark
 ↪  true onset
    Duration: 2 * time.Millisecond,
    Keys:     []triggers.EventKey{{Code: "corr", Value: 1}, {Code: "rt",
 ↪  Value: 423}},
})

ns.StopRecording()
```

Timing note: a network event marker is convenient (it carries a rich label and needs no wiring) but its latency is bounded by the TCP round-trip, not the sub-millisecond edge of a TTL pulse. For hard sub-millisecond EEG synchrony, prefer a `Pulse` on a TTL device for the onset marker and use NetStation events for the accompanying metadata. The driver always advertises the `QNTEL` (Intel/little-endian) handshake and encodes every field little-endian, so it is portable regardless of the machine it runs on. See `tests/test_netstation` for a runnable session.

Never leave a recording open. The output `.mff` is written by NetStation Acquisition — no ECI command names it, chooses its format or finalizes it, so nothing here can change the file's format. What *does* depend on this client is whether the recording is closed properly. A bundle that contains `Acquiring.xml` and no `info.xml`, whose events and EEG signal are both unreadable, is EGI's documented signature of an acquisition that never completed its recording. Two rules follow:

- Check the error from `Close` (as above). It reports a refused stop instead of swallowing it, and every ECI command's acknowledgement is now validated — a host that answers `F` (refused) or `R` (not ready to record) produces an error rather than a silent success.
- Do not call `log.Fatal` between `StartRecording` and `Close`. `log.Fatal` exits without running deferred functions, so the host keeps acquiring. Put the session in a `run() error` function and fail from `main` after the deferred `Close` has run — `tests/test_netstation/main.go` shows the pattern, including a Ctrl-C trap.

If a recording was left open, EGI's File Validator utility (shipped with Net Station 5.4, under Applications ▸ EGI ▸ Utilities) removes `Acquiring.xml` and makes the file openable again.

7.18.11.2 BEL video recorder (labelled camera over TCP/IP)

`VideoRecorder` is the client for the NeuroSpin EEG-room **behavioral video recorder** ("BEL_video"): a camera PC films the participant, burns event labels (subject id, trial, condition, ...) into the footage, and saves an AVI. This type starts/stops that recording and pushes the labels over TCP — the capture and encoding stay on the camera PC. It is not an eye tracker: there is no gaze, pupil, or calibration data, only a labelled video.

```
vr, err := triggers.NewVideoRecorder("192.168.8.212") // default port 55113
if err != nil { log.Fatal(err) }
defer vr.Close()            // stops recording + disconnects

vr.Start()
vr.SetSubject("bb0012025")  // "NIP:<id>" — names the output file
vr.Label("TRL", "001")      // "KEY:VALUE" overlay, shown until the next
 ↪  label / timeout
vr.Label("CND", "007")
vr.Stop()
```

Unlike NetStation, the recorder's protocol has **no framing and no acknowledgement** — it is fire-and-forget, and the server reads one socket buffer per captured

video frame. Two messages sent too close together can therefore arrive coalesced and be mis-parsed, so `VideoRecorder` waits a short `SendGap` (default 50 ms; tune with `triggers.WithVRSendGap`, or set 0 to disable) after each message. This makes the labels unsuitable for precise stimulus timing — treat them as monitoring/annotation metadata, not event markers. See `tests/test_videorecorder` for a runnable session.

Device	Type	Role	Connection	Ack?
EGI NetStation	`NetStation`	EEG event markers + recording control + clock sync	TCP/IP (ECI, port 55513)	yes (1 byte/command)
BEL_video	`VideoRecorder`	Labelled video recording control	TCP/IP (port 55113)	no (fire-and-forget)

7.19 18. Display Compositor Bypass

A compositing window manager (compositor) intercepts every application frame and blends it with other on-screen elements before sending the result to the display. This introduces additional latency and jitter that are problematic for millisecond-accurate stimulus presentation. The degree to which `goxpyriment` can avoid this overhead depends on the operating system.

On **Linux with X11**, SDL3 automatically sets the `_NET_WM_BYPASS_COMPOSITOR` window property when the application enters fullscreen mode. Compliant compositors (KWin, Mutter, Compton) respond by *unredirecting* the fullscreen window — routing its framebuffer directly to the display scan-out pipeline without compositing. The result is presentation latency below 1 ms, comparable to a compositor-free session. On **Linux with Wayland**, the compositor is always the intermediary for frame delivery, but modern compositors (KWin 5.21+, GNOME Mutter 44+) support *direct scan-out* for fullscreen applications, achieving similar low-latency behavior in practice. The best achievable timing on Linux is obtained by running without any display server: setting the environment variable `SDL_VIDEODRIVER=kmsdrm` before launching the experiment directs SDL3 to communicate with the Linux KMS/DRM subsystem directly, bypassing X11 and Wayland entirely. This configuration — typically used from a virtual terminal (`Ctrl+Alt+F2`) — is recommended for the most demanding timing requirements such as single-frame subliminal stimulation or high-frequency RSVP.

On **Windows**, SDL3's fullscreen mode creates an exclusive fullscreen surface that bypasses the Desktop Window Manager (DWM), again yielding frame-interval jitter typically below 0.3 ms (one standard deviation), comparable to Linux.

macOS is the exception. Apple's Metal rendering pipeline always delivers frames to the display through the WindowServer compositor, and no third-party application can

bypass it — even in fullscreen.

These platform differences can be quantified empirically with the `display` and `av` sub-tests of the bundled `Timing-Tests` suite.

7.20 19. Variable Refresh Rate (VRR) Stimulus Presentation

Standard displays refresh at a fixed rate (e.g., 60 Hz), which quantises every stimulus duration to multiples of the frame period (16.67 ms at 60 Hz). A researcher wishing to present a stimulus for 10 ms, 17 ms, or 23 ms cannot do so on such a display without relying on temporal dithering or hardware frame-blending techniques that introduce their own artefacts. Variable Refresh Rate (VRR) technology — marketed as AMD FreeSync, NVIDIA G-Sync, or the VESA Adaptive-Sync standard — removes this constraint by allowing the display to hold each frame for an arbitrary duration and refresh only when instructed to do so by the GPU.

`goxpyriment` exposes VRR through a single API call: `exp.Screen.SetVSync(0)` disables the VSYNC block, so that every subsequent `SDL_RenderPresent` call returns immediately. The following pattern then achieves arbitrary stimulus durations with duration error typically below 0.5 ms (one standard deviation):

1. Draw the stimulus and call `screen.FlipTS()`, which presents the frame and records `onsetNS` (an SDL nanosecond timestamp) immediately after `Present()` returns.
2. Busy-wait for the desired duration d: spin until `time.Now() >= onset + d`. The last ~500 µs use a CPU spin loop (already used in the trigger timing routines) to eliminate OS scheduling jitter.
3. Draw the blank screen, call `FlipTS` again to record `offsetNS`.

The actual software-controlled duration is `(offsetNS - onsetNS) / 1e6` ms. On a VRR-capable display this equals the true on-screen duration plus a small, constant hardware pipeline latency (GPU scan-out and panel response, typically 1–5 ms) that can be calibrated once with a photodiode using the `frames` timing sub-test of the `Timing-Tests` suite.

Every VRR panel supports VRR only within a finite refresh-rate window (e.g., 48–144 Hz, corresponding to frame durations of 6.9–20.8 ms for a typical gaming monitor). Durations outside this window cause the panel to revert to fixed-rate behaviour, re-introducing quantisation. The `Timing-Tests` suite includes a dedicated `vrr` sub-test that sweeps target durations from 1 ms to a user-specified maximum in 1 ms steps and reports duration errors at each step, directly revealing the panel's VRR window in the output data.

On a display without VRR support, disabling VSYNC produces frame tearing. The `vrr` sub-test detects this automatically: on a non-VRR display the duration errors cluster at multiples of the frame period rather than being uniformly small, providing an

unambiguous diagnostic.

7.21 20. Putting It All Together

Here is a skeleton that illustrates how the concepts compose in a realistic experiment:

```
package main

import (
    "fmt"
    "log"

    "github.com/chrplr/goxpyriment/control"
    "github.com/chrplr/goxpyriment/design"
    "github.com/chrplr/goxpyriment/stimuli"
)

const (
    NReps           = 10
    FixationDuration = 500
    ResponseTimeout  = 3000
)

type trialDef struct {
    word  string
    color control.Color
    name  string
}

func main() {
    exp := control.NewExperimentFromFlags("Color Word Task", control.Black,
↪   control.White, 36)
    defer exp.End()

    exp.AddDataVariableNames([]string{"trial", "word", "ink", "key", "rt_
↪   ms", "correct"})

    // Build trial list
    words  := []string{"RED", "GREEN", "BLUE"}
    colors := []control.Color{control.Red, control.Green, control.Blue}
    names  := []string{"red", "green", "blue"}
    keys   := []control.Keycode{control.K_R, control.K_G, control.K_B}

    var trials []trialDef
    for _, word := range words {
        for j := range colors {
            trials = append(trials, trialDef{word, colors[j], names[j]})
```

```
        }
    }
    for i := 0; i < NReps-1; i++ {
        trials = append(trials, trials[:9]...)
    }
    design.ShuffleList(trials)

    // Preload stimuli
    fixation := stimuli.NewFixCross(20, 2, control.White)

    err := exp.Run(func() error {
        // Preload fixation cross (it will be used every trial)
        stimuli.PreloadVisualOnScreen(exp.Screen, fixation)

        exp.ShowInstructions(
            "Name the INK COLOR of each word.\n\n" +
            "R = Red   G = Green   B = Blue\n\n" +
            "Press SPACE to start.",
        )

        for i, t := range trials {
            // Fixation
            exp.ShowTimed(fixation, FixationDuration)

            // Stimulus + Response
            stim := stimuli.NewTextLine(t.word, 0, 0, t.color)
            key, rt, _ := exp.ShowAndGetRT(stim, keys, ResponseTimeout)

            // Score
            correct := false
            for j, k := range keys {
                if key == k && names[j] == t.name {
                    correct = true
                    break
                }
            }

            exp.Data.Add(i, t.word, t.name, key, rt, correct)
            fmt.Printf("trial %3d  %s/%s  rt=%dms  %v\n", i, t.word, t.name,
↪   rt, correct)

            if !correct {
                exp.Audio.PlayBuzzer()
            }

            // Release texture (new TextLine each trial)
            stim.Unload()
```

```
            exp.Blank(500)
        }

        return control.EndLoop
    })

    if err != nil && !control.IsEndLoop(err) {
        log.Fatalf("experiment error: %v", err)
    }
}
```

Key patterns illustrated:

- `NewExperimentFromFlags` + `defer exp.End()` at the top.
- `exp.AddDataVariableNames` before the run loop.
- `design.ShuffleList` for trial randomization.
- `stimuli.PreloadVisualOnScreen` inside `exp.Run` for timing-sensitive stimuli.
- `exp.Show` for single stimuli; `exp.Blank` for ITIs.
- `exp.Keyboard.WaitKeysRT` for response + RT.
- `stim.Unload()` for dynamically created stimuli.
- `exp.Audio.PlayBuzzer()` for feedback.
- `return control.EndLoop` at the end.

For the complete function signatures and type definitions, see the API Reference. For more worked examples, browse the `examples/` directory.

Chapter 8

goxpyriment API Reference

This guide documents the complete public API of the `goxpyriment` framework, organized by package.

Note that you can access the source code documentation by running `pkgsite` and opening http://localhost:8080. To install pkgsite:

```
go install golang.org/x/pkgsite/cmd/pkgsite@latest
```

See Viewing the documentation locally for details (and for serving this site offline with zensical).

8.1 Package Overview

```
control/      ← experiment lifecycle and orchestration (start here)
stimuli/      ← visual and audio stimulus objects
media/        ← multi-movie playback with hardware-verified display onsets
apparatus/    ← SDL window/renderer, keyboard, mouse, gamepad, gamma
 ↪  corrector
results/      ← experiment data file and output file
design/       ← trial/block structure and randomization
clock/        ← timing utilities
geometry/     ← coordinate conversion helpers
triggers/     ← hardware trigger devices (EEG sync, etc.)
```

8.2 Package `control`

Import: `github.com/chrplr/goxpyriment/control`

8.2.1 Boilerplate

Every experiment starts the same way:

```
exp := control.NewExperimentFromFlags("My Experiment", control.Black,
 ↪  control.White, 32)
defer exp.End()

err := exp.Run(func() error {
    // trial loop
    return control.EndLoop
})
if err != nil && !control.IsEndLoop(err) {
    log.Fatalf("experiment error: %v", err)
}
```

8.2.2 Pre-experiment Setup Dialog

`GetParticipantInfo` opens a graphical SDL window **before** the experiment starts to collect participant demographics, monitor properties, and display preferences. Call it before `NewExperiment` / `NewExperimentFromFlags`.

```
fields := append(control.StandardFields, control.FullscreenField)
info, err := control.GetParticipantInfo("My Experiment", fields)
if err != nil {
    log.Fatalf("setup cancelled: %v", err) // user pressed Escape or closed
 ↪  window
}
```

8.2.2.1 Types

```
// FieldType selects how a field is rendered.
type FieldType int
const (
    FieldText     FieldType = iota // text input box (default)
    FieldCheckbox                  // tick-box; value is "true" or "false"
)

// InfoField describes one entry in the dialog.
type InfoField struct {
    Name    string    // key in the returned map
    Label   string    // displayed label
    Default string    // initial value
    Type    FieldType // FieldText (default) or FieldCheckbox
}
```

8.2.2.2 Pre-built field sets

Variable	Fields
`control.ParticipantFields`	`subject_id`, `age`, `gender`, `handedness`
`control.MonitorFields`	`screen_width_cm`, `viewing_distance_cm`, `refresh_rate_hz`
`control.StandardFields`	`ParticipantFields` + `MonitorFields`
`control.FullscreenField`	Checkbox: `fullscreen` (`"true"` / `"false"`)

8.2.2.3 Function

Function	Description
`GetParticipantInfo(title string, fields []InfoField) (map[string]string, error)`	Shows the dialog and returns collected values. Returns `ErrCancelled` if the user closes or presses Escape without confirming.

8.2.2.4 Session persistence

All values except `subject_id` are saved to `~/.cache/goxpyriment/last_session.json` on OK and pre-filled on the next run. `subject_id` is always reset.

8.2.2.5 Using the fullscreen checkbox and persisting to the data file

```
info, err := control.GetParticipantInfo("My Experiment", fields)
// ...
fullscreen := info["fullscreen"] == "true"
width, height := 0, 0
if !fullscreen {
    width, height = 1024, 768
}
exp := control.NewExperiment("My Experiment", width, height, fullscreen,
    control.Black, control.White, 32)

// Set Info (and SubjectID) BEFORE Initialize — they are written to the info
 ↪  file automatically
exp.SubjectID, _ = strconv.Atoi(info["subject_id"])
exp.Info = info

if err := exp.Initialize(); err != nil { log.Fatal(err) }
defer exp.End()
```

`Initialize()` writes a `--PARTICIPANT INFO` block to the companion `-info.txt` file whenever `exp.Info` is non-nil at that point. No explicit call to `WriteParticipantInfo` is needed.

8.2.2.6 Sentinel error

```
control.ErrCancelled  // returned when the user cancels the dialog
```

8.2.3 Constructor Functions

Function	Description
`NewExperimentFromFlags(name string, bg, fg Color, fontSize float32, extra ...InfoField) *Experiment`	Creates and fully initializes an experiment from `-w` (windowed 1024×768), `-d N` (display index, -1 = primary), and `-s N` (subject ID) command-line flags. Calls `log.Fatal` on error. **This is the preferred entry point.** Any `extra` fields are appended to the session-setup dialog that opens when `-s` is absent, and read back from `exp.Info` (nil when the dialog does not open, so keep a flag as the fallback); an extra field named like one of the four built-ins replaces it.
`NewExperiment(name string, width, height int, fullscreen bool, bg, fg Color, fontSize float32) *Experiment`	Lower-level constructor; call `Initialize()` before use.

8.2.4 Lifecycle Methods

Method	Description
`exp.Initialize() error`	Initializes SDL, audio, window, renderer, font, and data file.
`exp.End()`	Cleans up all resources. Always `defer exp.End()` immediately after construction.
`exp.Run(logic func() error) error`	Runs the main trial loop on the SDL main thread. Return `control.EndLoop` to exit cleanly.
`exp.HideCursor() error`	Hides the mouse cursor. `Initialize()` already does this — only needed to hide it again after a `ShowCursor` call.

Method	Description
`exp.ShowCursor() error`	Makes the mouse cursor visible. Call it after `Initialize()` in mouse-driven paradigms; the cursor is hidden by default so it never sits over the stimuli. Equivalent to setting `exp.CursorVisible = true` before `Initialize()`. `GetParticipantInfo` shows the cursor for its own dialog regardless.

8.2.5 Presentation Methods

Method	Description
`exp.Show(stim VisualStimulus) error`	Clear → draw → flip. The standard one-call stimulus presentation.
`exp.ShowTS(stim VisualStimulus) (uint64, error)`	Clear → draw → flip, and return the SDL nanosecond timestamp captured immediately after the VSYNC flip. Use with `GetKeyEventTS` for hardware-precision RT measurement.
`exp.ShowTimed(stim VisualStimulus, durationMs int) error`	`Show(stim)` + `Wait(durationMs)` in one call. For fixation crosses, cues, and passive stimulus viewing.
`exp.ShowFrames(stim VisualStimulus, n int) (uint64, error)`	Hold the stimulus for exactly `n` display frames, returning the onset timestamp of the first flip. The stimulus is redrawn every frame — see *Holding a stimulus for a fixed number of frames* below.
`exp.BlankFrames(n int) (uint64, error)`	Frame-locked counterpart of `Blank(ms)`: clear and hold blank for exactly `n` frames, returning the timestamp of the first flip (i.e. the previous stimulus's offset).
`exp.ShowAndGetRT(stim VisualStimulus, keys []Keycode, timeoutMs int) (Keycode, int64, error)`	Clears stale keyboard events, shows stim with hardware-precise onset timing, waits for a key, and returns `(key, rtMs, error)`. Pass `timeoutMs = -1` for no timeout; returns `(0, 0, nil)` on timeout. This is the canonical single-stimulus RT measurement call.
`exp.ShowEndMessage(message string) error`	Display a centered completion message and wait for any key. For end-of-experiment screens. Laid out by `FittedTextBox`.
`exp.ShowInstructions(text string) error`	Display centered text and wait for spacebar. Laid out by `FittedTextBox`.

Method	Description
`exp.FittedTextBox(text string) *stimuli.TextBox`	A centered `TextBox` wrapped to the drawing area and rendered at the largest point size — never above the default — at which the whole block fits. See *Laying out a block of text* below.
`exp.DrawArea() (w, h float32)`	Size of the coordinate space stimuli are drawn in (the logical resolution). What layout code should measure against.
`exp.Blank(ms int) error`	Clear and flip screen, then wait `ms` milliseconds.
`exp.Wait(ms int) error`	Wait `ms` ms while pumping SDL events (ESC-abortable).
`exp.ShowSplash(waitForKey bool) error`	Show experiment name + version splash.
`exp.Flip() error`	Present the backbuffer and hold to the frame boundary (one call = one display frame).

8.2.6 Input

Method	Description
`exp.Keyboard`	`*apparatus.Keyboard` — see Keyboard section
`exp.Mouse`	`*apparatus.Mouse` — see Mouse section
`exp.PollEvents(handle func(sdl.Event) bool) EventState`	Process all pending SDL events; optionally forward to a handler. Returns `EventState` including nanosecond timestamps.
`exp.HandleEvents() (Keycode, uint32, error)`	Convenience wrapper: returns `(key, mouseButton, error)`.

`EventState` now includes SDL event timestamps:

```
type EventState struct {
    LastKey            sdl.Keycode
    LastMouseButton    uint32
    LastKeyTimestamp   uint64  // SDL nanosecond timestamp of the last key
 ↪  event
    LastMouseTimestamp uint64  // SDL nanosecond timestamp of the last mouse
 ↪  event
    QuitRequested      bool
}
```

8.2.7 Design and Data

Method	Description
`exp.AddDataVariableNames(names []string)`	Register CSV column names for the data file.
`exp.Data.Add(values ...interface{})`	Append a data row. Subject ID is prepended automatically.
`exp.AddBlock(b *design.Block, copies int)`	Add trial blocks to the experiment.
`exp.ShuffleBlocks()`	Randomize block presentation order.
`exp.AddBWSFactor(name string, conditions []interface{})`	Register a between-subjects factor for Latin-square counterbalancing.
`exp.GetPermutedBWSFactorCondition(name string) interface{}`	Return this subject's condition for a BWS factor.
`exp.Design`	*`design.Experiment` — full design object
`exp.Info`	`map[string]string` — values from `GetParticipantInfo`; set before `Initialize()` to persist them automatically to the `-info.txt` file
`exp.CursorVisible`	`bool` — whether the mouse pointer is shown over the experiment window. Defaults to `false`: `Initialize()` hides the cursor. Set it to `true` before `Initialize()` (or call `ShowCursor()` after) for mouse-driven paradigms.

8.2.8 Font and Display

Method	Description
`exp.LoadFont(path string, size float32) error`	Load a TTF font from file.
`exp.LoadFontFromMemory(data []byte, size float32) error`	Load a TTF font from a byte slice.
`exp.SetVSync(vsync int) error`	Toggle vertical sync (1 = on, 0 = off).
`exp.SetLogicalSize(w, h int32) error`	Set device-independent logical resolution.
`exp.SetOutputDirectory(dir string)`	Override default data file directory (`~/goxpy_data`).

8.2.9 Gamma Correction

Standard monitors apply a power-law transfer function L(V) = k·(V/255)^γ ($\gamma \approx 2.2$ for sRGB displays). Equal steps in RGB values do **not** produce equal steps in physical luminance. Use `SetGamma` to enable inverse-gamma correction.

Method	Description
`exp.SetGamma(gamma float64)`	Install a uniform inverse-gamma corrector. Call once after `Initialize()`.
`exp.CorrectColor(c sdl.Color) sdl.Color`	Apply gamma correction to a color. Returns c unchanged when no corrector is set.
`exp.GammaCorrector`	`*apparatus.GammaCorrector` — set directly for per-channel calibration.

```
// Uniform gamma (typical sRGB monitor)
exp.SetGamma(2.2)

// Per-channel gamma (from photometer measurements)
exp.GammaCorrector = apparatus.NewGammaCorrector(2.1, 2.2, 2.3)

// Use in trial loop — specify colors in linear luminance space (0–255)
disk := stimuli.NewFilledCircle(exp.CorrectColor(control.RGB(128, 128,
 ↪  128)), radius)
```

The `apparatus.GammaCorrector` type is also available directly:

```
gc := apparatus.NewGammaCorrectorUniform(2.2)
corrected := gc.CorrectColor(sdl.Color{R: 128, G: 128, B: 128, A: 255})
// corrected.R ≈ 186 — the physical digital value for 50% luminance on γ=2.2
```

8.2.10 Colors, Types, and Constants

```
// Named colors
control.Black, White, Red, Green, Blue, Yellow, Magenta, Cyan
control.Gray, DarkGray, LightGray

// Type aliases (so you only need to import "control")
type Color   = sdl.Color
type Keycode = sdl.Keycode
type FPoint  = sdl.FPoint
type FRect   = sdl.FRect

// Constructors
control.RGB(r, g, b uint8) Color
control.RGBA(r, g, b, a uint8) Color
control.Point(x, y float32) FPoint
control.Origin() FPoint  // returns (0, 0)

// Font helpers
control.FontFromFile(path string, size float32) (*ttf.Font, error)
control.FontFromMemory(data []byte, size float32) (*ttf.Font, error)
```

```
// Loop control
control.EndLoop          // sentinel error: return from Run callback to exit
control.IsEndLoop(err)   // test whether an error is the EndLoop sentinel

// Keyboard codes
control.K_SPACE, K_ESCAPE, K_RETURN, K_BACKSPACE, K_TAB
control.K_UP, K_DOWN, K_LEFT, K_RIGHT
control.K_A … K_Z                                    // full alphabet
control.K_0 … K_9                                    // digit row
control.K_KP_0 … K_KP_9, K_KP_ENTER, K_KP_PLUS, K_KP_MINUS   // numeric
 ↪  keypad
control.K_MINUS, K_PLUS, K_EQUALS, K_LEFTBRACKET, K_RIGHTBRACKET  //
 ↪  punctuation

// Mouse buttons
control.BUTTON_LEFT, BUTTON_RIGHT
```

> **Tip:** The full alphabet, digit row, keypad, and common punctuation keys are re-exported, so experiment code never needs to import `go-sdl3` just to name a key. For a rarely-used code not listed in `defaults.go`, import `go-sdl3/sdl` directly and use `sdl.K_…`.

8.2.11 Audio

```
exp.AudioDevice  // sdl.AudioDeviceID — pass to Sound.PreloadDevice()

// Top-level helper (call before NewExperiment)
control.SetAudioSampleFrames(frames int)  // set audio buffer size
 ↪  (256–2048)
```

8.2.12 Microphone

```
exp.Microphone  // *apparatus.Microphone — nil until OpenMicrophone is
 ↪  called

// Open the default recording device (nil = F32LE mono 44100 Hz)
exp.OpenMicrophone(spec *sdl.AudioSpec) error

// Closed automatically by exp.End()
```

8.3 Package `stimuli`

Import: `github.com/chrplr/goxpyriment/stimuli`

8.3.1 Interfaces

```
type Stimulus interface {
    Present(screen *apparatus.Screen, clear, update bool) error
    Preload() error
    Unload() error
}

type VisualStimulus interface {
    Stimulus
    Draw(screen *apparatus.Screen) error
    GetPosition() sdl.FPoint
    SetPosition(pos sdl.FPoint)
}
```

GPU textures are **lazily allocated** on the first `Draw`. To force early allocation (for timing-sensitive code), use:

```
stimuli.PreloadVisualOnScreen(screen, stim)     // single stimulus
stimuli.PreloadAllVisual(screen, []VisualStimulus{...})  // slice
```

8.3.2 Visual Stimuli

8.3.2.1 Text

Constructor	Description
`NewTextLine(text string, x, y float32, color Color) *TextLine`	Single line of text.
`NewTextBox(text string, width int32, pos FPoint, color Color) *TextBox`	Word-wrapped multi-line text.

Both support a `Font *ttf.Font` field — set it to override the screen default.

8.3.2.2 Shapes

Constructor	Description
`NewFixCross(size, lineWidth float32, color Color) *FixCross`	Fixation cross centered at (0, 0).
`NewCircle(radius float32, color Color) *Circle`	Filled circle.
`NewRectangle(cx, cy, w, h float32, color Color) *Rectangle`	Filled rectangle.
`NewLine(x1, y1, x2, y2 float32, color Color) *Line`	Line segment.

8.3.2.3 Images and Video

Constructor/Function	Description
`NewPicture(filePath string, x, y float32) *Picture`	Image loaded from file (PNG, JPG, BMP…).
`NewPictureFromMemory(data []byte, x, y float32) *Picture`	Image loaded from embedded bytes.
`NewSpriteSheet(filePath string) *SpriteSheet`	One image holding many stimuli, sharing a single GPU texture.
`NewSpriteSheetFromMemory(data []byte) *SpriteSheet`	Same, from embedded bytes (required for the browser build).
`(*SpriteSheet) Grid(screen, cols, rows int) ([]*Sprite, error)`	Cut into `cols*rows` equal cells, row-major.
`(*SpriteSheet) GridWithSpacing(screen, cols, rows int, margin, spacing float32) ([]*Sprite, error)`	As `Grid`, for sheets with a border or gutters.
`(*SpriteSheet) Sprites(screen, clips []sdl.FRect) ([]*Sprite, error)`	Explicit source rectangles, for irregular sheets.
`(*SpriteSheet) Unload() error`	Destroys the shared texture. `Sprite.Unload` is a no-op by design.
`PlayGv(screen, path string, x, y float32) ([]UserEvent, error)`	Play a `.gv` (LZ4-compressed RGBA) video file, VSYNC-locked.
`NewGvVideo(path string) (*GvVideo, error)`	Open a `.gv` file for frame-by-frame access.

8.3.2.4 Psychophysics Stimuli

Constructor	Description
`NewGaborPatch(sigma, theta, lambda, phase, psi, gamma float64, bgColor Color, size float32) *GaborPatch`	Static Gabor patch. `theta` in degrees, `lambda` = spatial wavelength in pixels.
`NewDotCloud(radius float32, bgColor, dotColor Color) *DotCloud`	Static random-dot cloud. Call `Make(nDots, dotRadius, gap)` to populate.
`NewRDS(imgSize, innerSize [2]int, shift, gap, scale int) *RDS`	Random-dot stereogram (side-by-side pair).
`NewVisualMask(w, h, dotW, dotH float32, bgColor, dotColor Color, pct int) *VisualMask`	Random-dot masking stimulus. `pct` = dot fill percentage 0–100.

8.3.2.5 Composite / Interactive

Constructor	Description
`NewThermometerDisplay(size FPoint, nSegments int, state, goal float32) *ThermometerDisplay`	Segmented progress bar. `State` and `Goal` in 0–100.
`NewChoiceGrid(choices []string, maxSelect int, prompt string) *ChoiceGrid`	Multiple-choice button grid (mouse + keyboard). See below.
`NewTextInput(msg string, pos FPoint, boxW float32, bgColor, frameColor, textColor Color) *TextInput`	Free-text keyboard input box. Call `ti.Get(screen, keyboard)`.
`NewMenu(items []string) *Menu`	Numbered keyboard-navigable list. Call `m.Get(screen, keyboard, initialSel)`.

8.3.3 ChoiceGrid

```
cg := stimuli.NewChoiceGrid(choices, maxSelect, prompt)
cg.Cols = 7       // optional: set column count (0 = auto)

selections, err := cg.Get(exp.Screen, exp.Keyboard)
// selections is a []string preserving selection order
```

- `MaxSelect > 0`: auto-submits after N selections.
- `MaxSelect == 0`: participant presses ENTER or SPACE to submit.
- BACKSPACE removes the last selection.
- Both mouse click and matching keypress (single-char labels) activate buttons.

8.3.4 Menu

```
m := stimuli.NewMenu([]string{"Option A", "Option B", "Option C"})
m.Pos = sdl.FPoint{X: 0, Y: 0}    // optional: reposition (default = screen
 ↪  center)
m.HighlightColor = control.Yellow  // optional: override highlight color

idx, err := m.Get(exp.Screen, exp.Keyboard, 0)  // 0 = initially highlight
 ↪  first item
// idx is 0-based; -1 + sdl.EndLoop on ESC/quit
```

Navigation: UP/DOWN arrows move the highlight; ENTER or SPACE confirms; number keys 1–9 (0 for tenth) select and confirm directly. The selected item is shown in `HighlightColor` with a > prefix; others use `TextColor`. `LineSpacing` controls vertical item spacing (0 = auto from font height).

8.3.5 Animated / VSYNC-locked Loops

All three functions disable GC, lock to VSYNC, and return `(MotionResult, error)`.

```
type MotionResult struct {
    Key    sdl.Keycode // interrupt key pressed (0 if none)
    Button uint8       // mouse button pressed (0 if none)
    RTms   int64       // ms from first frame to response (or total duration
 ↪  on timeout)
}
```

Function	Description
`PresentMovingDotCloud(screen, nDots int, dotRadius, cloudRadius float32, center FPoint, speedPxPerSec float32, maxDurationMs int64, interruptKeys []Keycode, catchMouse bool, dotColor, bgColor Color) (MotionResult, error)`	Animated random-dot cloud. Each dot moves at a fixed speed and respawns when it exits the boundary.
`PresentMovingGrating(screen, width, height float32, center FPoint, orientation, spatialFreq, temporalFreq, contrast, bgLuminance float64, maxDurationMs int64, interruptKeys []Keycode, catchMouse bool) (MotionResult, error)`	Drifting sinusoidal grating in a rectangular aperture.
`PresentMovingGabor(screen, size float32, sigma float64, center FPoint, orientation, spatialFreq, temporalFreq, contrast, bgLuminance float64, maxDurationMs int64, interruptKeys []Keycode, catchMouse bool) (MotionResult, error)`	Drifting Gabor patch with Gaussian envelope (alpha-blended edges).

Spatial frequency is in **cycles per pixel** (e.g. `0.05` = one cycle every 20 px). Temporal frequency is in **Hz**. Orientation is in **degrees from horizontal** (0° = vertical bars drifting right).

8.3.6 Stimulus Streams (High-Precision RSVP)

Stream functions disable GC, lock every onset and offset to a VSYNC boundary, and return `([]UserEvent, []TimingLog, error)`.

```
type UserEvent struct {
    Event       sdl.Event     // raw SDL event (KeyboardEvent,
 ↪  MouseButtonEvent, …)
    Timestamp   time.Duration // time relative to stream start (Go clock, ms
 ↪  precision)
    TimestampNS uint64        // SDL3 hardware event timestamp, nanoseconds
 ↪  (same clock as Screen.FlipTS)
}
```

```
type TimingLog struct {
    Index        int
    TargetOn     time.Duration
    ActualOnset  time.Duration // DIAGNOSTIC (Go clock): first-frame draw,
 ↪  stream-relative; not for RT
    ActualOffset time.Duration // DIAGNOSTIC (Go clock): after last
 ↪  on-frame, stream-relative; not for RT
    OnsetNS      uint64        // AUTHORITATIVE: SDL3 ns timestamp
 ↪  (sdl.TicksNS clock) of the stimulus onset
    OffsetNS     uint64        // AUTHORITATIVE: SDL3 ns timestamp
 ↪  (sdl.TicksNS clock) of the stimulus offset
}
```

The `OnsetNS`/`OffsetNS` pair is the authoritative timing record: both are on the SDL3 nanosecond clock (`sdl.TicksNS()`) — the same clock that stamps input events (`UserEvent.TimestampNS`, `Keyboard.GetKeyEventTS`) and VSYNC flips (`Screen.FlipTS`). A reaction time or a displayed duration is a plain subtraction within this one clock: `int64(event.TimestampNS - l.OnsetNS)`, or `int64(l.OffsetNS - l.OnsetNS)`.

`ActualOnset`/`ActualOffset` are Go-clock (`time.Now`) **diagnostics only**, kept for coarse pacing checks. They live on a different timebase (different origin) from the NS fields and from input events, so they must never be subtracted from an event timestamp or logged as a canonical onset/offset. Use the NS fields for any real timing.

`OnsetNS` is zero only when a stimulus was never displayed (zero on-frames); `OffsetNS` is the VSYNC timestamp of the flip that takes the stimulus down, and for the final element of a contiguous stream (which has no ISI flip of its own) it is synthesised from the onset plus the on-frame count so that it, too, is on the SDL clock rather than one frame early.

`UserEvent.TimestampNS` and `TimingLog.OnsetNS`/`OffsetNS` are all on the SDL3 nanosecond clock, so reaction times measured during a stream can be computed with full hardware precision:

```
for _, ev := range events {
    if ev.Event.Type == sdl.EVENT_KEY_DOWN {
        // Find the stimulus that was on-screen when the key was pressed
        for _, l := range logs {
            if ev.TimestampNS >= l.OnsetNS && ev.TimestampNS < l.OffsetNS {
                rtNS := int64(ev.TimestampNS - l.OnsetNS)
                fmt.Printf("RT from stimulus %d: %d ms\n", l.Index, rtNS/1_
 ↪  000_000)
            }
        }
    }
}
```

8.3.6.1 Searching event lists

Function	Description
`FirstKeyPress(events []UserEvent, key sdl.Keycode) (UserEvent, bool)`	Returns the first `KEY_DOWN` event matching key from the slice, plus a found flag.

```
if ev, ok := stimuli.FirstKeyPress(events, sdl.K_SPACE); ok {
    fmt.Printf("Space pressed at %d ms\n", ev.Timestamp.Milliseconds())
}
```

8.3.6.2 Visual Streams

```
// RSVP text stream — simplest entry point
events, logs, err := stimuli.PresentStreamOfText(
    exp.Screen, words, durationOn, durationOff, x, y, color,
)

// Image/mixed stream
elements := stimuli.MakeRegularVisualStream(stims, durationOn, durationOff)
events, logs, err := stimuli.PresentStreamOfImages(exp.Screen, elements, x,
 ↪  y)

// Irregular timing
elements, err := stimuli.MakeVisualStream(stims, onsetMs, durationMs)
events, logs, err := stimuli.PresentStreamOfImages(exp.Screen, elements, x,
 ↪  y)
```

8.3.6.3 Audio Streams

```
// Regular timing
elements := stimuli.MakeRegularSoundStream(sounds, durationOn, durationOff)
events, logs, err := stimuli.PlayStreamOfSounds(elements)

// Irregular timing
elements, err := stimuli.MakeSoundStream(sounds, onsetMs, durationMs)
events, logs, err := stimuli.PlayStreamOfSounds(elements)
```

`sounds` is `[]stimuli.AudioPlayable` — satisfied by both `*Sound` and `*Tone`. The returned `TimingLog` carries `OnsetNS`/`OffsetNS` on the SDL3 clock (captured at the `Play()` instant and the end of the on-phase), so audio-stream onsets are directly comparable with input-event timestamps, same as visual streams.

8.3.6.4 Mixed Streams

`PresentStreamOfStimuli` presents a heterogeneous, **sequential** stream that freely mixes visual and audio stimulus types (anything satisfying the broad `Stimulus` interface). It uses the same VSYNC-locked, GC-disabled loop as `PresentStreamOfImages` (which now delegates to it).

```
elements := []stimuli.StreamElement{
    {Stimulus: stimuli.NewTextLine("Ready?", 0, 0, color), DurationOn: 600 *
     ↪  time.Millisecond, DurationOff: 200 * time.Millisecond},
    {Stimulus: pic,  DurationOn: 800 * time.Millisecond}, // held while the
     ↪  tone plays
    {Stimulus: tone, DurationOn: 400 * time.Millisecond, DurationOff: 200 *
     ↪  time.Millisecond},
}
events, logs, err := stimuli.PresentStreamOfStimuli(exp.Screen, elements,
 ↪  x, y)

// Builders mirror the visual/sound ones but take []stimuli.Stimulus:
elements = stimuli.MakeRegularStream(stims, durationOn, durationOff)
elements, err = stimuli.MakeStream(stims, onsetMs, durationMs)
```

Per-element semantics:

- **Visual** elements (those satisfying `VisualStimulus`) are centered on `(x, y)` and redrawn every frame for `DurationOn`, then blanked for `DurationOff`.
- **Audio / non-visual** elements (and a `nil` Stimulus) are triggered once, right after the slot's first VSYNC flip, and **do not clear the screen** — the previous frame is held for the whole slot. Place a visual just before a sound to keep it on screen while the sound plays. `TimingLog.OnsetNS` is the VSYNC reference at which the audio was triggered.

Audio elements must already be bound to the device via `PreloadDevice(exp.AudioDevice)` before the call (same precondition as `PlayStreamOfSounds`). For pure-audio streams prefer `PlayStreamOfSounds`, whose sub-frame sleep-polling gives finer timing than 60 Hz frame quantization.

8.3.6.5 Per-frame callback

`PresentStreamOfStimuliFunc` adds an optional `FrameCallback` invoked once per frame, **just before each flip**. Use it to draw a persistent overlay (trial counter, frame border, fixation) on top of the stimulus, or to run real-time logic such as firing feedback the instant a response window lapses — things the plain stream functions cannot express.

```
header := stimuli.NewTextLine("3 / 40", 0, -300, control.White)
cb := func(ctx stimuli.FrameContext) error {
    _ = header.Draw(ctx.Screen)           // overlay, rendered over the
 ↪  stimulus
    if ctx.OnPhase && ctx.FirstFrame {    // onset of element ctx.Index
        // real-time feedback using ctx.NowNS / ctx.Events;
```

```
        // return sdl.EndLoop to stop early, any other error to abort
    }
    return nil
}
events, logs, err := stimuli.PresentStreamOfStimuliFunc(exp.Screen,
 ↪  elements, x, y, cb)
```

`FrameContext` fields: `Screen`, `Index`, `Frame`, `OnPhase`, `FirstFrame`, `NowNS` (pre-flip `sdl.TicksNS()`), `Elapsed`, `Events` (through the previous frame). Passing `nil` for the callback is identical to `PresentStreamOfStimuli`.

8.3.6.6 Per-stimulus onset trigger (post-flip hook)

To emit a hardware TTL — or run any action — aligned to each stimulus **onset**, use `PresentStreamOfStimuliHooks`, which adds an optional `OnsetCallback` to the per-frame callback:

```
func PresentStreamOfStimuliHooks(
    screen *apparatus.Screen, elements []StreamElement, x, y float32,
    onFrame FrameCallback, onOnset OnsetCallback,
) ([]UserEvent, []TimingLog, error)

type OnsetCallback func(index int, onsetNS uint64) error
```

`onOnset` fires once per element with a stimulus onset, **immediately after** the VSYNC flip that turns it on (for a held audio element whose `DurationOn` rounds to zero frames, after the first-off-frame flip that triggers it). `index` matches the returned `TimingLog` slice and `onsetNS` equals `TimingLog[index].OnsetNS`.

This is the **post-flip** counterpart to `FrameCallback`: `FrameCallback` runs one frame earlier (pre-flip, at GPU-submission time), whereas `OnsetCallback` runs on the photon side of the frame boundary, so a trigger emitted here coincides with the logged onset instead of leading it by a frame.

```
dev := triggers.NewParallelPort("/dev/parport0") // any
 ↪  triggers.OutputTTLDevice
if err := dev.Open(); err != nil {
    log.Fatal(err)
}
defer dev.Close()

onset := func(index int, onsetNS uint64) error {
    return dev.Send(codes[index])        // per-stimulus code, at the
     ↪  displayed onset
}
clear := func(ctx stimuli.FrameContext) error {
    if !ctx.OnPhase && ctx.FirstFrame {  // drop the lines at the start of
     ↪  each ISI
        return dev.AllLow()
```

```
    }
    return nil
}
events, logs, err := stimuli.PresentStreamOfStimuliHooks(exp.Screen,
 ↪  elements, 0, 0, clear, onset)
```

The hook runs on the timing-critical path with GC disabled, between the onset flip and the next frame, so it **must be short and non-blocking**: emit an edge (`SetHigh` / `Send`) here and clear it from a later `FrameCallback` or after the stream — never `time.Sleep` or a blocking `Pulse`, or a frame is missed. Elements with a `nil` Stimulus (pure delay / hold slots) have no onset and do not fire it. Either callback may be `nil`; `PresentStreamOfStimuliFunc` is exactly `PresentStreamOfStimuliHooks(..., onFrame, nil)`.

For pure-audio streams the equivalent is `PlayStreamOfSoundsHook(elements, onOnset)`, which fires `onOnset` once per non-nil `Sound`, immediately after `Play()`, with the same SDL-clock `onsetNS` (`== TimingLog[index].OnsetNS`). `PlayStreamOfSounds` is `PlayStreamOfSoundsHook(elements, nil)`.

8.3.7 Audio Stimuli

```
// WAV file
snd := stimuli.NewSound(filePath)
snd.PreloadDevice(exp.AudioDevice)
snd.Play()
snd.Wait()                                      // block until done
snd.PlaySegment(onset, offset, rampSec)        // time-delimited segment with
 ↪  optional fade

// Embedded WAV
snd := stimuli.NewSoundFromMemory(data)

// Procedural tone
tone := stimuli.NewTone(frequency, duration, volume)   // duration:
 ↪  time.Duration; volume: 0–255
tone.PreloadDevice(exp.AudioDevice)
tone.Play()

// One-shot helper (no preload needed)
stimuli.PlaySoundFromMemory(exp.AudioDevice, data)

// Embedded feedback sounds (via assets_embed)
import "github.com/chrplr/goxpyriment/assets_embed"
stimuli.PlaySoundFromMemory(exp.AudioDevice, assets_embed.BuzzerWav)
stimuli.PlaySoundFromMemory(exp.AudioDevice, assets_embed.CorrectWav)
```

8.4 Package `apparatus` and `results`

Import: `github.com/chrplr/goxpyriment/apparatus` (screen, input, gamma) Import: `github.com/chrplr/goxpyriment/results` (data file)

In normal experiments you access `apparatus` types through `exp.Screen`, `exp.Keyboard`, `exp.Mouse`, and `exp.Data`. Direct use of `apparatus` is only needed when writing custom stimulus types.

8.4.1 Screen

All stimulus positions use a **center-origin coordinate system**: `(0, 0)` is the screen center; positive Y is upward.

`screen.Width` and `screen.Height` are the **logical** drawing space — the coordinate system stimuli are positioned in — and move with `SetLogicalSize`. They are what layout code should measure against; a width computed from anything else lands somewhere other than where `CenterToSDL` puts it. For *physical* pixels, as in a degrees-of-visual-angle calculation, ask `screen.Renderer.CurrentOutputSize()` instead. In fullscreen with no logical size set the two are the same, which is what makes the distinction easy to miss.

```
screen.CenterToSDL(x, y float32) (float32, float32)  // convert to SDL
 ↪  top-left coords
screen.CenteredRect(pos FPoint, w, h float32) *FRect  // SDL dest rect for a
 ↪  w×h texture centered at pos
screen.MousePosition() (float32, float32)             // current cursor in
 ↪  center coords
screen.Clear() error                                  // fill with
 ↪  background color
screen.Update() error                                 // present + hold to
 ↪  the frame boundary
screen.Flip() error                                   // alias for Update
screen.FlipTS() (uint64, error)                       // present + return
 ↪  SDL nanosecond timestamp after flip
screen.FrameDuration() time.Duration                  // nominal frame
 ↪  duration (falls back to 60 Hz)
screen.CalibrateRefresh(n int) (time.Duration, error)  // measured frame
 ↪  period, pacing bypassed
screen.SetLogicalSize(w, h int32) error               // also updates
 ↪  screen.Width/Height
screen.SetVSync(vsync int) error
screen.DisplayInfo() apparatus.DisplayInfo                  // monitor
 ↪  properties
screen.Destroy()
```

8.4.2 Laying out a block of text

`ShowInstructions`, `ShowEndMessage` and `FittedTextBox` size a text screen by measuring it, not by a fixed fraction of the window. Two things go wrong when a wrap width is written as a constant share of the screen:

- **The column count moves with the display.** A wrap width in pixels divided by a font in points is a number of characters, and it changes per machine: at 28 pt in the default monospace font, 80 % of a 1024-pixel-wide screen is 48 characters and 80 % of a 1920-pixel one is 90. Instruction text hand-wrapped in the source at the usual 55–70 columns therefore survives on the author's screen and is re-broken on a narrower one, each over-long line becoming a full line plus a short orphan.
- **Nothing checks the height.** A screen that has grown a few lines simply runs off the bottom edge, with no error and nothing in the log.

`FittedTextBox` wraps at `0.92 × DrawArea().w` and then picks the largest point size, up to `exp.DefaultFontSize`, at which the rendered block fits within `0.90 × DrawArea().h`. Where it can do so without shrinking below `keepBreaksFloor` (three quarters) of that size, it prefers a size at which no line the author wrote has to be re-wrapped — so hand-wrapped text keeps its own breaks on any display, while a paragraph written as one long line still wraps, as intended. Below `control.MinTextFontSize` (11 pt) it stops and logs a warning: a screen that will not fit at 11 pt is too long, not too big.

The fitted font belongs to the experiment and is closed by `End()`; a caller of `FittedTextBox` only has to `Unload()` the box.

`FlipTS` returns `sdl.TicksNS()` captured immediately after the flip. This timestamp is on the same nanosecond clock as SDL3 event timestamps, so `int64(event.Timestamp - onsetNS)` gives hardware-precision reaction time without any polling latency.

`Update` (and therefore `Flip`, `FlipTS`, `Show`, `ShowTS`) presents **and then holds to the frame boundary**, so one call always occupies exactly one display frame. This is unconditional because `SDL_RenderPresent` cannot be trusted to block until the retrace — under triple/mailbox buffering it returns immediately — and the wait costs nothing where the driver does block. Pacing is skipped when VSync is off.

`CalibrateRefresh` bypasses that pacing and presents directly, so it reports the *unaided* driver behaviour; compare it against `FrameDuration`. `Initialize` runs it over 60 frames at startup and writes `sys.refresh_nominal_hz` / `sys.refresh_measured_hz` into the session's `-info.txt`, warning if they differ by more than 10%. Measured slower than nominal means frames are being dropped before the panel, which pacing cannot fix.

8.4.2.1 Holding a stimulus for a fixed number of frames

There is no "wait for *n* VSYNC edges" call, and there cannot be a useful one: SDL3 exposes no `SDL_WaitVBlank`, so the only way to stay locked to the display is to present a frame — and a frame carrying no draw calls is not reliably scanned out under a compositor (see *Never present a frame with no draw calls* in `apparatus/CLAUDE.md`). A hold must therefore **redraw its content once per frame**:

```
for i := 0; i < n; i++ {
    screen.Clear()
    stim.Draw(screen)
    screen.Flip()
}
```

`exp.ShowFrames(stim, n)` and `exp.BlankFrames(n)` wrap that loop and return the onset timestamp of the first flip:

```
onsetNS, _ := exp.ShowFrames(rect, 10)   // 10 frames on screen
offsetNS, _ := exp.BlankFrames(10)       // 10 frames blank
exp.Data.Add(onsetNS, offsetNS)
```

There is no "wait for n frames" call that skips the redraw. A hold must redraw its content once per frame: SDL invalidates the backbuffer on every present, so re-clearing with whatever draw colour was last set paints the screen in that colour instead of holding the stimulus. Use the two calls above.

8.4.3 Keyboard

```
key, err := exp.Keyboard.Wait()                                    // any key
key, err := exp.Keyboard.WaitKey(control.K_SPACE)                  // specific
 ↪  key
key, err := exp.Keyboard.WaitKeys(keys, timeoutMS)                  // first
 ↪  of several keys (-1 = no timeout)
key, rt, err := exp.Keyboard.WaitKeysRT(keys, timeoutMS)           // with RT
 ↪  in ms from call site
key, ts, err := exp.Keyboard.GetKeyEventTS(keys, timeoutMS)        // with
 ↪  SDL event timestamp (nanoseconds)
events, err := exp.Keyboard.GetKeyEventsTS(keys, timeoutMS)        // first
 ↪  key + 50 ms window; for bilateral responses
events, err := exp.Keyboard.CollectKeyEventsTS(keys, durationMS)   // all
 ↪  keys during full fixed window
key, err := exp.Keyboard.Check()                                   //
 ↪  non-blocking poll
held := exp.Keyboard.IsPressed(key)                                // true if
 ↪  key is physically held now
upTS, err := exp.Keyboard.WaitKeyReleaseTS(key, timeoutMS)         // wait
 ↪  for KEY_UP; returns hardware timestamp
exp.Keyboard.Clear()                                              // drain SDL
 ↪  event queue
```

`WaitKeys` and `WaitKeysRT` return `0, nil` on timeout; return `sdl.EndLoop` on ESC or window close.

`IsPressed` queries SDL's scancode state array — no event queue involvement. `WaitKeyReleaseTS` returns the KEY_UP hardware timestamp so that `upTS - downTS` gives nanosecond-precision press duration.

`GetKeyEventsTS` — two-phase collection for bilateral responses. The timeout governs how long to wait for the *first* key. After the first key arrives, the function waits an additional 50 ms for any second key before returning. This extra window is necessary because human "simultaneous" bilateral presses (e.g. both hands at once) arrive 10–50 ms apart — a non-blocking drain after the first key would miss the second key almost every time. Use `GetKeyEventTS` for ordinary single-key trials to avoid the 50 ms overhead.

`GetKeyEventTS` returns the SDL3 `KeyboardEvent.Timestamp` field — the nanosecond time at which the hardware key-down event was generated, on the same clock as `sdl.TicksNS()` and `Screen.FlipTS()`. This allows computing reaction time from any specific stimulus onset without manual arithmetic:

```
onset, _ := exp.ShowTS(stim1)    // nanoseconds at VSYNC flip
exp.Wait(500)
exp.ShowTS(stim2)
key, eventTS, _ := exp.Keyboard.GetKeyEventTS(responseKeys, -1)
rtToStim1 := int64(eventTS - onset)  // nanoseconds
```

8.4.4 Mouse

```
x, y := exp.Mouse.Position()                                  // current position
 ↪  (center coords)
btn, err := exp.Mouse.WaitPress()                              // block until
 ↪  button pressed
btn, rt, err := exp.Mouse.WaitPressRT(timeoutMS)               // with RT in ms
 ↪  from call site
btn, ts, err := exp.Mouse.GetPressEventTS(timeoutMS)           // with SDL event
 ↪  timestamp (nanoseconds)
btn, err := exp.Mouse.Check()                                 // non-blocking poll
held := exp.Mouse.IsPressed(sdl.BUTTON_LEFT)                   // true if button
 ↪  is physically held now
upTS, err := exp.Mouse.WaitButtonReleaseTS(btn, timeoutMS) // wait for
 ↪  MOUSE_BUTTON_UP; hardware timestamp
exp.Mouse.ShowCursor(show bool) error
```

`WaitPressRT` mirrors `Keyboard.WaitKeysRT`: reaction time is measured in milliseconds from the call site. `GetPressEventTS` returns the SDL3 hardware event timestamp in nanoseconds, suitable for use with `ShowTS`. `IsPressed` and `WaitButtonReleaseTS` mirror the keyboard's `IsPressed` and `WaitKeyReleaseTS`.

8.4.5 GamePad

```
pads, err := apparatus.GetGamePads()                                    //
 ↪  enumerate connected gamepads
defer pads[0].Close()
```

```
btn, err := pads[0].WaitPress()                                    // block until
 ↪  any button
btn, ts, err := pads[0].GetPressEventTS(timeoutMS)                  // with SDL
 ↪  event timestamp (nanoseconds)
```

`GetPressEventTS` returns the `GamepadButtonEvent.Timestamp` field — same nanosecond clock as `Screen.FlipTS` and keyboard/mouse event timestamps.

8.4.6 Unified Input — `WaitAnyEventTS`

When the response device is not fixed in advance (keyboard *or* mouse click), use the method on `Experiment`:

```
// Accept F or J key, or any mouse button, timeout after 3 s
ev, err := exp.WaitAnyEventTS(
    []control.Keycode{control.K_F, control.K_J},
    true,   // catchMouse
    3000,
)
```

Returns an `apparatus.InputEvent`:

```
type InputEvent struct {
    Device        apparatus.DeviceKind     // DeviceKeyboard | DeviceMouse |
 ↪  DeviceGamepad
    Key           sdl.Keycode       // non-zero for keyboard events
    Button        uint32             // non-zero for mouse events
    GamepadButton sdl.GamepadButton // non-zero for gamepad events
    TimestampNS   uint64             // SDL3 hardware timestamp, nanoseconds
}
```

`TimestampNS` is on the same clock as `ShowTS`, so RT computation is identical regardless of device:

```
onset, _ := exp.ShowTS(stim)
ev, _ := exp.WaitAnyEventTS(keys, true, -1)
rtNS := int64(ev.TimestampNS - onset)
```

Pass `keys = nil` to accept any key. Pass `catchMouse = false` to ignore the mouse. On timeout, returns a zero `InputEvent` and `nil` error. On ESC or quit, returns `sdl.EndLoop`.

8.4.7 ResponseDevice

`ResponseDevice` is a unified interface over all participant-input hardware — SDL-event-driven devices (keyboard, mouse, gamepad) **and** polled TTL devices (MEGTTLBox, DLPIO8). It is the recommended abstraction when the experiment design does not commit to a specific input modality.

```
type ResponseDevice interface {
    WaitResponse(ctx context.Context) (Response, error)
    DrainResponses(ctx context.Context) error
    Close() error
}

type Response struct {
    Source  apparatus.DeviceKind  // DeviceKeyboard | DeviceMouse |
 ↪  DeviceGamepad | DeviceTTL
    Code    uint32         // SDL Keycode, mouse button, gamepad button, or
 ↪  TTL bitmask
    RT      time.Duration  // elapsed from WaitResponse call to detection
    Precise bool           // true = hardware event timestamp; false =
 ↪  software poll
}
```

`Response.Precise` distinguishes two timing regimes:

Device	Precise	RT origin
Keyboard, Mouse, Gamepad	`true`	SDL3 hardware event timestamp (nanosecond)
MEGTTLBox, DLPIO8	`false`	`time.Now()` at poll detection (±poll interval, ~5 ms)

Construct wrappers with the provided adapters:

```
// SDL-event-driven devices
rd := &apparatus.KeyboardResponseDevice{KB: exp.Keyboard}
rd := &apparatus.MouseResponseDevice{M: exp.Mouse}
rd := &apparatus.GamepadResponseDevice{GP: pad}

// Polled TTL device (MEGTTLBox, DLPIO8, or any type with
 ↪  ReadAll/DrainInputs)
box, _ := triggers.NewMEGTTLBox("/dev/ttyACM0")
rd := apparatus.NewTTLResponseDevice(box, 5*time.Millisecond)
```

Usage in a trial loop:

```
onset, _ := exp.ShowTS(stim)
_ = rd.DrainResponses(ctx)
resp, err := rd.WaitResponse(ctx)
// resp.RT is always valid; resp.Precise tells you whether to trust
 ↪  nanosecond accuracy
```

8.4.8 Microphone

```
// Construction
mic, err := apparatus.NewMicrophone(spec)                          // default
 ↪  recording device
mic, err := apparatus.NewMicrophoneFromDevice(devID, spec)          //
 ↪  specific device
apparatus.DefaultRecordingSpec() sdl.AudioSpec                      // F32LE
 ↪  mono 44100 Hz
devices, err := apparatus.GetRecordingDevices() []sdl.AudioDeviceID

// Fields
mic.DeviceID  sdl.AudioDeviceID
mic.Stream    *sdl.AudioStream
mic.Spec      sdl.AudioSpec

// Methods
captureStartNS, err := mic.StartCapture()    // flush buffer, resume device,
 ↪  return TicksNS()
err := mic.StopCapture()                     // pause device
n, err := mic.ReadSamples(buf []byte)       // read PCM bytes (non-blocking)
n, err := mic.Available()                    // bytes waiting to be read
mic.Close()                                  // destroy stream
```

In normal experiments, open the microphone via `exp.OpenMicrophone(spec)` (see control package); `exp.Microphone` is then available throughout the experiment and closed by `exp.End()`.

8.4.9 VoiceKey

Detects voice onset by computing per-window RMS over F32LE PCM samples.

```
vk := apparatus.NewVoiceKey(mic, threshold float32, windowSz int) *VoiceKey
// threshold: RMS amplitude (0–1). Recommended starting value: 0.02–0.05.
// windowSz:  samples per RMS window. 0 → 128 (≈ 2.9 ms at 44100 Hz).

captureStartNS, err := vk.Arm()   // flush mic buffer, start capture, return
 ↪  TicksNS()

onsetNS, pcm, err := vk.WaitOnset(captureStartNS uint64, timeoutMS int)
 ↪  (uint64, []byte, error)
// Returns: onset timestamp (sdl.TicksNS() domain), all PCM bytes captured,
 ↪  nil or ErrVoiceTimeout.
```

`onsetNS` and the `imageOnsetNS` returned by `exp.ShowTS` or `screen.FlipTS` are on the same SDL3 nanosecond clock:

```
captureStartNS, _ := vk.Arm()
imageOnsetNS, _   := exp.ShowTS(picture)
```

```
onsetNS, pcm, _   := vk.WaitOnset(captureStartNS, 2000)
rtMs := int64(onsetNS - imageOnsetNS) / 1_000_000
```

Sentinel error: `apparatus.ErrVoiceTimeout`

Post-hoc helpers (for re-analysing saved WAV files):

```
sampleIdx := apparatus.ScanOnset(pcm []byte, threshold float32, windowSz
 ↪  int) int
// Returns the sample index of the first onset window, or -1 if not found.

onsetNS := apparatus.SampleOnsetNS(captureStartNS uint64, sampleIndex,
 ↪  sampleRate int) uint64
// Converts a sample index back to a nanosecond timestamp.

names, err := apparatus.DeviceNames() map[sdl.AudioDeviceID]string
// Returns all recording devices and their human-readable names.
```

8.4.10 WriteWAV

```
apparatus.WriteWAV(path string, spec sdl.AudioSpec, data []byte) error
```

Saves raw PCM bytes as a standard RIFF/WAV file. Supports `AUDIO_F32LE`, `AUDIO_S16LE`, `AUDIO_S32LE`, and `AUDIO_U8`. Writes atomically via a temp file + rename.

8.4.11 DataFile

```
exp.Data.Add(field1, field2, ...)              // append a data row
exp.Data.AddVariableNames([]string{...})       // write column header
exp.Data.WriteDisplayInfo(info)                // append display metadata to
 ↪  the info file
exp.Data.WriteHostInfo(sysinfo.Host())         // append machine/OS metadata
 ↪  to the info file
exp.Data.WriteParticipantInfo(info)            // append --PARTICIPANT INFO
 ↪  block to the info file (called automatically by Initialize when exp.Info
 ↪  is set)
exp.Data.WriteEndTime()                        // append end time + duration
 ↪  to the info file
```

Two files are written to `~/goxpy_data/` for each session:

File	Contents
`<expname>_sub-<NNN>_date-<YYYYMMDD>-<HHMMSS>.csv`	Pure CSV data rows — directly importable by Excel, R, or pandas

File	Contents
`<expname>_sub-<NNN>_date-<YYYYMMDD>-<HHMMSS>-info.txt`	#-prefixed metadata: start/end time, hostname, OS, framework version, display and audio configuration, participant info

---

8.5 Package `media`

Import: `github.com/chrplr/goxpyriment/media`

Multi-movie playback with shared master-clock synchronisation, look-ahead frame conditions, and post-vsync display events for hardware- trigger alignment. Adds `MovieManager` (per-experiment owner of the clock and movie set), `Movie` (per-movie state with PsyScope-Tahoe semantics), and the `media/present` subpackage of OS-level vsync backends. Pure Go, `.gv` movies only, silent.

Full reference (precision contracts, platform support, mapping to PsyScope movie commands): see MediaMovies.md.

8.5.1 Quick API summary

```
mgr := media.NewMovieManager(exp.Screen)
defer mgr.Close()

gv, _ := gvvideo.LoadGVVideo("clip.gv")
mov, _ := media.NewMovie(mgr, gv,
    media.WithTag("M1"),
    media.WithRepeat(-1),
    media.WithSize(640, 360),
    media.WithPosition(sdl.FPoint{X: -200, Y: 0}),
)
defer mov.Close()

// Look-ahead — fires from inside DrawWithoutFlip, BEFORE the frame is
 ↪  presented.
mov.OnAt(media.Frame(186), func(o media.Onset) { /* ... */ })

// Hardware-verified — fires from NotifyFlipped with the OS vsync timestamp.
mov.OnAtDisplay(media.Frame(186), func(o media.Onset) {
    _ = ttl.Pulse(0, 5*time.Millisecond)
})

// Display[Onset/Offset] — fires when the named tag changes visibility.
mgr.OnDisplayOnset("M1", func(o media.Onset) { /* ... */ })

// Atomic command burst (PsyScope script-line equivalent):
```

```
mgr.BeginBurst()
movieA.Pause(); movieB.Pause(); movieC.Pause()
mgr.EndBurst()

// Per-frame, mixed compositing:
exp.Screen.Clear()
mgr.DrawWithoutFlip()           // decode + render movies
fix.Present(exp.Screen, false, false)  // overlay
ts, _ := exp.Screen.FlipTS()
mgr.NotifyFlipped(ts)
```

8.5.2 Onset precision summary

Source	Where	Meaning
`LookAhead`	`OnAt` / `OnDone` callbacks	`sdl.TicksNS()` at fire time, pre-vsync
`VsyncEstimated`	`OnAtDisplay` / `Display*` callbacks (fallback timer)	Post-`Present` `FlipTS`, ~vsync-period
`HardwareVerified`	`OnAtDisplay` / `Display*` callbacks (Stage 5 timer)	OS-measured first-pixel-visible, sub-ms

8.5.3 Platform status

Platform	Backend	Precision
macOS	`CVDisplayLink` (CoreVideo via purego)	`HardwareVerified`
Linux	`DRM_IOCTL_WAIT_VBLANK` (`/dev/dri/cardN`, `video` group)	`HardwareVerified`
Windows	(not yet — fallback)	`VsyncEstimated`
Other	(fallback)	`VsyncEstimated`

See MediaMovies.md §8 for the Windows roadmap and §7 for the trigger-synchrony contract (the wire pulse arrives ~0.5–5 ms after `Onset.TimestampNS`; calibrate the constant offset with a photodiode for sub-ms wire-side alignment).

8.6 Package `design`

Import: `github.com/chrplr/goxpyriment/design`

8.6.1 Data Structures

```
// Trial — one experimental trial
trial := design.NewTrial()
trial.SetFactor("condition", "congruent")
trial.GetFactor("condition")   // → "congruent"
trial.Copy()                   // deep copy

// Block — a sequence of trials
block := design.NewBlock("Practice")
block.SetFactor("type", "practice")
block.AddTrial(trial, copies, randomPosition)
block.ShuffleTrials()

// Experiment design (separate from control.Experiment)
exp.Design  // *design.Experiment — contains Blocks, DataVariableNames, etc.
```

8.6.2 Randomization

```
design.RandInt(a, b int) int                          // random int in [a, b]
design.RandElement(list []T) T                        // random element
 ↪  (generic)
design.CoinFlip(headBias float64) bool                // weighted coin flip
design.RandNorm(a, b float64) float64                 // truncated normal in
 ↪  [a, b]
design.ShuffleList(list []T)                         // in-place Fisher-Yates
 ↪  shuffle (generic)
design.MakeMultipliedShuffledList(list []T, n int) []T  // n shuffled
 ↪  copies concatenated
design.RandIntSequence(first, last int) []int         // shuffled range
 ↪  [first, last]
```

8.6.3 Latin Square (Between-Subjects Counterbalancing)

```
// Registration
exp.AddBWSFactor("handedness", []interface{}{"left", "right"})

// At runtime — returns the condition for the current subject
condition := exp.GetPermutedBWSFactorCondition("handedness")

// Low-level
square, err := design.LatinSquare(elements, design.PBalancedLatinSquare)
square, err := design.LatinSquareInts(n, design.PCycledLatinSquare)
// permutation types: design.PBalancedLatinSquare, PCycledLatinSquare,
 ↪  PRandom
```

8.7 Package `clock`

Import: `github.com/chrplr/goxpyriment/clock`

```
clock.Wait(ms int)                      // block for ms milliseconds
clock.GetTime() int64                   // ms since package first used
clock.GetTimeNS() int64                // nanoseconds since package first used

c := clock.NewClock()                   // clock relative to "now"
c.NowMillis() int64                     // ms elapsed
c.NowNanos() int64                      // nanoseconds elapsed
c.Now() time.Duration
c.Reset()                               // restart reference
c.SleepUntil(target time.Duration)     // sleep until target offset (returns
 ↪  immediately if past)
```

> **Note:** Prefer `exp.Wait(ms)` over `clock.Wait(ms)` in experiment code — `exp.Wait` pumps SDL events and detects ESC.
>
> **Clock domains:** `GetTimeNS()` and `NowNanos()` use the Go monotonic clock (`time.Since`). SDL event timestamps from `Screen.FlipTS`, `GetKeyEventTS`, `GetPressEventTS`, and `WaitAnyEventTS` use `sdl.TicksNS()`. The two clocks have different origins and **must not be subtracted from each other** for reaction-time computation. Use the SDL-based functions exclusively for RT measurement.

8.8 Package `geometry`

Import: `github.com/chrplr/goxpyriment/geometry`

```
geometry.GetDistance(p1, p2 sdl.FPoint) float32
geometry.CartesianToPolar(x, y float32) (radius, angleDeg float32)
geometry.PolarToCartesian(radius, angleDeg float32) (x, y float32)
geometry.DegreeToRadian(deg float32) float64
```

8.9 Package `triggers`

Import: `github.com/chrplr/goxpyriment/triggers`

Provides hardware TTL signal output (EEG/MEG trigger codes) and TTL input (response pads wired over serial). Lines are **0-indexed (0–7)** throughout; bit N of a bitmask corresponds to line N.

8.9.1 Interfaces

```
// OutputTTLDevice — send trigger codes to recording equipment.
type OutputTTLDevice interface {
    Send(mask byte) error                    // set all 8 lines from bitmask
    SetHigh(line int) error                  // drive line HIGH (0-indexed)
    SetLow(line int) error                   // drive line LOW  (0-indexed)
    Pulse(line int, d time.Duration) error   // HIGH for d, then LOW (blocks)
    AllLow() error                           // all lines LOW
    Close() error                            // AllLow + release port
}

// InputTTLDevice — read TTL inputs from response hardware.
type InputTTLDevice interface {
    ReadAll() (byte, error)                                          //
 ↪  bitmask of all lines
    ReadLine(line int) (byte, error)                                // 0
 ↪  or 1 (0-indexed)
    WaitForInput(ctx context.Context) (mask byte, rt time.Duration, err
 ↪  error)
    DrainInputs(ctx context.Context) error
    Close() error
}
```

8.9.2 DLPIO8 (DLP-IO8-G, USB-CDC serial)

Implements both `OutputTTLDevice` and `InputTTLDevice`.

```
// Auto-detect (recommended): returns NullOutputTTLDevice if not found, no
 ↪  error.
out, portName, err := triggers.AutoDetectDLPIO8()
defer out.Close()
out.Pulse(0, 10*time.Millisecond)   // 10 ms pulse on line 0

// Manual:
d, err := triggers.NewDLPIO8("/dev/ttyUSB0")
defer d.Close()
d.Send(0b00000101)                  // lines 0 and 2 HIGH
mask, err := d.ReadAll()            // bitmask of all 8 input lines
mask, rt, err := d.WaitForInput(ctx)
```

8.9.3 DLPIO20 (DLP-IO20, USB-CDC serial)

Implements both `OutputTTLDevice` and `InputTTLDevice`. 5 V logic (the T4 is 3.3 V).

> **Untested on hardware** — written from the datasheet rev 1.1. Bring it up with `tests/test_dlpio20` and a multimeter before relying on it.

The device has 17 usable digital channels but the TTL interfaces are 8-bit, so interface

lines 0–7 address a *window* of 8 channels — `AN0–AN7` for output and `AN8–AN13`, `RB6`, `RB7` for input by default. Channels outside the windows stay reachable through the channel-level methods.

```
// Auto-detect (recommended): returns NullOutputTTLDevice if not found, no
 ↪  error.
out, portName, err := triggers.AutoDetectDLPIO20()
defer out.Close()

// Manual, with a remapped output window:
d, err := triggers.NewDLPIO20("/dev/ttyUSB0",
    triggers.WithIO20OutputChannels(
        triggers.IO20_AN0, triggers.IO20_AN1, triggers.IO20_AN2,
 ↪  triggers.IO20_AN3,
        triggers.IO20_AN4, triggers.IO20_AN5, triggers.IO20_AN6,
 ↪  triggers.IO20_AN7),
    triggers.WithIO20PollInterval(5*time.Millisecond),
)
defer d.Close()

d.Send(0b00000101)                      // lines 0 and 2 → AN0, AN2
d.Pulse(0, 5*time.Millisecond)
mask, err := d.ReadAll()                // bitmask of the 8 input-window
 ↪  channels

// Any of the 20 channels, in or out of the windows:
d.SetChannelHigh(triggers.IO20_RA4)
v, err := d.ReadChannel(triggers.IO20_AN12)
```

Channel constants are `IO20_AN0–IO20_AN13`, `IO20_RA4`, `IO20_P5–IO20_P7` (relay drivers — not TTL, not readable), `IO20_RB6`, `IO20_RB7`.

Timing: the device has no write-all command, so `Send` issues one 5-byte packet per line — the 8 lines change over ~3.5 ms rather than simultaneously. Use `SetHigh` on a single line for trigger onsets. On Linux, lower the FTDI latency timer (16 ms default) before timing-sensitive reads: `echo 1 | sudo tee /sys/bus/usb-serial/devices/ttyUSB0/latency_timer`.

8.9.4 MEGTTLBox (NeuroSpin Arduino Mega TTL box)

Implements both `OutputTTLDevice` and `InputTTLDevice`. Provides 8 TTL output lines (D30–D37) and 8 TTL input lines for a FORP response pad (D22–D29).

```
box, err := triggers.NewMEGTTLBox("/dev/ttyACM0",
    triggers.WithResetDelay(2*time.Second),   // wait for Arduino boot
 ↪  (default 2 s)
    triggers.WithPollInterval(5*time.Millisecond),
)
defer box.Close()
```

```
// Output
box.Pulse(0, 5*time.Millisecond)    // pulse line 0
box.PulseMask(0b00000011, 5*time.Millisecond)  // pulse lines 0 and 1
box.Send(0b00000001)                // set line 0 HIGH, all others LOW

// Input (FORP response pad)
_ = box.DrainInputs(ctx)            // clear latched presses from previous
 ↪  trial
mask, rt, err := box.WaitForInput(ctx)
buttons := triggers.DecodeMask(mask)  // []FORPButton
```

`FORPButton` constants (also serve as 0-indexed line numbers for bitmask operations):

```
triggers.FORPLeftBlue    // 0
triggers.FORPLeftYellow  // 1
triggers.FORPLeftGreen   // 2
triggers.FORPLeftRed     // 3
triggers.FORPRightBlue   // 4
triggers.FORPRightYellow // 5
triggers.FORPRightGreen  // 6
triggers.FORPRightRed    // 7
```

8.9.5 FT232HTrigger (Adafruit FT232H, Linux)

Implements both `OutputTTLDevice` and `InputTTLDevice`. Pure-Go driver via Linux usbfs — no libftdi or D2XX installation required.

Wiring: AD0–AD7 (D-bus) → 8 TTL output lines; AC0–AC7 (C-bus) → 8 TTL input lines.

```
box, err := triggers.NewFT232H(
    triggers.WithFT232HPollInterval(5*time.Millisecond), // optional;
 ↪  default 5 ms
)
if err != nil { log.Fatal(err) }
defer box.Close()

box.Send(0b00000001)              // AD0 HIGH (persistent)
box.Pulse(0, 5*time.Millisecond)  // AD0: HIGH for 5 ms, then LOW
box.AllLow()

_ = box.DrainInputs(ctx)
mask, rt, err := box.WaitForInput(ctx)

// Auto-detect (falls back to NullOutputTTLDevice if no device found):
out, err := triggers.AutoDetectFT232H()
defer out.Close()
```

Prerequisites (Linux): - The `ftdi_sio` kernel module must not hold the device: `sudo rmmod ftdi_sio` - User needs rw access to `/dev/bus/usb/BBB/DDD`. Recommended udev

rule: `ACTION=="add", SUBSYSTEM=="usb", ATTRS{idVendor}=="0403", ATTRS{idProduct}=="6014", MODE="0666", GROUP="plugdev"`

8.9.6 LinuxGPIOTrigger (Raspberry Pi and other Linux SBCs)

Implements both `OutputTTLDevice` and `InputTTLDevice`. Uses the Linux GPIO character device v2 API (kernel ≥ 5.10) — no external libraries required.

Wiring: any 8 GPIO pins for output, any other 8 for input. Pin numbers are chip-relative offsets (= BCM numbers on Raspberry Pi).

```
box, err := triggers.NewLinuxGPIOTrigger(
    triggers.WithGPIOOutputPins([8]int{17, 27, 22, 5, 6, 13, 19, 26}),
    triggers.WithGPIOInputPins([8]int{12, 16, 20, 21, 4, 25, 24, 23}),
    triggers.WithGPIOChip("/dev/gpiochip0"),           // optional; this is
 ↪  the default
    triggers.WithGPIOPollInterval(5*time.Millisecond), // optional; default
 ↪  5 ms
)
if err != nil { log.Fatal(err) }
defer box.Close()

box.Send(0b00000001)              // pin 17 HIGH
box.Pulse(0, 5*time.Millisecond)  // pin 17: HIGH for 5 ms, then LOW

_ = box.DrainInputs(ctx)
mask, rt, err := box.WaitForInput(ctx)
```

Output-only and input-only configurations are both valid — omit `WithGPIOOutputPins` or `WithGPIOInputPins` as needed.

Prerequisites: - Linux kernel ≥ 5.10 - User in the `gpio` group: `sudo usermod -aG gpio $USER`

8.9.7 LabJackT4 (Modbus TCP)

Implements both `OutputTTLDevice` and `InputTTLDevice`. Pure-Go Modbus TCP driver — no LabJack SDK or system library required.

Wiring: the T4's digital lines are DIO4–DIO19. Output lines 0–7 → DIO4–DIO11 (FIO4–FIO7 screw terminals + EIO0–EIO3 on the DB15); input lines 0–7 → DIO12–DIO19 (EIO4–EIO7 + CIO0–CIO3). DIO0–DIO3 are the T4's dedicated analog inputs AIN0–AIN3 and cannot be used as digital lines. DIO4–DIO11 are *flexible I/O* that power up as analog inputs; `NewLabJackT4` switches them to digital mode (`DIO_ANALOG_ENABLE`) before setting directions.

```
box, err := triggers.NewLabJackT4("192.168.1.100",
    triggers.WithT4PollInterval(5*time.Millisecond), // optional; default 5
 ↪  ms
    triggers.WithT4Timeout(1*time.Second),           // optional; default 1
 ↪  s
```

```
    triggers.WithT4UnitID(1),                          // optional; default 1
    triggers.WithT4OutputBase(4),                       // optional; DIO of
 ↪  output line 0
    triggers.WithT4InputBase(12),                       // optional; DIO of
 ↪  input line 0
)
if err != nil { log.Fatal(err) }
defer box.Close()

box.Send(0b00000001)              // output line 0 (FIO4) HIGH
box.Pulse(0, 5*time.Millisecond)  // FIO4: HIGH for 5 ms, then LOW

_ = box.DrainInputs(ctx)
mask, rt, err := box.WaitForInput(ctx)
```

The host string may include the port number (`"192.168.1.100:502"`); default Modbus port 502 is appended automatically if omitted.

8.9.8 ParallelPort (Linux LPT)

Implements `OutputTTLDevice`.

```
ports := triggers.AvailableParallelPorts()      // scans /dev/parport0..3
pp := triggers.NewParallelPort("/dev/parport0")
if err := pp.Open(); err != nil { log.Fatal(err) }
defer pp.Close()
pp.Send(0b00000111)                             // lines 0,1,2 HIGH
pp.Pulse(0, 10*time.Millisecond)
status, _ := pp.ReadStatus()                  // Linux only: status register
```

Prerequisites: `sudo modprobe ppdev`; user in the `lp` group.

8.9.9 Null devices

`NullOutputTTLDevice` and `NullInputTTLDevice` are silent no-ops, safe to call without hardware. `AutoDetectDLPIO8` and `AutoDetectFT232H` return a `NullOutputTTLDevice` when no device is found.

8.10 Package `assets_embed`

Import: `github.com/chrplr/goxpyriment/assets_embed`

Provides embedded default assets as `[]byte` slices, ready for `FontFromMemory` or `PlaySoundFromMemory`:

```
assets_embed.InconsolataFont  []byte  // default monospace TTF font
assets_embed.BuzzerWav        []byte  // error/incorrect feedback sound
```

```
assets_embed.CorrectWav       []byte  // correct/reward feedback sound
```

8.11 Coordinate System

All stimulus positions are in **screen-center coordinates**:

- `(0, 0)` = screen center
- Positive X = right; positive Y = **up** (not down)
- `screen.CenterToSDL(x, y)` converts to SDL's top-left origin

```
          +Y (up)
           |
 -X ------+------ +X
           |
          -Y (down)
```

8.12 Typical Experiment Structure

```
package main

import (
    "log"
    "github.com/chrplr/goxpyriment/control"
    "github.com/chrplr/goxpyriment/design"
    "github.com/chrplr/goxpyriment/stimuli"
)

func main() {
    exp := control.NewExperimentFromFlags("My Experiment", control.Black,
↪   control.White, 32)
    defer exp.End()

    exp.AddDataVariableNames([]string{"block", "trial", "condition", "key",
↪   "rt_ms"})

    // Build design
    block := design.NewBlock("main")
    for _, cond := range []string{"left", "right"} {
        t := design.NewTrial()
        t.SetFactor("condition", cond)
        block.AddTrial(t, 10, false)
    }
    block.ShuffleTrials()
```

```
    exp.AddBlock(block, 1)

    err := exp.Run(func() error {
        exp.ShowInstructions("Press F for left, J for right.\n\nPress SPACE
↪   to start.")

        for bi, blk := range exp.Design.Blocks {
            for ti, trial := range blk.Trials {
                cond := trial.GetFactor("condition").(string)

                exp.Show(stimuli.NewFixCross(20, 2, control.White))
                exp.Wait(500)

                stim := stimuli.NewTextLine(cond, 0, 0, control.White)
                exp.Show(stim)

                key, rt, err := exp.Keyboard.WaitKeysRT(
                    []control.Keycode{control.K_F, control.K_J}, 3000,
                )
                if control.IsEndLoop(err) {
                    return control.EndLoop
                }

                exp.Data.Add(bi, ti, cond, key, rt)
                exp.Blank(500)
            }
        }
        return control.EndLoop
    })
    if err != nil && !control.IsEndLoop(err) {
        log.Fatalf("experiment error: %v", err)
    }
}
```

Chapter 9

Migrating to goxpyriment

This guide is for researchers who already know one of the mainstream experiment libraries and want to get productive in goxpyriment quickly. Each section maps the concepts and idioms of one tool to their goxpyriment equivalents, highlights the most important differences, and provides side-by-side code for a representative trial sequence.

Before reading this guide, complete the Getting Started tutorial. It introduces the coordinate system, the rendering model, and the trial loop idiom that all three sections below assume.

9.1 From Expyriment (Python)

Goxpyriment was directly inspired by Expyriment and shares its philosophy: a single experiment object owns all resources, stimuli are objects with a `present`/`Show` method, and data are written to a structured file automatically. The mapping is the closest of the three tools covered here.

9.1.1 Concept map

Expyriment (Python)	goxpyriment (Go)	Notes
`design.Experiment("name")`	`control.NewExperimentFromFlags("name", bg, fg, fontSize)`	Also parses `-w`, `-d`, `-s` flags.
`control.initialize(exp)`	*(called inside `NewExperimentFromFlags`)*	SDL, audio, font, and data file initialized automatically.

Expyriment (Python)	goxpyriment (Go)	Notes
`control.end(exp)`	`defer exp.End()`	Always defer immediately after construction.
`exp.clock.wait(1000)`	`exp.Wait(1000)`	Both pump the event loop; both abort on ESC.
`exp.clock.reset()`	`c := clock.NewClock()` then `c.Reset()`	`clock.NewClock()` starts a new reference; `c.Reset()` restarts it.
`exp.clock.time`	`c.NowMillis()`	Returns `int64` milliseconds.
`stim.present()`	`exp.Show(stim)`	Both do clear → draw → flip in one call.
`stimuli.TextLine("Hello", (0,0), exp.screen)`	`stimuli.NewTextLine("Hello", 0, 0, control.White)`	Position is center-relative in both (Expyriment center = (0,0), Y+ = up).
`stimuli.Circle(radius=50)`	`stimuli.NewCircle(50, color)`	
`stimuli.FixCross(size=30)`	`stimuli.NewFixCross(30, lineWidth, color)`	
`stimuli.Picture("img.png")`	`stimuli.NewPicture("img.png", x, y)`	
`exp.keyboard.wait()`	`exp.Keyboard.Wait()`	Returns `(Keycode, error)`.
`exp.keyboard.wait_for_keys([K_F, K_J])`	`exp.Keyboard.WaitKeys([]Keycode{K_F, K_J}, timeout)`	`timeout = -1` means no timeout.
`exp.keyboard.wait_for_keys(...)` RT	`exp.Keyboard.WaitKeysRT(keys, timeout)`	Returns `(key, rtMs, error)`.
`exp.data_variable_names = [...]`	`exp.AddDataVariableNames([]string{...})`	`subject_id` is prepended automatically — do not include it.
`exp.data.add([v1, v2, ...])`	`exp.Data.Add(v1, v2, ...)`	
`design.Block()`	`design.NewBlock("name")`	
`design.Trial()`	`design.NewTrial()`	
`trial.set_factor("cond", "target")`	`trial.SetFactor("cond", "target")`	
`trial.get_factor("cond")`	`trial.GetFactor("cond")`	
`block.add_trials([t1, t2])`	`block.AddTrial(t, copies, randomPosition)`	
`block.shuffle_trials()`	`block.ShuffleTrials()`	

`io.DataFile`	`exp.Data (*io.DataFile)`	Opened automatically; output to `~/goxpy_data/`. `.csv` is the same CSV-with-comments format.

9.1.2 Side-by-side: simple RT trial

Expyriment

```
import expyriment
from expyriment import design, control, stimuli, io

exp = design.Experiment("SimpleRT")
control.initialize(exp)

fix   = stimuli.FixCross()
target = stimuli.Circle(radius=30, colour=(255,255,255))

exp.data_variable_names = ["rt"]

fix.present()
exp.clock.wait(500)
target.present()
key, rt = exp.keyboard.wait()
exp.data.add([rt])

control.end(exp)
```

goxpyriment

```
package main

import (
    "log"
    "github.com/chrplr/goxpyriment/control"
    "github.com/chrplr/goxpyriment/stimuli"
)

func main() {
    exp := control.NewExperimentFromFlags("SimpleRT", control.Black,
 ↪  control.White, 32)
    defer exp.End()

    fix    := stimuli.NewFixCross(30, 3, control.White)
    target := stimuli.NewCircle(30, control.White)

    exp.AddDataVariableNames([]string{"rt"})
```

```
    err := exp.Run(func() error {
        exp.ShowTimed(fix, 500)
        exp.Show(target)
        _, rt, err := exp.Keyboard.WaitKeysRT(nil, -1)
        if err != nil {
            return err
        }
        exp.Data.Add(rt)
        return control.EndLoop
    })
    if err != nil && !control.IsEndLoop(err) {
        log.Fatal(err)
    }
}
```

9.1.3 Key differences

The run loop. In Expyriment the trial logic runs in `main()` directly. In goxpyriment all SDL calls must run on the thread that owns the window; `exp.Run(func() error { ...})` enforces this. Return `control.EndLoop` to exit cleanly.

Clock domains. Expyriment's `exp.clock.time` measures from `control.initialize`. `clock.NewClock()` measures from whenever you create it. For stimulus-onset-locked RT, use `exp.ShowTS + exp.Keyboard.GetKeyEventTS` (SDL nanosecond timestamps) rather than the `clock` package — see UserManual §6.

`.csv` files. The format is compatible: plain CSV with #-prefixed metadata lines. Existing Python scripts that read Expyriment data files with `pandas.read_csv(..., comment='#')` will read goxpyriment output without modification.

Adaptive staircases. Expyriment has no built-in staircase. Goxpyriment provides `staircase.NewUpDown` (Levitt 1971) and `staircase.NewQuest` (Watson & Pelli 1983), including a `Runner` for interleaved designs. Import `github.com/chrplr/goxpyriment/staircase`.

RSVP streams. Expyriment's `stimuli.extras.StreamingTextDisplay` is replaced by `stimuli.PresentStreamOfText` / `stimuli.PresentStreamOfImages`, which are VSYNC-locked, disable GC, and return a full `TimingLog` per item with nanosecond onset/offset timestamps.

9.2 From PsychoPy (Python)

PsychoPy Coder mode and goxpyriment are structurally similar: both give you a window object, stimulus objects, and explicit flip calls. The main adjustments are coordinate convention, the rendering model, and how timing is measured.

9.2.1 Concept map

PsychoPy (Python)	goxpyriment (Go)	Notes
`visual.Window(size=[1024,768], units='pix')`	`control.NewExperimentFromFlags(...)` with `-w` flag	Pass `-w` for a 1024×768 window; default is fullscreen.
`win.flip()`	`exp.Flip()`	Both are VSYNC-locked. Usually use `exp.Show(stim)` instead.
`win.color = (-1,-1,-1)`	`control.Black` (background passed to constructor)	
`core.wait(0.5)`	`exp.Wait(500)`	PsychoPy uses seconds; goxpyriment uses milliseconds.
`core.Clock()` / `clock.getTime()`	`clock.NewClock()` / `c.NowMillis()`	See clock-domain note below.
`core.CountdownTimer(t)`	`c.SleepUntil(target)`	
`visual.TextStim(win, text="Hello")`	`stimuli.NewTextLine("Hello", x, y, color)`	
`visual.TextStim(..., wrapWidth=800)`	`stimuli.NewTextBox(text, width, pos, color)`	Word-wrapped.
`visual.Circle(win, radius=0.5, units='pix')`	`stimuli.NewCircle(radius, color)`	
`visual.Rect(win, width=100, height=50)`	`stimuli.NewRectangle(cx, cy, w, h, color)`	
`visual.ImageStim(win, image='img.png')`	`stimuli.NewPicture("img.png", x, y)`	
`visual.GratingStim(win, sf=0.05, ori=45)`	`stimuli.NewGaborPatch(sigma, theta, lambda, ...)`	See spatial/temporal frequency note below.
`stim.draw(); win.flip()`	`exp.Show(stim)`	
`stim.setPos((x, y))`	`stim.SetPosition(control.Point(x, y))`	
`event.waitKeys(keyList=['f','j'])`	`exp.Keyboard.WaitKeys(keys, timeout)`	
`event.waitKeys(maxWait=3)`	`exp.Keyboard.WaitKeys(keys, 3000)`	Timeout in ms.
`clock.getTime()` for RT	`exp.Keyboard.WaitKeysRT(keys, timeout)`	Returns RT in ms from call site.
`win.callOnFlip(clock.reset)` + `event.waitKeys`	`exp.ShowAndGetRT(stim, keys, timeout)`	Hardware-precise RT in ms; no callOnFlip needed.
`data.TrialHandler(trialList, nReps)`	`design.NewBlock(...)` + `block.AddTrial(t, copies, true)`	
`data.ExperimentHandler(...)`	`exp.Data`	

PsychoPy (Python)	goxpyriment (Go)	Notes
`thisExp.addData("rt", rt)`	`exp.Data.Add(rt)`	
`data.QuestHandler(startVal, ...)`	`staircase.NewQuest(cfg)`	
`data.StairHandler(startVal, ...)`	`staircase.NewUpDown(cfg)`	
`sound.Sound('A', secs=0.5)`	`stimuli.NewTone(frequency, duration, volume)`	
`sound.Sound('beep.wav')`	`stimuli.NewSound("beep.wav")`	

9.2.2 Side-by-side: simple RT trial

PsychoPy

```
from psychopy import visual, core, event

win   = visual.Window([1024,768], color='black', units='pix')
fix   = visual.TextStim(win, text='+', height=30, color='white')
target = visual.Circle(win, radius=30, fillColor='white')

trial_clock = core.Clock()

fix.draw(); win.flip()
core.wait(0.5)

trial_clock.reset()
target.draw(); win.flip()
keys = event.waitKeys(keyList=['f','j'], timeStamped=trial_clock)
key, rt = keys[0]  # rt is in seconds

win.close()
```

goxpyriment

```
package main

import (
    "log"
    "github.com/chrplr/goxpyriment/control"
    "github.com/chrplr/goxpyriment/stimuli"
)

func main() {
    exp := control.NewExperimentFromFlags("SimpleRT", control.Black,
 ↪  control.White, 32)
    defer exp.End()
```

```
    fix    := stimuli.NewFixCross(30, 3, control.White)
    target := stimuli.NewCircle(30, control.White)

    err := exp.Run(func() error {
        exp.ShowTimed(fix, 500)
        key, rtMs, _ := exp.ShowAndGetRT(target,
            []control.Keycode{control.K_F, control.K_J}, -1,
        )
        _ = key
        _ = rtMs
        return control.EndLoop
    })
    if err != nil && !control.IsEndLoop(err) {
        log.Fatal(err)
    }
}
```

9.2.3 Key differences

Units and coordinate system. PsychoPy supports multiple unit systems (norm, pix, deg, cm). Goxpyriment always uses pixels, with (0, 0) at the screen center, +X right, +Y **up** (same as PsychoPy's `units='pix'`). For visual-angle calculations, use the `units` package: `units.NewMonitor(widthCm, distanceCm, widthPx)` then `mon.DegToPx(deg)`.

Time units. PsychoPy uses seconds throughout (`core.wait(0.5)`, `clock.getTime()` returns float seconds). Goxpyriment uses **milliseconds** for `exp.Wait` and `WaitKeysRT`, and **nanoseconds** for hardware event timestamps (`ShowTS`, `GetKeyEventTS`). Divide nanosecond differences by `1_000_000` to get milliseconds.

`callOnFlip` is not needed. PsychoPy's `win.callOnFlip(clock.reset)` is a workaround for the fact that `flip()` and the clock reset are separate operations. `exp.ShowTS(stim)` captures the SDL nanosecond timestamp atomically at the VSYNC flip — no workaround needed.

Builder vs Coder. PsychoPy Builder generates Python Coder scripts. There is no Builder equivalent in goxpyriment; all experiments are written in code.

Spatial frequency. PsychoPy's `GratingStim` spatial frequency is in cycles/degree (when `units='deg'`). Goxpyriment's `spatialFreq` is in **cycles per pixel**. Convert: `sf_cpx = sf_cpd / mon.DegToPx(1)`.

TrialHandler nReps. PsychoPy's `TrialHandler` accepts a list of condition dicts and repeats them. The equivalent in goxpyriment is `block.AddTrial(t, nReps, randomPosition)` where `randomPosition=true` shuffles. For multiple condition lists, build one block per block and call `exp.ShuffleBlocks()` if needed.

Psychtoolbox-style timestamps. PsychoPy's `logging.flush()` and `win.recordFrameIntervals` give post-hoc timing diagnostics. In goxpyriment, `PresentStreamOfImages` returns a `[]TimingLog` per item. Use the SDL-clock `OnsetNS`/`OffsetNS` for timing

analysis (they share the clock of input-event timestamps); `ActualOnset`/`ActualOffset` are Go-clock diagnostics only and must not be subtracted from event timestamps.

9.3 From Psychtoolbox (MATLAB)

Psychtoolbox-3 (PTB) and goxpyriment share a lower-level philosophy: you open a window, draw into it, flip explicitly, and get a VBL timestamp back. The key differences are language ergonomics, the data-file model, and the adaptive staircase API.

9.3.1 Concept map

Psychtoolbox (MATLAB)	goxpyriment (Go)	Notes
`Screen('OpenWindow', screenNum, bgColor)`	`control.NewExperimentFromFlags(...)` `defer exp.End()`	Also initializes audio, font, and data file.
`Screen('CloseAll')`		
`Screen('Flip', win)`	`exp.Flip()`	Both return a timestamp.
`Screen('Flip', win)` → VBL timestamp	`exp.ShowTS(stim)` → `uint64` nanoseconds	PTB returns seconds (64-bit float); goxpyriment returns nanoseconds (uint64).
`Screen('FillRect', win, color, rect)`	`stimuli.NewRectangle(cx, cy, w, h, color)` then `exp.Show(rect)`	
`Screen('DrawText', win, text, x, y, color)`	`stimuli.NewTextLine(text, x, y, color)` then `exp.Show(tl)`	
`Screen('MakeTexture', win, img)`	`stimuli.NewPicture("path", x, y)`	Texture is lazily uploaded on first `Draw`.
`Screen('DrawTexture', win, tex)`	`exp.Show(pic)`	
`Screen('MakeTexture', win, sheet)` (one texture for many stimuli)	`stimuli.NewSpriteSheet("sheet.png")` or `NewSpriteSheetFromMemory(data)`	The PTB idiom of packing a condition into one image.
`Screen('DrawTexture', win, tex, sourceRect)`	`sprites, _ := sheet.Grid(exp.Screen, cols, rows)` then `exp.Show(sprites[i])`	`sourceRect` becomes the sprite's `Clip`. One upload for the whole set.
`Screen('DrawLine', ...)`	`stimuli.NewLine(x1, y1, x2, y2, color)` then `exp.Show(line)`	
`DrawFormattedText(win, text, 'center', 'center', color)`	`exp.ShowInstructions(text)`	Centered, waits for spacebar.

Psychtoolbox (MATLAB)	goxpyriment (Go)	Notes
`WaitSecs(secs)`	`exp.Wait(ms)`	PTB uses seconds; goxpyriment uses **milliseconds**.
`GetSecs()`	`clock.GetTime()` (ms) or `clock.GetTimeNS()` (ns)	See clock-domain note below.
`KbWait`	`exp.Keyboard.Wait()`	
`KbCheck`	`exp.Keyboard.Check()`	Non-blocking poll.
`KbStrokeWait` / `KbName`	`exp.Keyboard.WaitKeys(keys, timeout)`	Pass `nil` for any key.
RT via `GetSecs` before/after `KbWait`	`exp.Keyboard.WaitKeysRT(keys, timeout)` (ms)	Or `GetKeyEventTS` for nanosecond accuracy.
`PsychPortAudio('Open', ...)` / `'Start'`	`stimuli.NewSound(path)` + `snd.Play()`	
`MakeBeep(freq, dur, rate)`	`stimuli.NewTone(freq, duration, volume)`	
`Snd('Play', ...)`	`tone.Play()`	
`Quest('init', ...)`	`staircase.NewQuest(cfg)`	
`QuestUpdate(q, x, response)`	`sc.Update(correct)`	
`QuestQuantile(q)`	`sc.Intensity()`	
`QuestMean(q)`	`sc.Threshold()`	
Custom up-down staircase	`staircase.NewUpDown(cfg)`	Levitt (1971); 2-down-1-up built in.

9.3.2 Side-by-side: simple RT trial

Psychtoolbox

```
screenNum = max(Screen('Screens'));
[win, rect] = Screen('OpenWindow', screenNum, [0 0 0]);

cx = rect(3)/2;  cy = rect(4)/2;

% Fixation cross
Screen('DrawLine', win, [255 255 255], cx-15, cy, cx+15, cy, 3);
Screen('DrawLine', win, [255 255 255], cx, cy-15, cx, cy+15, 3);
Screen('Flip', win);
WaitSecs(0.5);

% Target circle
Screen('FrameOval', win, [255 255 255], [cx-30, cy-30, cx+30, cy+30], 3);
t0 = Screen('Flip', win);   % VBL timestamp in seconds

[~, t1, ~] = KbWait([], 2);
rt = (t1 - t0) * 1000;  % milliseconds
```

```
Screen('CloseAll');
```

goxpyriment

```
package main

import (
    "log"
    "github.com/chrplr/goxpyriment/control"
    "github.com/chrplr/goxpyriment/stimuli"
)

func main() {
    exp := control.NewExperimentFromFlags("SimpleRT", control.Black,
 ↪  control.White, 32)
    defer exp.End()

    fix    := stimuli.NewFixCross(30, 3, control.White)
    target := stimuli.NewCircle(30, control.White)

    err := exp.Run(func() error {
        exp.ShowTimed(fix, 500)
        _, rtMs, _ := exp.ShowAndGetRT(target, nil, -1)
        _ = rtMs
        return control.EndLoop
    })
    if err != nil && !control.IsEndLoop(err) {
        log.Fatal(err)
    }
}
```

9.3.3 Key differences

Time units. PTB uses **seconds** (double-precision float) throughout. Goxpyriment uses **milliseconds** for `exp.Wait` / `WaitKeysRT`, and **nanoseconds** for hardware event timestamps. To convert: multiply PTB seconds by 1000 to get ms, or 1,000,000,000 to get ns.

VBL timestamps. PTB's `Screen('Flip')` returns the VBL timestamp as `GetSecs` seconds. `exp.ShowTS(stim)` returns `sdl.TicksNS()` nanoseconds captured immediately after the VSYNC flip — a different clock origin. Do not mix SDL timestamps with `clock.GetTimeNS()` for RT calculation; use SDL timestamps exclusively (see UserManual §6).

Coordinate system. PTB uses a top-left origin (+Y down), consistent with screen pixel conventions. Goxpyriment uses a **center origin (+Y up)**, same as PsychoPy `pix` units. A stimulus at (0, 0) appears at screen center; a stimulus at (0, 200) appears 200 pixels **above** center.

Drawing model. PTB requires explicit `DrawLine`/`FillRect`/etc calls per frame and a separate `Flip`. Goxpyriment encapsulates each stimulus as an object; `exp.Show(stim)` does clear → draw → flip in one call. For multi-stimulus frames, draw each one (`stim.Draw(exp.Screen)`) then call `exp.Flip()` explicitly.

Data files. PTB has no built-in data file; researchers typically write custom CSV writers or use `fopen`/`fprintf`. Goxpyriment writes a structured `.csv` file automatically. Declare column names once with `exp.AddDataVariableNames`; append rows with `exp.Data.Add(...)`.

Quest staircase. PTB's `Quest` functions are a set of global procedures (`QuestCreate`, `QuestUpdate`, `QuestMean`). `staircase.NewQuest(cfg)` is an object that implements the same Bayesian update; call `sc.Intensity()`, `sc.Update(correct)`, `sc.Threshold()`. The parameters (`tGuess`, `tGuessSd`, `pThreshold`, `beta`, `delta`, `gamma`) map directly.

EEG triggers. PTB sends triggers via `lptwrite` (Windows parallel port). Goxpyriment provides `triggers.NewDLPIO8`, `triggers.NewMEGTTLBox`, and `triggers.NewParallelPort`, all implementing the same `OutputTTLDevice` interface with `Send(mask)`, `Pulse(line, duration)`, and `AllLow()`. For labs that mark the EEG stream over the network instead of with a TTL line, `triggers.NewNetStation(host)` speaks the EGI NetStation ECI protocol (`Synchronize`, `StartRecording`, `SendEvent("STIM")`); and `triggers.NewVideoRecorder(host)` drives the "BEL_video" participant-video recorder (`Start`, `Label("TRL", "001")`, `Stop`). Both are TCP clients — richer than a TTL pulse but bounded by network latency, so keep a `Pulse` for sub-millisecond onset markers. See UserManual §17.

No Flip scheduling. PTB's `Screen('Flip', win, when)` allows scheduling a flip at a specific VBL. Goxpyriment does not support scheduled flips; instead, use `clock.SleepUntil(target)` before `exp.Show` to achieve frame-accurate onset scheduling, or use the stream functions (`PresentStreamOfImages`) which handle VSYNC-locked scheduling internally.

9.4 Common gotchas (all three tools)

ESC handling. Goxpyriment treats ESC as a universal experiment abort. `exp.Wait`, `exp.Keyboard.WaitKeys`, and similar functions return `control.EndLoop` when ESC is pressed. Propagate this error upward from `exp.Run`'s callback; `control.IsEndLoop(err)` tests for it. There is no equivalent of PTB's `ListenChar`/`CharAvail` or PsychoPy's `event.clearEvents()` — use `exp.Keyboard.Clear()` to flush stale key presses before a new trial.

GPU texture allocation. The first time any visual stimulus is drawn, its SDL texture is allocated on the GPU. In all three tools the first presentation can be slower than subsequent ones. In goxpyriment, call `stimuli.PreloadVisualOnScreen(exp.Screen, stim)` (or `stimuli.PreloadAllVisual`) before the critical section to force allocation during an instruction screen, not during a timed trial.

RSVP / rapid stimulus sequences. Go's garbage collector can pause execution mid-sequence. The stream functions (`PresentStreamOfImages`, `PresentStreamOfText`)

disable GC automatically for the duration of the stream. Do not implement your own VSYNC-locked loop without also disabling GC (`debug.SetGCPercent(-1)`).

Single binary distribution. Unlike Python (Expyriment, PsychoPy) and MATLAB (PTB), goxpyriment experiments compile to a single self-contained binary. Run `go build .` in the experiment directory; the result runs on any machine with the same OS and architecture without any runtime installation. For cross-platform distribution, use `GOOS=windows GOARCH=amd64 go build .` etc. — see Installation.

Chapter 10

goxpyriment compared with PsychoPy

PsychoPy is the established, most widely used platform for creating and running psychological experiments. This page compares it with goxpyriment feature by feature, so that you can judge whether goxpyriment fits your paradigm — and, where it does not, what you would be giving up.

The comparison is based on the PsychoPy documentation sources (psychopy/psychopy-docs, `release` branch) and on the current state of this repository.

Summary in one sentence: PsychoPy optimises for *accessibility and breadth* (a GUI builder, online deployment, support for nearly every laboratory device); goxpyriment optimises for *deployability and timing determinism* (one static binary, no interpreter, no garbage collection inside the presentation loop).

If you are porting existing PsychoPy code, read the Migration Guide — it gives a concept map and side-by-side code for a representative trial sequence.

10.1 The structural difference

PsychoPy is three layers stacked on top of each other:

1. **Builder** — a graphical editor in which an experiment is assembled from routines, loops, and ~45 drag-and-drop components. It compiles to Python *or* to JavaScript.
2. **Coder** — the `psychopy.*` Python library, which the Builder generates calls into and which you can use directly.
3. **PsychoJS / Pavlovia** — a JavaScript runtime plus a hosting platform, with integrations for Prolific, MTurk, Qualtrics, and Sona.

goxpyriment is only the middle layer: a Go library. There is no graphical builder and no hosting platform. That single fact explains most of the gaps listed below — and most of the advantages.

10.2 Where goxpyriment matches or has an edge

Aspect	PsychoPy	goxpyriment
Deployment	Python environment plus wheels, or Pavlovia hosting	A single self-contained binary. SDL3, SDL3_ttf and SDL3_image are embedded as compressed blobs and loaded at startup — nothing to install on the target machine
Presentation loop	Python, subject to the GIL and to interpreter overhead; Psychtoolbox/ioHub back-ends are used for precision	VSYNC-locked loops with the garbage collector disabled (`debug.SetGCPercent(-1)`) for the duration of the critical section
Clock	`psychopy.clock`, `core.getTime`	`clock` package built on SDL's high-resolution performance counter
RSVP and rapid streams	Written by hand as a frame-by-frame loop	A dedicated high-precision path, `stimuli.PresentStreamOfImages`, with variants for images, text, tones, mixed streams, and per-frame callbacks
Adaptive procedures	`StairHandler`, `QuestHandler`	`staircase.UpDown` (Levitt 1971) and `staircase.Quest` (Watson & Pelli 1983) — the same two families
Unit conversion	`psychopy.monitors`, `deg`/`cm`/`norm` units	`units` package: pixels ↔ degrees ↔ cm through a `Monitor` struct
Counterbalancing	`CounterbalanceRoutine` (a recent addition)	`design.AddBWSFactor` / `GetPermutedBWSFactorCondition` — Latin-square between-subject assignment, present from the start
Gamma correction	`monitors` calibration	`apparatus.Gamma`
Data output	`.csv` / `.psydat` / Excel	`.csv` with a #-prefixed metadata header, written through a buffered output file

Aspect	PsychoPy	goxpyriment
System reporting	`psychopy.info`	`sysinfo` package: CPU, GPU, audio, memory, machine — including a WMI path on Windows
Hardware triggers	Parallel, serial, LabJack, Cedrus, EGI, BrainProducts, TriggerBox	Parallel port (Linux LPT), DLP-IO8, FT232H, Linux GPIO (Raspberry Pi and other SBCs), LabJack T4 over Modbus TCP, NeuroSpin MEG TTL box, generic serial
Networked recording control	Egi/`iohub`, `emulator`, MQTT/OSC components	EGI NetStation event markers over TCP/IP (ECI), plus a "BEL_video" participant-video recorder client
Worked examples	A demos folder	~50 complete paradigms in `examples/`, plus a `tests/` suite of programs whose output is analysed to verify timing and hardware behaviour

The core of an experiment — window, text, images, shapes, gratings, dot fields, keyboard and mouse input, sound, trial structure, randomisation, staircases, data files — is covered by both.

10.3 Where PsychoPy is ahead

10.3.1 1. Eye tracking

This is the single largest gap. PsychoPy's **ioHub** provides a vendor-neutral API over EyeLink, Tobii, GazePoint and Pupil Labs, with Builder routines for calibration and validation and a region-of-interest component.

goxpyriment has no eye-tracking support at all — no gaze stream, pupillometry, calibration, or region-of-interest components. (It *does* ship a `triggers.VideoRecorder` client for a networked participant-video recorder, but that only records and labels footage of the participant; it is not gaze tracking and returns no eye position.)

10.3.2 2. Breadth of visual stimuli

PsychoPy's `visual` module documents 42 classes. Notable ones without a goxpyriment equivalent:

- **`ElementArrayStim`** — thousands of elements drawn in a single call. Required for Glass patterns, large-scale multiple-object tracking, and texture-defined stimuli. Probably the most consequential omission after eye tracking.
- **`Aperture`** — restricting drawing to an arbitrary region.
- **`RadialStim`**, **`NoiseStim`**, **`SecondOrder`** — specialised psychophysical textures.
- **`Form`**, **`Slider`**, **`RatingScale`** — questionnaire and rating widgets. goxpyriment offers `stimuli.ChoiceGrid`, `stimuli.ThermometerDisplay` and `stimuli.TextInput`, which cover some but not all of this ground.
- **`BufferImageStim`** — snapshotting a rendered screen for fast redisplay.
- **`windowWarp`** — spherical and cylindrical warping for domes and curved screens.
- The entire 3D/VR branch: `rift`, `SphereStim`, `ObjMeshStim`, `PhongMaterial`, `LightSource`, `SceneSkybox`.

10.3.3 3. Online experiments

PsychoPy's `online/` documentation covers Pavlovia synchronisation, participant-recruitment integrations (Prolific, MTurk, Qualtrics, Sona), credit and statistics dashboards, and compatibility checking.

goxpyriment compiles to WebAssembly and runs in the browser (see WASM.md), so the *runtime* half of this exists. What does not exist is the surrounding service: hosting, recruitment, and server-side data collection are left to you.

10.3.4 4. Video

PsychoPy plays whatever ffmpeg can decode. goxpyriment plays MPEG-1 directly (pure Go, with MP2 audio) and requires conversion to `.gv` for timing-critical clips — `cmd/gv-convert` does this, needing no external tools for MPEG-1 input and ffmpeg for anything else. See UserManual §15.

The `.gv` requirement is a deliberate trade: independent per-frame blocks remove the decode-cost asymmetry between I- and B-frames, which is what makes H.264 onset timing jitter periodically. But it is still friction, the files are roughly an order of magnitude larger, and it rules out playing arbitrary participant-supplied media at measured onsets.

10.3.5 5. Ecosystem and tooling

- A **plugin architecture** for third-party components.
- The **`Session`** API for running several experiments under one process.
- **Alerts and linting** that catch common design errors before a run.
- A **logging framework** with configurable levels.
- A substantial **teaching corpus** and community.

10.3.6 6. Camera and speech

PsychoPy has a `CameraComponent`, Google Speech integration, and audio/visual validator routines. goxpyriment provides `apparatus.Microphone`, `apparatus.VoiceKey` and WAV writing, but no camera capture and no speech recognition.

10.3.7 7. The Builder itself

For readers who do not program, the Builder is not a convenience — it is the only way in. goxpyriment requires writing Go.

10.4 Choosing between them

Reach for PsychoPy when you need eye tracking; when the paradigm depends on large element arrays, apertures, or 3D/VR; when the study runs online with recruitment-platform integration; when arbitrary video must be played; or when the person building the experiment does not program.

Reach for goxpyriment when the experiment must be distributed as a binary that runs on a machine you do not control and cannot install software on; when timing determinism matters more than stimulus breadth (rapid serial presentation, tight audio-visual synchronisation, hardware-triggered acquisition); when Python environment drift has been a recurring problem; or when you are already comfortable in Go and want the compiler to catch design errors before the participant arrives.

The two are not mutually exclusive. A common pattern is to prototype in PsychoPy and re-implement the timing-critical parts in goxpyriment once the design is settled.

10.5 See also

- Migration Guide — concept map and side-by-side code for Expyriment, PsychoPy and Psychtoolbox
- Timing Tests — measured presentation timing on real hardware
- Gallery of Examples — the full catalogue of examples and tests
- WASM build — running experiments in the browser

Chapter 11

Timing-Tests: A Guide for Researchers

This document explains how to use the `Timing-Tests` program to characterise the timing behaviour of your computer before running a psychophysical experiment.

> **Quick reference** — flags, equipment, and one-line test descriptions are in `tests/Timing-Tests/README.md`. This document explains the *why*.

> For the EEG/MEG question specifically — how consistent the delay is between a TTL trigger and the physical stimulus — see Minimising trigger-to-stimulus jitter, which gives the measured decomposition and the two things that matter most.

11.1 Why timing matters

In a psychology experiment every stimulus has an intended onset and offset time. Whether you are presenting a word for exactly 100 ms, playing a tone that should coincide with a visual flash, or measuring a reaction time to the nearest millisecond, you are trusting the computer to do two things correctly:

1. **Present stimuli when you ask it to.** If you ask for a 100 ms word, does the word actually appear on screen for 100 ms?
2. **Record timestamps accurately.** If you record the time of a key press, is that the time the key was physically depressed, or is it delayed by polling?

Neither is guaranteed. Both depend on your operating system, graphics driver, audio driver, and hardware. This suite lets you measure them on *your specific machine*, so you can report them in your methods section and fix problems before data collection begins.

11.2 Read this before you trust any number

The statistics these tests print come from software timestamps. They record when goxpyriment *believes* a flip happened — not what reached the panel.

This is not a theoretical caveat. In July 2026 a presentation bug on a GNOME/Wayland machine left the display showing stale frames for *seconds at a time*, while the program's own numbers stayed textbook-perfect throughout: bright-phase duration 199.915 ± 0.16 ms against a 200 ms target, cycle period 500.05 ± 2.45 ms. Nothing in the console output hinted at a problem. It was visible only with a photodiode. (Cause and fix are documented in `apparatus/CLAUDE.md`; the regression test is `tests/test_clear_only_frames`.)

So:

- **Software statistics tell you about the software.** They are genuinely useful for detecting dropped frames, scheduler jitter, and audio-buffer effects.
- **A photodiode and a trigger box tell you about the experiment.** Only they can confirm that light actually changed when you said it would.

If you are validating a rig for publication, measure it with hardware. The `av` test is built for exactly that.

11.3 The six tests

Two groups: what runs on any machine, and what needs equipment.

```
No hardware needed
  check    Go/no-go: can this machine display and make noise at all?
  display  What is my true refresh rate, and how stable are frame intervals?
  latency  How long does SDL hold audio before it sounds?

Photodiode and/or trigger box
  av       THE stimulus timing test — visual + audio + TTL, every cycle
  vrr      Can I present arbitrary durations, not just multiples of a frame?
  rt       How precise is my reaction-time measurement?
```

Two related tests live in their own directories: `tests/test_gv_sync` (`.gv` playback synchronisation) and `tests/test_dlpio8` (DLP-IO8-G square-wave characterisation).

This document assumes you have built the program:

```
go build -o Timing-Tests ./tests/Timing-Tests
```

11.4 Recording system information (`-sysinfo`)

Before running anything, capture a snapshot of the machine. `-sysinfo` prints it and exits — it opens no window, though it does initialise SDL briefly to enumerate displays and audio devices:

```
Timing-Tests -sysinfo
```

```
Machine:    product: Precision 5490  System: Dell Inc.  Type: laptop
System:     Host: mylab-pc  Kernel: 7.0.0-28-generic x86_64  Uptime: 3h 12m
            OS: Ubuntu 26.04  Shell: bash 5.2.37  Desktop: ubuntu:GNOME
CPU:        Model: Intel(R) Core(TM) Ultra 7 165H  Info: 16 cores / 22
 ↪  threads
Memory:     RAM: total: 30.04 GiB  used: 6.21 GiB (20.7%)
Graphics:   Card: Intel Corporation Meteor Lake-P [Intel Arc Graphics]
 ↪  Driver: i915
Audio:      Server: PipeWire  v: 1.4.7
Displays:   [0] Built-in display  1920x1200  60.040 Hz  bounds 0,0 1920x1200
 ↪  [primary]
            video driver: wayland   (the [N] above is the -d N value)
Audio out:  [0] Speaker
            driver: pipewire
Vblank:     default: onsets are timestamped the moment SDL_RenderPresent()
 ↪  returns.
            Where the driver blocks on the retrace, that moment IS the
             ↪  retrace,
            so the timestamp is a hardware instant. On a frame where the
             ↪  present
            returned early instead, the timestamp comes from the pacing
             ↪  schedule.
            available if asked for with GOXPY_VBLANK=on: Linux DRM vblank
             ↪  (card /dev/dri/card1, crtc 0 driving eDP-1 1920x1200@60.0386
             ↪  Hz, DRM_IOCTL_WAIT_VBLANK, hardware-verified)
```

11.4.1 The `Vblank` line

It reports two independent things: where this run's onset timestamps will come from, and whether the machine has a kernel vblank clock at all.

By default `FlipTS` returns whichever anchor the driver made available that frame: the present's own return where `SDL_RenderPresent` blocks on the retrace, and a timestamp derived from the pacing schedule where it does not (the run's `Frame pacing` block says which — see The `Frame pacing` block). `GOXPY_VBLANK=on` instead anchors every frame on the kernel's vblank stamps — **off by default, and there is currently no reason to turn it on.**

Measured with a photodiode against a TTL, 1010 cycles per run, on a Raspberry Pi 4 (V3D/kmsdrm) and a Radeon Pro W5700 (radeonsi, X11 exclusive fullscreen):

arm	flip → photon slope	one-frame errors
pacing schedule (5 runs)	−1.62 … +0.12 ppm	none in any run
kernel vblank (1 run)	+0.48 ppm	none

The two are indistinguishable and both are flat to well under a ppm, so the default is the one with five runs behind it on two machines rather than one.

The case the vblank anchor exists for is a host whose nominal refresh rate is badly wrong, where a schedule advancing on it walks away from the panel. Neither machine above is that host — the W5700's nominal is 5.9 ppm from true, or 0.2 ms over an eight-minute block. If you have a rig where the two disagree by much more, this is the switch to try, and `sys vblank_resolution:` in the run's `-info.txt` is how to check it behaved.

If you enable it, read that line. It reports how each frame's vblank was resolved, and it is not decoration: on the W5700 the flip query beat the vblank interrupt on **31.3 % of frames**, and identifying which vblank a frame belongs to is the whole job. A wrong answer there is exactly one frame out, which on a frame grid still looks like a perfectly regular train — so nothing in the timestamps themselves would tell you.

The same line ends with the check that the vblanks came from the **right display**:

```
sys vblank_resolution: frames=30000 … measured=59.9514 Hz nominal=59.9506
↪  Hz (frame period -13 ppm, matches the display)
```

`measured` is the cadence of the vblanks the run actually read, taken from the kernel's own stamps and divided by the vblank *count* so a dropped frame cannot stretch it. `nominal` is the display the experiment drew on. On a single-head machine the two agree by construction and there is nothing to think about. On a laptop with an external monitor they are the sentence worth reading, because the two heads run different clocks and the vblank ioctl names a pipe by index, not by monitor.

That mistake was made and measured. On a Precision 5490 driving a U2720Q on DP-1, with the internal panel also lit, the backend read the internal panel's CRTC — 60.0386 Hz against the external monitor's 59.9514, **1449 ppm apart**. The presents were flawless (the photodiode saw an unbroken 30-frames-per-cycle lock with 3 µs stability for 8.4 minutes) but the recorded onsets walked 24.2 µs per frame until they were a whole frame out and jumped back — 44 times in the run, a flip→photon lag sawtoothing across a full frame, and `onset_source` reporting `hardware-verified` in all 1010 rows. It was the right kind of hardware and the wrong piece of it.

The backend now reads the card's mode resources and picks the CRTC whose programmed mode is the display you are presenting to, names that head in the `Vblank` line (`crtc 1 driving DP-1 2560x1440@59.9514 Hz`), refuses to start if no lit CRTC matches — falling back to the default present-return anchoring, which cannot pick the wrong display — and keeps checking the live cadence for the whole run. If it ever says `WRONG DISPLAY`, the onsets in that file are on another monitor's grid and should not be used.

Check this line **before** committing to a photodiode capture rather than afterwards

in the run's `-info.txt`: a machine with no backend and a run that never opted in produce identical data, and an A/B whose two arms ran the same way looks like a null result rather than a switch that never took.

Keep it alongside your results:

```
Timing-Tests -sysinfo > sysinfo-$(hostname)-$(date +%Y%m%d).txt
```

Reviewers routinely ask for CPU model, OS version, graphics driver, and audio server. Every data file also carries this in its `-info.txt` header, including the **display the window actually opened on** — read it from there rather than assuming, especially on a multi-monitor machine.

11.5 Understanding the display refresh cycle

Most monitors refresh at a fixed rate — typically 60 Hz (a new frame every 16.67 ms), 120 Hz, or 144 Hz. Your graphics driver synchronises presentation to this cycle (VSYNC). Two consequences:

- **Durations are quantised to multiples of the frame period.** At 60 Hz you can show a stimulus for 16.67, 33.33, or 50 ms — but not 25 ms. Plan accordingly, or see the `vrr` test.
- **Onset is not when you call the flip; it is the next VSYNC**, anywhere from 0 to 16.67 ms later. goxpyriment keeps that offset constant once the pipeline is warm, but the first frames should be excluded — `-warmup` does this.

11.5.1 And the screen does not change all at once

An LCD paints top to bottom. On the rig this suite was developed against, a square at the bottom of the screen lights **13.15 ms** after one at the top — close to a full frame period. Measured on a 1280×1024 @ 60.02 Hz panel, that is **15.96 µs per pixel row**.

This matters more than it sounds. A stimulus at screen centre appears ~6.6 ms after one at the top. If your photodiode sits in a corner but your stimulus is central, your measured onset is biased by that amount — a systematic error larger than most of the jitter people worry about.

The `av` test draws five squares (four corners plus centre) precisely so you can measure this on your own display. Put a photodiode on a top square and another on a bottom one; the difference is your panel's scan-out gradient. Two squares on the same row are only microseconds apart and serve as a sanity check. Each display needs its own figure.

11.6 No hardware needed

11.6.1 `check` — does anything work at all

```
Timing-Tests -test check
```

Flashes a white screen for a second, then plays two sounds. If you do not see the flash or hear both sounds, stop and fix that first. Common causes: the window opened on the wrong monitor (`-d`), the audio device is muted, or the default output is not your speakers.

Measures nothing. It exists so you do not waste a session discovering the hardware was not plugged in.

11.6.2 `display` — true refresh rate and frame stability

```
Timing-Tests -test display -duration-s 30
```

Flips a uniform screen for `-duration-s` and reports the distribution of frame-to-frame intervals, plus the refresh rate implied by their mean. Use that measured rate for `-hz` in later tests rather than assuming 60.

What to look for: a tight distribution centred on your nominal frame period. A long right tail means dropped frames — a compositor, a background process, or power management. A *bimodal* distribution usually means the display is not running at the rate it reports.

11.6.2.1 The `Frame pacing` block

Below the interval statistics the test prints a second block. **It does not report dropped or missed frames** — every present is counted in it, and a frame that never reached the panel would show up in the intervals above, not here.

It answers a different question: *what were this run's flip timestamps anchored to?* `SDL_RenderPresent` is supposed to block until the retrace, but under triple or mailbox buffering it queues the frame and returns immediately. When it does, `Screen.Update` holds the frame to the boundary itself — and the onset it reports is then a *scheduled* time rather than a hardware one. The counts say which happened:

line	meaning	the onset `FlipTS` reports
`blocked`	present covered the frame and returned at, or just inside, the boundary	the present's own return — a hardware instant
`early`	the part of `blocked` that returned *just* inside the nominal boundary	unchanged: still the present's return

line	meaning	the onset `FlipTS` reports
`paced`	present came back with most of the frame left, so `Update` held it	the scheduled boundary — synthesised
`vblank`	a kernel vblank timestamp was available (`GOXPY_VBLANK=on`)	the vblank — measured

A typical good result, on an Intel/Mesa laptop under Wayland:

```
── Frame pacing ──────────────────────────────
  presents: 1798   (frame = 16.661 ms)
  blocked : 1798 (100.0 %)  — present carried the frame; its return is the
   ↪  onset
            of which 1776 came back inside the nominal boundary by mean
            0.676 ms, max 1.140 ms — the phase offset between the nominal
            frame grid and the panel's, not a wait.
  paced   : 0 (0.0 %)  — returned early; Update held to the schedule
  verdict : the driver blocks. Flip timestamps carry the display's own
            instant, and cannot drift against the panel.
```

The early count is not a fault, and a large one is not a warning. A blocking present returns one *panel* period after the last one, while the boundary it is compared against is one *nominal* period after — and the two grids are never in exact phase. On the machine above the gap was 0.676 ms of a 16.661 ms frame, meaning present had blocked for 15.99 ms of every frame. It is a constant offset, re-established by the hardware on every frame, so it cannot accumulate and it does not enter any duration or reaction time you compute (both are *differences* between timestamps, and the offset cancels).

What each verdict means for your experiment:

- **`the driver blocks`** — nothing to do. Onsets are hardware-anchored.
- **`onsets come from the kernel's vblank timestamp`** — nothing to do; this is the most accurate configuration available.
- **`the driver does NOT block`** — your frames are still correctly paced and your *durations* are fine, but the onsets are stamped with a schedule running at the nominal refresh rate. If that rate is wrong, absolute onsets slide against the panel over long blocks. Compare the estimated refresh rate printed above against the nominal one, run `timing-drift` on a photodiode capture before quoting absolute onsets, and consider fullscreen or a session without a compositor (see Improving timing on your system).
- **`MIXED`** — some frames took each path. Worth a photodiode check before quoting absolute onsets.

For a rig whose durations and reaction times matter to a few milliseconds, only the third verdict asks anything of you, and even then it bounds a *slow slide* over minutes, not per-trial error.

11.6.3 `latency` — audio pipeline delay

```
Timing-Tests -test latency
Timing-Tests -test latency -audio-frames 256 -drain-reps 20
```

Plays tones of known duration and spin-polls until the audio stream has drained, measuring how long SDL holds PCM beyond the nominal duration. That difference is your audio output latency.

No hardware needed — but note it measures the *software* pipeline, not speaker-to-microphone delay. For true acoustic onset, use `av` with a microphone.

11.7 Photodiode and/or trigger box

11.7.1 `av` — the stimulus timing test

This is the one that matters. Every cycle presents all three modalities at once:

- **Visual** — five squares (four corners + centre) bright for `-frames-on` frames, then dark for `-frames-off` frames
- **Audio** — a tone lasting `frames-on × frame-period`, synchronised to the visual onset or offset from it by `-soa-ms`
- **TTL** — a `-trigger-ms` pulse at the visual onset, on the device chosen by `-trigger-device` (see below)

```
# 200 ms on, 300 ms off at 60 Hz, 100 cycles — the defaults
Timing-Tests -test av -cycles 100

# Visual only, no hardware at all
Timing-Tests -test av -no-sound -no-ttl

# Audio leading the flash by 50 ms
Timing-Tests -test av -soa-ms -50
```

`-no-sound` and `-no-ttl` drop a modality. This is how one test covers what used to be three: `-no-sound` is a pure visual-onset test, and the audio channel of a recording gives tone-onset jitter over a long session.

What to record. A photodiode on one square, the TTL line on a second channel, a microphone if you are checking audio. The quantity of interest is the *offset* between the TTL edge and the luminance step, and how stable it is across cycles. A constant offset is a calibration you subtract; a varying one is jitter you cannot.

What the console tells you. Bright-phase duration, cycle period, and (with sound) audio-queued minus visual onset. All software-side — cross-checks, not ground truth. See the warning at the top of this document.

Reference numbers. Dell Precision 5490, external 1280×1024 @ 60.02 Hz display under Wayland: 100/100 cycles clean, TTL→photodiode 29.878 ± 0.628 ms (from cy-

cle 10 on), top-to-bottom gradient 13.150 ± 0.221 ms, ~5 dropped frames per 100 cycles. Same machine on a bare console with `SDL_VIDEODRIVER=kmsdrm`: 58/58 cycles, TTL→photodiode 21.51 ± 1.19 ms, **zero** dropped frames. Compositing costs roughly one extra frame of latency and a few percent of dropped frames.

Expect a warm-up transient. The first six cycles on that rig ran 36–39 ms before settling to ~27 ms. `-warmup 10` excludes them from the test's own statistics, but offline analysis tools do not know about it — quote steady-state numbers, and say which cycles you used.

Audio onsets are quantised by the buffer. With `-audio-frames 256` at 44100 Hz, tone onsets land on 5.8 ms steps. This appears as a lattice in the inter-onset intervals and is not jitter — it is the buffer. Halving the buffer halves the step, at the cost of underrun risk.

11.7.2 Choosing a TTL output device

`-trigger-device` selects among five back-ends. They are **not** interchangeable. The trigger is fired immediately after the flip returns, so whatever the write costs falls between the flip and the TTL edge — and it shows up in the trial-to-trial *spread* of that interval, not as a constant offset you could subtract afterwards.

`-trigger-device`	Path to the pin	Logic	Extra flags
`dlpio8` (default)	USB serial (FTDI)	5 V	`-port`
`parallel`	`ioctl` on ppdev	5 V	`-parallel-port`
`gpio`	`ioctl` on a GPIO chip	**3.3 V**	`-gpio-chip`, `-gpio-pins`
`ft232h`	USB bulk transfer (MPSSE via usbfs)	**3.3 V**	— (first device found)
`labjackt4`	Modbus TCP over the network	**3.3 V**	`-labjack-host` (required)

`parallel` and `gpio` are both a local `ioctl` and carry none of the USB frame scheduling that governs the DLP-IO8's output latency (see the DLP-IO8 section below — the FTDI latency timer does *not* fix that). Prefer `parallel` on a desktop that still has an LPT port, and `gpio` on a single-board computer. `ft232h` and `labjackt4` are for a rig that has neither: both put a link back between the flip and the edge — USB for the FT232H, an Ethernet round trip for the T4 — so they are a fallback, not an improvement on the two `ioctl` paths.

How much each costs here has **not been measured**; the ordering above is what the transport implies, not a result. The TTL channel of the recording is the only thing that settles it for a given machine.

```
Timing-Tests -test av -trigger-device parallel                    # LPT,
 ↪  auto-detect
Timing-Tests -test av -trigger-device parallel -parallel-port /dev/parport1
Timing-Tests -test av -trigger-device gpio                        #
 ↪  /dev/gpiochip0, default pins
```

```
Timing-Tests -test av -trigger-device gpio -gpio-chip /dev/gpiochip4
 ↪  -trigger-pin 2
Timing-Tests -test av -trigger-device ft232h                        # first
 ↪  FT232H on the bus
Timing-Tests -test av -trigger-device labjackt4 -labjack-host 192.168.1.100
```

`-trigger-pin` is 1–8 on all five, but it names a different thing on each. Read what the program prints at start-up and probe *that* pin:

- **dlpio8** — the number printed on the terminal block.
- **parallel** — a data line: pin 1 is D0, which is **DB25 pin 2** (D0–D7 are DB25 pins 2–9). Ground is any of DB25 pins 18–25.
- **gpio** — a *position in `-gpio-pins`*, not a line number. With the default list `17,27,22,5,6,13,19,26`, pin 1 is **BCM 17**.
- **ft232h** — a D-bus line counted from 0: pin 1 is **AD0**, pin 8 is AD7. The C-bus (AC0–AC7) is the input side and is not driven here.
- **labjackt4** — a position in DIO4–DIO11: pin 1 is **DIO4**, the screw terminal marked **FIO4**; pin 8 is DIO11 = EIO3 on the DB15. DIO0–DIO3 are the analog inputs AIN0–AIN3 and cannot be driven, which is why the group starts at DIO4.

Prerequisites on Linux: `parallel` needs `sudo modprobe ppdev` and membership of the `lp` group; `gpio` needs kernel $\geq$ 5.10 and membership of the `gpio` group; `ft232h` is Linux-only and needs `ftdi_sio` unloaded (`sudo rmmod ftdi_sio`) plus rw access to `/dev/bus/usb/BBB/DDD` (udev rule or the `plugdev` group); `labjackt4` needs the T4 reachable on TCP port 502. `usermod` changes need a re-login. Verify the wiring with `tests/test_parallel_port`, `tests/test_linuxgpio`, `tests/test_ft232h` or `tests/test_labjackt4` before spending a long capture.

Whichever device is used is written into the results header as `trigger=…`, because they do not yield the same onset-vs-TTL figure. A session that does not record it cannot be compared with one that does.

> **A device that fails to open does not stop the run.** It logs a warning, disables triggers, and continues — so the visual measurement is still obtained. The trap is that the run then exits *successfully* with no TTL in the trace. Under `BBTK_CAPTURE=1` that spends a capture window on a recording with an empty TTL channel. Watch the start-up lines.

11.7.3 Recording it automatically

`tests/Timing-Tests/run-timing-tests.sh` can drive `bbtk-capture` so the stimulus always falls inside the capture window, without two terminals and a stopwatch:

```
BBTK_CAPTURE=1 BBTK_CAPTURE_BIN=~/00_git/bbtkv3/_build/bbtk-capture \
    ./run-timing-tests.sh av-visual
```

It launches the capture, blocks until the device is *actually* recording, runs the stimulus, then waits for the capture to download and save. Only the photodiode steps are wrapped. `BBTK_MARGIN_S` (default 8) sets how much is recorded either side of the stimulus.

The waiting matters: `bbtk-capture` needs 11–40 s between launch and the device recording — fixed command pacing plus an internal-memory erase whose duration depends on whether the box needs a full format — and its own "Capturing events…" message appears several seconds *before* that instant. The script synchronises on the `BBTK-CAPTURE-READY` line emitted the moment the device is armed. Budget that startup for every recorded step.

11.7.4 Microphone coupling

If you are measuring audio, **max the output volume and place the BBTK microphone within a few millimetres of the speaker membrane.** Anything less and events are silently dropped: there is no warning, and the run looks like it worked.

Two checks on the resulting `-events.csv`:

- **N(Mic1) should equal N(TTLin1).** Far fewer means bad coupling, not bad timing — a 197-cycle run yielding 6 Mic events is a placement problem.
- **Mic duration should be close to `frames-on × frame period`** (200 ms at the defaults). A 20 ms Mic pulse for a 200 ms tone means the threshold is catching only the loudest fraction of the tone.

11.7.5 `vrr` — arbitrary stimulus durations

```
Timing-Tests -test vrr -vrr-max-ms 50 -cycles 5
```

11.7.5.1 Why fixed refresh is a problem

On a 60 Hz monitor every duration is a multiple of 16.67 ms. You can show a word for 16.67, 33.33, or 50 ms, but not 20 or 25 ms. At 120 Hz the quantum shrinks to 8.33 ms — better, but still coarse for subliminal work where you may want 10, 12, or 15 ms.

Variable Refresh Rate — AMD FreeSync, NVIDIA G-Sync, or the underlying VESA Adaptive-Sync standard — removes the quantisation. The display holds the current frame for exactly as long as you ask, then refreshes when told. Durations become controlled by your software timer rather than the display clock.

11.7.5.2 How the test works

goxpyriment normally presents with VSync enabled, blocking until the next fixed edge. The `vrr` test switches the renderer to vsync=0 for the duration of the test (restored on exit), then sweeps target durations from 1 ms to `-vrr-max-ms` in 1 ms steps, holding each with a busy-wait and recording what was achieved.

11.7.5.3 Reading the result

- **On a working VRR display:** duration error stays small and roughly constant across the whole sweep.

- **On a fixed-refresh display:** the error is *periodic* — near zero at multiples of the frame period, rising in between, because the panel can only change on its own schedule. A sawtooth in the per-step output means you do not have VRR, whatever the monitor's box claimed.

The test prints one line per target plus an aggregate over the whole sweep.

11.7.5.4 Enabling VRR on Linux

```
# Does the connector support it? Look for "vrr_capable: 1"
sudo cat /sys/kernel/debug/dri/0/*/vrr_capable
```

Confirm with the test itself rather than trusting the setting — that is the point of having it.

11.7.6 `rt` — reaction-time timestamp precision

```
Timing-Tests -test rt -cycles 50
```

Flashes the screen at a jittered interval and waits for a key press, recording the SDL3 hardware event timestamp against the flip timestamp. This characterises the *input* path: how much delay and jitter your keyboard, USB stack, and event queue add.

Press as fast as you can, or better, use a solenoid — you are characterising the machine, and human variability swamps it. The useful number is the SD, not the mean.

11.8 Interpreting results: what is "good"?

Metric	Excellent	Acceptable	Problematic
Frame-interval SD (`display`)	< 0.1 ms	< 0.5 ms	> 1 ms
Frames > 0.5 ms late (`display`)	< 1 %	< 5 %	> 10 %
Audio pipeline latency (`latency`)	< 15 ms	< 30 ms	> 50 ms
Bright-phase duration SD (`av`)	< 0.2 ms	< 1 ms	> 2 ms
TTL→photodiode SD (`av`, hardware)	< 0.5 ms	< 2 ms	> 5 ms
Dropped frames per 100 cycles (`av`)	0	< 5	> 10
VRR duration error SD (`vrr`)	< 0.1 ms	< 0.5 ms	> 1 ms (or periodic → no VRR)
RT SD (`rt`)	< 3 ms	< 10 ms	> 20 ms

Metric	Excellent	Acceptable	Problematic
Pacing verdict (`display`)	vblank, or blocks	MIXED	does NOT block

The hardware rows are the ones worth quoting in a methods section.

The pacing row grades *what the onsets are anchored to*, not frame quality, and it is the one row where "problematic" still leaves durations and reaction times intact — read The `Frame pacing` block before acting on it.

But an SD alone cannot grade a TTL→photodiode series, and the row above is the one most likely to mislead. A delay that slides steadily across a run has an SD like any other, and a large one — yet nothing about it is jitter, it cannot be averaged away, and every within-channel statistic in the BBTK report looks perfect while it happens. Measured on a Raspberry Pi 4: SD 4.07 ms, of which **99.8 % was a monotonic 13.9 ms ramp** and 0.18 ms was real scatter. Run `timing-drift` (below) before reading this row.

11.9 `timing-drift` — is the flip timestamp tracking the panel?

The two `av` output files each look fine on their own; the failure only exists *between* them. This tool joins them and reports the slope first:

```
make timing-drift
./_build/timing-drift bbtk-av-001-events.csv Timing-Tests_sub-000_date-…csv
```

It prints, for TTL→light and then for flip-timestamp→light:

- the **slope**, in µs/cycle and ppm — the number that matters;
- the **de-trended SD**, the real trial-to-trial scatter once the ramp is gone;
- how much of the variance the ramp accounts for;
- one-frame jumps, and the panel's **true frame period** fitted from the photon train, scored against both rates the framework recorded.

The host and the instrument run off different crystals — tens of ppm apart before anything interesting has happened — so the tool fits that clock ratio out of the trigger-versus-flip relation rather than assuming it away. Events are paired by nearest neighbour within half a cycle, not by index, because a single spurious detection shifts every later pair by a whole cycle and turns the series into a constant near minus one period.

Every slope is fitted over the longest stretch containing no dropped frame, and the `fitted over` line says which cycles those were. This is not tidying: a frame-sized step at index k of n points biases a least-squares slope by $6 \cdot h \cdot m \cdot k/n^3$ (with $m = n-k$), which at the midpoint is $1.5 \cdot h/n$. Over a 1000-cycle run at 60 Hz that is 25 µs/cycle — **50 ppm from a single dropped frame in eight minutes**, ten times the effect these captures exist to resolve, and it does not average away with more cycles because the

lever arm grows with *n* too. On a short capture the bias is larger still: 20 cycles with one drop puts it above 2000 ppm.

The `one-frame jumps` count is printed separately, and a run with more than a few of them is not a usable capture however clean the surviving stretch looks — find the load first.

Read the verdict line. `DRIFT DOMINATES` means the flip timestamps and the panel are on different clocks and no absolute onset from that run can be trusted; `NO DRIFT, but … scatter` means the timestamps track the panel and the spread is genuine.

11.9.1 What a validated machine looks like

Two machines, measured with a BBTK v3 photodiode over 1010 cycles each on 2026-08-17, after the flip-timestamp anchoring was corrected:

	Raspberry Pi 4 (V3D/kmsdrm)	Radeon Pro W5700 (radeonsi/X11)
flip → photons, slope	**+0.01 ppm**	**+0.13 ppm**
total drift over the run	0.006 ms / 8.3 min	0.066 ms / 8.3 min
de-trended scatter	0.128 ms	0.278 ms
one-frame jumps	0	0
TTL → light	23.43 ms, SD 0.075 ms (GPIO, kmsdrm)	54.88 ms, SD 0.296 ms (DLP-IO8, **X11**)
TTL → light, same box on kmsdrm	—	**22.55 ms, SD 0.057 ms**
audio-visual lag, 1024 frames	49.88 ms, SD 6.17 ms	23.34 ms, SD 6.54 ms

The TTL row is about the display stack, not the trigger device. The obvious reading — the Pi's GPIO writes through a local ioctl while the W5700's DLP-IO8 crosses a USB link — is wrong, and the third row is the control that shows it. Stopping the X server and running the same machine, panel, photodiode position and DLP-IO8 on bare kmsdrm moved the onset from 54.88 to 22.55 ms and the scatter from 0.296 to 0.057 ms. A USB serial trigger under kmsdrm beats a GPIO trigger under kmsdrm on scatter; the link was never the problem.

The audio row is the same story in reverse — the W5700 drives a USB audio interface and the Pi its on-board codec, which is worth 26 ms of constant latency — while the *jitter* is one audio buffer period over root twelve on both, 6.16 ms predicted at 1024 frames. Nothing about the machine changes it.

11.9.1.1 The display stack is worth two frames of latency and five times the jitter

W5700, everything else identical	onset latency	scatter
X11, exclusive fullscreen	~55 ms	0.296 ms
kmsdrm, bare console	~22 ms	0.057 ms

Measured 2026-08-17, 481 and 286 trials, one variable changed. 32.4 ms is almost exactly two frames, which is what an X11 Present path plus a driver swapchain queues ahead; on kmsdrm the application page-flips straight to the CRTC.

The panel is not involved: its 10-90% transition measured 16.29 ms on the Pi and 17.28 ms here, with the same asymmetry between halves, so the same monitor is behaving the same way on both machines and the latency is upstream of it.

None of this is visible from inside the program. Both configurations report `class=blocking`, sub-ppm drift against the panel, and frame intervals with sub-millisecond scatter. An experiment that quotes an onset time would be 32 ms wrong under X11 and would have no way to know. If absolute latency matters — EEG, MEG, anything cross-modal — run the stimulus on a console without a display server, and measure it rather than assuming either number.

For the drift row, the W5700 is the more informative case. Its GPU and CPU keep independent crystals, so its loop cadence runs **−4.20 ppm** off nominal; before the anchoring was corrected its flip timestamps advanced on the CPU clock at the nominal rate while the panel ran on the GPU clock, and that difference showed up as a measured **−4.73 ppm** of drift. The two numbers are the same quantity. That the cadence is still −4.20 ppm while the drift is now +0.13 is the evidence that onsets follow the panel rather than the schedule.

11.9.2 Read the run's `sys pacing:` line alongside it

`-test av` and `-test display` both record which branch their presents took, in the `-info.txt` beside `sys vblank_backend`:

```
# sys pacing: presents=30000 blocked=30000 early=29610 paced=0
 ↪  vblank_held=0 (0.0 % paced) … class=blocking
# sys pacing: presents=30000 blocked=0 early=0 paced=30000 (100.0 % paced)
 ↪  wait_mean=16.141 ms … class=not-blocking
```

A drift-free result means two different things in those two cases. **`class=blocking`** means `SDL_RenderPresent` returned at the retrace and every frame re-anchored to the hardware — the pacing schedule was barely used, so the run says little about how accurate that schedule is. **`class=not-blocking`** means the schedule *was* what advanced the clock, and a flat result there is evidence about the schedule itself. `class=vblank` is a third case again: the onsets came from kernel vblank stamps and neither the driver nor the schedule was the reference.

The other fields: `early` is the subset of `blocked` that returned just inside the nominal boundary (see The `Frame pacing` block — normal, and `early_mean` is the phase between the nominal and panel frame grids); `hold_frac` is the share of a frame period the loop

spent holding against a synthesised boundary, averaged over every present, and is what `class` is derived from.

Without this line the cases are indistinguishable after the fact, and a set of captures can end up unable to answer the question it was run for.

11.10 Loading data in Python

Each run writes two files to `~/goxpy_data/`, or to `-outdir` if given (`run-timing-tests.sh` points it at the session directory so everything from one session stays together):

- `Timing-Tests_sub-000_date-<YYYYMMDD>-<HHMMSS>.csv` — data rows, no comments
- `...-info.txt` — session metadata: start/end time, hostname, OS, and the display and audio configuration actually in use

```
import pandas as pd

df = pd.read_csv("~/goxpy_data/Timing-Tests_sub-000_
 ↪  date-20260728-091541.csv")

# av: drop the warm-up cycles before quoting anything
steady = df[df.cycle >= 10]
print(steady.bright_duration_ms.describe())
print(steady.period_ms.std())
```

The `av` test writes one row per cycle: `cycle`, `t_visual_before_ms`, `t_visual_after_ms`, `bright_duration_ms`, `period_ms`, `t_audio_queued_ms`, `soa_intended_ms`, `soa_actual_ms`.

Always read display parameters from the run's own `-info.txt`. On a multi-monitor machine the displays differ in resolution, refresh rate and pixel density, and the scan-out calibration is per-display. The header records which one the window actually opened on.

11.11 Improving timing on your system

Linux

- **Disable the compositor.** By far the largest single improvement. Start from a virtual terminal (`Ctrl+Alt+F2`) with the window system stopped (`systemctl stop gdm`) and `SDL_VIDEODRIVER=kmsdrm`. On the reference rig this took dropped frames from ~5 % to zero and removed a frame of latency. Failing that, use a plain window manager (i3, openbox).

- Disable CPU frequency scaling: `cpupower frequency-set -g performance`.
- **Run with real-time scheduling.** Grant your user the privilege once (below); after that, goxpyriment programs — Timing-Tests included — request priority 50 at startup on their own, so nothing needs prefixing:

```
Timing-Tests -test display -duration-s 30       # asks for real-time
 ↪  itself
Timing-Tests -no-realtime -test display         # opt out
chrt -f 50 some-other-program                  # for anything that does
 ↪  not
```

  This matters most when a program is launched by clicking its icon, where there is no command line to prefix. If the grant is missing the program says so and continues at normal priority rather than refusing to run, and the `Sched:` line in its system report records which it got. The privilege comes from a file you add to `/etc/security/limits.d/` — **not** from editing `/etc/security/limits.conf`, which is package-managed and can be replaced on upgrade, taking your setting with it. See Setting priority under Linux for the procedure: create a group, drop in `/etc/security/limits.d/99-goxpyriment.conf`, add yourself, log out and back in.

  Two traps worth knowing before you spend a session on it:

  - **The limit is inherited from your login session.** A new terminal will not pick up the change; only a full logout and login will. Check `ulimit -r` before trusting a run — if it still says `0`, the grant is not in effect and `chrt` will simply fail.
  - **Editing the file without `sudo` fails silently in some editors.** Confirm the text actually landed, with `grep rtprio /etc/security/limits.d/*.conf`, rather than assuming the save worked.

  Do not reuse an existing group such as `audio` for this. Its rtprio grant belongs to another package (jackd), which can revoke it on reconfigure — and a run that quietly loses real-time priority looks exactly like a run that had it.

  The priority must not exceed the `rtprio` value granted in the setup above. Asking for more fails with "Operation not permitted" – the same error as having no grant at all, so it is easily misdiagnosed. The setup document grants 50, hence `-f 50` here; keep the two numbers equal.

  You *can* skip the setup with `sudo chrt -f 50 Timing-Tests …`, but the test then runs as root: data files land owned by root and it picks up root's environment, not yours. Fine for a one-off check, wrong for collecting data.

- Consider a real-time kernel — see https://ubuntu.com/blog/enable-real-time-ubuntu.
- **Lower the FTDI latency timer** if you read responses over a USB serial device — see the DLP-IO8 section below. Note it does *not* reduce the latency of *sending* a trigger; that is USB frame scheduling, which the timer does not touch.
- **Send triggers off the USB bus if you can.** Since the timer above cannot

help with output, the fix is a different device: `-trigger-device parallel` on a machine with an LPT port, or `-trigger-device gpio` on a single-board computer. Both write via one `ioctl`. See Choosing a TTL output device.

Windows

- Disable "Hardware-Accelerated GPU Scheduling" in Display Settings if you see high frame jitter.

11.12 Audio buffer size

The hardware audio buffer is controlled by the SDL hint `SDL_AUDIO_DEVICE_SAMPLE_FRAMES`, exposed as `-audio-frames`. It must be set before the audio device opens.

```
# Default (platform-dependent, often 512–2048 frames)
Timing-Tests -test latency

# Aggressive low-latency (~5.8 ms at 44100 Hz)
Timing-Tests -test latency -audio-frames 256 -drain-reps 20

# Conservative (~46 ms, stable anywhere)
Timing-Tests -test latency -audio-frames 2048
```

On startup the program prints the actual device format:

```
audio: 44100 Hz  2 ch  256 sample frames (~5.8 ms latency)
```

The buffer sets the floor on audio-onset precision: onsets can only land on buffer boundaries, so 256 frames at 44100 Hz quantises them to 5.8 ms steps. Use `latency` at several sizes to find the smallest that drains stably (low SD), then use that everywhere — including in your actual experiment.

11.12.1 Choosing the buffer: jitter against glitches

"Stable" above means more than a low SD in `latency`, and the two ends fail in opposite ways. Measured on a Raspberry Pi 4 (PipeWire, 48 kHz) on 2026-08-17, capturing the Pi's line output directly with an Analog Discovery 3 at 100 kS/s and taking the onset from a running-RMS envelope:

driver	`-audio-frames`	period	n	median AV lag	SD	`period/sqrt(12)`	torn tones
PipeWire	2048	42.67 ms	60	101.50 ms	12.28 ms	12.32 ms	none
PipeWire	**1024**	21.33 ms	481	**49.88 ms**	**6.17 ms**	6.16 ms	**none**
PipeWire	512	10.67 ms	500	27.30 ms	7.28 ms*	3.08 ms	10 (2.0 %)

driver	`-audio-frames`	period	n	median AV lag	SD	`period/sqrt(12)`	torn tones
PipeWire	256	5.33 ms	483	9.43 ms	2.36 ms	1.54 ms	100 (20.7 %)
ALSA direct	512	10.67 ms	463	4.95 ms	6.74 ms	3.08 ms	**463 (100 %)**

* inflated by a mid-run escalation, below; its first half scatters by 3.10 ms.

1024 is the recommended setting on this hardware — the only one that is both clean and stable. Below it the stack degrades quickly, and at 512 it does something worse than degrade.

11.12.1.1 A large buffer costs jitter, and the jitter is a beat, not noise

The tone is handed over on time and then waits for the queue, so the audio onset scatters across one buffer period — matching `period/sqrt(12)` to within 0.2 % at 1024 and 2048. But it is not random. The tone's position in the buffer grid advances by `cycle mod period` every trial and wraps, a deterministic sawtooth:

```
cycle 499.830 mod buffer 10.667 = 9.163 ms
  predicted  +1.503 ms per trial, wrapping by -9.163
  observed   +1.500 ms per trial (n=161), wraps -9.170 (n=58)
```

Agreement to 7 us. Two consequences: a per-trial lag can be **predicted** from the cycle, the period and the trial index rather than merely bounded; and a cycle chosen as a whole multiple of the buffer period removes the beat altogether. At 512, a 501.33 ms cycle (47 x 10.667) would hold the lag constant instead of spreading it over 10.7 ms.

11.12.1.2 A small buffer costs glitches — and PipeWire moves the goalposts

At 512, ten of the first 245 tones were torn (gaps to 37.7 ms), and then at trial 245 the lag stepped by **+10.85 ms — one buffer period — and the tearing stopped dead**, with no glitch in the remaining 255 tones. That is the audio server adding a period to the graph after repeated underruns.

So a small buffer is not a setting the machine holds: it underruns, relocates your latency mid-run, and the naive linear fit through that step reports a 15 ms "drift" that does not exist. Anything measured across such a step is two populations averaged together.

11.12.1.3 The floor is the hardware, not the audio server

It is tempting to read the rows above as PipeWire's fault and reach for a "direct" path. That was tested: PipeWire stopped (services *and* sockets, or socket activation restarts it), `SDL_AUDIODRIVER=alsa` and `SDL_AUDIO_ALSA_DEFAULT_PLAYBACK_DEVICE=hw:2,0` to

bypass the `default` device, which on a PipeWire system is the PipeWire plugin and fails with `ENOTSUPP` once the server is gone.

Direct ALSA at 512 gave the lowest latency of anything measured — a median lag of **4.95 ms**, with some trials negative, the sound reaching the jack before the panel was 10 % lit. It was also completely unusable: **every one of 463 tones was torn**, typically five times each, the envelope collapsing to about 1 % of plateau every 30-40 ms and recovering.

So the Pi cannot sustain a 10.7 ms buffer, and the audio server was never the constraint. PipeWire's mid-run escalation at 512 was it discovering the same hardware floor and working around it, which is why its glitch rate there was 2 % rather than 100 %. A server that adapts is a nuisance for a measurement and a kindness for an experiment; the fix is to sit above the floor deliberately rather than to remove the thing that was hiding it.

Restart the server afterwards (`systemctl --user start pipewire.socket pipewire-pulse.socket wireplumber.service`), and delete any `~/.asoundrc` written for the test, or the machine will bypass PipeWire from then on.

11.12.1.4 An instrument's detector can set the scale of what it reports

The 512 case was first measured through a BBTK microphone channel, which reported 23 % of tones torn by gaps whose median was 22.3 ms. The electrical capture finds the tearing to be real but 2.0 % of tones with gaps from 0.5 to 37.7 ms, mostly around 2 ms. A threshold detector watching an acoustic envelope needs the signal to recover past its threshold before it calls the tone present again, so short electrical glitches read as long acoustic gaps.

The rate differed too, and plausibly for a real reason: the BBTK session ran `bbtk-capture` on the same Pi, competing for it. Take the lesson as: **a second instrument measuring by a different route is the only way to learn what the first one is adding**, and prefer the route with fewer transducers in it when the question is about software.

What survived every recheck: **scheduling priority is not involved.** The 2x2 of `512/2048` against `-realtime-priority 50/0` scratched at 512 under both policies and was clean at 2048 under both.

Neither failure appears in anything the host prints — in these same runs the software-side SOA read **0.080 ms +- 0.035**, because handing the tone over on time is the part the software does correctly.

Check underruns at the source rather than by ear: run `pw-top` over ssh (a fullscreen test covers the console) and watch the **ERR** column.

11.13 Walkthrough: a Raspberry Pi, GPIO triggers, and BBTK capture

A full `run-timing-tests.sh` session on a Raspberry Pi, driving the TTL from the GPIO header instead of a USB trigger box and recording it on a Black Box ToolKit. The GPIO path matters here: it removes the USB link from between the flip and the TTL edge, so what remains in the onset-vs-TTL spread is the Pi's display path rather than a serial transaction.

> **None of the steps below have been run on a Pi by the author of this section.** The procedure follows from the flags and the wiring; the numbers it produces are what the session is *for*. Treat any figure quoted from another machine in this document as inapplicable until you have measured this one.

11.13.1 1. Find the GPIO chip that owns the 40-pin header

Do not assume `/dev/gpiochip0`. Which chip carries the header varies by Pi model and kernel — on some combinations the header is *not* chip 0, and an unrelated chip is. Ask the system:

```
sudo apt install gpiod        # once
gpiodetect                    # lists chips, line counts, and labels
gpioinfo | less               # per-line names; the header lines are named
 ↪  GPIOnn
```

Pick the chip whose lines are the header's, and pass it as `GPIO_CHIP`. Getting this wrong is not merely inert — another chip's lines may drive real board hardware.

11.13.2 2. Grant access and verify the wiring

```
sudo usermod -aG gpio $USER   # then log out and back in
go run ./tests/test_linuxgpio -chip /dev/gpiochip0 -out
 ↪  17,27,22,5,6,13,19,26
```

Confirm on a scope or logic analyser that the pin you intend to use actually toggles **before** going any further. `test_linuxgpio` exists precisely so that a wiring fault is found here rather than in a 1000-cycle capture.

11.13.3 3. Wire the TTL and the photodiode

- **TTL** — the header pin for the line you chose, into a BBTK TTL input. With the default `GPIO_PINS`, `TRIGGER_PIN=1` is **BCM 17** (physical pin 11). Ground to any header ground pin. `Timing-Tests` prints the BCM number it resolved to at start-up — probe that one, not the flag value.
- **Photodiode** — on the **top-left** stimulus square, at the square's centre. The panel scans top-to-bottom, so the bottom squares light nearly a frame later; a second diode on a bottom square measures that gradient.

3.3 V is not TTL. Pi GPIO swings 0–3.3 V. Whether a given BBTK input latches at 3.3 V is a property of that unit — check it with a 20-cycle run and confirm `N(TTLin1)` equals the cycle count before committing to 1000 cycles. If it does not latch, a 3.3 V→5 V level shifter on the trigger line is the fix; do not feed 5 V back into a Pi input.

11.13.4 4. Get off the desktop

This is worth more than any flag in this document. On the reference hardware the compositor accounted for essentially all of the measured jitter. Run from a bare console with no desktop session:

```
sudo systemctl set-default multi-user.target && sudo reboot   # revert with
↪  graphical.target
```

Then, in the console, use the KMS/DRM video driver:

```
export SDL_VIDEODRIVER=kmsdrm
```

Confirm the refresh rate the Pi is actually driving before trusting any duration — `-frames-on 12` is 200 ms only at 60 Hz:

```
./Timing-Tests -test display -duration-s 30
```

11.13.5 5. Run the session

```
cd tests/Timing-Tests
go build .

TRIGGER_DEVICE=gpio \
GPIO_CHIP=/dev/gpiochip0 \
GPIO_PINS=17,27,22,5,6,13,19,26 \
TRIGGER_PIN=1 \
REFRESH_HZ=60 \
BBTK_CAPTURE=1 \
BBTK_CAPTURE_BIN=~/00_git/bbtkv3/_build/bbtk-capture \
    ./run-timing-tests.sh
```

Check the banner before answering the first prompt — it echoes the trigger device, chip and pins, the tone frequency, and the presented/analysed cycle split. If it says `dlpio8`, the environment variable did not take.

Set `BBTK_PORT` if the capture tool cannot find the box on its own. Budget 11–40 s of device setup per recorded step, and remember every `av` step is ~8½ minutes of stimulus at the defaults.

To pilot the wiring cheaply first, shorten the run rather than skipping it:

```
CYCLES=30 WARMUP=5 TRIGGER_DEVICE=gpio ./run-timing-tests.sh av
```

11.13.6 6. Check the capture before believing it

On the resulting `-events.csv`:

- **`N(TTLin1)` should equal the number of cycles presented** (`CYCLES`, including the warm-up ones — they fire the TTL too). Fewer means the 3.3 V swing is not latching, not that triggers were dropped.
- **`N(Opto1)` should equal `N(TTLin1)`.** A shortfall is photodiode placement or threshold, not timing.
- **Opto duration ≈ 200 ms** at the defaults. Note that BBTK smoothing adds roughly 20 ms to raw durations — read `CorrectedDuration`, and do not interpret the raw figure as panel behaviour.
- **The results header should read `trigger=gpio chip=… line=17 …`.** If it says `trigger=none`, the chip failed to open and the run continued without triggers; the capture's TTL channel is empty and the step must be re-run.

11.13.7 7. What the Pi will not fix

A previous Pi session measured a TTL→Opto lag of **38.5 ms**, against the 4–7 ms range Bridges et al. (2020) report across packages. A local GPIO write removes the USB serial link from that figure, but not the Pi's HDMI/display path, which is the more likely explanation for a lag of that size. Expect the GPIO change to tighten the *spread*; do not expect it to close the *lag*. Distinguishing the two requires the raster-position and display-path measurements, not a different trigger.

11.14 Related tests

- `tests/test_gv_sync` — `.gv` video playback: does `PlayGvFunc` put a frame on screen when it says it does?
- `tests/test_linuxgpio` — GPIO character-device output/input, used to verify header wiring before a session.
- `tests/test_parallel_port` — LPT data lines via ppdev, the equivalent check for a parallel-port trigger.
- `tests/test_dlpio8` — square-wave output for characterising the trigger box itself against an oscilloscope.
- `tests/test_clear_only_frames` — regression test for the compositor presentation bug described at the top of this document.
- `tests/test_vsync_blocking` — does `SDL_RenderPresent` block on VSYNC on this platform, or return immediately?

11.15 DLP-IO8

Driver: `triggers/dlpio8.go`. Full protocol notes and raw data at https://github.com/chrplr/dlp-io8-g; the timing figures below were measured there with a Siglent

SDS1104X-E and are repeated because they change how you should use the device.

11.15.1 Lower the FTDI latency timer before reading anything

The DLP-IO8 is an FTDI device, and the `ftdi_sio` driver defaults to a **16 ms latency timer**: the chip holds a partly-filled buffer that long before sending it to the host. A poll the module answers instantly still takes 16 ms to come back.

Measured, n=300 per setting, for an 8-channel read:

`latency_timer`	round trip	poll rate
16 (default)	**15.98 ms**	**63 Hz**
4	3.99 ms	251 Hz
1	1.01 ms	995 Hz

The relationship is exactly `round trip = latency_timer`, so the module's own processing is negligible and the whole cost is driver batching. A polling loop gets the worst case rather than the average: waiting for each reply synchronises the loop to the timer and pays the full 16 ms every iteration.

```
echo 1 | sudo tee /sys/bus/usb-serial/devices/ttyUSB0/latency_timer
```

That reverts on replug. To make it stick:

```
# /etc/udev/rules.d/99-ftdi-latency.rules
SUBSYSTEM=="usb-serial", DRIVERS=="ftdi_sio", ATTR{latency_timer}="1"
```

The rule applies to every FTDI serial device on the machine, not just this one. That is usually what you want, but it is worth knowing it is system-wide.

It does nothing for sending triggers. Output latency is governed by USB frame scheduling. Do not expect this setting to make a trigger arrive sooner.

11.15.2 A multi-bit code is not atomic

There is no multi-channel command: every command is one ASCII byte affecting one line. `Send(mask)` therefore emits eight bytes which the module acts on as they arrive, and **the port takes ~610 µs to settle**, showing partly-updated values throughout. Measured n=99: 86.2 µs per byte, 609.5 µs from the first line to the eighth.

Against a system sampling at 1 kHz that is about 61 % of a sample period, so a code change is sampled mid-transition roughly three times in five and recorded as a value that was never sent.

Use one line per event type, pulsed. A single line is one command byte, so there is no skew at all, and eight lines still distinguish eight event types. Reserve `Send` for a multi-bit code for the cases where the acquisition reads the code milliseconds after the onset edge rather than latching it at the edge, or where a strobe line is raised last once the code has settled.

11.15.3 Pulse width is only as good as your scheduling

The device has no pulse timer, so a pulse is two host writes and the width inherits host scheduling in full. Measured n=50 per width:

host state	median error	spread
idle	−10 to −20 µs	≤ 120 µs
under CPU load	up to +1.85 ms	up to 4.75 ms

The host's own busy-wait interval degrades by the same amount, tracking the wire to within 80 µs — so the cause is the scheduler descheduling the process, not the USB path or the device. See Setting priority under Linux; that is the fix, and it is a different mechanism from the latency timer above.

11.16 Parallel port and GPIO alternatives

Both avoid the USB path entirely: a write is one `ioctl`, not a USB serial transaction subject to frame scheduling. On hardware that offers either, this is the cheapest available improvement to onset-vs-TTL precision.

In the timing tests, select them with `-trigger-device` (see Choosing a TTL output device):

```
Timing-Tests -test av -trigger-device parallel
Timing-Tests -test av -trigger-device gpio -gpio-chip /dev/gpiochip0
```

or, for a whole session, `TRIGGER_DEVICE=parallel` / `TRIGGER_DEVICE=gpio` with `run-timing-tests.sh`.

In your own experiment code:

```
// Parallel port (LPT), 5 V. Needs `sudo modprobe ppdev` and the `lp` group.
p := triggers.NewParallelPort("/dev/parport0")
if err := p.Open(); err != nil { log.Fatal(err) }
defer p.Close()
p.Send(0x01)                    // all 8 data lines at once, D0 = DB25 pin 2

// GPIO character device (Raspberry Pi, Rock Pi, …), 3.3 V. Needs kernel >=
 ↪  5.10
// and the `gpio` group. Check which chip owns the header with `gpiodetect`.
g, err := triggers.NewLinuxGPIOTrigger(
    triggers.WithGPIOChip("/dev/gpiochip0"),
    triggers.WithGPIOOutputPins([8]int{17, 27, 22, 5, 6, 13, 19, 26}),
)
if err != nil { log.Fatal(err) }
defer g.Close()
g.Pulse(0, 5*time.Millisecond)    // line 0 = the first pin in the array =
 ↪  BCM 17
```

`Send(mask)` sets all 8 lines simultaneously on both — unlike the DLP-IO8, where a multi-bit code is written one byte per line and takes ~610 µs to settle.

Note the voltage difference: parallel is 5 V, GPIO is 3.3 V. Confirm your acquisition system latches at 3.3 V before relying on the GPIO path.

Chapter 12

Media — Multi-Movie Playback with Hardware-Verified Display Onsets

This document describes the `media/` package added to goxpyriment for multi-movie playback synchronised to a shared master clock, with hardware-verified display-onset events suitable for wiring to external EEG / MEG / TMS triggers.

12.1 Table of Contents

12.2 1. Scope and design

12.2.1 What `media/` is

A new top-level package alongside `stimuli/` that adds **multi-movie** playback with shared timing. It is the home for everything related to synchronising several video streams against one another and to external hardware (EEG / MEG / TMS triggers, photodiodes).

It imports the semantics of PsyScope Tahoe's movie commands — `MovieDo[ PLAY/PAUSE/STOP/SET/GET ]` actions and `Movie[Done/At/AtDisplay]` + `Display[Onset/Offset]` conditions — adapted to a Go API.

12.2.2 Hard scope choices

- **`.gv` movies only.** Decoding is delegated to the existing pure-Go `funatsufumiya/go-gv-video` library (LZ4-compressed RGBA frames, random access). No ffmpeg, no MPEG-1, no H.264.
- **Silent.** Movies have no audio. The package does not load, decode, mix, or schedule audio of any kind.
- **Pure Go, no cgo.** SDL3 is loaded via `purego` (already used by `go-sdl3`); the macOS Stage 5 backend uses purego for CoreVideo / libSystem; the Linux Stage 5 backend uses the standard `syscall` package.
- **Coexists with `stimuli`.** `stimuli.GvVideo` and `stimuli.PlayGv` remain the single-movie convenience path. `media.Movie` is added alongside for the multi-movie / sync use cases.

12.2.3 What problems it solves

Problem	Solution
Multiple AVPlayer-style movies drift apart over hundreds of pause/resume cycles	One `MasterClock` shared by every `Movie`; positions computed from a single clock reading per draw cycle. Zero cumulative drift by construction.
Three sequential `Pause` calls reading the clock at slightly different times	`BeginBurst()` / `EndBurst()` freezes `MasterClock.Now` so all commands inside the burst observe the same value.
`Movie[At "f:N"]` needs to fire ~one vsync *before* the target frame is displayed so an overlay can land on the same vsync	`OnAt(Target, fn)` evaluates from inside the per-frame decode loop, *before* `FlipTS`.
External EEG trigger must be aligned to actual first-pixel-visible time, not to when our `Present` call returned	Stage 5 `media/present` package; on macOS uses `CVDisplayLink`, on Linux uses `DRM_IOCTL_WAIT_VBLANK`, both report OS-measured vsync.
Need to know post-hoc which exact vsync each trigger landed on	`Onset.TimestampNS` is in the same nanosecond reference as `sdl.TicksNS()` and SDL keyboard event timestamps; subtract directly to compute reaction times.

12.3 2. Quick start

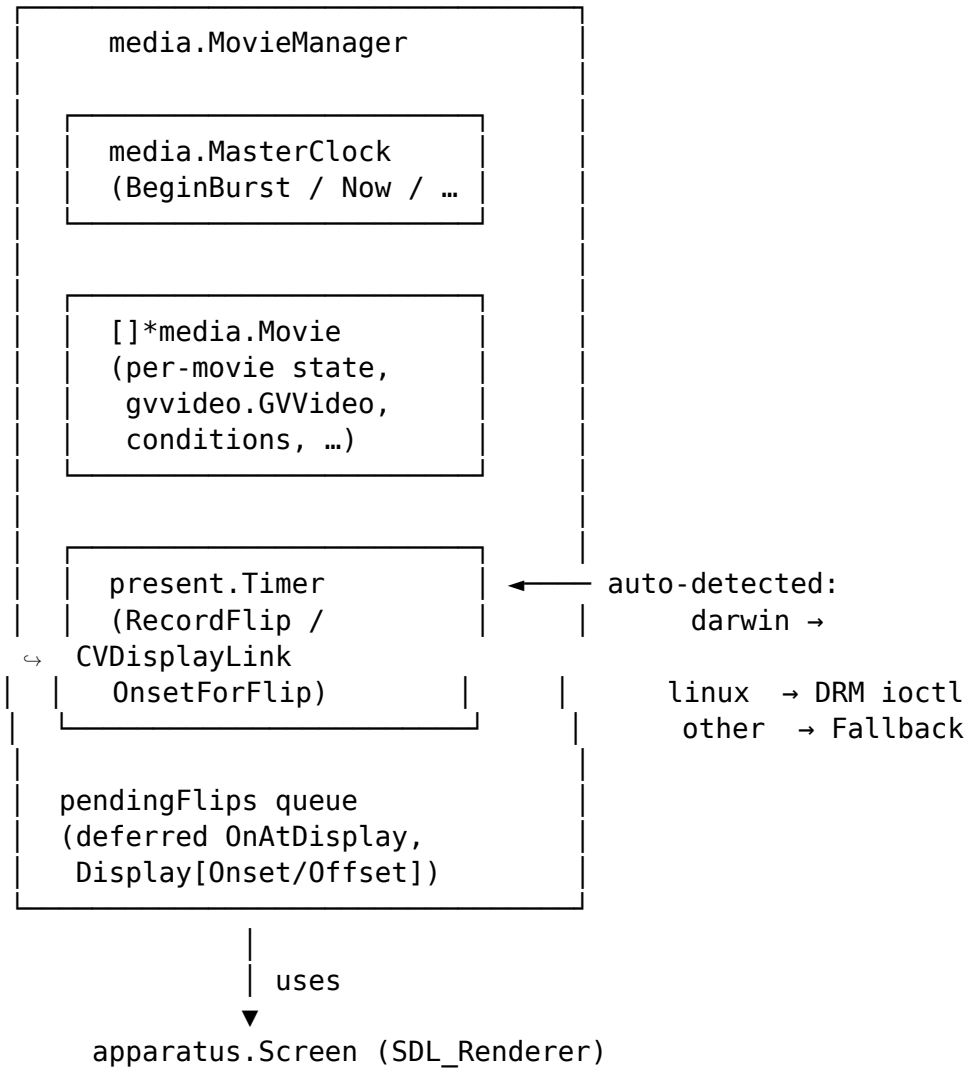
A complete two-movie experiment in ~25 lines, including hardware trigger wiring (silently no-ops if no DLP-IO8 is plugged in):

```
package main

import (
    "log"
    "time"

    "github.com/funatsufumiya/go-gv-video/gvvideo"

    "github.com/chrplr/goxpyriment/control"
    "github.com/chrplr/goxpyriment/media"
    "github.com/chrplr/goxpyriment/triggers"
)

func main() {
    exp := control.NewExperimentFromFlags("Demo", control.Black,
↪   control.White, 32)
    defer exp.End()

    gvL, _ := gvvideo.LoadGVVideo("clipL.gv")
    gvR, _ := gvvideo.LoadGVVideo("clipR.gv")

    mgr := media.NewMovieManager(exp.Screen)
    defer mgr.Close()

    left, _  := media.NewMovie(mgr, gvL, media.WithTag("L"),
↪   media.WithRepeat(-1))
    right, _ := media.NewMovie(mgr, gvR, media.WithTag("R"),
↪   media.WithRepeat(-1))
    defer left.Close()
    defer right.Close()

    ttl, _, _ := triggers.AutoDetectDLPIO8()
    defer ttl.Close()

    left.OnAtDisplay(media.Frame(60), func(o media.Onset) {
        log.Printf("frame 60 displayed @ %d (%s)", o.TimestampNS, o.Source)
        _ = ttl.Pulse(0, 5*time.Millisecond)
    })

    mgr.BeginBurst(); left.Play(); right.Play(); mgr.EndBurst()

    exp.Run(func() error {
        _, err := mgr.Draw()
        return err
```

```
    })
}
```

A more comprehensive walkthrough with composited overlays, burst pause, seek-back, and `Display[Onset/Offset]` events lives in examples/demo_sync_two_gv_movies/.

12.4 3. Architecture in one diagram



Per-frame call site (mixed compositing form):

```
exp.Screen.Clear()              // background
mgr.DrawWithoutFlip()           // decode + render all movies, fire OnAt
 ↪  look-ahead
otherStim.Draw(exp.Screen)      // overlay text / fixation cross / …
```

```
ts, _ := exp.Screen.FlipTS()    // present to display
mgr.NotifyFlipped(ts)           // fire OnAtDisplay + Display[Onset/Offset]
```

For movie-only frames the all-in-one `mgr.Draw()` collapses these into one call.

12.5 4. API reference (package `media`)

12.5.1 4.1 `MovieManager`

The per-experiment owner of the master clock and the registered movies.

```
func NewMovieManager(screen *apparatus.Screen, opts ...ManagerOption)
 ↪  *MovieManager
```

Constructor. Auto-detects the best `present.Timer` for the platform unless overridden via `WithPresentTimer`. Logs the chosen backend at construction:

```
media: present backend: macOS CVDisplayLink (CoreVideo, hardware-verified)
 ↪  (precision=hardware-verified)
```

```
type ManagerOption func(*MovieManager)

func WithPresentTimer(t present.Timer) ManagerOption
```

Override the auto-detected timer. Pass `present.NewFallback()` to force vsync-estimated mode for testing or to silence the macOS `CVDisplayLink` background thread.

12.5.1.1 Lifecycle

Method	Purpose
`(*MovieManager).Close() error`	Stops the macOS CVDisplayLink thread / closes the Linux DRM fd. Idempotent. **Defer this before `exp.End()`.**
`(*MovieManager).Clock() *MasterClock`	Returns the shared master clock (call `Reset()` at trial boundaries if needed).
`(*MovieManager).LogicalSize() (float32, float32)`	Returns the renderer's coordinate-space size (HiDPI-correct). Use for layout instead of `Screen.Size()`.
`(*MovieManager).LastFlipTS() uint64`	Most recent `FlipTS` passed to `NotifyFlipped`.

12.5.1.2 Movie set

Method	Purpose
`Add(m *Movie)`	Implicitly called by `NewMovie`.
`Remove(m *Movie)`	Deregister a movie.
`Movies() []*Movie`	Snapshot of registered movies in registration order.

12.5.1.3 Burst (atomic command groups)

Method	Purpose
`BeginBurst()`	Freeze `MasterClock.Now` at its current value (nestable).
`EndBurst()`	Decrement freeze counter; clock thaws when it reaches zero.

`MovieManager.Draw` / `DrawWithoutFlip` already wrap their per-frame body in an internal burst, so all movies on a frame share one `MasterClock.Now`. Use the public `BeginBurst` / `EndBurst` to make a sequence of script-style command calls atomic across multiple movies.

12.5.1.4 Per-frame

Method	Purpose
`Draw() (uint64, error)`	All-in-one: clear → decode → render → flip → notify. Returns the post-flip SDL ticks. Movie-only frames.
`DrawWithoutFlip() error`	Decode + render, no clear, no Present. Caller composites other stimuli, then calls `Screen.FlipTS()` and `NotifyFlipped(ts)`.
`NotifyFlipped(ts uint64)`	Publish post-vsync events: advance each movie's `currentFrameIndex`, fire `OnAtDisplay`, fire `Display[Onset/Offset]`. Defers callbacks until the matching hardware-verified vsync timestamp arrives (see §7.4).

12.5.1.5 Display[Onset/Offset]

Method	Purpose
`RegisterStimulusName(name string, visible func() bool) func()`	Declare a non-movie stimulus by name. Visibility predicate is sampled in `DrawWithoutFlip`. Returns an unregister function. (Movies are auto-registered with their tag.)
`OnDisplayOnset(name string, fn func(Onset)) func()`	Fires when `name` transitions from not-visible to visible across two flips. Returns unsubscribe.
`OnDisplayOffset(name string, fn func(Onset)) func()`	Fires when `name` transitions from visible to not-visible. Returns unsubscribe.

12.5.2 4.2 `Movie`

Per-movie state, mirrors `PSMovieDecoder` in PsyScope Tahoe.

```
func NewMovie(mgr *MovieManager, gv *gvvideo.GVVideo, opts ...Option)
 ↪  (*Movie, error)
```

Wraps an already-loaded `gvvideo.GVVideo` and registers it with the manager. The caller retains ownership of `gv` and is responsible for closing the underlying file (matches `stimuli.GvVideo.Unload`).

12.5.2.1 Construction options

```
type Option func(*Movie)

WithTag(tag string)              // identifier; required for
 ↪  Display[Onset/Offset]
WithRate(r float64)             // playback rate; 1.0 = normal, must be > 0
WithRepeat(n int)               // -1 = infinite, 1 = play once (default)
WithFromTime(d time.Duration)   // initial playhead within movie media-time
WithScale(s ScaleMode)          // ScaleNone | ScaleFit | ScaleFill
WithSize(w, h float32)          // explicit destination size; overrides
 ↪  WithScale
WithPosition(p sdl.FPoint)      // center-relative draw position
WithFadeIn(d time.Duration)     // linear opacity 0→1 ramp over d effective
 ↪  time
WithPersistent(b bool)          // last frame stays on screen after Done
```

12.5.2.2 Playback control

Method	Effect
`Play()`	Start (first call) or resume (after `Pause`). Fresh start anchors playhead at `WithFromTime`. Idempotent.
`Pause()`	Freeze at current frame.
`PauseWithLoop(window time.Duration)`	Pause and bounce within ±
`Stop()`	Halt, release GPU texture, reset state. After `Stop` the movie can be `Play`-ed again from `WithFromTime`.
`Close()`	Release GPU resources. Does not close the underlying `gvvideo.GVVideo`.

12.5.2.3 Property access

Method	Returns
`Time() time.Duration`	Effective media time (rate-scaled, cumulative across loops).
`Frame() int`	Cumulative 1-based displayed-frame index (advanced by `NotifyFlipped`). 0 before first frame.
`FPS() float64`	Frames per second from the `.gv` header.
`FrameCount() int`	Frames per loop iteration.
`LoopCounter() int`	0-based loop iteration currently playing.
`Repeat() int`	Configured loop count (-1 = infinite).
`Width(), Height() float32`	Native frame dimensions.
`IsActive(), IsPaused(), IsDone() bool`	State queries.

12.5.2.4 Property mutation

Method	Effect
`SetRate(r float64) error`	Change playback rate; preserves current effective time. r ≤ 0 returns an error.
`SeekTime(d time.Duration) error`	Jump to elapsed media time `d`. Resets condition fired-flags so re-pass over the same target re-fires.
`SeekFrame(n int) error`	Jump to cumulative 1-based frame `n`.

12.5.2.5 Conditions / callbacks

```
// Sealed Target type — only these three constructors:
type Frame int                  // 1-based cumulative target
type AtTime time.Duration       // elapsed media time target
type Done struct{}              // end-of-file target
```

```
// Look-ahead (fires from DrawWithoutFlip, BEFORE the target frame is
 ↪  presented):
OnAt(target Target, fn func(Onset)) func()    // returns unsubscribe
OnDone(fn func(Onset)) func()                 // sugar for OnAt(Done{}, ...)
DoneCh() <-chan Onset                         // channel form for OnDone

// Hardware-verified (fires from NotifyFlipped, AFTER the target frame
 ↪  appears on screen):
OnAtDisplay(target Target, fn func(Onset)) func()
```

The look-ahead vs hardware-verified distinction is critical for trigger alignment — see §7.

12.5.3 4.3 `MasterClock`

Monotonic media clock with freeze-during-burst support.

```
func NewMasterClock() *MasterClock

(*MasterClock).Now() time.Duration   // monotonic since last Reset /
 ↪  construction
(*MasterClock).Reset()                // back to zero
(*MasterClock).BeginBurst()           // freeze Now at current value
 ↪  (nestable)
(*MasterClock).EndBurst()            // unfreeze when freeze count hits zero
(*MasterClock).Frozen() bool          // currently inside a burst?
```

12.5.4 4.4 Onset events

```
type Onset struct {
    TimestampNS uint64       // SDL ticks; same reference as sdl.TicksNS /
 ↪  FlipTS
    Frame       int          // cumulative 1-based; 0 if not applicable
    Source      OnsetSource  // precision class
    Movie       *Movie       // nil for generic Display events of non-movie
 ↪  stimuli
    Name        string       // movie tag for movies; user name for others
}

type OnsetSource int

const (
    VsyncEstimated   OnsetSource = iota  // post-Present FlipTS,
 ↪  vsync-period precision
    HardwareVerified                     // OS-measured first-pixel-visible
 ↪  (sub-ms)
```

```
    LookAhead                           // pre-vsync, fired from decode loop
)
```

12.5.5 4.5 Time-spec helpers

For users translating PsyScope scripts to Go:

```
ParseTimeSpec("s:3,ms:100")  // → 3.1 * time.Second, nil
ParseFrameSpec("f:186")      // → 186, nil
ParseFrameSpec("frame:1")    // → 1, nil
```

Native API uses `time.Duration` and `int` directly — these helpers are a convenience for porting existing scripts.

12.6 5. API reference (package `media/present`)

The presentation-timer subpackage. Each backend reports OS-measured vsync timestamps that pair with `Screen.FlipTS()` values to give hardware-verified display-onset times.

```
type Timer interface {
    RecordFlip(flipTS uint64)
    OnsetForFlip(flipTS uint64) (timestamp uint64, source OnsetSource, ok
    ↪ bool)
    Precision() OnsetSource
    Close() error
    Description() string
}

func AutoDetect(screen *apparatus.Screen) Timer  // platform-best, never nil
func NewFallback() Timer                          // always-available no-op
```

`MovieManager` calls `RecordFlip(ts)` immediately after each `FlipTS`, then `OnsetForFlip(ts)` to read back the matching hardware-verified vsync timestamp. If `ok=false`, the manager defers and retries on the next `NotifyFlipped` (the OS may publish vsync timestamps asynchronously).

Most users never construct a `Timer` directly — `NewMovieManager` handles auto-detection. Use `WithPresentTimer(present.NewFallback())` in tests if you need deterministic vsync-estimated behaviour.

12.7 6. Multi-movie synchronisation: what is guaranteed

12.7.1 6.1 Mechanism

Every per-frame body opens an internal burst, reads `MasterClock.Now` exactly once, and computes each movie's effective media time from that single value:

```
// Inside MovieManager.DrawWithoutFlip:
mgr.clock.BeginBurst()
defer mgr.clock.EndBurst()
now := mgr.clock.Now()          // ← single read for this frame

for _, m := range movies {
    eff := m.effectiveLocked(now)   // each movie computes its own offset
    target := int(eff.Seconds() * m.fps) + 1
    // … decode + render …
}
```

Because every movie computes from the same `now`, the **inter-movie spread within one frame is zero by construction**. There is no rounding error to accumulate. Pause/resume cycles update each movie's `mediaTimeOffset` independently but always by the same delta when issued inside the same burst.

12.7.2 6.2 What that means in numbers

Measurement	Result	Notes
Inter-movie spread at `Play`	**0 ms**	All movies anchored to one `MasterClock.Now` reading inside `BeginBurst/EndBurst`.
Inter-movie spread at first frame on screen	**0 ms**	All movies render in one `Renderer.Present` call → one vsync → one display refresh.
Cumulative drift after 500 pause/resume cycles	**0 ms**	Mathematical: pause/resume only changes per-movie `mediaTimeOffset`; drift would require a per-movie clock, which the design forbids.
Same-vsync rendering	All visible movies share the same vsync	Single `Renderer.Present` call → atomic page flip.

This matches PsyScope Tahoe's measured numbers (see `MultiMovieSyncStrategy.md` §8 in that repo).

12.7.3 6.3 The atomic-burst pattern

Use `mgr.BeginBurst() / mgr.EndBurst()` around any sequence of script- style commands that should observe the same clock:

```
mgr.BeginBurst()
left.Pause()      // captures MasterClock.Now at burst-start time T
right.Pause()     // captures the same T
center.Pause()    // captures the same T
mgr.EndBurst()
```

Without the burst, three sequential `Pause` calls would each read `MasterClock.Now` a few microseconds apart. The burst ensures all three movies pause at the exact same media time, regardless of scheduling jitter between the calls.

12.7.4 6.4 What is *not* synchronised

- **Rendering completion across multiple Renderers.** This package uses one renderer (the `apparatus.Screen.Renderer`), so this is moot. If you build a multi-window experiment, each window has its own `MovieManager` and they synchronise only as well as the OS schedules their flips.
- **Decode latency variation.** Different movies may take different times to LZ4-decompress a frame. The decode happens before the shared `Renderer.Present` call, so a slow decode would push the entire frame past its target vsync. Prefer movies of similar resolution and bitrate; avoid per-frame allocations (the decode buffer is reused).
- **Audio.** Audio is out of scope (see §1).

12.8 7. External-trigger synchronisation with frames on screen

This section is the most important one for EEG / MEG / TMS experiments and the question most users will have. Read it carefully.

12.8.1 7.1 The two timestamps that matter

For a callback registered via `mov.OnAtDisplay(media.Frame(N), fn)`, two distinct timestamps are involved:

Quantity	What it is	Source	Precision
`Onset.TimestampNS`	When frame N's first pixel begins scanning out of the display controller	OS vsync API (CVDisplayLink / DRM_IOCTL_ WAIT_VBLANK)	**Sub-ms** vs scanout start (Stage 5 platforms)

Quantity	What it is	Source	Precision
Wire-arrival of TTL pulse	When the EEG box sees the trigger line go HIGH	DLP-IO8 USB / parallel port / MEGTTLBox + your callback latency	`TimestampNS + 0.5 –5 ms` (constant + small jitter)

`Onset.TimestampNS` is what you want for **post-hoc analysis**: it is in the same nanosecond reference frame as `sdl.TicksNS()` and SDL3 keyboard event timestamps, so reaction times computed as `keyEvent.Timestamp - onset.TimestampNS` are sub-ms accurate.

The wire-arrival is what arrives at your EEG amplifier's TTL input. It lags `TimestampNS` by a small chain of latencies.

12.8.2 7.2 The wire-arrival delay

Breakdown of the ~0.5–5 ms gap between `TimestampNS` and the TTL line going HIGH:

Stage	Typical	Notes
OS publishes vsync timestamp	0 µs (Linux DRM, sync ioctl) to ~1 ms (macOS, async CV callback may defer)	See §7.4
Manager dispatches the callback	~1–100 µs	Mutex + closure invocation
`ttl.Pulse` writes to the device	0.5–2 ms (DLP-IO8 USB-CDC); <1 µs (parallel port); ~1 ms (MEGTTLBox 115200 baud)	Bus + firmware
Total	**~0.5 ms (parallel) to ~5 ms (USB)**	Stable / calibratable

The delay is **constant and stable**. Calibrate once with a photodiode + TTL recording (see §7.5) and subtract the measured offset in your analysis pipeline.

12.8.3 7.3 LookAhead vs HardwareVerified vs VsyncEstimated

The three `OnsetSource` values describe the precision class of `Onset.TimestampNS`:

Source	Where it fires	TimestampNS meaning	Use case
`LookAhead`	Inside `DrawWithoutFlip`, before `Present`	`sdl.TicksNS()` at fire time (NOT a vsync)	Set state for the same-vsync compositor (e.g., "show overlay this frame")
`VsyncEstimated`	After `Present`, in `NotifyFlipped`	Post-`Present` `FlipTS` value	Vsync-period precision (~16 ms at 60 Hz). Default fallback when no Stage 5 backend is available.
`HardwareVerified`	After `Present`, in `NotifyFlipped`, possibly deferred to next flip	OS-measured first-pixel-visible time	Sub-ms precision vs scanout start. Available on macOS + Linux.

Callback	Source on Stage 5 platforms	Source on fallback platforms
`mov.OnAt(t, fn)`	`LookAhead`	`LookAhead`
`mov.OnDone(fn)`	`LookAhead`	`LookAhead`
`mov.OnAtDisplay(t, fn)`	`HardwareVerified`	`VsyncEstimated`
`mgr.OnDisplayOnset(name, fn)`	`HardwareVerified`	`VsyncEstimated`
`mgr.OnDisplayOffset(name, fn)`	`HardwareVerified`	`VsyncEstimated`

When the timer is `VsyncEstimated`, the manager logs a one-time warning on the first display callback registration. When the timer is `HardwareVerified`, the warning is suppressed.

12.8.4 7.4 The deferral pattern (macOS specifically)

On macOS, the `CVDisplayLink` callback fires on a background thread asynchronously to `Present`. In the rare race where the callback hasn't yet published a vsync timestamp by the time `NotifyFlipped(ts)` runs, the manager:

1. Stashes this flip's callbacks in `pendingFlips`.
2. Continues normally (no error).
3. On the *next* `NotifyFlipped` (one vsync later, ~16 ms wall-clock), tries `OnsetForFlip(stashed_ts)` again. The CVDisplayLink callback has surely fired by now, so the lookup succeeds and the callbacks fire — with the **correct** hardware-verified timestamp for the stashed flip.

The trigger pulse therefore arrives ~16 ms after the visual onset in this race case (the wire is one vsync late), but the `Onset.TimestampNS` you receive is still sub-ms accurate for *frame N's actual visual onset*. Post-hoc analysis is unaffected; only the real-time wire timing is delayed by one vsync.

If a stash sits in the queue for more than `~3 × frame period` (~50 ms at 60 Hz), the manager gives up waiting and fires the callbacks with `(FlipTS, VsyncEstimated)` as a fallback, logging a single warning. This handles compositor stalls and display-link issues without silently inaccurate timestamps.

On Linux this race does not happen: `RecordFlip` queries DRM synchronously right after `FlipTS`, the kernel returns the matching vblank's count and timestamp before `OnsetForFlip` is called, and the callback fires immediately on the same `NotifyFlipped`.

12.8.5 7.5 Calibration

For sub-ms wire-side alignment to visual onset, you must measure the constant delay of your specific hardware:

1. Display a flashing white square in the corner of the screen.
2. Place a photodiode on the square; route both the photodiode signal and the TTL line to a recording amplifier with sub-ms sampling (most EEG amplifiers can do this directly).
3. Measure `delta = wire_HIGH_time − photodiode_first_light` over many trials. Take the median.
4. In your data analysis, subtract `delta` from every TTL marker timestamp.

After calibration, wire-side alignment to visual onset is bounded by hardware jitter (typically tens of µs) plus your measurement uncertainty. The `tests/Timing-Tests/` framework in this repository can drive this calibration measurement automatically.

12.8.6 7.6 What is *not* measured

The Stage 5 backends report when the **display controller** starts scanning out, not when LCD photons actually emit. Two layers of photon-side latency are invisible to any software-only path on these platforms:

- **Panel response** — 1–10 ms typical for IPS / VA / OLED panels at 60 Hz, longer for cheap monitors with image processing turned on ("game mode" usually disables this). Fixed offset per monitor.
- **Scanout phase** — at 60 Hz, the top of the screen receives its new pixels at vsync; the bottom receives them ~16 ms later (full scanout = one frame period). The reported timestamp is **vsync** = start-of-scanout, so a stimulus at the bottom of the screen has up to a frame period of additional latency before its photons emit. Place timing-critical stimuli near the top, or use a photodiode at their actual location.

These limits also apply to Apple's `addPresentedHandler` and Vulkan's `VK_EXT_present_timing` — they are intrinsic to the display pipeline, not specific to this implementation. Photon-precise onset still requires a photodiode.

12.8.7 7.7 Recommended pattern

```
// At experiment start:
mov.OnAtDisplay(media.Frame(N), func(o media.Onset) {
    // Always log the hardware-verified timestamp, even if you also pulse a
     ↪  wire:
    exp.Data.Add(trialID, "stim_onset_ns", o.TimestampNS, o.Source.String())

    // Pulse the trigger immediately. The pulse arrives ~0.5–5 ms after
     ↪  o.TimestampNS;
    // calibrate the constant and subtract in analysis.
    _ = ttl.Pulse(0, 5*time.Millisecond)
})
```

Two records per event: the timestamp (hardware-verified, sub-ms) goes into the data file; the wire pulse goes into the EEG amplifier. Align post-hoc by subtracting the calibrated wire-arrival delay.

12.9 8. Platform support (Stage 5)

Platform	Backend	Precision	Status	Prerequisites
macOS (Apple Silicon + Intel)	`CVDisplayLink` via `purego` (CoreVideo + libSystem)	`HardwareVerified`, sub-ms	**Shipped**	None — frameworks are part of the OS
Linux (X11 + Wayland)	`DRM_IOCTL_WAIT_VBLANK` via `syscall` on `/dev/dri/cardN`	`HardwareVerified`, sub-ms	**Shipped**	User must be in `video` group (default on most distros). At least one DRM device.
Windows	(none yet)	`VsyncEstimated` only	**Not yet shipped**	—
FreeBSD / OpenBSD / others	(none)	`VsyncEstimated` only	Falls back to `Fallback`	—

The startup log line confirms the chosen backend:

```
media: present backend: macOS CVDisplayLink (CoreVideo, hardware-verified)
 ↪  (precision=hardware-verified)
media: present backend: Linux DRM vblank (DRM_IOCTL_WAIT_VBLANK,
 ↪  hardware-verified) (precision=hardware-verified)
media: present backend: vsync-estimated (post-Present FlipTS, no OS
 ↪  integration) (precision=vsync-estimated)
```

If auto-detection fails (e.g., Linux without `video` group, missing `/dev/dri/cardN`), the manager logs the failure once and falls back gracefully:

```
present: Linux DRM vblank unavailable (open /dev/dri/cardN: permission
 ↪  denied); falling back to vsync-estimated
media: present backend: vsync-estimated (post-Present FlipTS, no OS
 ↪  integration) (precision=vsync-estimated)
```

12.9.1 8.1 What Windows users get today

Windows builds compile cleanly (the `media/present/autodetect_other.go` file matches `!darwin && !linux`) and use the `VsyncEstimated` fallback. This means:

- All multi-movie synchronisation works (§6 guarantees are independent of Stage 5).
- All `Movie[At]` look-ahead callbacks work (LookAhead is software-only, always available).
- All `OnAtDisplay` / `Display[Onset/Offset]` callbacks fire — but with `Source: VsyncEstimated`, accurate to vsync-period (~16 ms at 60 Hz) rather than sub-ms.
- A one-time warning is logged at the first display-callback registration:

  ```
  media: display-onset precision is vsync-period (~16.666ms at this
   ↪  refresh rate). Install a sub-ms presentation backend (Stage 5:
   ↪  Vulkan VK_EXT_present_timing or Metal addPresentedHandler) for
   ↪  hardware-verified timing.
  ```

Workaround for Windows users today: photodiode calibration (§7.5) using the `VsyncEstimated` timestamp as a reference. The constant offset includes one extra source of jitter (the ~vsync-period uncertainty), but for low-trial-rate studies this is often acceptable.

12.9.2 8.2 What needs to be done for Windows

To bring Windows up to parity with macOS / Linux, a Stage 5 backend needs to be added at `media/present/dxgi_windows.go` (next to the existing `cvdisplaylink_darwin.go` and `drm_linux.go`). The most promising APIs:

API	Pros	Cons
`IDXGISwapChain::GetFrameStatistics` (DXGI 1.0+)	Available on every Windows since Vista. Returns `PresentRefreshCount` + `PresentQPCTime`.	DXGI swapchain owned by SDL_Renderer. We can't construct our own without taking over rendering.
`DwmGetCompositionTimingInfo`	Returns `qpcCompose` and `qpcVBlank` for the desktop window manager. Doesn't require swapchain ownership.	DWM-specific (composition mode); behaviour with fullscreen exclusive may differ.

API	Pros	Cons
`D3DKMTWaitForVerticalBlankEvent` + `D3DKMTGetScanLine` (gdi32)	Direct kernel-mode access; very precise.	Undocumented headers, requires `libloaderapi` workaround for the kernel32 entrypoint.

Recommended path: **`DwmGetCompositionTimingInfo`** via purego on `dwmapi.dll`. It's the cleanest API that doesn't require swapchain ownership and works alongside SDL_Renderer.

Skeleton:

```
//go:build windows

package present

// Use purego.Dlopen("dwmapi.dll") to get a handle, then bind:
//   DwmGetCompositionTimingInfo(HWND, *DWM_TIMING_INFO) HRESULT
// where DWM_TIMING_INFO contains qpcVBlank (QueryPerformanceCounter
// units) for the most recent vblank. Convert QPC ticks to SDL ticks
// via QueryPerformanceFrequency at startup.

func newWindowsBackend(screen *apparatus.Screen) (Timer, error) { ... }
```

Estimated effort: ~250 lines of Go (similar shape to the existing macOS backend), plus ~100 lines of tests (purego symbol-loading, QPC → SDL-ticks conversion, struct layout pinning).

Until that lands, Windows experiments needing sub-ms wire alignment should rely on photodiode calibration (§7.5).

12.10 9. Mapping to PsyScope movie commands

Users porting PsyScript-based experiments will recognise this table. Volume-related rows are explicitly unsupported (audio is out of scope — see §1).

PsyScope construct	goxpyriment Go API
`EventType Movie, Stimulus "clip.gv", MovieTag M1`	`gv, _ := gvvideo.LoadGVVideo("clip.gv"); media.NewMovie(mgr, gv, media.WithTag("M1"))`
`MovieRate 1.0`	`media.WithRate(1.0)` / `mov.SetRate(...)`
`Repeat 3` / `Repeat -1`	`media.WithRepeat(3)` / `WithRepeat(-1)`
`FromTime "s:5"`	`media.WithFromTime(5*time.Second)`
`Flags Scale_X Scale_Y`	`media.WithScale(media.ScaleFit)`
`FadeIn 1000`	`media.WithFadeIn(1*time.Second)`

PsyScope construct	goxpyriment Go API
`ClearType NO_CLEAR`	`media.WithPersistent(true)`
`MovieDo[ PLAY ]`	`mov.Play()`
`MovieDo[ PAUSE ]`	`mov.Pause()`
`MovieDo[ PAUSE ... LoopRegion=ms:N ]`	`mov.PauseWithLoop(±N*time.Millisecond)`
`MovieDo[ STOP ]`	`mov.Stop()`
`MovieDo[ SET Rate=R ]`	`mov.SetRate(R)`
`MovieDo[ SET Time=... ]`	`mov.SeekTime(...)`
`MovieDo[ SET Frame=N ]`	`mov.SeekFrame(N)`
`MovieDo[ GET Time/TimeMs/Frame/FrameCount/FPS/Counter/Repeat ]`	`mov.Time()`, `Frame()`, `FPS()`, `FrameCount()`, `LoopCounter()`, `Repeat()`
`MovieDo[ SET Volume=... ]`, `GET Volume`	**Not supported** (audio out of scope)
`Movie[ Done ]`	`mov.OnDone(fn)` or `<-mov.DoneCh()`
`Movie[ At THISMOVIE "f:N" ]`	`mov.OnAt(media.Frame(N), fn)`
`Movie[ At THISMOVIE "s:T,ms:U" ]`	`mov.OnAt(media.AtTime(d), fn)`
`Movie[ AtDisplay THISMOVIE "f:N" ]`	`mov.OnAtDisplay(media.Frame(N), fn)`
`Display[ Onset ]` / `Display[ Offset ]`	`mgr.OnDisplayOnset(name, fn)` / `OnDisplayOffset(name, fn)`
`SerialOut[ "0xFF" ]` action	call inside the OnDisplay/OnAtDisplay callback into `triggers.OutputTTLDevice`
`THISMOVIE` keyword	irrelevant — the closure already captures `mov`
`Instances: "-1"`	irrelevant — Go calls execute once

Burst-pause atomicity (the goxpyriment equivalent of three sequential `MovieDo[ PAUSE ... ]` script lines hitting the same draw cycle):

```
mgr.BeginBurst()
movieA.Pause(); movieB.Pause(); movieC.Pause()
mgr.EndBurst()
```

12.11 10. References

In this repository:

- `examples/demo_sync_two_gv_movies/` — comprehensive walkthrough example
- `tests/Timing-Tests/` — empirical timing-measurement framework

External / upstream:

- PsyScope Tahoe `MultiMovieSyncStrategy.md` — describes the original Apple-side architecture this package mirrors (CMTimebase, AVAssetReader, Metal addPresentedHandler).
- PsyScope Tahoe `CrossPlatformAnalysis.md` — feasibility study that led to the present `media/` package.

- `VK_EXT_present_timing` Vulkan spec — the future direct-Vulkan path (not yet implemented; would replace `SDL_Renderer` entirely).
- Apple `CVDisplayLink` reference — the API the macOS backend uses.
- Linux DRM uAPI `drm_wait_vblank` — the API the Linux backend uses.

Chapter 13

Packaging Your GoXpyriment Experiment

This guide explains how to package and distribute your `goxpyriment` experiments as standalone binary executables and installers (e.g., `.exe`, `.deb`, `.rpm`, `.zip`) for different platforms and architectures.

> To split **one example from this repo's `examples/`** into its own published GitHub repository — with release CI and an optional in-browser (WebAssembly) build on GitHub Pages — see Publishing an example as a standalone repository.

13.1 Prerequisites

1. **GoReleaser**: The primary tool for automating the build and release process. Install it from goreleaser.com.
2. **No SDL3 installation needed**: `goxpyriment` uses `github.com/Zyko0/go-sdl3/bin/binsdl` which embeds pre-built SDL3 and SDL3_ttf libraries for Linux, macOS, and Windows (both amd64 and arm64) directly inside the Go binary. There is nothing extra to bundle or install.

13.2 Project Structure Recommendation

For a distributable experiment, keep your assets organized:

```
my_experiment/
├── assets/             # Large files (videos, high-res images)
├── assets_embed/       # Small files (fonts, icons, sound effects)
├── main.go             # Your experiment logic
├── go.mod
└── .goreleaser.yaml    # GoReleaser configuration
```

13.3 Step 1: Handling Assets

13.3.1 A. Embedding (Small Assets)

For small files (fonts, small sounds), use Go's `//go:embed` directive. This includes the assets directly inside the binary.

```
//go:embed assets_embed/font.ttf
var MyFont []byte
// ...
exp.LoadFontFromMemory(MyFont, 32)
```

13.3.2 B. Bundling (Very Large Assets)

For assets that are too large to embed in the binary (e.g., video files), bundle them alongside the executable and look for them relative to the executable path at runtime. For images, CSVs, audio clips, and similar small-to-medium files, prefer embedding (`//go:embed`) over bundling.

13.4 Step 2: GoReleaser Configuration

Create a `.goreleaser.yaml` in your project root. Here is a template based on the `retinotopy` example:

```
version: 2

before:
  hooks:
    - go mod tidy

builds:
  - id: experiment
    env:
      - CGO_ENABLED=0  # Required for easy cross-compilation with purego
    goos:
      - linux
      - windows
      - darwin
    goarch:
      - amd64
      - arm64
    main: ./main.go
    binary: my_experiment

archives:
  - id: default
    name_template: "{{ .ProjectName }}_{{ .Os }}_{{ .Arch }}"
    formats: [zip]
```

```
    files:
      - assets/**/*
      - README.md

nfpms:
  - id: linux-packages
    package_name: my-experiment
    maintainer: Your Name <you@example.com>
    description: My Awesome Behavioral Experiment
    formats:
      - deb
      - rpm
    contents:
      - src: assets/**/*
        dst: /usr/share/my-experiment/assets
```

13.5 Step 3: Building

13.5.1 Local Snapshot Build

To test the packaging locally without creating a GitHub release:

```
goreleaser build --snapshot --clean
```

The resulting binaries and packages will be in the `dist/` folder.

13.5.2 Official Release

1. Commit your changes.
2. Tag your repository: `git tag -a v1.0.0 -m "First release"`
3. Push the tag: `git push origin v1.0.0`
4. Run GoReleaser (or use a GitHub Action like `retinotopy_build.yml`):

```
goreleaser release --clean
```

13.6 Platform-Specific Notes

13.6.1 Windows

SDL3 is embedded in the binary by `binsdl`, so users can simply unzip and run the experiment without installing anything extra.

13.6.2 Linux

Generating `.deb` or `.rpm` files via `nfpms` (included in GoReleaser) allows users to install your experiment using `sudo apt install ./my_experiment.deb`.

13.6.3 macOS

For macOS, GoReleaser creates a binary. To create a proper `.app` bundle or `.dmg`, you may need additional tools like `gon` for notarization or custom scripts to create the folder structure: `MyExperiment.app/Contents/MacOS/my_experiment`.

13.7 Using GitHub Actions

You can automate this entire process by using the `retinotopy_build.yml` workflow. Whenever you push a new version tag (e.g., `v1.2.3`), GitHub will automatically build all versions and attach them to a new Release page on your repository.

Chapter 14

Publishing an example as a standalone repository

This guide turns one directory from `examples/` into its own self-contained GitHub repository, so a colleague can:

- download a **prebuilt binary** for Linux/macOS/Windows and run it, and
- optionally **play it in the browser** (a WebAssembly build on GitHub Pages).

It is the process used for the two reference repos — copy either as a template:

Repo	Kind	Browser build?
`chrplr/Language-Localizer-French-audio`	audio experiment, ships `.wav` assets	no
`chrplr/Rush-Hour`	mouse experiment, everything embedded	yes (Pages)

> **Just need to hand off a zip, not publish a repo?** Use the Makefile wrapper from the repo root:
>
> ```
> make share-<ExampleName>                    # → _
>  ↪  build/share/<ExampleName>/
> make share-<ExampleName> VERSION=v0.12.5     # pin a specific
>  ↪  goxpyriment release
> ```
>
> That exports a standalone *module* (buildable on its own with `go run .`, no workspace) to zip and email — see `examples/README.md`. This guide goes further: it turns that output into a *published repository* with release CI and an optional in-browser build. Start from `share.sh` (Step 1), then add the pieces below.

14.1 When to use it

Publish an example this way when you want an external audience to run it without the goxpyriment workspace. Only publish a **browser** build for paradigms that tolerate browser timing (choice RT, memory, attention, puzzles) — not ones that need sub-millisecond onset control (rapid RSVP, subliminal priming). See WASM.md.

14.2 Prerequisites

- The `gh` CLI, authenticated (`gh auth status`).
- For a **browser** build: the example must embed *all* its assets with `//go:embed` (no filesystem exists in the browser), and it should avoid APIs still stubbed on js — see WASM.md.
- Use a goxpyriment release that contains the browser fixes (≥ **v0.12.5**): the js `Save()` no-op (one download at the end, not per block) and the mouse path fix (`RenderCoordinatesFromWindow`). Earlier releases crash or spam downloads in the browser.

14.3 Step 1 — Generate the standalone module

From the goxpyriment repo root, export the example with the latest release pinned (or pass a specific version):

```
bash examples/share.sh <ExampleName> <version> <output-parent-dir>
# e.g.
bash examples/share.sh Rush-Hour v0.12.5 ~/00_git
# → ~/00_git/Rush-Hour/  with a generated go.mod/go.sum requiring
 ↪  goxpyriment v0.12.5
```

`share.sh` copies the directory, drops the workspace `go.mod`/`go.work` coupling, writes a fresh `go.mod` requiring the *published* goxpyriment, and runs `go mod tidy`. The result builds on its own with `go build .`.

`make share-<ExampleName>` is the convenient wrapper, but it always writes to `_build/share/<ExampleName>/`; call `share.sh` directly with the third argument when you want the module created straight where the repo will live (e.g. `~/00_git`).

14.4 Step 2 — Prune goxpyriment-internal files

The copy still contains files that only make sense inside the monorepo. Remove:

```
cd ~/00_git/<ExampleName>
rm -rf .claude meta.yaml description.md   # gallery/agent metadata, not
 ↪  needed standalone
```

Keep the `.go` sources, embedded assets, tests, and `README.md`.

14.5 Step 3 — Add the repo scaffolding

Add these files (copy and adapt from the reference repos):

- **`LICENSE`** — pick a license. goxpyriment itself is Apache-2.0, but **you are its author**, so you may license your example however you like (Rush-Hour is MIT). If you change the license, update the per-file header comments to match.
- **`build.sh`** / **`build.bat`** — a one-command local build (`CGO_ENABLED=0 go build -trimpath -ldflags="-s -w" -o <Name> .`). These let a colleague build a *locally-signed* binary that avoids OS gatekeeper warnings.
- **`.gitignore`** — ignore the built binary. Note `go build .` produces a binary named after the **module** (lowercase, e.g. `rush-hour`) *and* `build.sh` produces the `-o` name (e.g. `Rush-Hour`); ignore **both**, plus `dist/` and `_build/`.
- **`README.md`** — adapt the example's README: change `go run ./examples/<Name>` to `go run .`, and add "Binaries" (Releases) and, if applicable, "Play in your browser" sections. The reference READMEs include the standard notes about unsigned macOS/Windows binaries.
- For a **browser** build only: **`web/index.html`** (copy Rush-Hour's; it wires the canvas, `sdl.js`, `wasm_exec.js`, and `main.wasm`).

14.6 Step 4 — Add the CI workflows

14.6.1 `.github/workflows/release.yml` (native binaries)

Cross-compiles Linux/macOS/Windows on every push and, on a `v*` tag, publishes a GitHub Release with archives + `SHA256SUMS`. `CGO_ENABLED=0`; go-sdl3 is pure Go and embeds SDL for every platform, so all three cross-compile from Linux. This path uses **upstream** go-sdl3 — no `replace`. Copy Rush-Hour's verbatim and change the binary name.

14.6.2 `.github/workflows/pages.yml` (browser build — optional)

Builds the WebAssembly bundle and deploys it to GitHub Pages. Two things make it work (copy Rush-Hour's file and change nothing but the repo name in comments):

1. It checks out the **`chrplr/go-sdl3-wasm`** fork (branch **`wasm-render-fixes`**), which supplies the js/wasm SDL bindings and the `wasmsdl` bundler (ships `sdl.js` + `sdl.wasm`).
2. It runs `go mod edit -replace github.com/Zyko0/go-sdl3=../go-sdl3-wasm` **in CI only** (never committed), so the browser build uses the fork while native/release builds keep using upstream go-sdl3.

   Gotcha — branch ref, not pinned commit. `pages.yml` tracks the fork's *branch HEAD*, whereas goxpyriment's own `go.mod` pins a specific fork pseudo-version. A fix that lives only in goxpyriment's vendored copy (or an uncommitted fork edit) will **not** reach your repo's browser build until it is committed and pushed to `wasm-render-fixes`. This is the trap that broke Rush-Hour's first deploy.

14.7 Step 5 — Create the GitHub repo and enable Pages

```
cd ~/00_git/<ExampleName>
git init -b main && git add -A && git commit -m "Initial commit"

gh repo create chrplr/<ExampleName> --public --source=. --remote=origin
 ↪  --push

# Browser build only: enable Pages with GitHub Actions as the source
gh api -X POST repos/chrplr/<ExampleName>/pages -f build_type=workflow
```

The first push runs both workflows. The Pages URL is `https://chrplr.github.io/<ExampleName>/`.

14.8 Step 6 — Cut a release

Binaries are published only on a version tag:

```
git tag v0.1.0 && git push origin v0.1.0   # → release.yml builds +
 ↪  publishes the Release
```

Native binaries are independent of the browser build — tag whenever the source is ready.

14.9 Updating the pinned goxpyriment later

When a newer goxpyriment fixes something you need, re-pin and redeploy:

```
go get github.com/chrplr/goxpyriment@vX.Y.Z && go mod tidy
git commit -am "Bump goxpyriment to vX.Y.Z" && git push   # redeploys Pages
```

> **Gotcha — fresh tags and the checksum DB.** Right after you push a goxpyriment tag, `sum.golang.org` may not have indexed it yet and `go get` fails with a `500`. Fetch straight from the tag once with `GOPROXY=direct GOSUMDB=off go get github.com/chrplr/goxpyriment@vX.Y.Z`; the correct hashes land in your `go.sum`, so CI (which reads `go.sum`) is fine.

14.10 Verifying a browser build

Serve the bundle and open it in a real, focused, foreground tab:

```
# from the goxpyriment repo, before you split it out:
make wasm-<ExampleName>-serve      # http://localhost:8080/?s=1

# or the built standalone repo's Pages URL after deploy
```

Sanity checks: the experiment renders and responds; a normal end triggers **one** `.csv` + `-info.txt` download (not one per block); on a crash you see the error overlay and **no** download (see WASM.md). Check the browser console for panics — each names the file:line of any still-stubbed binding.

14.11 See also

- `examples/README.md` — `make share-NAME` (the zip-and-email path)
- WASM.md — how the browser build works, timing, and known gaps
- How_to_package_your_experiment.md — the GoReleaser alternative for native binaries

Chapter 15

Running goxpyriment experiments in a web browser (WASM)

Status (2026-07-13): a real experiment runs end-to-end in the browser. `examples/parity_decision` — instructions screen, fixation cross, 10 RT trials with keyboard responses, feedback logic, and CSV + info-file download — was run to completion in headless Chrome, driven by DevTools-protocol key events. The data files that Chrome downloaded are well-formed goxpyriment output with correct browser metadata (`video_driver: emscripten`). See What remains for the open items.

> An earlier version of this document described a manual Emscripten workflow (hand-built `SDL3.js`, an `EXPORTED_FUNCTIONS` list, a custom `index.html`). That workflow is **obsolete**, replaced by the `wasmsdl` bundler described below. A condensed record of the old recipe is kept in the appendix as a reference for rebuilding the SDL wasm blob.

15.1 Architecture

Two WASM runtimes cooperate in the same page:

1. **`sdl.wasm`** — SDL3 + SDL3_ttf + SDL3_image + SDL3_mixer (with a Web Audio backend) compiled by Emscripten into a single module, loaded by `sdl.js`.
2. **`main.wasm`** — the Go experiment binary compiled with `GOOS=js GOARCH=wasm`, loaded by Go's `wasm_exec.js`.

The Go side talks to the Emscripten side through `go-sdl3`'s js bindings (`syscall/js` calls into `Module._SDL_*`), sharing one Emscripten heap.

15.1.1 Where the pieces live

Piece	Location
go-sdl3 fork with the js/wasm target	github.com/chrplr/go-sdl3-wasm, branch **wasm-render-fixes** (local clone: ~/00_git/go-sdl3-wasm). The module path is unchanged (github.com/Zyko0/go-sdl3), and goxpyriment's go.mod has a replace pointing at a pinned pseudo-version of the fork; vendor/ is kept in sync with GOWORK=off go mod vendor
Prebuilt sdl.js / sdl.wasm + bundler	cmd/wasmsdl in the fork — embeds the blobs and an index.html, and builds/serves a complete browser bundle. Rebuild recipe: .docker/emscripten-build/Dockerfile in the fork
goxpyriment js platform code	control/platform_js.go (URL-parameter flags, no dialog, audio device open), apparatus/screen_newscreen_js.go (canvas window), apparatus/screen_present_js.go (RAF-synced flips), results/output_file_wasm.go (CSV → browser download) — all build tag js
Export-list generator	cmd/gen-wasm-exports — scans go-sdl3's js bindings + goxpyriment's own calls **as compiled for GOOS=js** (files excluded by build constraints, and triggers/, are not counted), emits wasm/exported_functions.json for emcc -sEXPORTED_FUNCTIONS=@…, and **lists the go-sdl3 calls whose js bindings are still panic-stubs** (the remaining-work list)

15.1.2 The go-sdl3 replace — what dependents need to know

goxpyriment's go.mod pins the fork with a line of the form

```
replace github.com/Zyko0/go-sdl3 => github.com/chrplr/go-sdl3-wasm
 ↪  v0.1.2-0.<timestamp>-<hash>
```

(the authoritative, current pseudo-version is the one in go.mod at the repo root — it is bumped whenever the fork changes).

Go applies replace directives **only in the main module**, so a project that imports goxpyriment as a dependency does *not* inherit this line. Such a project resolves upstream Zyko0/go-sdl3 — fine on desktop (behaviour is unchanged there), but the browser build will hit panic("not implemented on js") stubs. **To build your own experiment for the browser, copy the current replace line from goxpyriment's go.mod into your project's go.mod.** (It works because the fork keeps the upstream module

path.)

When developing the fork itself, point the replace at the local clone instead (`go mod edit -replace github.com/Zyko0/go-sdl3=/home/cp983411/00_git/go-sdl3-wasm`, then `GOWORK=off go mod vendor`), and restore the pinned GitHub pseudo-version before committing (`git log -1 --format=%cd-%h --date=format:%Y%m%d%H%M%S` in the fork gives the timestamp/hash for a fresh pseudo-version; the base tag is `v0.1.1`, so the form is `v0.1.2-0.<timestamp>-<12-char-hash>`).

15.2 Building and serving an example

From the repo root:

```
# Build a self-contained bundle (index.html + sdl.js + sdl.wasm + main.wasm
# + wasm_exec.js) into _build/wasm/parity_decision/
make wasm-parity_decision

# Or build + serve on http://localhost:8080 in one step, then open
# http://localhost:8080/?s=1
make wasm-parity_decision-serve
```

The targets run the `wasmsdl` bundler out of the pinned go-sdl3 fork (`go run github.com/Zyko0/go-sdl3/cmd/wasmsdl …`), so they need no local fork clone and no Emscripten install. `wasmsdl serve` also sends the COOP/COEP headers that unlock the high-resolution browser clock (see “Timing in the browser”).

WASM cannot be loaded from `file://` URLs; for the `build` output any static server works (`python3 -m http.server`), but plain servers don’t send the COOP/COEP headers — timestamps then tick at ~100 µs instead of ~5 µs.

15.2.1 Session settings via URL parameters

There is no command line in a browser. On `GOOS=js`, `control.NewExperimentFromFlags` synthesizes `os.Args` from the page URL’s query string, so the standard flags — and any experiment-specific flags the program registered — work unchanged:

`http://localhost:8080/?s=3&w` equivalent to: `-s 3 -w`

Keys that don’t match a registered flag are ignored with a console note. The participant-info dialog never opens in the browser (it needs its own SDL window); when `s` is absent the subject ID defaults to 0, exactly like `-headless`.

15.2.2 Headless verification (no display needed)

```
cd /tmp/hello_bundle && python3 -m http.server 8123 &
google-chrome --headless=new --use-gl=swiftshader --no-sandbox \
  --window-size=1100,850 --virtual-time-budget=6000 \
  --screenshot=/tmp/hello.png --enable-logging=stderr \
  "http://localhost:8123/?s=1" 2>&1 | grep "INFO:CONSOLE"
```

Go panics (with full stack traces) appear as `INFO:CONSOLE` lines; the screenshot shows what rendered. Each remaining stub panics with its binding's file:line, which tells you exactly what to un-stub next.

15.3 What works today (verified 2026-07-13)

- `GOOS=js GOARCH=wasm go build` for the library (all packages except `triggers/`, which needs a serial port and stays desktop-only) and examples.
- `wasmsdl build` bundles a goxpyriment example; the page boots both WASM modules.
- Full `control.Experiment` initialization: SDL + TTF init, canvas window + renderer, embedded font via the in-memory IOStream path (`IOFromDynamicMem`/`Write`/`Seek`), keyboard/mouse wiring, system- and display-info gathering, data-file creation.
- Text rendering (`stimuli.TextBox`/`TextLine` with wrap alignment and string metrics), fixation cross, `SetLogicalSize` letterboxing.
- The full trial machinery inside `exp.Run`: `ShowInstructions`, `Blank`/`ShowTimed`/`Wait`, `ShowAndGetRT` with hardware-timestamp RTs, `exp.Data.Add`, ESC/quit handling via `pumpFrame` (`HasEvent`/`GetKeyboardState`).
- Keyboard events reach SDL from the canvas; RTs come back as plausible millisecond values (timestamp *granularity* not yet measured — see below).
- `results` output: at experiment end the browser downloads the `.csv` and `-info.txt` files (verified: files land in the download directory intact).
- Audio playback: the buzzer/feedback sounds, `PlaySync`/`PlayAsync`, and `Tone` all work (see "Audio in the browser" below); confirmed by ear in an interactive session, not just headlessly.

15.3.1 The key design points

- **Blocking experiment code works in the browser — but only on the main goroutine.** Go's wasm scheduler parks a blocked main goroutine and returns to the JS event loop, so DOM events fire and SDL sees them. Code that blocks inside a `js.FuncOf` callback instead wedges the tab (the scheduler spins in `findRunnable`). Consequently `sdl.RunLoop` on js paces frames with `requestAnimationFrame` but runs the experiment logic on the goroutine that called `Run`; the RAF callback only signals a channel. (`SDL_Delay` is a no-op on js — it would busy-wait; use `time.Sleep`.)
- **Assets must be embedded.** There is no filesystem; `//go:embed` + `FontFromMemory` / `sdl.IOFromBytes` is the portable path (and was already the recommended pattern on desktop). Path-based loaders fail on js.

15.4 What remains for a complete port

All five phases of the roadmap are **done**: branch consolidation and vendor/fork unification; IOStream + info bindings, URL-parameter setup, demo_hello_world rendering; `parity_decision` running end-to-end (keyboard trials, RTs, CSV download) under the redesigned main-goroutine `RunLoop`; timestamp granularity measured and fixed; flips

aligned to `requestAnimationFrame` with measured 60.00 Hz pacing and zero dropped frames; audio playback; and packaging (`make wasm-NAME` / `wasm-NAME-serve` targets, obsolete artifacts removed, `GOOS=js` builds in CI via `.github/workflows/go-build.yml`).

Open decision: PR the fork's js/wasm work upstream to `Zyko0/go-sdl3`, or keep maintaining the fork.

Known gaps:

- Fullscreen and multi-monitor are meaningless in a canvas; the js `NewScreen` always opens 1024×768 (or the requested size). A "resize canvas to viewport" option would be nicer for participants.
- Less-common APIs are still stubbed (gamepad/joystick enumeration, audio recording, `WaitEvent`/`WaitEventTimeout`, `SetRenderLogicalPresentation`, video playback). They panic with a clear console message when hit.
- `GetParticipantInfo` (the SDL dialog) cannot run on js; experiments that call it directly need an HTML-form alternative or URL parameters.

15.5 Audio in the browser

Audio playback works (verified 2026-07-13): `platformInitAudio` on js opens the default playback device exactly like on desktop, and the whole goxpyriment audio API — `exp.Audio.PlayBuzzer/PlayCorrect/PlaySync/PlayAsync`, `Sound.PreloadDevice/Play/Wait`, `Tone` — runs on SDL's Emscripten Web Audio backend. Verified in `parity_decision`: the device opens as 48 kHz stereo F32LE with a 2048-frame buffer, the AudioContext reaches `running`, and the buzzer feedback plays on incorrect trials.

Browser specifics:

- **First sound needs a user gesture.** Browsers create the AudioContext suspended; SDL's Emscripten backend resumes it automatically on the first click or keypress. Any experiment that starts with a "press SPACE" screen satisfies this without extra code. Sounds triggered *before* any interaction stay silent until the first one.
- **Latency is higher than desktop.** The Emscripten device reports a 2048-frame buffer at 48 kHz ≈ 43 ms; `SetAudioSampleFrames` (an SDL hint) is not honoured the same way by the Web Audio backend. Treat audio onset timing as approximate in the browser; `PlaySyncedWithFlip`'s desktop guarantees do not transfer.
- If the device cannot be opened, the experiment continues with a silent no-op `AudioManager` instead of crashing (browser audio support varies).

15.6 Timing in the browser

15.6.1 Clock / timestamp resolution (measured 2026-07-13, headless Chrome)

SDL timestamps in the browser (`sdl.TicksNS` and input-event timestamps, i.e. what `ShowAndGetRT` / `GetKeyEventTS` use) tick at:

Configuration	Resolution
plain page	~100 µs
cross-origin isolated page (COOP/COEP headers)	~5 µs

`wasmsdl serve` sends the COOP/COEP headers, so served experiments get the ~5 µs clock; any other hosting needs `Cross-Origin-Opener-Policy: same-origin` and `Cross-Origin-Embedder-Policy: require-corp` to match.

These numbers required a fix in the fork (commit `5eb7455`): out of the box, **every SDL timestamp was quantized to 1 ms**, because SDL3's `SDL_GetPerformanceCounter` probes `CLOCK_MONOTONIC_RAW`, Emscripten's WASI shim rejects that clock id, and SDL silently falls back to `gettimeofday = Date.now()`. The fork's `sdl.js` asset patches `emscripten_date_now` to `performance.timeOrigin + performance.now()`, and the Dockerfile re-applies the patch on rebuild.

Note: Go's `time.Now()` on js still has 1 ms resolution (the wall clock is `Date.now()`). This is one more reason for the existing rule: RTs come from the SDL event clock, never wall-clock deltas.

15.6.2 Frame pacing (measured 2026-07-13, headless Chrome)

On js, `Screen.Update()` (and therefore `Flip`/`FlipTS`/`Show`) presents to the canvas and then parks until the browser's next `requestAnimationFrame` tick — the browser's VSYNC equivalent (implemented in `apparatus/screen_present_js.go` on top of `sdl.WaitAnimationFrame` in the fork). This is required for correctness, not just pacing: canvas updates are only composited when the page yields, and the desktop busy-wait would freeze the tab. A 250 ms fallback tick prevents deadlock in background tabs (where browsers throttle RAF) — but run experiments in a **focused, foreground tab**.

Measured flip-loop intervals (299 frames, alternating full-screen colors):

Loop	mean	SD	min–max	dropped frames
`Update` loop	16.666 ms (60.00 Hz)	0.12 ms	16.1–17.3 ms	0
`Flip` loop	16.666 ms (60.00 Hz)	0.11 ms	16.3–17.1 ms	0

Caveat: headless Chrome ticks a virtual 60 Hz compositor; on real hardware the rate follows the display (e.g. 120 Hz laptop panels) and jitter depends on system load. The hardware-locked VSYNC timing available on desktop (DRM ioctl, CoreVideo) is not available in a browser, and there is no photodiode validation of actual pixel onset yet. Experiments that require sub-millisecond stimulus onset precision (rapid RSVP, subliminal priming) should be run natively. Cognitive tasks (choice RT, memory, attention) work fine.

15.7 Notes for developers

- **Un-stubbing a go-sdl3 js binding** is usually just deleting the `panic("not implemented on js")` first line, *but audit the marshalling*: the fork's `CLAUDE.md` documents the rules (opaque handles vs input structs vs out-params, `float32` passed directly, `size_t` as i32 not BigInt, u64 returns via `internal.GetInt64`, struct returns via `internal.NewObject`, `HEAPU8` fetched fresh).
- **`Module._SDL_Foo is not a function`** at runtime means the C symbol is missing from the `sdl.wasm` export list — regenerate the list with `go run ./cmd/gen-wasm-exports/` and relink the blob (`.docker/emscripten-build/Dockerfile` in the fork).
- **After changing the fork**, re-sync goxpyriment's vendor tree: `GOWORK=off go mod vendor` at the repo root.

15.8 Appendix — historical manual Emscripten recipe

The early-2026 attempt built the SDL blob by hand. Kept here (condensed) as a reference; the current blob is built by `.docker/emscripten-build/Dockerfile` in the fork and additionally bundles SDL3_image and SDL3_mixer.

```
# Emscripten SDK
git clone https://github.com/emscripten-core/emsdk.git && cd emsdk
./emsdk install latest && ./emsdk activate latest && source ./emsdk_env.sh

# SDL3 (static)
git clone https://github.com/libsdl-org/SDL.git -b release-3.2.x SDL3-src
emcmake cmake -S SDL3-src -B SDL3-wasm -DCMAKE_BUILD_TYPE=Release \
  -DSDL_SHARED=OFF -DSDL_STATIC=ON -DSDL_TESTS=OFF -DSDL_EXAMPLES=OFF
emmake make -C SDL3-wasm -j$(nproc)
cmake --install SDL3-wasm --prefix "$PREFIX"

# FreeType, then SDL3_ttf against the same prefix
emcmake cmake -S freetype-src -B freetype-wasm -DCMAKE_BUILD_TYPE=Release
emmake make -C freetype-wasm -j$(nproc) && cmake --install freetype-wasm
 ↪  --prefix "$PREFIX"
emcmake cmake -S SDL3_ttf-src -B SDL3_ttf-wasm -DCMAKE_BUILD_TYPE=Release \
  -DCMAKE_PREFIX_PATH="$PREFIX" -DSDL3TTF_HARFBUZZ=OFF -DSDL3TTF_
   ↪  SAMPLES=OFF
emmake make -C SDL3_ttf-wasm -j$(nproc) && cmake --install SDL3_ttf-wasm
 ↪  --prefix "$PREFIX"

# Link one module; export list generated by cmd/gen-wasm-exports
emcc -o sdl.js "$PREFIX"/lib/libSDL3.a "$PREFIX"/lib/libSDL3_ttf.a
 ↪  "$PREFIX"/lib/libfreetype.a \
  -s MODULARIZE=0 -s ENVIRONMENT=web -s ALLOW_TABLE_GROWTH=1 \
  -s EXPORTED_RUNTIME_
   ↪  METHODS='["HEAPU8","stackAlloc","stackSave","stackRestore","getValue
   ↪  ","setValue","stringToUTF8OnStack","UTF8ToString","addFunction"]' \
```

```
  -s EXPORTED_FUNCTIONS=@wasm/exported_functions.json
```

Pitfalls recorded at the time: use `-DCMAKE_PREFIX_PATH` (where `SDL3Config.cmake` was installed), not `-DSDL3_DIR` at the raw build dir; SDL3_ttf pins a minimum SDL3 version — if they mismatch, rebuild SDL3 from the tag SDL3_ttf requires; all libraries must be linked into **one** module because the Go bindings share a single Emscripten heap; force-link static archives (`-Wl,--whole-archive`) or `EXPORT_ALL` drops unreferenced symbols.

Chapter 16

Fullscreen + HIGH_PIXEL_DENSITY issue

16.1 Symptom

On **macOS** (Retina), running an experiment in fullscreen mode shows a gray screen after the initial setup UI. Windowed mode (`-w`) works correctly. The issue was reported for the `retinotopy` example but likely affects any experiment that uses hardcoded pixel dimensions.

16.2 Root cause analysis

16.2.1 The asymmetry in `apparatus/NewScreen`

In `apparatus/screen.go`, the fullscreen and windowed paths differ in one critical flag:

Fullscreen path:

```
window, err := sdl.CreateWindow(title, 0, 0, sdl.WINDOW_HIGH_PIXEL_DENSITY)
// ...
w, h, _ := window.SizeInPixels()
return &Screen{..., Width: int(w), Height: int(h)}
```

Windowed path:

```
window, renderer, err := sdl.CreateWindowAndRenderer(title, width, height,
 ↪  sdl.WINDOW_HIDDEN)
// no WINDOW_HIGH_PIXEL_DENSITY
return &Screen{..., Width: width, Height: height}
```

16.2.2 What `SDL_WINDOW_HIGH_PIXEL_DENSITY` does

Without it, SDL3 follows the OS logical coordinate space: - On a macOS Retina display at 2560×1600 logical points, SDL gives a 2560×1600 renderer — the OS compositor

silently doubles to 5120×3200 physical pixels. - On standard Linux (pixel density = 1), there is no difference.

With it, SDL3 bypasses logical scaling and gives raw physical pixels: - Same Retina display: you get a 5120×3200 renderer directly. - All sizing and coordinate code must work in physical pixels.

16.2.3 Consequence

Mode	Flag present	`Screen.Width/Height`	`CenterToSDL`
Windowed	No	logical pixels (e.g. 1024×768)	correct
Fullscreen	Yes	physical pixels (e.g. 5120×3200 on Retina)	correct

`CenterToSDL` uses `renderer.RenderOutputSize()`, so it is always internally consistent. The problem is **any code that uses hardcoded pixel counts**.

In `retinotopy/main.go`:

```
const WindowWidth  = 768
const WindowHeight = 768

// Texture created at hardcoded logical size:
tex, err := r.Exp.Screen.Renderer.CreateTexture(
    apparatus.PIXELFORMAT_RGBA32,
    apparatus.TEXTUREACCESS_STREAMING,
    WindowWidth, WindowHeight,   // ← wrong in fullscreen HiDPI
)
r.PixelBuffer = make([]byte, WindowWidth*WindowHeight*4)
// ...
r.CombinedTexture.Update(nil, r.PixelBuffer, WindowWidth*4)
```

On a Retina Mac in fullscreen the renderer expects physical pixels (e.g. 5120×3200) but the texture and pixel buffer are 768×768. The stimulus either renders at a tiny fraction of the screen or lands outside the viewport entirely.

16.3 Why the Pi issue was different

The Pi gray screen was caused by a different bug: `binsdl.Load()` tried to extract and load an x86-64 `libSDL3.so` on an ARM64 system, which fails immediately with:

```
binsdl: couldn't sdl.LoadLibrary: /tmp/.../libSDL3.so.0: cannot open shared
 ↪  object file
```

This was fixed by adding build-tag-selected loader files: - `control/sdlload_embedded.go` (`//go:build !linux || !arm64`) — uses `binsdl/binttf/binimg.Load()` - `control/sdlload_system.go` (`//go:build linux && arm64`) — calls `sdl.LoadLibrary(sdl.Path())` against the system-installed SDL3

After that fix the Pi works correctly in both windowed and fullscreen — confirming the Pi had no HiDPI issue (pixel density = 1 on a standard monitor).

16.4 The Mac issue — implemented fix

The Mac fullscreen gray screen has been addressed with a framework-level fix in `apparatus/screen.go`.

16.4.1 Approach — `SetLogicalPresentation` (implemented)

`WINDOW_HIGH_PIXEL_DENSITY` is **kept** (it is needed to get correct centering on Linux). After the renderer is created, the framework immediately queries the logical (OS) pixel dimensions via `window.Size()` and calls:

```
renderer.SetLogicalPresentation(logW, logH, sdl.LOGICAL_PRESENTATION_
↪  STRETCH)
```

SDL3 then maps all drawing commands from logical coordinates (e.g. 2560×1600 on a Retina display) to the full physical resolution (e.g. 5120×3200) transparently. The `Screen` struct stores the logical `Width`/`Height`, which is what experiment code should use for coordinate math. `CenterToSDL` and `MousePosition` already use the `LogicalSize` field, so no changes are needed in experiment code.

Platform	`window.Size()`	`window.SizeInPixels()`	Renderer logical space
Linux standard (Pi, desktop)	1920×1080	1920×1080	1920×1080 — no change
macOS Retina 2560×1600	2560×1600	5120×3200	2560×1600 — SDL upscales

Because `Screen.Width/Height` are now always in logical pixels and `SetLogicalPresentation` is active, experiments using hardcoded pixel values (like the retinotopy texture at 768×768) work correctly on all platforms — SDL scales the output to fill the screen without the experiment needing to know the physical pixel count.

Chapter 17

frame-syncing

To ensure your software has perfect pacing on as many systems as possible, you should follow this hierarchy of detection in your Vulkan renderer: * Priority Method Hardware/OS Support * Primary VK_EXT_present_timing Best for 2026+ systems (NVIDIA, AMD, Intel). * Secondary VK_KHR_present_waitGood for simpler “wait until displayed” logic; widely supported. * Legacy VK_GOOGLE_display_timingKeep as a fallback for older Android-based or mobile Linux setups. * System Wayland wp_ presentationIf running natively on Wayland, use the presentation-time protocol.

Chapter 18

Minimising trigger-to-stimulus jitter (EEG / MEG)

In EEG and MEG the TTL trigger is the timestamp. Everything downstream — epoching, averaging, latency measurement — is referred to it. So the question is not how quickly the stimulus appears after the trigger, but **how *consistently* it does**.

That distinction decides what you have to fix:

- **A constant offset is harmless.** If the screen always lights 22 ms after the trigger, subtract 22 ms in the analysis and nothing is lost.
- **Jitter is not recoverable.** A trigger-to-stimulus delay that varies from trial to trial smears every average by that amount, and no analysis can undo it because nothing recorded says which trial was late.

Everything below is aimed at the second quantity.

18.1 The short version

0. **Get off the compositor.** Bare Xorg with a plain non-compositing window manager measured **sixteen times** better than the same binary in a Wayland session — 0.083 ms against 1.34 — and a bare KMS/DRM console 0.113 ms. Nothing else here comes close.
1. **Never sleep for an inter-trial interval. Count frames.**
2. **Fire the trigger synchronously, on the flip thread, immediately after the flip returns** — not from a goroutine.
3. **Discard the first several trials.** They are measurably different.
4. **Run at real-time priority.** Measured worth: it halves the jitter, 2.34 ms to 1.32 ms over five minutes. Read the warning about busy-waits in Setting priority under Linux.
5. **One TTL line per event type.** A multi-bit code on a DLP-IO8 takes ~610 µs to settle and will be sampled mid-transition.
6. **Measure it on your own rig.** The numbers below are one host and two panels; yours will differ.

Done properly, at real-time priority and **without a compositor**, this gives **sd 0.083 ms over a five-minute run** on the hardware tested — an order of magnitude better than the best figure in the published cross-package comparison. In a Wayland session the same binary gives 1.34 ms, and at normal priority 2.34 ms. None of it is ever the display, which put every one of 2951 stimuli measured on an exact frame boundary. See “The jitter is in the timestamp, not in the display”.

18.2 Where the time actually goes

Measured on a Linux 7.0 laptop, 60 Hz panel, DLP-IO8 trigger and a photodiode on the screen, both recorded in one acquisition on an Analog Discovery 3 so the instrument’s clock cancels:

stage	contribution
`ShowTS` returns → TTL write issued	**25 µs** (p95 33 µs)
DLP-IO8 host write → edge on the wire	tens of µs (bounded ≤ 0.79 ms)
flip → photons on the panel	**20–40 ms**, and this is where all the variance is
panel black→white transition (10–90%)	**5.5–6.5 ms**

The software path is three orders of magnitude below the display. Nothing in your experiment code, the Go runtime, or the trigger box is what limits you.

18.3 Why sleeping is the mistake

An inter-trial interval written as `time.Sleep(300 * time.Millisecond)` is not a whole number of frames: at 60 Hz it is 18.01 of them. Each trial therefore starts at a slightly different point in the frame cycle, and the stimulus flip lands at a drifting phase. Measured over 35 trials with a sleep-based ITI:

	sleep ITI	frame-counted ITI
trigger→light spread	**14.1 ms**	**3.4 ms**
sd	2.80 ms	1.13 ms
host clock vs display clock	drifts 0.27 ms per trial	—

(Both from 35-trial runs. Over five minutes the frame-counted figure is worse — sd 2.3 ms — for the reason in “Measure over a realistic run length” below. The comparison between the two ITI styles stands; the absolute value does not.)

The failure mode is worth recognising because it does **not** look like noise. The trigger-to-light delay walks smoothly across a run — 27 ms at the start, 41 ms in the middle, back to 37 ms — as the host’s trial period slides past the display’s frame period. Two things follow:

- **A short pilot will not reveal it.** Ten trials sample a small arc of the walk and look tight.
- **Within-channel checks will not reveal it either.** Trigger-to-trigger and flash-to-flash intervals are both perfectly stable while their *relative* phase drifts. If your timing check measures only interval regularity, it will pass.

Count frames instead:

```
// Inter-trial interval as 18 blank frames -- exactly 300 ms at 60 Hz, and
// exactly a whole number of frames at any refresh rate.
for f := 0; f < 18; f++ {
    exp.Screen.ClearAndUpdate()   // Update blocks on VSYNC
}
```

This keeps the display pipeline continuously fed, so the stimulus flip stays vsync-locked. `tests/Timing-Tests` has always done this; it is why its numbers have always been good.

18.4 Fire the trigger synchronously

```
onset, err := exp.ShowTS(stim)   // presents and timestamps the flip
if err != nil { return err }
trig.SetHigh(line)               // next statement, same thread, nothing
 ↪  between
```

Measured this way the gap between the flip timestamp and the trigger write is **25 µs**. Launching the same call as `go triggers.FireTrigger(...)` hands it to the Go scheduler instead; at normal priority under CPU load that path has been measured at **+0.73 ms with about 1 ms of spread**. That is forty times the synchronous figure, for no benefit at the scale that matters here.

`tests/Timing-Tests` fires from a goroutine deliberately, to keep its own measurement loop unblocked. Do not copy that pattern into an experiment.

18.5 What is left, and why it is quantised

After the two fixes above, the residual is not a smear — it is **whole frames**. Measured over 34 consecutive intervals in a frame-counted run, every single one was an exact multiple of the frame period: mostly 20 frames, occasionally 21. None were in between.

So the display is genuinely vsync-locked, and the remaining variability is a trial occasionally slipping one frame later. At 60 Hz that is 16.7 ms when it happens; at 120 Hz, 8.3 ms.

This matters because a quantised error is **detectable**, and a smeared one is not. `ShowTS` returns the flip timestamp, so an experiment can log it and find the slips afterwards:

```
onset, _ := exp.ShowTS(stim)
if prev != 0 {
    frames := float64(onset-prev) / float64(frameDur)
    if math.Abs(frames-math.Round(frames)) > 0.25 || math.Round(frames) !=
     ↪  expected {
        // this trial's stimulus was a frame late: mark it, or drop it
    }
}
prev = onset
```

For EEG and MEG this is the difference between an unknown error and a known one. Log the flip timestamp on every trial and put it in the data file.

18.6 Measure over a realistic run length, not a pilot

A short pilot can tell you the timing is excellent, and it will be wrong.: Measure over the length of a real block (5~10min).

18.7 The jitter is in the timestamp, not in the display

This is the most useful thing on this page, and it took a five-minute run to see. Recording the trigger and the photodiode on one instrument gives three event trains, and asking whether each one falls on a whole number of frame periods separates them completely:

train, intervals between consecutive trials	off a whole frame by > 1 ms
photons on the panel	**0 of 597 (0.0 %)**
TTL edge on the wire	~5 % of trials, up to 6.8 ms
the host's own `ShowTS` timestamp	**82 of 638 (12.9 %)**, up to 6.7 ms

The panel is exact. Photodiode intervals sit on a whole number of frames with a median error of **5 µs** and a worst case of 169 µs, across 553 intervals of exactly 30 frames, 37 of 31 and 7 of 32. The implied frame period, 16.6557 ms, is **60.0395 Hz** against the 60.0400 Hz SDL reports for the panel — agreement to 8 ppm, from an instrument that knows nothing about the display.

What wanders is the moment the software believes the flip happened. And it wanders in one direction: across the 594 intervals that were exactly 30 frames, the host timestamp was never early (minimum −0.01 ms) and was late by as much as **+6.12 ms**. `Update()` returned, `ShowTS` stamped the clock, and the photons had already been on their way for several milliseconds.

The trigger inherits that error, because it is fired off the flip's return. So the 2.3 ms of trigger-to-stimulus jitter is not the display and not the trigger box:

The stimulus appears on an exact frame boundary every time. The TTL that is supposed to mark it is occasionally several milliseconds late.

Two things follow, and they point in opposite directions from the usual advice:

- **Do not treat `ShowTS`'s return as a photon timestamp.** It is right to within microseconds most of the time and several milliseconds wrong about one trial in eight, with no indication of which.
- **The error is bounded and reconstructible.** Because the photons are on an exact grid, the true onsets of a whole block can be recovered by fitting that grid to the flip timestamps, which is not possible for genuinely random noise.

18.7.1 The display stack is worth sixteen times more than anything else

Three runs differing only in the display stack — same binary, same `chrt -f 50`, same synchronous trigger, same night:

	mean	**sd**	full range, ~590 trials	flips > 1 ms off the frame grid
Wayland session	21.75 ms	1.344 ms	18.83–36.74	7.1 %
KMS/DRM, no display server	18.91 ms	**0.113 ms**	18.58–19.13	**0 of 590**
Bare Xorg + openbox, exclusive fullscreen	35.74 ms	**0.083 ms**	**35.52–35.95**	**0 of 580**

Sixteen times better, and the mechanism is gone rather than reduced. Off a compositor, not one flip in ~590 lands more than 1 ms off the frame grid; in a Wayland session one trial in fifteen does. An entire five-minute Xorg run fits inside a 0.43 ms band.

This is the single largest effect on this page. Real-time priority is worth 1.8×; the display stack is worth 16×.

Xorg is the steadiest and the latest, which is not a contradiction — the two are different quantities. The gap is exactly one frame:

```
Xorg − KMS/DRM = 16.826 ms = 1.010 frames
```

One more buffer in the pipeline, dead constant. So goxpyriment emits its trigger one frame before the photons on bare hardware and two frames before them under X, and neither is the compositor's doing — what the compositor added was the *variance* around it. That is consistent with `Update()` returning when a page flip is queued at one vblank while the content becomes visible at a later one.

For EEG and MEG the distinction is the whole point: **a constant 19 or 36 ms is subtracted in analysis and costs nothing; 1.3 ms of scatter around it cannot be.** Measure your offset once per rig, record it, subtract it.

A worthwhile aside on instruments. Across the Wayland and KMS/DRM runs the BBTK reproduced the AD3's sd to three significant figures. On the Xorg run it does not — 1.55 ms against 0.083 — because seven trials read 24–28 ms where the AD3 saw nothing below 35.52 in any of 581. Its optical threshold was never calibrated against this panel, and a threshold high on a 5.5 ms ramp crosses erratically when the final luminance wobbles. Agreement between instruments is worth a great deal right up to the point where one of them is misconfigured, and that point is not announced.

18.7.2 Removing the compositor is worth twelve times more than anything else

The same protocol again, in a bare Linux console with `SDL_VIDEODRIVER=kmsdrm` and no display server at all:

five minutes each	Wayland, FIFO 50	**KMS/DRM console, FIFO 50**
trigger→light sd, AD3	1.344 ms	**0.113 ms**
trigger→light sd, BBTK	1.35 ms	**0.17 ms**
p05–p95	3.12 ms	**0.38 ms**
full range over 590 trials	18.8–36.7 ms	**18.58–19.13 ms**
largest trial-to-trial step	16.7 ms	**0.275 ms**
`ShowTS` > 1 ms off the frame grid	6.4 %	**0 of 640**

Twelve times better, and the mechanism is gone rather than reduced: not one flip in 640 lands more than 1 ms off the frame grid, where a Wayland session misses on one trial in fifteen. The whole five-minute run fits in a 0.55 ms band.

This is the single largest effect on this page. Real-time priority is worth 1.8×; the compositor is worth 12×.

It also means the earlier sections diagnosed the mechanism correctly and blamed the wrong layer for the offset. Removing the compositor removed the *variance* around the one-frame lag but not the lag itself, which only fell 2.8 ms, from 21.75 to 18.91. goxpyriment emits its trigger about one frame before the photons either way. That residual is consistent with `Update()` returning when the flip is queued at one vblank while the content appears at the next — 16.66 ms plus about 2 ms of scanout down to the patch.

For EEG and MEG the distinction is the whole point: a constant 19 ms is subtracted in analysis and costs nothing, and 1.3 ms of scatter around it cannot be.

See the mega-study comparison — on a bare console this beats every configuration in the published table; under Wayland it is near the bottom of it.

18.7.3 Real-time priority halves it — and that is measured, not assumed

A one-sided, several-millisecond lateness in returning from a vsync wait is what preemption looks like. The run above was at normal priority, so the same protocol was repeated changing nothing but `chrt -f 50`:

five minutes, same rig, same night	SCHED_OTHER	SCHED_FIFO 50
trigger→light sd, **AD3**	2.342 ms	**1.320 ms**
trigger→light sd, **BBTK v3**	2.33 ms	**1.32 ms**
p05–p95 spread (BBTK)	7.2 ms	**3.0 ms**
`ShowTS` timestamps > 1 ms off the frame grid	12.85 %	**6.43 %**
TTL edges > 1 ms off the frame grid	13.07 %	**6.28 %**
photons > 1 ms off the frame grid	**0.00 %**	**0.00 %**
trial-to-trial steps > 3 ms	30 of 597	**10 of 589**
mean delay	21.18 ms	20.96 ms

Real-time scheduling **halves the jitter** and leaves the mean where it was, which is the signature of removing a delay that only ever happened sometimes. Both instruments agree on the improved figure to three significant figures, as they did on the worse one.

The off-grid fractions for `ShowTS` and for the TTL fall together — 12.85 → 6.43 and 13.07 → 6.28 — which confirms the trigger is simply following the flip timestamp and contributes nothing of its own. And the photons stay exactly where they were: perfectly frame-locked in both conditions. Scheduling never touched the display, only the software's knowledge of it.

It is not a complete fix. 6.4 % of flips are still late, and 1.32 ms is not 1.32 µs. Whatever remains is not answered here.

! `tests/Timing-Tests` does not request real-time priority, so its own numbers — including the SCHED_OTHER column above — are at normal priority unless you launch it under `chrt`. It builds its experiment with `control.NewExperiment`, and the elevation lives in `control.NewExperimentFromFlags`, so nothing is attempted and nothing is logged. Experiments built the normal way, through `NewExperimentFromFlags`, do ask. Check `sched_policy` in your data file's header rather than assuming either way, and see Setting priority under Linux.

18.8 Discard the first trials — they are genuinely different

The first few trials after a run starts have a longer and more variable delay than everything that follows, and they dominate the summary statistics if left in. Same recording, the only difference being how many leading trials are excluded:

	all trials	discarding 10
trigger→light sd	1.127 ms	**0.384 ms**
spread	3.44 ms	**1.15 ms**

Three times the sd, from the first handful of trials. On the BBTK recording of the same test the raw sequence starts at 32.25 ms and settles to about 25 ms within four trials, so the transient is real and not an artifact of one instrument.

`tests/Timing-Tests` already discards ten cycles (`-warmup`). An experiment should do the equivalent: present some warm-up trials before the first one that counts, or mark the early trials in the data so the analysis can drop them.

18.9 Two instruments agree

Everything above was measured with an Analog Discovery 3. The same `Timing-Tests` run was also recorded with a Black Box ToolKit v3 — a different sensor, a different front end and its own clock — with both instruments on the same trigger line:

five-minute run, ~598 trials	AD3 (1 µs resolution)	BBTK v3 (250 µs)
trigger→light **sd**	**2.340 ms**	**2.327 ms**
spread	24.74 ms	24.75 ms
p05 – p95	17.1 – 24.5 ms	19.8 – 27.0 ms
mean	21.18 ms	23.84 ms

The two instruments agree on the variability to three decimal places while differing by 2.7 ms in the mean. That is exactly the expected pattern: the offset depends on where each instrument's threshold sits on the panel's 5.5 ms ramp, and is constant; the jitter does not depend on the threshold at all. It is also the strongest evidence available that the number is real, since the two share nothing but the signal.

Over a short window the BBTK looks worse than the AD3 (sd 0.79 against 0.38 ms across thirteen trials) because a 0.43 ms mean step is only 1.7 of its 250 µs quanta. Over a realistic run that resolution difference is irrelevant — the quantity being measured is ten times its resolution.

18.10 How this compares with other packages

Bridges et al. (2020) measured this same quantity — trigger pulse to pixels changing — for PsychoPy, PsychToolBox, Presentation, E-Prime, OpenSesame and Expyriment, on a 60 Hz panel. Their best is 0.18 ms and their worst is 4.82 ms.

goxpyriment measures **0.083 ms on bare Xorg** — the stack they actually ran, and twice as good as the best figure in their table — and **1.32 ms under Wayland**, worse

than thirteen of the fourteen. Same binary, same machine, same night. That column describes a display stack at least as much as it describes a package.

The lag runs the other way and belongs in the same breath: 35.7 ms on bare Xorg against 2.35–7.10 ms for every Linux and Windows package they measured. PsychToolBox's flip returns at scanout; goxpyriment's returns two frames early. Constant, so correctable, but real. See the mega-study comparison.

18.11 The floor you cannot code around

Frame quantisation. A stimulus can only appear when the panel scans it out. The only way to reduce this is a faster panel: 16.7 ms of quantisation at 60 Hz becomes 4.2 ms at 240 Hz.

Panel transition time. Measured 10–90% on two unrelated LCDs, one external 4K and one laptop panel: **6500 µs and 5571 µs**. A multi-millisecond black-to-white transition appears to be a property of the technology, not of one bad monitor. It is also why quoting an onset to better than a millisecond is not meaningful without saying which point on the ramp you mean.

Scanout position. The top of the panel lights nearly a frame before the bottom. Put the photodiode — and the stimulus that matters — near the top, and record where it was.

18.12 The honest assessment for EEG / MEG

With a frame-counted ITI, a synchronous trigger and warm-up trials discarded, the trigger-to-stimulus delay on the hardware tested has **sd 2.3 ms across a five-minute block**, on a mean of about 21 ms that is panel-specific. Two independent instruments agree on that figure.

Where that jitter lives matters more than its size. The panel puts the stimulus on an exact frame boundary on every one of 1188 trials measured; the software is late to notice on 6 % of them at real-time priority and 13 % at normal priority. The jitter is in the timestamp, not in the photons.

Whether this is good enough depends on the paradigm. For ERP components with latencies of tens of milliseconds, averaged over many trials, 1.3 ms of onset jitter is a small smear. For anything resolving fine temporal structure — early auditory components, phase measures, single-trial latency — measure it on your own rig before relying on it.

Unlike a genuinely noisy display, this one has somewhere to go. Real-time priority is worth a factor of two and is one flag. Beyond that, the frame grid is exact enough — 6 µs — to reconstruct the true onsets after the fact, which is possible precisely because the residual is not random.

The three ways to make it worse are all in this page and all avoidable: sleeping for the ITI costs a factor of twelve in spread, firing the trigger from a goroutine costs a

factor of forty in the host term, and keeping the warm-up trials costs a factor of three in sd. None of them announce themselves in the data.

The robust answer for both cases is the same one used in MEG labs generally: **record a photodiode alongside the TTL** and use the photodiode as the onset in analysis. That converts every term on this page — offset, quantisation, panel rise — into a measured per-trial quantity rather than an assumption.

18.13 Verify it on your own hardware

```
# Does this display block on VSYNC, and at what rate?
go run ./tests/test_vsync_blocking

# Do the flip timestamps drift against the display? (no photodiode needed)
go run ./tests/test_vblank_drift

# Trigger against a photodiode, one AD3 acquisition, per-trial gap logged
go run ./tests/test_photodiode_latency -s 1 -isi-frames 18 -diode all

# The same with the established harness
go run ./tests/Timing-Tests -test av -no-sound

# Two trigger devices against each other, on one schedule and one timebase
go run ./tests/test_triggers -device parallel:pin=1 -device dlpio8:pin=1
```

`tests/test_vblank_drift` is the one to run first, because it needs no hardware at all. It compares `FlipTS` against the kernel's own DRM vblank timestamps — an independent clock on the display, playing a photodiode's role minus the photons — and reports the drift in ppm. It cannot measure the *offset* between the flip and the photons (scanout position and panel rise are outside it), but it can tell you whether that offset is constant, which is the part no host-side statistic can otherwise reach. On a Precision 5490 its refresh estimate agreed with a BBTK photodiode to 1.3 ppm.

`tests/test_photodiode_latency` logs the flip timestamp, the trigger timestamp and the gap between them for every trial, so the host-side contribution is in the data rather than assumed. Its README works through the arithmetic of converting an instrument's trigger-to-light interval into the quantity you actually want.

`tests/test_triggers` answers a different question: *which* trigger device. The figures above come from one device at a time, and comparing runs made hours apart on different clocks is exactly the comparison this document warns about everywhere else. That test pulses two or more devices on **one absolute schedule** — a locked thread each, all waiting on the same deadline — so an oscilloscope reads the difference between their edges directly and its own clock cancels. It records when the host issued each write as well, which is what separates a device that is slow from a program that was late. Its `-sequential` mode measures the other thing worth knowing: what a blocking USB write costs the device queued behind it, which is what experiment code pulsing two boxes in a row actually does.

Chapter 19

goxpyriment against the timing mega-study

Bridges D, Pitiot A, MacAskill MR, Peirce JW (2020). *The timing mega-study: comparing a range of experiment generators, both lab-based and online.* PeerJ 8:e9414. DOI 10.7717/peerj.9414

Their Table 2 gives visual onset, visual duration, audio onset and audiovisual synchrony for PsychoPy, PsychToolBox, Presentation, E-Prime, OpenSesame and Expyriment across three operating systems. This page puts goxpyriment's measured numbers next to it, on the visual measures.

Short answer: the display stack decides this, not the package. In a Wayland session goxpyriment is worse than thirteen of their fourteen lab configurations. On bare Xorg — the stack they actually measured — it is better than all fourteen, by a factor of two, at sd 0.083 ms. The same binary, the same machine, the same night.

The protocol mapping — which flags reproduce their trial structure — lives in `paper/megastudy/megastudy_timing_tests.md`, which is not tracked because that directory holds the paper PDF.

See also Minimising trigger-to-stimulus jitter, which is where the mechanism is worked out in detail.

One machine, one panel, three five-minute runs. Read the caveats at the end before quoting any of this: the paper's own authors decline to generalise across machines, and so should we.

19.1 Conditions

Built-in 2560×1600 laptop panel at **60.0400 Hz** (SDL) / **60.0395 Hz** (measured from the photodiode train — 8 ppm apart), Mesa, Intel Arc (MTL). 12 frames on, 18 off, 640 cycles, fullscreen, photodiode on the top-left patch. An Analog Discovery 3 at 200 kS/s recorded the TTL and the photodiode in **one acquisition**, so the instrument's clock

cancels; a Black Box ToolKit v3 was on the same trigger line throughout. `onset` is the photodiode's 10 % crossing.

Five runs. A→C vary the software while holding the stack fixed; C, D and E vary only the stack.

	harness / stack	n	**sd, AD3**	mean, AD3	steps > 1 ms	TTL > 1 ms off the frame grid
A	Wayland, normal priority, goroutine trigger	598	2.342 ms	21.18	82	13.07 %
B	Wayland, FIFO 50, goroutine	590	1.320 ms	20.96	37	6.28 %
C	Wayland, FIFO 50, syn-chronous	590	1.344 ms	21.75	43	7.13 %
D	**KMS/DRM**, no display server	591	**0.113 ms**	18.91	**0**	**0.00 %**
E	**Bare Xorg + openbox**, exclusive fullscreen	581	**0.083 ms**	35.74	**0**	**0.00 %**

Photons were more than 1 ms off a whole frame period in **0.00 %** of intervals in all five runs — median error 6 µs, worst case 169 µs, 2951 trials. The panel was never the variable.

Three software hypotheses were tested against the fixed Wayland stack. Real-time priority is worth 1.8×. Firing the trigger synchronously rather than from a goroutine is worth **nothing** — 1.344 against 1.320, in the wrong direction; `test_photodiode_latency` logs its own flip-to-trigger gap and puts the entire host-side path at **median 11 µs, max 38 µs**. Changing the display stack is worth **16×**.

19.2 The stack decides it, and it is not a single axis

only the stack differs (all FIFO 50, synchronous trigger)	mean	sd	full range
Wayland session	21.75 ms	1.344 ms	18.83–36.74
KMS/DRM, no display server	18.91 ms	0.113 ms	18.58–19.13
Bare Xorg + openbox	**35.74 ms**	**0.083 ms**	**35.52–35.95**

Xorg is the **steadiest and the latest**. The lag difference is not approximate:

```
Xorg − KMS/DRM = 16.826 ms = 1.010 frames
```

One whole extra buffer in the pipeline, and dead constant. Neither Xorg nor KMS/DRM puts a single flip more than 1 ms off the frame grid, in 580 and 590 intervals respectively; Wayland misses on one trial in fifteen.

So the answer to "where does X11 fall between the other two" is: neither between nor beyond. It is a different trade — best-in-class stability bought with an extra frame of latency.

19.3 Against Table 2

Their "visual onset" is *"the difference between the occurrence of the trigger pulse and the pixels changing on the LCD screen"* — the same quantity, their monitor is also 60 Hz, and **bare Xorg is the stack they ran**, which makes E the honest row to compare.

configuration	onset Var (ms)
goxpyriment — bare Xorg (E)	**0.083**
goxpyriment — KMS/DRM console (D)	**0.113**
PsychToolBox Ubuntu · E-Prime Win10	0.18
PsychToolBox Win10 · Expyriment Win10	0.19
PsychoPy Ubuntu · Presentation Win10	0.34
PsychoPy Win10	0.35
PsychToolBox macOS	0.41
OpenSesame Ubuntu	0.50
PsychoPy macOS	0.55
OpenSesame Win10 · Expyriment Ubuntu	0.72 / 0.73
OpenSesame macOS	0.79
goxpyriment — Wayland (B, C)	**1.32 / 1.34**
goxpyriment — Wayland, normal priority (A)	**2.34**
Expyriment macOS	4.82

On the stack they measured, goxpyriment has **the best onset precision in the table**, twice as good as the best published figure. In a Wayland session the same binary is

worse than thirteen of the fourteen. Nobody's row in that column is a property of their package alone.

The lag tells the opposite story and should be reported alongside:

	visual onset lag
PsychToolBox / PsychoPy / E-Prime, Linux & Windows	2.35 – 7.10 ms
PsychoPy macOS · PsychToolBox macOS (the 10.13 buffering bug)	18.24 / 21.52 ms
Expyriment macOS — their worst	29.02 ms
goxpyriment, bare Xorg	**35.74 ms**

PsychToolBox's flip returns essentially *at* scanout. goxpyriment's returns two frames early, and on KMS/DRM one frame early. **That is a software property, not a hardware one, and it is the one number here that looks reducible.** It costs nothing scientifically — a constant offset subtracts out of any analysis — but it is a real difference in how deep the swap chain runs. See `TODO.md`.

Visual duration Var: 3.30 / 3.25 ms under Wayland, from 2.6–3.0 % of trials running exactly one frame long and nothing else; excluding those, 0.15 ms. Against their Ubuntu rows (PTB 0.15, PsychoPy 1.19, Expyriment 8.31, OpenSesame 9.16) that is mid-pack under a compositor.

19.4 Threshold sensitivity, and where the second instrument stops being usable

The BBTK's Opto1 threshold was left at its default and never calibrated against this panel. Sweeping the AD3's onset level across the panel's whole rise shows what a threshold can and cannot affect:

onset level	Wayland (C): mean / sd	KMS/DRM (D): mean / sd
5 %	21.14 / 1.344	18.31 / 0.116
10 %	21.75 / 1.344	18.91 / 0.113
50 %	24.70 / 1.344	21.86 / 0.102
90 %	27.22 / 1.345	24.38 / 0.094

A 6 ms sweep moves the **mean by 6 ms and the sd by at most 0.02 ms**. So every precision figure here is threshold-insensitive and every lag figure is not — quote a lag only with its level attached, which is why `extract-onsets.py` records it in the file.

For A–D the BBTK independently reproduced the AD3's sd to three significant figures (2.33/2.342, 1.32/1.320, 1.35/1.344, 0.17/0.113 — the last at its 250 µs quantisation floor). **For E it does not, and should not be quoted.** It reports sd 1.55 ms against the AD3's 0.083, because seven trials read 24–28 ms where the AD3 saw nothing below 35.52 ms in any of 581. An uncalibrated threshold sitting high on a 5.5 ms

ramp crosses erratically when the final luminance wobbles, and this is the condition where that finally showed. Calibrating Opto1 against this panel would be needed before the BBTK can corroborate E, and before any BBTK lag here can be set beside a published BBTK lag.

19.5 What this licenses saying

That goxpyriment's own timing code is not a limiting factor anywhere: the host-side trigger path is under 40 µs in every run, and on the stack the published study used, the trigger-to-photon interval is stable to **83 µs over five minutes** — better than any configuration in that study.

That a Wayland session costs a factor of sixteen in onset precision on this machine, and is the dominant term for anyone doing EEG or MEG with a compositor running, whatever package they use.

That goxpyriment carries one to two frames more presentation latency than PsychToolBox does. Constant, correctable, and worth fixing anyway.

One machine, one panel. The study's own authors decline to generalise across machines, and so should this.

Here are the step-by-step instructions to set up the `goxpyriment` group with those high-priority privileges.

Steps 1-5 cover real-time priority. Step 6 covers the `input` and `video` groups, which a machine needs before it can read keyboards or drive the display from a bare console — the configuration the timing measurements recommend, and the one where their absence first shows — and `lp`, for a machine that triggers through a parallel port. Running from a virtual console covers that configuration itself: getting there, the display mode you actually get, and what a second lit monitor does to the timing.

If you are here because of EEG or MEG trigger timing, read Minimising trigger-to-stimulus jitter as well: real-time priority is necessary but nowhere near sufficient, and the dominant term is not the one most people expect.

On another platform? See Setting priority under Windows or Setting priority under macOS. Both use a different mechanism, and neither has been measured — this page is the only one whose figures come from real runs, and most of its *reasoning* transfers even where the commands do not.

Chapter 20

Step 1: Create the Group

First, you need to create the group in the system database.

```
sudo groupadd goxpyriment
```

20.0.1 Step 2: Create the Limits Configuration

Linux stores these specific "privilege" rules in `/etc/security/limits.d/`. You should create a new file specifically for your group so it doesn't mess with other system settings.

1. **Open a new config file:**

   ```
   sudo nano /etc/security/limits.d/99-goxpyriment.conf
   ```

2. **Paste the following lines into the file:**

   ```
   @goxpyriment - nice -20
   @goxpyriment - rtprio 50
   @goxpyriment - memlock unlimited
   ```

3. **Save and Exit:** Press `Ctrl + O`, then `Enter` to save, and `Ctrl + X` to exit.

4. **Check the text actually landed:**

   ```
   grep rtprio /etc/security/limits.d/*.conf
   ```

   Worth the two seconds. An editor opened *without* `sudo` — or a graphical editor that cannot write to `/etc` — may fail to save without making it obvious. The only later symptom is `ulimit -r` still returning `0` after a re-login, which is easy to misread as "the limits system doesn't work" rather than "the file was never written".

   Why a new file rather than `/etc/security/limits.conf`? That file belongs to the `libpam-modules` package and can be replaced on upgrade, silently taking your setting with it. Anything you add under `/etc/security/limits.d/` is yours and survives.

Why not simply join an existing group such as `audio`? Tempting, because on many systems `audio` already carries an rtprio grant. But that grant is installed by the jackd package and can be revoked by `dpkg-reconfigure -p high jackd2`, and it hands out far more than you need (`rtprio 95`, `memlock unlimited`). The real objection is subtler: an experiment that has quietly lost real-time priority behaves exactly like one that still has it, right up until you look at the timing data. A group of your own cannot be switched off by another package's maintainer script.

20.0.2 Step 3: Add Yourself (and others) to the Group

Simply creating the group isn't enough; you have to tell Linux which users belong to it. Replace `$USER` with a specific username if you are adding someone else.

```
sudo usermod -aG goxpyriment $USER
```

20.0.3 Step 4: Apply the Changes

Important: Linux only checks group memberships and limits when a user **logs in**.

- You **must** log out of your Linux session and log back in.
- Alternatively, you can run `su - $USER` in your terminal to start a sub-shell with the new permissions for testing.

20.0.4 Step 5: Actually Use It

The grant only makes real-time priority *available* — nothing runs at it until something asks.

goxpyriment programs ask for themselves. `Experiment.Initialize()` requests priority 50 at startup, so once Steps 1-4 are done there is nothing further to do — including when the program is launched by clicking its icon, where no command-line prefix is possible. If the grant is not in place it says so and continues at normal priority rather than refusing to run:

```
real-time scheduling not obtained, continuing at normal priority: real-time
scheduling is not permitted for this user (RLIMIT_RTPRIO is 0). ...
```

Two flags control it:

```
./my-experiment -no-realtime                # do not ask at all
./my-experiment -realtime-priority 20       # ask for something other than 50
```

A program with no flags — one built with `NewExperiment` + `Initialize` rather than `NewExperimentFromFlags` — sets the field directly instead:

```
exp := control.NewExperiment(...)
exp.RealTimePriority = 0     // decline; the -no-realtime equivalent
```

Every run records what it ended up with, as `sys sched_policy` in its `-info.txt`. Read it from there rather than assuming: a run at `SCHED_OTHER` and one at `SCHED_FIFO` are

not comparable, so a study should not mix them.

For anything else, or to override, use `chrt`:

```
chrt -f 50 ./some-other-program
```

`-f` selects SCHED_FIFO and `50` is the priority. **It must not exceed the `rtprio` value granted in Step 2** — asking for more fails with "Operation not permitted", which is the same error you get when the grant is missing entirely, so it is easy to misdiagnose. Keep the two numbers equal unless you have a reason not to.

The policy is inherited by child processes, so one `chrt` covers the whole run. It does not persist to your shell or to the next command; that is deliberate. You *can* make a shell real-time with `chrt -f -p 50 $$`, but do not: every command you then type runs above most system threads, and a mistyped one is very hard to interrupt.

20.0.5 Step 6: The other groups the same machine usually needs

Real-time priority is not the only permission a stimulus machine wants, and the other two bite in exactly the configuration the timing measurements recommend: running without a display server, from a virtual console.

```
sudo usermod -aG input $USER      # /dev/input/event*  — keyboards, mice,
↪  gamepads
sudo usermod -aG video $USER      # /dev/dri/card*     — KMS/DRM output and
↪  vblank
```

Both need a full logout and login, the same as Step 4.

Why it only shows up on the console. In a desktop session `systemd-logind` attaches an ACL granting the active user access to that seat's devices — the `+` at the end of the permission bits is the ACL:

```
crw-rw----+ 1 root video  226,   0 /dev/dri/card0
crw-rw---- 1 root input   13,  64 /dev/input/event0
```

So everything works in a desktop session and the same program run from a bare VT finds the nodes closed to it. Group membership is not seat-dependent and covers both.

`input` — without it SDL prints a line per device it cannot open while it enumerates for joysticks:

```
Error: could not open /dev/input/event3
```

Harmless in itself, and a visual-only run is unaffected. It matters for **reaction times**: with evdev unavailable SDL falls back to reading the console tty, which is not the same input path as the one a desktop run uses. We have not measured the difference — which is the reason to remove the ambiguity rather than to reason about it. Never compare RT distributions between a console run and a desktop run without checking both used the same path.

> **!** A user in `input` can read every keystroke on the machine, in every session, including other users'. On a dedicated stimulus box that is a fair trade; on

a shared machine it is not, and the alternative is to make sure the VT login creates a proper logind session so the ACLs are applied (`loginctl session-status` should show it active on `seat0`).

`video` — needed to open `/dev/dri/card*`, which is both how SDL drives the display under `kmsdrm` and how the vblank backend reads hardware timestamps (`GOXPY_VBLANK=on`, see `vblank/drm_linux.go`). If GPU rendering then fails while output works, add `render` as well: the two are separate nodes with separate groups.

```
crw-rw----+ 1 root video  226,   0 /dev/dri/card0        # video
crw-rw----+ 1 root render 226, 128 /dev/dri/renderD128   # render
```

`audio` — usually **not** needed, despite looking like the same case. The nodes are protected identically:

```
crw-rw----+ 1 root audio  116,   8 /dev/snd/pcmC0D0p
```

but the resemblance stops there. SDL does not normally open them: it talks to PipeWire or PulseAudio over a socket in your own runtime directory, and the sound server holds the device. So audio keeps working from a console where the display and the keyboard do not, and joining `audio` changes nothing.

It matters only if you bypass the server and open ALSA directly (`SDL_AUDIODRIVER=alsa` with no server running) — which our own measurements argue against on other grounds: on a Raspberry Pi 4 with the server stopped and the device opened directly, **every one of 463 tones was torn**. If you are in that configuration, the missing group is not your biggest problem.

Do not join `audio` to obtain real-time priority either — see the note in Step 2 on why that grant is the wrong one to rely on. PipeWire gets its own real-time threads through RealtimeKit, not through the group.

`lp` — only on a machine that triggers through a **parallel port**, and it is two separate things wearing one name. Get both, or the port is unusable in a way that is hard to read as a permissions problem:

```
sudo usermod -aG lp $USER   # the GROUP lp — rw access to /dev/parport0
sudo rmmod lp               # the MODULE lp — the parallel printer driver
```

The group is the ordinary case: without it `/dev/parport0` cannot be opened and you get a plain permission error.

The module is the one that costs an afternoon. `lp` and `ppdev` can both be registered on one port, and `ParallelPort.Open`'s `PPCLAIM` then goes through the kernel's `parport_claim_or_block`. If `lp` is *holding* the port, that ioctl blocks in **uninterruptible sleep** — the process survives Ctrl-C and `kill -9` alike, and on the machine where this was diagnosed (2026-08-21, a PCIe LPT card) the desktop had to be powered off at the switch. It is intermittent, because `lp` holds the port only some of the time, so a run that works proves nothing.

`dmesg` says whether you are exposed, at boot:

```
[    2.503789] lp: driver loaded but no devices found      ← lp loads before
↪  the card probes
```

```
[    2.824137] parport0: PC-style at 0x3100, irq 16 [PCSPP,TRISTATE]
[    2.929160] lp0: using parport0 (interrupt-driven).     ← and then
 ↪  attaches to it
```

That last line is the one to grep for. To keep it away across reboots:

```
echo 'blacklist lp' | sudo tee /etc/modprobe.d/blacklist-lp.conf
```

Nothing is lost: `lp` is the parallel *printer* driver, and unloading it also leaves the port's IRQ unarmed, which ppdev writes never use. See `tests/test_parallel_port` for the diagnosis in full.

Verify the same way as the rest of this page — from the data, not from memory. A run that obtained the vblank clock says so in its `-info.txt`:

```
# sys vblank_backend: Linux DRM vblank (card /dev/dri/card2, crtc 0, ...)
```

and a run that could not falls back silently to a software-derived onset, which is a difference of several parts per million in a long block.

20.0.6 How to Verify it Worked

Once you've logged back in, you can verify that your Go program (or any process) will have these rights.

1. Check Group Membership: Run the command `groups`. You should see `goxpyriment` in the list.

2. Check the Memory Lock Limit: Run this command to see your current "Max locked memory" limit:

```
ulimit -l
```

If it says **`unlimited`**, you are good to go!

3. Check Real-Time Priority: Run:

```
ulimit -r
```

It should return **`50`**. If it still returns `0`, the grant is not in effect — either the file was never written, or you have not fully logged out and back in.

4. Check from inside your experiment. Any goxpyriment program's system report includes a `Sched:` line, so a recorded run carries its own evidence:

```
Sched:      policy: SCHED_FIFO  priority: 50  REAL-TIME
Sched:     policy: SCHED_OTHER  nice: 0  (real-time available up to 50, not
 ↪  used)
Sched:      policy: SCHED_OTHER  nice: 0  (real-time NOT available to this
 ↪  user)
```

Those three lines are three different situations with three different fixes, and they are worth being able to tell apart after the fact.

The **last** is this setup not being in place — go back to Step 2. The **middle** is the setup working but the program not having asked: for a goxpyriment program that means `-no-realtime` was passed or `RealTimePriority` was set to 0, since otherwise it asks on its own; for anything else it means no `chrt` prefix. Without the line in the data you would be left comparing timing distributions and guessing which had happened.

20.1 Running from a virtual console (VT)

Running with no display server at all — GDM/SDDM stopped, a bare virtual console, SDL on the `kmsdrm` driver — is the configuration this page's groups exist for, and it is the single largest improvement available on Linux. On a Radeon Pro W5700 with everything else held identical it took onset latency from ~55 ms to ~22 ms and the scatter from 0.296 ms to 0.057 ms (measured 2026-08-17 with a photodiode; see The display stack is worth two frames of latency and five times the jitter).

It also changes two things that a desktop session hides, and both have already cost a real capture.

20.1.1 Getting there

```
sudo systemctl stop gdm3          # or sddm, lightdm — whatever your machine
 ↪  runs
# Ctrl-Alt-F3, log in, then:
cd ~/my-experiment && go run .
```

You need the `input` and `video` groups from Step 6 — in a desktop session `systemd-logind` grants those devices by ACL, and the ACL does not follow you to a bare VT. If SDL still fails to go fullscreen, see SDL3 fullscreen in a Linux virtual console; the usual fix is `SDL_VIDEODRIVER=kmsdrm`.

20.1.2 The console's mode is the mode you get

goxpyriment **never changes the display mode.** On KMS/DRM it opens a fullscreen-desktop window and the fullscreen mode it passes is always the display's *current* mode, so resolution and refresh rate come from whatever the kernel already set on that console — not from the monitor's native mode, and not from what the desktop session was using before you stopped it.

That is easy to miss, because the number that changes is not the one you would watch. A Dell U2720Q (native 3840x2160) driven from a VT at **2560x1440** ran at **59.951 Hz** instead of the 59.997 Hz of its native mode — 750 ppm apart, and the monitor was upscaling every frame to its own panel for the whole run. Nothing in the experiment complained; the pacing schedule, the vblank clock and the `-hz` figure passed to the tone all simply worked from the mode that was actually set.

Read back what you got, from the run's own `-info.txt` rather than from memory:

```
# sys physical_resolution: 2560x1440 px
# d name: DP-1
# d refresh_rate_hz: 59.9500
# sys vblank_backend: Linux DRM vblank (card /dev/dri/card2, crtc 1 driving
 ↪  DP-1 2560x1440@59.9514 Hz, ...)
```

Before a run, the same questions from the console:

```
# Which heads are connected, and what modes each offers (first = preferred)
for c in /sys/class/drm/card*-*; do
  [ "$(cat $c/status)" = connected ] && { echo "== $c"; head -3 $c/modes; }
done

# What is actually set right now, with exact pixel clocks
sudo apt install libdrm-tests   # provides modetest; drm_info also works
modetest -c | grep -A2 'connector.*connected'
```

20.1.3 Pinning the mode

The console's mode is set by the kernel at boot, so the place to change it is the kernel command line — one entry per connector, named exactly as it appears in `/sys/class/drm/cardN-<NAME>`:

```
video=DP-1:3840x2160@60
```

Add it to `GRUB_CMDLINE_LINUX_DEFAULT` in `/etc/default/grub`, run `sudo update-grub`, reboot, and check `/proc/cmdline` and the `-info.txt` afterwards. Several connectors take several `video=` entries.

A monitor on USB-C is still `DP-N`. The DRM connector type names the *protocol*, not the plug: USB-C carries DisplayPort over Alt Mode, so a monitor on a USB-C–USB-C cable appears as `DP-1`, `DP-2`, … exactly like one on a full-size DisplayPort socket. There is no `USB-C-1`. On a Precision 5490 all four external outputs are USB-C/Thunderbolt and all four are exposed as `card2-DP-1` … `card2-DP-4`, one per port; each carries its own ACPI firmware node, so a given socket keeps its number across reboots. Plugging the same monitor into a *different* socket will change it — which is the reason to read the name off `/sys/class/drm` rather than remember it:

```
ls -d /sys/class/drm/card*-* | while read c; do
  echo "$(basename $c)  $(cat $c/status)"
done
```

One exception: a monitor reached through an MST hub or a docking station is a branch device, and its connector is created on hotplug rather than fixed in firmware. It has a `path` attribute (`cat /sys/class/drm/card*-DP-*/path`) where a directly-attached one has none, and a boot-time `video=` entry cannot be relied on to find it. For a timing rig, plug the stimulus monitor straight into the machine.

`modetest -s` can also set a mode, but it holds DRM master for as long as it runs and drops the mode when it exits, so it is a diagnostic rather than a way to prepare a run — SDL cannot open the device while it is held.

Whether the native mode is worth insisting on is a measurement, not a rule. What is certain is that a non-native mode makes the monitor scale, and that its refresh rate is a different number from the one on the box. The TTL→photon lag in the 1440p capture above ran 38–55 ms end to end; how much of that the scaler accounted for was not measured, because the native mode was never captured for comparison. If input lag matters to your paradigm, capture both.

20.1.4 One display, or name the one you mean

A VT lights every connected head, and that is where the second trap is. The vblank ioctl names a CRTC by index, not by monitor, and a laptop's internal panel and an external monitor run **different clocks** — on a Precision 5490 they were 1449 ppm apart. Reading the wrong one produced onsets that looked perfectly regular, reported themselves as `hardware-verified`, and walked a whole frame away from the photons and back 44 times in eight minutes.

goxpyriment now resolves the CRTC from the display it is presenting to and says which head it picked in `sys  vblank_backend`, and refuses to use a vblank clock it cannot match rather than timing the wrong monitor. Two things are still worth doing:

- Pass `-d N` deliberately and confirm `d name:` in the `-info.txt` is the monitor the participant is looking at.
- Read the end of `sys  vblank_resolution:` on any run with `GOXPY_VBLANK=on`. It compares the vblanks actually read against the display drawn on:

  ```
  sys vblank_resolution: frames=30000 … measured=59.9514 Hz
   ↪  nominal=59.9506 Hz (frame period -13 ppm, matches the display)
  ```

  If it says `WRONG DISPLAY`, the onsets in that file are on another monitor's grid.

The simplest way to have neither problem is to blank or unplug the head you are not using, so there is only one mode and one clock on the machine.

20.1.5 ! A Friendly "Warning"

Once this grant is in place, goxpyriment programs run at real-time priority **by default** — you no longer have to remember a `chrt` prefix, and equally you no longer get a reminder that you are asking for it. A busy loop (`for { }` with no sleep, or a spin-wait) is then running above the window manager and the input handling, and the OS will not interrupt it to let you click "Stop".

Two things bound the damage, and it is worth knowing which one you are relying on:

- **In-program elevation raises only one thread** — the experiment's own. The Go runtime's other threads, including the garbage collector, stay at normal priority, so a spinning goroutine occupies one core rather than all of them.
- **`chrt -f 50 prog` raises the whole process**, because the policy is set before `exec` and every thread inherits it. That is the more dangerous of the two, and the one this warning is mostly about.

Either way: keep a terminal with `top` or `htop` open while developing, and use `-no-realtime` when stepping through code in a debugger — a breakpoint hit on a real-time thread can leave the desktop unresponsive until the process is killed.

If a program is unkillable and real-time priority is *not* involved, suspect a different mechanism with the same symptom: a thread blocked in the kernel, in uninterruptible sleep, which no signal can reach. The parallel port has a known way of doing this — see `lp` in Step 6.

The kernel's real-time throttle (`/proc/sys/kernel/sched_rt_runtime_us`, 950 ms per second by default) is a backstop, not a licence: it stops a runaway task locking the machine completely, but a machine at 95 % real-time occupancy is not usable.

20.1.6 ! The throttle also wrecks the timing you asked for

That backstop is not only a protection against you — it is a hazard to the very thing real-time priority was requested for. Once a `SCHED_FIFO` thread reaches 100 % duty, the kernel stops it for the remaining 50 ms of the second, and that suspension lands wherever it lands: in the middle of a pulse, a frame, or a response window.

Measured on a 22-core Linux 7.0 host, with a pinned `SCHED_FIFO 50` thread spinning continuously:

host state	stalls	each	when they landed
idle	**0 in 20 s**	—	—
under `stress-ng --cpu 20`	**24 in 25 s**	51.0 ms	0.999, 2.000, 3.001, 4.002 s …

One per second, to the millisecond — the throttle period exactly. **Load is a necessary condition**, which is the worst way for a fault to behave: on an idle runqueue the kernel borrows unused real-time bandwidth from the other CPUs and the limit is never reached. So it does not happen on the quiet machine you develop on, and does happen on the loaded one you run participants on. A pulse train that spun through its inter-trial gaps at `chrt -f 50` lost up to **49.63 ms** on 23 of 1000 trials this way — about ten times the 4.75 ms worst-case spread that real-time priority was bought to remove in the first place.

The rule that avoids it: sleep, and spin only the last millisecond or two. A wait that sleeps most of its duration never approaches the limit, and the short spin at the end recovers the precision `time.Sleep` cannot give on its own. goxpyriment's frame pacing works this way (`apparatus.paceToFrame`), which takes a 60 Hz present loop from ~100 % duty to ~10 % with no loss of landing accuracy. If you write your own wait, write it the same way — a bare `for time.Now().Before(deadline) {}` over a whole trial is the shape that gets throttled.

If you genuinely need a thread at 100 % duty — a spin-wait on a panel fast enough that there is nothing left worth sleeping — the limit can be lifted:

```
sudo sysctl kernel.sched_rt_runtime_us=-1        # until reboot
```

That removes the backstop along with the throttle, so a runaway real-time loop will then hold a CPU with nothing left to take it back. Reasonable on a dedicated stimulus machine; not on a laptop you also read mail on.

20.1.7 ! The speed the CPU runs at is not fixed either

Real-time priority decides **when** your thread runs. It says nothing about how fast the core runs once it does, nor how long the core takes to wake up, and both default to saving power rather than responding quickly:

- **The frequency governor.** `powersave` and `schedutil` raise the clock only after they observe load, so the first work after an idle gap runs at a lower clock than the work after it.
- **Idle states.** A deeply idle core takes tens to hundreds of microseconds to come back. An experiment that sleeps between frames is idle by design — which is precisely the pattern that pays this cost, and the pattern the previous section tells you to adopt.

To pin the clock for a session:

```
sudo cpupower frequency-set -g performance     # package: linux-tools-common
 ↪  / cpupower
cpupower frequency-info | grep -i "current policy"
```

It does not survive a reboot. Make it persistent through your distribution's usual mechanism (a systemd unit, or `GOVERNOR=performance` in `/etc/default/cpufrequtils` on Debian and Ubuntu) rather than by remembering to type it, since the failure mode is silent.

Measured, on a Radeon Pro W5700 workstation driving a 4K panel over DisplayPort, 1010 cycles per run, onsets anchored on kernel DRM vblanks and verified against a photodiode. Two runs differing only in the governor:

governor	flip -> photons drift	shape
default (`schedutil`)	+11.8 ppm	flat at +0.3 ppm for 316 s, then +10.6 ppm
`performance`	**+0.41 ppm**	no change of regime; 0.059 ms scatter

The default-governor run is the instructive one. Its cadence did not degrade gradually: it held to 0.3 ppm for five minutes and then switched, cleanly, to 10.6 ppm and stayed there — with the *panel* holding its own period to 0.27 ppm across the same instant, so the display was not what moved. Nothing in the recorded configuration changed. A run like that reports excellent frame statistics throughout and would pass any host-side check.

That is the failure worth guarding against: not noise, which shows up in a standard deviation, but two halves of a session that are internally tidy and not comparable to each other. Pinning the governor removed it. The recorded `host cpu_mhz` went from 3600 of a 4800 maximum to 4500.

One pair of runs, one machine — enough to act on, not enough to call it the only mechanism.

Check it from the data rather than from memory. Every run records the clock it saw at start-up in its `-info.txt`:

```
# host cpu_mhz: 3600 (max 4800)
```

A run that starts well below its maximum was not on a pinned clock. As with `sys sched_policy`, the value is in the file so that two runs can be compared afterwards without anyone having to recall what the machine was doing.

Idle states are the untested half. Holding `/dev/cpu_dma_latency` open at 0 to keep cores out of deep idle is the usual next step, and we have not measured it; if you do, measure the pair before and after and add the numbers here. That is what the rest of this page is made of.

> **Note on the grant itself:** goxpyriment only uses `rtprio`. The `nice -20` and `memlock unlimited` lines in Step 2 are there because they are commonly wanted alongside it and cost nothing, not because anything here requires them. Granting `rtprio` alone is enough if you prefer the smaller privilege.

Chapter 21

Setting priority under Windows

If you are here because of EEG or MEG trigger timing, read Minimising trigger-to-stimulus jitter as well: priority is necessary but nowhere near sufficient, and the dominant term is not the one most people expect.

21.0.1 ! None of this has been measured on Windows

The Linux procedure in Setting priority under Linux was verified against `chrt` and `nice` on real hardware, and every figure in it comes from a run. **This page has not been.** goxpyriment's own Windows scheduling code (`sysinfo/scheduling_windows.go`) is likewise written against the documented Win32 API and never exercised — it says so at runtime, in the `Sched:` line of every report.

What follows is the mechanism and the procedure, not a result. Measure it before quoting a number.

21.1 The short version

From **`cmd.exe`** (not PowerShell — see below):

```
start /high my_experiment.exe
```

That is the recommendation. `/realtime` also exists and is discussed below, but it is **not** the default suggestion, for reasons that are worth reading before you use it.

21.2 Why a launcher prefix at all

On Linux, a goxpyriment program raises its own priority at startup — it calls `sysinfo.RaiseToRealTime` and needs no `chrt` prefix, which is what makes the Linux setup a one-time system configuration rather than a thing to remember.

On Windows it does not. `sysinfo/realtime_other.go` deliberately implements `raiseToRealTime` as an error on every non-Linux platform, so every Windows run logs this once at startup:

```
real-time scheduling not obtained, continuing at normal priority:
real-time scheduling is not implemented on windows
```

That message is expected on Windows and is not a fault. It is there because the alternative — a portable "raise priority" call that quietly does something different on each OS — would report success while giving you a different guarantee on every machine. Windows has no `SCHED_FIFO` and no POSIX priority to raise; it has *priority classes*, which are a different mechanism with different consequences.

So on Windows the priority decision is made **outside** the program, by however you launch it. `-no-realtime` and `-realtime-priority` have no effect there beyond suppressing the message.

21.3 Step 1: launch through `start`

`start` is a **`cmd.exe` built-in command**, and its priority switches are the simplest way to set a process's priority class at launch:

switch	priority class
`/low`	`IDLE_PRIORITY_CLASS`
`/belownormal`	`BELOW_NORMAL_PRIORITY_CLASS`
`/normal`	`NORMAL_PRIORITY_CLASS` (the default)
`/abovenormal`	`ABOVE_NORMAL_PRIORITY_CLASS`
`/high`	`HIGH_PRIORITY_CLASS`
`/realtime`	`REALTIME_PRIORITY_CLASS`

A shortcut or a `.bat` file next to the experiment is the practical way to make this reproducible, so that a run started by someone else gets the same treatment as one started by you:

```
@echo off
rem run_experiment.bat — always launch at HIGH, never at NORMAL by accident
start /high /wait my_experiment.exe %*
```

`/wait` keeps the console window attached so you can see the program's own log output, including the `Sched:` line.

21.3.1 ! In PowerShell, `start` is a different command

This is the trap most likely to catch you, because it fails *quietly*:

```
start /high my_experiment.exe        # NOT what you want in PowerShell
```

In PowerShell, `start` is an alias for `Start-Process`, which has no `/high` switch and no priority parameter at all. Depending on the version you will get an error about an unexpected argument, or `/high` will be passed through as an *argument to your program* — which goxpyriment's flag parser will reject, but a different program might silently ignore. Either way the priority class is `NORMAL`.

Two ways out. Call `cmd` explicitly:

```
cmd /c start /high my_experiment.exe
```

or start the process and set the class afterwards:

```
$p = Start-Process .\my_experiment.exe -PassThru
$p.PriorityClass = 'High'
```

The second raises the priority a moment *after* the process starts, so the very beginning of the run is at `NORMAL`. That is harmless for an experiment with an instructions screen and a keypress before the first trial, and not harmless for one that starts measuring immediately.

21.4 Step 2: decide about `/realtime` — the honest answer is "probably not"

`REALTIME_PRIORITY_CLASS` is not the Windows equivalent of Linux's `SCHED_FIFO 50`. It is considerably more aggressive, and Microsoft's own documentation warns against it: threads in this class run above most operating-system threads, including ones handling disk flushing and mouse and keyboard input. A busy loop there can make the machine unresponsive in a way that `/high` cannot.

goxpyriment's frame pacing sleeps most of each wait and spins only the last couple of milliseconds (`apparatus.paceToFrame`), specifically so that it does not sit at 100 % duty — so the worst case is less likely than it would be with a naive spin-wait. But "less likely" is the whole claim, and it is not one this project has tested on Windows.

Two further reasons to prefer `/high` as the default:

- **`/realtime` needs administrator privileges** (`SeIncreaseBasePriorityPrivilege`). Without them the process is created at `HIGH_PRIORITY_CLASS` instead — and **no error is reported**. You get the priority you would have got from `/high`, while believing you got something stronger. This is the same failure shape the Linux page warns about, where a missing `rtprio` grant produces a run that looks identical to one that has it.
- **`/high` is very likely where the return curve flattens.** The gap between `NORMAL` and `HIGH` is the one that stops ordinary background work preempting the experiment. The gap between `HIGH` and `REALTIME` mostly buys precedence over the OS itself, which is not what a dropped frame usually comes from.

If you do use `/realtime`, launch the shell as Administrator, and **check the `Sched:` line afterwards** rather than assuming it took.

21.5 Step 3: the thing that probably matters more — timer resolution

For millisecond stimulus timing on Windows, the system timer resolution is plausibly a larger term than the priority class, and it is the reason `sysinfo/scheduling_windows.go` collects it alongside the priority:

> The timer resolution matters more than the priority class for millisecond stimulus timing [...] the default tick has historically been ~15.6 ms, and a process that has not raised it cannot place an event to the millisecond however high its priority.

A process that has not requested a finer tick can have its sleeps rounded to the nearest ~15.6 ms, which is roughly one 60 Hz frame — no priority class rescues that. The Go runtime does request high-resolution timers on modern Windows, so in practice you are likely to see a much smaller number, but that is a runtime implementation detail rather than a guarantee, which is why goxpyriment **measures** it (`NtQueryTimerResolution`) instead of assuming it.

Read it off the report:

```
Sched:      policy: HIGH_PRIORITY_CLASS  nice: 0  [UNTESTED on Windows; no
 ↪  POSIX
           policy, priority class shown instead; timer resolution 0.500 ms]
```

If that figure reads ~15.6 ms rather than ~1 ms or below, **stop and fix that first** — no amount of priority will compensate, and any timing you record until you do is quantised to a frame.

21.6 Step 4: verify

Every goxpyriment program's system report carries the state it actually ran with, so a recorded run is self-labelling. Print it without running an experiment:

```
Timing-Tests -sysinfo
```

and look at the `Sched:` line:

what you see	what it means
`policy: NORMAL_PRIORITY_CLASS`	the prefix did not take — most likely the PowerShell trap above
`policy: HIGH_PRIORITY_CLASS`	`/high` worked

what you see	what it means
`policy: REALTIME_PRIORITY_CLASS … REAL-TIME`	`/realtime` worked *and* you had the privilege
`policy: unknown`	`GetPriorityClass` failed; report it as a bug

Note the asymmetry that makes checking worth the two seconds: asking for `/realtime` **without** administrator rights shows `HIGH_PRIORITY_CLASS` here, which is exactly what a successful `/high` shows. The line tells you what you got, not what you asked for.

The `[UNTESTED on Windows …]` note is printed deliberately and will stay until someone runs the verification below on real hardware.

21.7 How to find out whether it helped, on your machine

This is the part that turns the page from advice into a result, and it takes about ten minutes. Nothing above substitutes for it, because the answer depends on your GPU driver, your compositor settings and what else the machine is doing.

Run the frame-interval test twice, changing only the launcher:

```
Timing-Tests -test display -duration-s 300
start /high Timing-Tests -test display -duration-s 300
```

Then compare the two `-info.txt`/`.csv` pairs. What to read, in order:

1. **Dropped frames and the maximum interval** — the tail, not the average. A priority change moves the worst frames; it barely moves the median, so a comparison of means will show almost nothing even when the change helped.
2. **The SD of the frame intervals.**
3. **The `Sched:` line in each `-info.txt`**, which records which run was which so the two files cannot be mixed up afterwards.

Do it on a machine in the state you will actually run participants on. An idle machine hides scheduling problems — that is the single most common way this kind of comparison produces a false negative.

If you run it, the numbers would be genuinely valuable to this project; there are currently none for Windows.

21.8 Other Windows-specific things worth knowing

Listed as pointers rather than recommendations — none has been verified here:

- **Fullscreen optimizations.** Windows 10/11 may run an apparently exclusive fullscreen window through the desktop compositor instead. That is the same class of problem as the compositor issues documented for Linux in TimingTests.md, and it can be disabled per-executable in the file's Properties → Compatibility tab.
- **Variable refresh rate / G-Sync / FreeSync.** Turn it off on a stimulus machine. A panel that varies its own frame period defeats the assumption every frame-counted duration rests on.
- **Power plan.** "High performance" rather than "Balanced"; core parking and aggressive frequency scaling both add wake-up latency.
- **Windows Game Mode and background apps.** Both change scheduling in ways that are documented only loosely. Whatever you choose, keep it fixed across a study and record it — a setting that is invisible afterwards makes two runs incomparable.

21.9 See also

- Setting priority under Linux — the measured version of this page, and the one to read for the *reasoning*, most of which transfers even though the mechanism does not.
- Minimising trigger-to-stimulus jitter
- Timing tests — how to run the measurements above.

Chapter 22

Setting priority under macOS

If you are here because of EEG or MEG trigger timing, read Minimising trigger-to-stimulus jitter as well: priority is necessary but nowhere near sufficient, and the dominant term is not the one most people expect.

22.0.1 ! None of this has been measured on macOS

The Linux procedure in Setting priority under Linux was verified against `chrt` and `nice` on real hardware, and every figure in it comes from a run. **This page has not been.** goxpyriment's own Darwin scheduling code (`sysinfo/scheduling_darwin.go`) is likewise written against the documented BSD/Darwin APIs and never exercised — it says so at runtime, in the `Sched:` line of every report.

What follows is the mechanism and the procedure, not a result. Measure it before quoting a number.

22.1 The short version

```
sudo nice -n -20 ./my_experiment
```

This is worth doing on a dedicated stimulus machine, **with two caveats that are not cosmetic** — one about what it actually buys you, and one about what `sudo` does to your data files. Both are below. Read them before adopting this as a lab habit.

22.2 Why a launcher prefix at all

On Linux, a goxpyriment program raises its own priority at startup — it calls `sysinfo.RaiseToRealTime` and needs no `chrt` prefix.

On macOS it does not. `sysinfo/realtime_other.go` deliberately implements `raiseToRealTime` as an error on every non-Linux platform, so every macOS run logs this once at startup:

```
real-time scheduling not obtained, continuing at normal priority:
real-time scheduling is not implemented on darwin
```

That message is expected on macOS and is not a fault. The reason it is an error rather than a silent success is worth stating, because it is the same reason `nice` is a weaker lever here than it looks:

> macOS has `thread_policy_set`, but [...] the setting that matters most for stimulus timing is something else entirely [...] a per-thread time-constraint policy on macOS. Pretending a portable "raise priority" call covers them would produce a program that reports success while doing something different on each platform.

So on macOS the priority decision is made **outside** the program. `-no-realtime` and `-realtime-priority` have no effect there beyond suppressing the message.

22.3 Caveat 1: `nice` is not the lever that governs real-time behaviour on Darwin

This is the part where the recommendation should be qualified rather than simply repeated.

`nice -n -20` lowers the process's nice value, which raises its priority **within the timeshare scheduling band**. That is a real effect: it makes the experiment less likely to be preempted by other user-space work, which is exactly what you want on a machine that is doing anything else at all.

But on Darwin, "will this thread wake on time" is not primarily decided by the timeshare priority. It is decided by which *scheduling band* the thread is in, and the band that gives hard wake-up guarantees is entered by setting a **per-thread time-constraint policy** (`thread_policy_set` with `THREAD_TIME_CONSTRAINT_POLICY`) — the mechanism Core Audio uses for its render threads. A negative nice value does not move a thread into that band.

goxpyriment does not set such a policy. `sysinfo/scheduling_darwin.go` is explicit that it cannot even *see* one:

```
Sched:      policy: SCHED_OTHER (assumed)  nice: -20  [UNTESTED on macOS;
↪  cannot
           see per-thread THREAD_TIME_CONSTRAINT_POLICY, so RealTime is not
           authoritative]
```

The practical reading: **expect `nice` to reduce preemption by other processes, not to give the experiment thread a deadline guarantee.** It is a real improvement of the "keep other work out of the way" kind, in the same family as quitting Spotlight-

heavy applications before a session — and it is a smaller, softer effect than `chrt -f 50` is on Linux. Do not carry a Linux expectation across.

22.4 Caveat 2: `sudo` leaves your data files owned by root

This one will bite you, and it has nothing to do with timing.

`sudo nice -n -20 ./my_experiment` runs the **whole experiment** as root, not just the priority change. Every file it writes is then owned by `root`:

```
-rw-r--r--  1 root  wheel   14523  Aug 15 10:24 Timing-Tests_sub-000_….csv
```

Two consequences. First, the results in `~/goxpy_data/` are no longer yours to move, edit or delete without `sudo`. Second, and worse, if the directory itself ends up root-owned, a **later run without `sudo` may fail to write its data file** — after the participant has done the task.

Mitigations, in order of preference:

1. **Check the ownership after your first `sudo` run**, and fix it once:

   ```
   ls -l ~/goxpy_data | head
   sudo chown -R "$USER" ~/goxpy_data
   ```

2. **Point the run somewhere explicit** with `-outdir`, so a mistake is confined to one directory rather than to your whole data tree.
3. **Decide per-machine, not per-run.** If a dedicated stimulus Mac always runs with `sudo`, that is consistent and fine. Mixing the two on the same data directory is what causes trouble.

There is also an identity question worth knowing about: macOS privacy permissions (screen recording, input monitoring, accessibility) are granted per user and per application, so a program run as root is not necessarily the same subject as the same program run as you. If you are prompted for a permission you thought you had already granted, this is why.

22.5 Step 1: run it

```
sudo nice -n -20 ./my_experiment
```

`nice` needs root only for **negative** values; `nice -n 5` (lower priority) works unprivileged. `-20` is the strongest available; the scale runs −20 (most favourable) to +19.

`renice` can change a process after it has started, which avoids running the whole experiment as root:

```
./my_experiment &
sudo renice -n -20 -p $!
```

The trade is that the first fraction of a second runs at normal priority. Harmless for an experiment with an instructions screen before the first trial; not harmless for one that starts measuring immediately.

To make it reproducible across people in a lab, put it in a one-line script beside the experiment rather than relying on everyone remembering the prefix.

22.6 Step 2: verify

Every goxpyriment program's system report carries the state it actually ran with, so a recorded run is self-labelling:

```
Timing-Tests -sysinfo
```

Read the `nice:` field of the `Sched:` line — `-20` if the prefix took, `0` if it did not.

What the line **cannot** tell you is whether anything is in the time-constraint band, and it says so rather than guessing. `RealTime` stays `false` on macOS even if a thread does hold such a policy; a confident "not real-time" would be an answer this code is not in a position to give.

22.7 How to find out whether it helped, on your machine

This is the part that turns the page from advice into a result, and it takes about ten minutes.

Run the frame-interval test twice, changing only the prefix:

```
Timing-Tests -test display -duration-s 300
sudo nice -n -20 Timing-Tests -test display -duration-s 300
```

Then compare the two `-info.txt`/`.csv` pairs. What to read, in order:

1. **Dropped frames and the maximum interval** — the tail, not the average. A priority change moves the worst frames; it barely moves the median, so a comparison of means will show almost nothing even when the change helped.
2. **The SD of the frame intervals.**
3. **The `Sched:` line in each `-info.txt`**, which records which run was which so the two files cannot be mixed up afterwards.

Do it on a machine in the state you will actually run participants on. An idle Mac hides scheduling problems — that is the single most common way this kind of comparison

produces a false negative. Given caveat 1, a **small or absent** difference on an idle machine is the outcome to expect; the interesting comparison is under load.

If you run it, the numbers would be genuinely valuable to this project; there are currently none for macOS.

22.8 Other macOS-specific things worth knowing

Listed as pointers rather than recommendations — none has been verified here, and on current hardware several of these are plausibly larger terms than `nice`:

- **App Nap and Low Power Mode.** Both throttle background and power-constrained work. Low Power Mode on a laptop is the more likely of the two to affect a foreground fullscreen application, and it is a single toggle in System Settings → Battery. **Run on mains power.**
- **ProMotion / variable refresh rate.** MacBook Pro and Studio Display panels can vary their refresh rate. Pin the display to a fixed rate on a stimulus machine: a panel that changes its own frame period defeats the assumption every frame-counted duration rests on. This is the macOS instance of the problem `tests/Timing-Tests -test vrr` exists to probe.
- **Apple silicon efficiency cores.** A thread scheduled onto an E-core behaves differently from one on a P-core, and QoS class is what influences that placement — another respect in which the meaningful lever on macOS is QoS rather than nice.
- **Display mirroring and Sidecar.** Both add a compositing stage between your frame and the panel. Turn them off.

Whatever you choose, keep it fixed across a study and record it — a setting that is invisible afterwards makes two runs incomparable.

22.9 See also

- Setting priority under Linux — the measured version of this page, and the one to read for the *reasoning*, most of which transfers even though the mechanism does not.
- Minimising trigger-to-stimulus jitter
- Timing tests — how to run the measurements above.

Chapter 23

SDL3 Fullscreen in a Linux Virtual Console (TTY)

> This page is the narrow troubleshooting note for one symptom: SDL not going fullscreen on a bare console. For running experiments there properly — the groups it needs, why the console's display mode (not the monitor's native one) is the mode you get, how to pin it, and what a second lit head does to the vblank clock — see Running from a virtual console.

23.1 Problem

When running with no display server (GDM stopped, bare TTY), SDL3 may fail to switch to fullscreen — the console stays visible instead of the experiment window.

23.2 Root Cause

SDL3 auto-detects the video driver. Without X11 or Wayland running, it should fall back to the **KMS/DRM** (`kmsdrm`) driver, but auto-detection may pick a non-functional driver silently.

23.3 Diagnostics

Check which driver SDL3 selects and whether KMS is available:

```
# Explicit KMSDRM driver — should work in a bare TTY
SDL_VIDEODRIVER=kmsdrm go run main.go -test display

# Check if SDL3 was built with KMSDRM support (no display needed)
SDL_VIDEODRIVER=offscreen go run main.go -test display 2>&1 | head -20
```

If `SDL_VIDEODRIVER=kmsdrm` gives "no available video device", SDL3 was not built with KMSDRM support — you need to rebuild SDL3 with KMSDRM enabled, or install a

package that includes it.

23.4 Fix

23.4.1 Option 1 — environment variable (quickest)

```
SDL_VIDEODRIVER=kmsdrm go run main.go -test display
```

Or via the existing wrapper:

```
# e.g. a run_vt.sh wrapper
SDL_VIDEODRIVER=kmsdrm go run "$@"
```

23.4.2 Option 2 — programmatic hint in screen.go

In `apparatus/screen.go`, set the hint before `sdl.Init` when no display server is detected:

```
import "os"

// Before sdl.Init(...)
if os.Getenv("DISPLAY") == "" && os.Getenv("WAYLAND_DISPLAY") == "" {
    sdl.SetHint("SDL_VIDEODRIVER", "kmsdrm")
}
```

23.5 Notes

- `window.SetFullscreen(true)` after creation may be unreliable on KMSDRM; creating the window with fullscreen from the start (as `apparatus/screen.go` already does) is the right approach.
- On a machine that already has a compositor running, `SDL_VIDEODRIVER=wayland` is the right choice; `kmsdrm` is for a bare TTY with no compositor at all.
- Ensure the user running the program is in the `video` (and possibly `input`) group:

  ```
  sudo usermod -aG video,input $USER
  ```

  KMS/DRM access is permission-gated; without group membership SDL will fail to open the device.

Chapter 24

How `control/experiment.go` works

`control/experiment.go` defines the **`Experiment` facade** — the single object an experiment program interacts with. It owns every subsystem (screen, input, audio, data, design) and exposes thin convenience methods over them. Here's how it works, following the lifecycle:

24.1 1. The `Experiment` struct (the facade)

`Experiment` (lines 70–108) bundles pointers to the focused subsystem packages:

- `Screen` (`apparatus.Screen`) — window + renderer
- `Keyboard` / `Mouse` (`apparatus`) — input
- `Audio` / `AudioDevice` — sound
- `Data` (`results.DataFile`) — CSV logging
- `Design` (`design.Experiment`) — trial/block structure

Plus config (`BackgroundColor`, `Fullscreen`, `ScreenNumber`, fonts) and internal state: `event EventState`, `quitFlag` (atomic, set by the signal handler), and `endOnce` (ensures the data footer is written exactly once).

The doc comment (lines 47–54) states the design intent: **most methods are thin delegations** to subsystem packages, keeping the user API simple.

24.2 2. Construction & initialization

- **`NewExperiment(...)`** (148) — just fills the struct; no SDL yet.
- **`NewExperimentFromFlags(...)`** (177) — the standard entry point. Parses `-w` / `-d N` / `-s N`, builds the struct, then calls `Initialize()` and `log.Fatal`s on failure.
- **`Initialize()`** (476) does the real setup, in order:
  1. Loads the embedded SDL/TTF/IMG binaries (reusing loaders cached by `GetParticipantInfo` to avoid a double-load crash on macOS).
  2. `sdl.Init` + `ttf.Init`, audio device, then `apparatus.NewScreen`.
  3. Wires `Keyboard`/`Mouse` with **poll closures** (lines 522–549) — these call back into `e.PollEvents`, so all input flows through the experiment's single event

drain.
4. Loads the default Inconsolata font, creates the `DataFile`, writes system/display/participant metadata into its header.
5. Spawns a **signal handler goroutine** (595) for Ctrl-C/SIGTERM: it sets `quitFlag`, calls `finalizeData()`, and force-exits after 3s. It deliberately never calls SDL teardown (concurrent CGo from two threads → SIGSEGV).

24.3 3. The run loop

`Run(logic func() error)` (936) is the heart:

- `runtime.LockOSThread()` pins the goroutine — SDL video/render/events must all stay on one thread (the **single-thread contract**).
- It calls `sdl.RunLoop`, and each frame: recovers from `exitPanic`, calls `pumpFrame()`, then your `logic()` callback.
- Return `control.EndLoop` from `logic` to exit cleanly.

`pumpFrame()` (428) checks `quitFlag`, pumps OS events, and checks for QUIT/ESC — **without draining the queue**, so keyboard/mouse events keep their original hardware timestamps for `GetKeyEventTS`.

24.4 4. Event handling

`PollEvents(handler)` (630) is the one place SDL events are drained. It resets transient state, loops over events, records the first key/mouse/keyup of the cycle plus their nanosecond timestamps, and sets the **sticky** `QuitRequested` on ESC or window-close. The `Keyboard`/`Mouse` closures and `WaitAnyEventTS` all sit on top of this.

24.5 5. Convenience methods (the ergonomic layer)

These compose the common patterns so experiment code stays short:

- `Show` / `ShowTS` (208, 222) — clear+draw+flip; `ShowTS` returns the VSYNC flip timestamp for precise RT.
- `ShowTimed`, `ShowAndGetRT`, `ShowInstructions`, `ShowEndMessage`, `Blank`, `Wait` — each replaces a 2–3 line idiom (documented inline).
- `WaitAnyEventTS` (252) — blocks for any device event with a hardware timestamp.

The `exitPanic` sentinel (136) lets these methods abort to `Run`'s recover on ESC/quit without threading an error through every line of user code.

24.6 6. Teardown

`End()` (877) — deferred by the caller. Calls `finalizeData()` (writes the end-time footer + saves, guarded by `endOnce`), closes font/screen/audio/microphone, then `ttf.Quit`/`sdl.Quit`, and unloads the embedded library blobs in reverse order. `Fatal()` (910) is the post-init replacement for `log.Fatalf` that cleans up first.

The rest of the file is **delegation wrappers** — `AddBlock`, `ShuffleBlocks`, `AddBWSFactor`, etc. forward straight to `e.Design`; `SetVSync`, `Flip`, `LoadFont` forward to `e.Screen`. That's the facade pattern the doc comment describes: a simple surface, with the real logic living in each subsystem package.

Chapter 25

Library usage: reducing direct SDL use

The examples in `examples/` sometimes call the underlying **go-sdl3** (`sdl`) library directly. The goxpyriment library provides wrappers and re-exports so that typical experiment code can avoid importing `github.com/Zyko0/go-sdl3/sdl` for common operations.

25.1 Run-loop exit and error handling

- **Return** from your `exp.Run(...)` callback with `control.EndLoop` for normal exit (e.g. ESC or window close), instead of `sdl.EndLoop`.
- **Check** the error after `exp.Run(...)` with `control.IsEndLoop(err)` instead of comparing to `sdl.EndLoop`:

```
err := exp.Run(func() error {
    // ...
    return control.EndLoop
})
if err != nil && !control.IsEndLoop(err) {
    log.Fatal(err)
}
```

25.2 Key and button constants

Use **control** key and button constants so you don't need the `sdl` import for key names:

- Keys: `control.K_SPACE`, `control.K_ESCAPE`, `control.K_S`, `control.K_F`, `control.K_J`, `control.K_Q`, `control.K_R`, `control.K_G`, `control.K_B`, `control.K_Y`, `control.K_N`, `control.K_1`–`control.K_4`, `control.K_KP_1`–`control.K_KP_4`.
- Buttons: `control.BUTTON_LEFT`, `control.BUTTON_RIGHT`.
- Type: `control.Keycode` (alias for `sdl.Keycode`).

Example: `exp.Keyboard.WaitKeys([]sdl.Keycode{control.K_SPACE}, -1)` can stay as-is (Keycode is the same type), or use `[]control.Keycode{control.K_SPACE}` if you only import control.

25.3 Points and colors

- **Points:** use `control.Point(x, y)` or `control.Origin()` instead of constructing `sdl.FPoint{X: x, Y: y}` or `sdl.FPoint{X: 0, Y: 0}`.
- **Colors:** use `control.RGB(r, g, b)` or `control.RGBA(r, g, b, a)` for ad-hoc colors, and predefined colors like `control.DarkGray`, `control.LightGray`, `control.Black`, `control.White`, etc.

25.4 Screen

- Use **`exp.Screen.ClearAndUpdate()`** instead of calling `exp.Screen.Clear()` then `exp.Screen.Update()` when you clear the backbuffer and present in one step.

25.5 Timing

- Use **`clock.Wait(ms)`** for delays (milliseconds) instead of `sdl.Delay(ms)`. This keeps timing in one place and avoids pulling in SDL for the main loop throttle (e.g. `clock.Wait(10)` in event loops).

25.6 When you still need SDL

You will still need the `sdl` import for:

- Constructing **sdl.FRect** or other SDL types not yet wrapped (e.g. in `demo_play_two_videos` for destination rects).
- Any SDL API not yet exposed by goxpyriment (e.g. raw `sdl.PollEvent` in special cases like loading screens).
- Passing SDL types into stimuli that accept them (e.g. `sdl.Color` is still used by stimuli; control re-exports colors and RGB/RGBA helpers).

Over time, more of these can be moved behind goxpyriment APIs so that most experiments only need `control`, `io`, `misc`, and `stimuli`.

Chapter 26

A discussion between Gemini and Me about Go vs. Python (March 2026)

Me: Is it fair to say that Go is simpler than Python?

Gemini:

The short answer is **yes**, but it depends on how you define "simple." In the programming world, there is a famous distinction between **Easy** and **Simple**.

Python is generally considered **easier** (more intuitive for humans), while Go is arguably **simpler** (fewer moving parts and more predictable).

26.0.1 1. Minimalism: The "One Way" Philosophy

Go was designed by Google to be a "small" language. The entire language specification is roughly 50 pages, whereas Python's documentation is massive. * **Go:** Deliberately omits features like classes, inheritance, method overloading, and traditional exceptions. There is usually only one way to do something, which reduces "cognitive load" when reading someone else's code. * **Python:** Offers multiple paradigms (Object-Oriented, Functional, Imperative). While "The Zen of Python" says there should be one obvious way, in practice, you have list comprehensions, `map/filter`, decorators, and multiple ways to handle strings or concurrency.

26.0.2 2. Explicit vs. Implicit

Go favors being explicit, which makes the code more "simple" to reason about because there is no "magic" happening behind the scenes. * **Error Handling:** In Python, an exception might bubble up from a deep library call, and you might not know it's coming. In Go, you must explicitly check `if err != nil`. It's more verbose, but it's simpler to track the flow of data. * **Typing:** Python is **dynamically typed**, which is "easy" to write but can be "complex" to debug when a variable unexpectedly changes

from an integer to a string. Go is **statically typed**, catching these errors before the code even runs.

26.0.3 3. Tooling and Deployment

Go simplifies the "meta" part of programming—getting your code to actually run on a server. * **The Binary:** Go compiles into a single static binary. You don't need to worry if the server has the right version of Python, `pip`, or a `virtualenv` installed. You just drop the file and run it. * **Standard Library:** Go's standard library is powerful enough that you rarely need external dependencies for web servers or JSON processing. This avoids the "dependency hell" often found in Python or Node.js.

26.0.4 Comparison at a Glance

Feature	Python (Easy)	Go (Simple)
Syntax	Reads like English; very high-level.	C-like but cleaned up; very few keywords.
Learning Curve	Extremely low; great for first-time coders.	Slightly steeper at first, then plateaus quickly.
Concurrency	Complex (Asyncio, GIL, Threads).	Built-in and easy (Goroutines, Channels).
Maintenance	Can become "messy" in very large teams.	Highly consistent; easy to read others' code.

Which of Go and Python easier for AI coding agents?

In 2026, the answer depends on whether you are measuring "ease" by how fast a **human** can write the code or how effectively an **AI agent** can generate and maintain it.

While **Python** remains the gateway for beginners due to its massive ecosystem, **Go (Golang)** has emerged as the surprise winner for "agentic" workflows—where AI agents are the ones doing the heavy lifting.

26.0.5 Comparison at a Glance

Feature	**Python** (Human-First Ease)	**Go** (Agent-First Ease)
Learning Curve	Very Low (Reads like English)	Moderate (Strict but simple)
Ecosystem	**Dominant** (PyTorch, LangChain, etc.)	Growing (LangGraph-Go, Genkit)
Concurrency	Difficult (GIL limitations)	**Native** (Goroutines/Channels)

Feature	**Python** (Human-First Ease)	**Go** (Agent-First Ease)
Agent Reliability	Medium (Dynamic typing risks)	**High** (Strict types, clear errors)
Deployment	Complex (Virtual environments/Docker)	**Seamless** (Single binary)

26.1 1. Why Python is Easier for *Humans*

Python's "ease" comes from its **abstractions**. It allows you to write high-level logic without worrying about memory or types. * **The Ecosystem Advantage:** If you want to use a new model from Hugging Face or a specific RAG (Retrieval-Augmented Generation) technique, the Python library exists today. * **Rapid Prototyping:** You can go from an idea to a working "toy" agent in 10 lines of code. * **Community:** Almost every AI tutorial on the planet is written in Python, making troubleshooting a matter of a quick search.

26.2 2. Why Go is Easier for *AI Agents*

Recent data from 2025–2026 suggests that LLMs (like Claude or GPT) actually produce **fewer bugs** when writing Go compared to Python. * **Predictability:** Go has a "one right way" philosophy. There are fewer ways to do the same thing, which reduces the "entropy" in AI-generated code. * **Explicit Error Handling:** In Python, agents often "bury" errors in deep `try-except` blocks. In Go, the `if err != nil` pattern forces the agent to handle issues exactly where they happen. * **Type Safety:** Go's compiler catches "hallucinations" (like calling a function that doesn't exist or passing a string instead of an int) before the code ever runs. * **Concurrency for Multi-Agent Systems:** Modern AI setups often involve many agents talking at once. Go's **Goroutines** make managing these simultaneous "thoughts" much simpler and more performant than Python's `asyncio`.

Peer Tip: If you are just starting and want to see results in 5 minutes, go with **Python**. If you are building a production-grade system where an AI agent will help you write the code, **Go** will save you months of "debugging hell" later on.

Chapter 27

Viewing the documentation locally

There are two complementary sets of documentation, and you can serve either of them on your own machine:

- **The API / source-code reference** — generated from the Go source by pkgsite.
- **This documentation site** — the hand-written guides you are reading now, built with zensical.

27.1 API reference (pkgsite)

`pkgsite` serves the Go source documentation (every package, type, and function) as a local website.

Install it once:

```
go install golang.org/x/pkgsite/cmd/pkgsite@latest
```

Then run it from the repository root and open http://127.0.0.1:8080 in your browser:

```
cd goxpyriment
pkgsite
```

To jump straight to a package, append its import path, for example the `stimuli` package:

```
http://localhost:8080/github.com/chrplr/goxpyriment@v0.0.0/stimuli
```

27.2 This documentation site (zensical)

The guides on this site are built with zensical. Install it once (it is listed in `docs/requirements.txt`):

```
pip install -r docs/requirements.txt
```

(or follow the zensical get-started guide).

Then, **from the repository root**, start a live-reload preview at http://127.0.0.1:8000:

```
zensical serve        # or: make serve
```

To produce a static HTML build in `site/` instead:

```
zensical build --clean   # or: make docs
```

!!! note Run these commands from the repository root. zensical auto-discovers its configuration file (`zensical.toml`), which lives at the root next to the `Makefile`.